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**Optimal fiscal policy in an open interdependent world**

**Park, Walter Ginn, Ph.D.**

**Yale University, 1991**

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**Optimal Fiscal Policy  
in an Open Interdependent World**

A Dissertation  
Presented to the Faculty of the Graduate School  
of  
Yale University  
in Candidacy for the Degree of  
Doctor of Philosophy

by  
Walter Ginn Park  
December 1991

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## **ABSTRACT**

### **Optimal Fiscal Policy in an Open Interdependent World**

Walter Ginn Park

Yale University

1991

This dissertation studies the effects of exhaustive government expenditures on productivity and welfare in an open economy setting. How do public investment and public consumption contribute to productivity and enhance welfare? The dissertation explores issues relevant for the current debate on the optimal choice of fiscal expenditures and on the international coordination of fiscal spending.

Optimal fiscal policy design is characterized by a utilitarian planner who chooses fiscal expenditures to maximize either global or national social welfare. Public consumption is an argument in private utility functions and public capital is an argument in private sector production functions. Public capital is also modelled as a source of international technological spillovers, implying that global production possibilities are affected by national public investments.

The utilitarian planner framework, in which governments choose public consumption optimally to maximize social welfare, yields a permanent income hypothesis for public consumption. The hypothesis is tested empirically and is not rejected. Empirical evidence also supports the view that public infrastructural and research capital are important determinants of private sector productivity. In addition, public investments in research and development are found to generate international technological spillovers.

**This Dissertation is Dedicated to the Memory of my Father,  
Professor Young L. Park, 8th July 1929 - 24th October 1991.**

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## **Introduction and Summary of Dissertation**

The subject of this dissertation is the conduct and efficacy of fiscal policies in an open, interdependent world economy. Existing work has predominantly dealt with issues concerning taxation and public debt. In this dissertation I shall be primarily investigating 'exhaustive' government expenditures, such as public consumption and public investment. Moreover existing work has largely treated government behaviour as 'exogenous'. There is no explanation as to why certain fiscal policies are chosen and why governments reset their policies. In this dissertation I emphasize aspects of "optimal" fiscal policy choices. Both theoretical and empirical analyses are provided. I limit the scope of the analyses largely to long-term structural issues rather than to short-term stabilization issues.

I believe that research on fiscal spending in the context of an open-economy is quite apropos in light of recent developments in the industrialized world. Recent studies by the OECD indicate that governments have been growing in size, as proxied by the ratio of total government expenditures to gross domestic or national product<sup>1</sup>. There is some dispute as to whether the growth in 'size' is occurring at an alarming pace and as to what the 'optimal' size of government is or should be. Other studies, such as those by the U.S. Congressional Budget Office, emphasize that the composition of government spending is of fundamental importance, and indicate that what is alarming is the declining share of government spending devoted to public investment (in infrastructure and research and development (R&D))<sup>2</sup>. Research on the economic implications of these developments is therefore desired.

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<sup>1</sup> See Saunders-Klau (1985) and OECD (1987).

<sup>2</sup> See Congressional Budget Office (1987), (1991).

An open-economy investigation allows for an analysis of what are the common and differential factors behind these developments in the industrialized world. Moreover, nations are interdependent so that fiscal policies implemented by one economy affect others, either through 'pecuniary' channels (ie. price variables such as the exchange rate, interest rate, and terms of trade) or through 'technological' channels (ie. production or utility function channels). Thus the fiscal spending choices made by one government affect and are affected by the choices made by other national governments. In the case where public spending generates global technological spillovers, international fiscal policy coordination is typically required since independent (national) setting of public spending is likely to be inefficient from the point of view of global welfare. The recent fiscal developments referred to also have implications for international competitiveness, current account patterns, and world interest rates.

An emphasis on endogenous government behaviour sheds light on the economic impacts of government size which analyses based on exogenous government behaviour overlook. If government behaviour were endogenous, governments would choose fiscal spending optimally with respect to some welfare criterion. Changes in government spending would then be interpreted as optimal responses to changes in the demand for public sector activities. In other words, changes in government size would occur to meet private sector needs. In analyses where government behaviour is exogenous there is often no indication as to whether the private economy induces or desires a particular change in fiscal policy. What motivates governments to enact or alter fiscal spending plans is not spelled out. The distinction between these two specifications of

government behaviour (ie. exogenous or endogenous) is relevant for interpreting whether recent developments in government size are optimizing responses or are motivated by something else. Developing a framework for optimal fiscal policy is also useful for establishing a normative benchmark for the assessment of fiscal spending plans.

The dissertation is organized as follows: Chapter 1 is a literature critique of recent work on fiscal spending. Chapter 2 is a theoretical open economy analysis of fiscal policy, featuring optimizing governments and public investment as a source of global "technological" spillovers. Chapter 3 provides an empirical test of whether governments optimize with respect to their choices of public consumption. Chapter 4 tests for evidence that public investment is productive and that public R&D investment generates global technological spillovers. In the remainder of this introduction I shall summarize the main points of each chapter.

Chapter 1 begins with a brief review of government accounting. That private agents fully incorporate government accounts is a feature shared by several previous studies. A discussion of government accounting also helps to define the scope of government activities in a macroeconomic context. Researchers should appreciate the detailed composition of government outlays, revenues, and sources of financing. Researchers should also note that in an intertemporal context, spending and revenue plans must be expressed in present value terms. An important concept to develop is that of net worth. The present discounted value of the stream of government consumption spending is "bounded" by the public sector's net worth. Consequently, trends in government size must be compatible with

the government's net worth or else the government's spending program is not sustainable over time. Eventually the government must revise its plans or risk becoming insolvent if the present value of its future consumption program exceeds its net worth.

The chapter then turns to discussing the positive impacts of government spending as debated in the recent literature on the equilibrium approach to fiscal policy. This literature has debated whether temporary or permanent changes in government spending are more expansionary. The models typically assume high degrees of intertemporal substitution, especially of labour supply. In some models the stock of capital is ignored and in others investment is endogenous. In models that allow for capital accumulation, permanent changes in government purchases are more expansionary.

The equilibrium approach literature is silent on the normative aspects surrounding public spending. The literature contains no explanation as to why government spending changes occur and whether spending changes are desirable. What is the optimal size of government? Are the fiscal spending experiments conducted from points where government size is at or near the optimum point? A section of this chapter investigates the "welfare" impacts of changes in public spending under alternative assumptions about financing method, labour supply elasticity, and openness of the economy. The welfare impacts are conducted for a typical individual living in steady-state in an overlapping generations world. I find that the welfare impacts depend on whether the economy is initially dynamically efficient. If the economy operates initially near or at the "golden rule" point the welfare impacts are typically negligible.



Finally in this chapter I survey some empirical evidence on the effects of public investment and public consumption. Much interesting empirical work has been done and remains to be done. It is too early to judge whether the evidence confirms the appropriateness of the equilibrium approach to fiscal policy.

In Chapter 2 I study the open-economy aspects of fiscal policy in a model featuring public investment as a source of international "technological" externality. National public capital not only contributes to national productivity but also generates positive spillovers for foreign productivity. In the existing open-economy literature the primary channels of international transmission of fiscal policy are "pecuniary" variables, such as the terms of trade, interest rate, and exchange rate. I therefore stress alternative channels. The chapter also allows for both exogenous and endogenous government behaviour. I specify two types of endogenous governments: a national planner and a global planner. The national planner chooses public investment to maximize national welfare subject to a national resource constraint while the global planner chooses public investment to maximize world welfare subject to the relevant constraints. For each type of planner there exist schemes for setting optimal user or rental charges for the services of public capital.

I consider both the small open-economy and two-country cases. Steady-state analyses and dynamic simulations are both provided. Throughout I make comparisons between productive public investment and unproductive government purchases. For a small open-economy, unproductive public consumption spending tends to lower long run private consumption and has an uncertain effect on the current account. Public

investment tends to increase long run private consumption and also has an uncertain effect on the current account. Public investment works to boost private output and wealth. However, there may come a point where the larger stocks of public capital and private capital (stimulated by public investment) absorb a larger share of steady-state resources because of depreciation. As long as the depreciation rates are not too high, a greater share of resources will go to private consumption rather than to the maintenance of the steady-state stocks of capital.

In the two-country section, I assess the impacts that domestic expansions in public capital and public consumption have on both domestic and foreign private consumption. Increased domestic public consumption raises long run foreign private consumption if the domestic economy turns into a long run debtor as a result of its fiscal action. Increased domestic public investment, if no cross-country spillovers exist, typically turns the domestic economy into a long run creditor and causes foreign long run private consumption to fall. If spillovers exist, long run foreign private consumption typically rises by more than the rise in long run domestic private consumption. The reason is that foreigners do not bear any of the burden of financing the increased stock of domestic public capital which essentially is an international public good or input. Finally I examine debt policies and tax reforms under national and global planning. In the endogenous government framework of this chapter, the stocks of public capital can be varied only through the government's financing instruments. I find that a globally coordinated short run tax increase (followed by a long run tax decrease) or a globally coordinated shift towards consumption taxation will increase foreign and domestic long run private consumption (assuming the economies are initially dynamically efficient).

Chapter 3 tests empirically whether governments choose their public consumption plans optimally. I assume that governments are utilitarian. Their objectives are to choose a stream of public consumption expenditures that maximizes national welfare subject to a national resource constraint. A national welfare function is represented by a weighted aggregate of individual utility functions (of private agents alive and to be born in a nation). Private agents derive utility from both public and private consumption. The model yields a permanent income theory for government consumption. The theoretical restrictions on the data are then tested.

To carry out the test, I follow the methodology of the 'overidentification' tests used in studies of private consumption behaviour. In contrast to private agents, a government should better "mimic" the assumptions required for the permanent income hypothesis to hold: namely the government is a representative national, infinitely-lived agent, who faces no liquidity constraints. For five OECD countries I find that in the majority of test cases I cannot reject the permanent income hypothesis for government consumption. I have also carried out several tests of specification. For example I test whether private and public consumption are non-separable in the private utility function and find that they are best represented as separable. I also test whether public consumption substitutes for private consumption and find that the parameter which measures substitutability is weak. I also examine various functional forms: the constant relative risk aversion, logarithmic, and quadratic utility functions.

An implication of the chapter is that hypotheses about growing government size (for example Wagner's Hypothesis) should focus on

permanent income variables rather than current income. Thus another task undertaken in the chapter is to reexamine Wagner's hypothesis - ie. that the scale of government activity tends to expand relative to the national economy as national income increases. Previous studies compute the elasticity of the share of government spending in GNP with respect to per capita GNP, and argue that the hypothesis is confirmed if the elasticity significantly exceeds unity. Existing findings using this procedure are mixed as to whether the hypothesis can be supported. I provide instead two measures of permanent income: (i) one is an exponentially weighted average of current and past levels of income; (ii) another is that constant level of income whose present discounted value equals the present discounted value of future expected income. Neither measure provides any support for Wagner's Law regarding public consumption. Future tests using government transfer payments may provide support for the hypothesis, given the well documented expansions in this component of government expenditures. However, transfer payments may be 'driven' more by current (transitory) income rather than permanent income.

Chapter 4 investigates empirically whether fiscal expenditures generate international "technological" spillovers. Public investments in research and development (R&D) and infrastructural capital are seen to expand national and global production possibilities. The importance of technological channels for the international transmission of fiscal policy is emphasized in Chapter 2. The empirical analyses are based on the aggregate production function and growth accounting framework. Both U.S. data and a panel data set of OECD economies are investigated.

Each country's private output is a function of global R&D capital stocks, which are constructed by aggregating individual stocks of national R&D capital (expressed in a common currency unit). Each nation in the sample is considered to be the unit of analysis. Cross national spillovers are captured in the globally available stocks of R&D capital to which individual nations contribute through national investments.

The evidence is favourable to the view that public investments in R&D and infrastructure are productive and that global technological spillovers are present. The chapter argues that in the case of R&D, public R&D investment influences private sector productivity growth primarily by stimulating private R&D investment. That is, public R&D capital is significant when it enters the production function without private R&D capital, but loses its significance once private R&D capital is introduced. This indicates that the more important determinant of private sector productivity is private R&D. Since public R&D is found to stimulate private R&D investment, it is argued that the positive impact of public R&D on private sector output is indirect. Consequently when public R&D enters the production function without private R&D, the public R&D variable acts as a proxy for private R&D.

I also breakdown public R&D by socioeconomic category. I find that public R&D in certain types of research, for example Space and Defense research, despite their being technologically intense fields, to be unproductive. The reason, I argue, is that these types of research are not immediately relevant for private sector output, as say agricultural, industrial, and infrastructural research investments are. The by-products of Space and Defense research do, however, stimulate private research

endeavours which in turn lead to improvements in private sector productivity.

The chapter also contains a battery of empirical tests. I test for returns to scale, non-stationarities, reverse causality, heterogeneity in the panel data set, and common factor restrictions (imposed by first-difference formulations such as the growth accounting equation). I also control for the influence of distortionary taxes. I find that the assumption of constant returns to scale in private capital and labour cannot be rejected, that the variables I work with exhibit non-stationarities, that the causality is typically from public capital to productivity and not the reverse, that the assumption of homogeneity within the OECD economies can be rejected, and that the common factor restrictions cannot be rejected. Finally, the introduction of distortionary taxes does not diminish the significance of the public investment variables. The existing literature has stressed that an increase in public investment, if accompanied by higher distortionary taxes, would have a disincentive effect on the economy and offset some, if not all, of the beneficial effects of public investment. In the results I obtain, public investment variables still retain their significance on net.

In conclusion, I hope that this dissertation impresses the point that future research efforts cannot afford to neglect the several important channels through which fiscal spending affects the real economy - that is, in enhancing welfare and contributing to productivity. The evidence obtained is supportive of the view that fiscal spending directly affects private utility and production (at home and abroad through international spillovers).

Financial policies (ie. debt-taxation) should be sufficiently flexible to permit optimal fiscal spending programmes to be implemented.

A natural task for further research is to address how independent sovereign fiscal authorities could implement or decentralize globally optimal solutions. More work in the area of international public finance seems to be needed in order to guide us with the appropriate welfare principles governing international decision-making and allocation. It would also be worthwhile to add business cycles, unemployment, and stabilization issues to the research agenda. Future work could investigate how fiscal spending interacts with business cycles. Finally, income distribution, as a government policy objective, is a topic that is scarcely dealt with in the current macroeconomic literature. Future work could investigate how fiscal policy has altered (or failed to alter) the distribution of income over time. In all of these future studies it would be highly desirable to include international comparisons.

**Chapter 1:**  
**Fiscal Spending in Intertemporal Equilibrium Models**



## **I. Introduction**

The equilibrium approach to fiscal policy has developed into a paradigm for recent research on various aspects of fiscal spending in both closed and open economies. This chapter critiques this recent approach.

The equilibrium approach to fiscal policy is derived from the general equilibrium approach to macroeconomics (or new classical macroeconomics) where prices are flexible and all the various markets are in equilibrium. In earlier studies there were some markets in which either 'quantity' was demand-determined and excess supply rationed or 'quantity' was supply-determined and excess demand rationed. In the equilibrium approach models all markets clear. All agents (households, firms, government) obey intertemporal budget constraints, and their choices over spending and saving are constrained only by permanent income. All agents can redistribute resources, consumption spending, and work effort intertemporally without restriction. All private agents are forward-looking and maximize utility. Government behaviour is typically exogenous, though in some cases governments can choose certain instruments optimally, such as distortionary taxes.

The real business cycle (RBC) research program also emerged from the general equilibrium approach to macroeconomics. While the real business cycle literature focuses on 'supply' shocks driving movements in real variables, the equilibrium approach to fiscal policy focuses on 'demand' shocks, specifically government spending shocks. Both the real business cycle literature and equilibrium approach to fiscal policy literature abstract from monetary or nominal variables.

A key feature of the equilibrium approach to fiscal policy models is that they exhibit Ricardian Equivalence - that is, for a given level and composition of government spending the substitution of budget deficits for (lump-sum) taxes has no real effects. The models feature economies with zero birth rates where private agents with infinite horizons fully internalize the actions of government. Private agents fully incorporate the lifetime government budget constraint into their own budget constraints. Real effects arise instead from changes in the level, composition, and timing of government spending and from changes in the composition and timing of distortionary taxes. This de-emphasis on deficit-finance has produced two broad types of research agenda. In the 'positive' literature there is a debate as to whether permanent or temporary changes in fiscal spending are more expansionary<sup>1</sup>. In the 'normative' literature the primary focus is not on public spending but on distortionary taxes, the choices of which are guided by the principle that the tax distortions be smoothed intertemporally<sup>2</sup>. The result is that tax rates are kept constant amidst temporary shocks to government spending or output, and adjusted if the shocks are permanent. Budgetary deficits/surpluses are determined residually (or occur passively) in this framework.

Included in my critique of the literature are what I believe to be the following limitations. First, the demand-supply distinction which separates the equilibrium approach to fiscal policy and RBC schools is not clear. Certain types of public spending, such as productive public capital expenditures, have "technological" effects. In this and several other

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<sup>1</sup> See Hall (1980), Barro (1981). These and other papers are reviewed in Section III.

<sup>2</sup> See Barro (1979), Kydland-Prescott (1980), and Lucas-Stokey (1983).

possible instances public policy can affect both aggregate demand and supply. Secondly, the normative analysis in the equilibrium approach is incomplete. Government spending is frequently treated exogenously. Alternative specifications could be that the public sector provides public capital services optimally to encourage long run growth and public goods services to maximize social welfare. There is no explanation as to why governments choose to vary spending, and whether the level or composition of government spending is initially optimal or suboptimal, for this would make a substantial difference in the welfare implications. If, moreover, government behaviour with respect to spending were endogenous, the interpretation of the results in the 'positive' analyses would be affected. Government purchases would be seen as responding to the real economy through a "feedback" rule rather than propagating impulses in the real economy. A third limitation concerns the literature's maintained assumption that government financing does not matter, when it may matter. In instances where there are effective demand failures and capital market imperfections, or in intergenerational settings where private agents possess no bequest motives, government financing does matter. In these and other instances the equilibrium approach to fiscal policy suffers from Type II errors and offers no guidance as to what the appropriate fiscal policy responses should be.

This chapter is organized as follows. Given the importance of the assumption in the equilibrium approach literature that private agents see through the "veil" of the intertemporal government budget constraint, I begin by reviewing the basic elements of government accounting in Section II. Section III discusses the 'positive' aspects of fiscal spending - that is, the output and real interest rate effects of temporary versus permanent changes

in fiscal spending. Section IV discusses the 'welfare' impacts of fiscal spending not addressed in the 'positive' literature. Section V reviews some of the empirical findings, and section VI concludes.

## **II. Government Accounts**

A critical assumption maintained in the intertemporal equilibrium approach to fiscal policy is that private agents internalize the government budget constraint and public sector balance sheet. In this section I review briefly the basic elements and principles behind the budget identities and comprehensive balance sheets<sup>1</sup>.

An important concept to develop is the "net worth" of the government sector. The value of the public sector's net worth (NW) defines the present value of current and future government consumption spending, and provides information about "government size" over the long term. The balance sheet identity is given by:

### **(1) Present Value of Government Consumption**

$$\begin{aligned} &= \text{Public Sector Net Worth} = \text{Public Sector Assets} - \text{Public Sector} \\ &\hspace{15em} \text{Liabilities} \end{aligned}$$

where the public sector's assets consist of real assets (such as the value of the stock of public capital, the value of public sector property rights in land and natural resources), financial assets (such as gold and net foreign exchange reserves), and implicit assets (such as the present value of future

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<sup>1</sup> This section is based on Buiter (1984).

taxes net of transfers, the present value of future seignorage, and the present value of future public sector capital formation). On the liabilities side are net interest-bearing public debt and the stock of base money.

To derive the comprehensive balance sheet in (1), I use the following government budget constraint (GBC) obtained by consolidating the Treasury and Central Bank flow accounts:

$$(2) \quad c_g(t) + \dot{k}_g(t) - \tau(t) + i(t)\frac{B(t)}{p(t)} - i^*(t)\frac{e(t)F^*(t)}{p(t)} - \rho_k(t)k_g(t) - \rho_R(t)R(t) \\ = p_R(t)\dot{R}(t) + \frac{\dot{M}(t) + \dot{B}(t) - e(t)\dot{F}^*(t)}{p(t)}$$

where  $c_g$  is real public consumption,  $k_g$  the real stock of public capital,  $R$  the number of public sector shares in natural resource property rights,  $\tau$  tax revenue net of transfers,  $B$  the nominal stock of government debt<sup>1</sup>,  $F^*$  the nominal stock of foreign assets, and  $M$  the stock of high-powered money.  $p$  is the price level,  $i$ ,  $i^*$  are the domestic and foreign nominal interest rates respectively,  $e$  the nominal exchange rate,  $\rho_k$  the real net of depreciation return per unit of public capital,  $\rho_R$  the real return per share in  $R$ ,  $p_k$  the real price of a unit of public capital assuming that it remains in the public sector, and  $p_R$  the real price of a unit of  $R$ . Note that public capital is not valued at the replacement cost of unity which is the value in an alternative (private) use. (Time derivatives are given by "dots" over variables).

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<sup>1</sup> For convenience I ignore consols and indexed debt.

The RHS of (2) is the public sector financial deficit, where the conventional measure of it is adjusted by placing asset sales ( $p_R \dot{R}$ ) on the RHS (since it is a source of funds) rather than on the LHS.

Assuming equalized anticipated real rates of return on non-money assets<sup>1</sup> and a transversality condition<sup>2</sup>, we can integrate equation (2) forward to yield equation (3) below which restates the identity in (1):

$$(3) \quad G(t) = NW(t)$$

where

$$(3a) \quad NW(t) = p_R(t)R(t) + p_K(t)k_g(t) + \left[ \frac{e(t)F^*(t) - M(t) - B(t)}{p(t)} \right] + T(t) + \Pi(t) + J(t)$$

and

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<sup>1</sup> That is,

$$r(t) = i(t) - \frac{\dot{p}(t)}{p(t)} = \frac{p_R(t) + \dot{p}_R(t)}{p_R(t)} = \frac{p_K(t) + \dot{p}_K(t)}{p_K(t)} \quad \text{and} \quad i(t) = i^*(t) - \frac{\dot{e}(t)}{e(t)}$$

<sup>2</sup>

Given by:

$$\lim_{T \Rightarrow \infty} \left[ \frac{M(T) + B(T) - e(T)F^*(T)}{p(T)} - p_R(T)R(T) - p_K(T)k_g(T) \right] e^{-\int_t^T r(u)du} = 0$$

$$(3b) \ G(t) = \int_t^\infty c_g(s) e^{-\int_t^s r(u) du} ds$$

$$(3c) \ T(t) = \int_t^\infty \tau(s) e^{-\int_t^s r(u) du} ds$$

$$(3d) \ \Pi(t) = S(t) + \frac{M(t)}{p(t)}$$

$$(3e) \ S(t) = \int_t^\infty \frac{\dot{M}(s)}{M(s)} \frac{M(s)}{p(s)} e^{-\int_t^s r(u) du} ds$$

$$(3f) \ J(t) = \int_t^\infty (p_K(s) - 1) \dot{k}_g(s) e^{-\int_t^s r(u) du} ds$$

$$(3g) \ p_K(t) = \int_t^\infty (\rho_K(s) e^{-\int_t^s r(u) du} ds$$

$$(3h) \ p_R(t) = \int_t^\infty (\rho_R(s) e^{-\int_t^s r(u) du} ds$$

$G$  is the present value of the government's future consumption program,  $T$  the present value of future taxes net of transfers, and  $\Pi$  the real capitalized value of the state's monopoly right to issue notes,  $S$  the present value of future seignorage, and  $J$  the present value of the government's future public investment program<sup>1</sup>.

The first two terms in (3a) represent the public sector's real assets, the middle term in brackets represents the public sector's financial assets

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<sup>1</sup> In the presence of distortionary taxes (eg. interest income taxes) the appropriate discount rate is the after-tax rate of interest.

less liabilities, and the last three terms ( $T$ ,  $\Pi$ , and  $J$ ) represent the public sector's implicit assets. Note that the  $J$  term is positive - ie. adds to net worth - to the extent that the shadow price of government capital exceeds the opportunity cost of 1 (that is,  $p_k > 1$ ). It must also be noted that the  $p_k$ 's represent "cash" returns only. A comparison between the cash rate of return to public investment and the marginal cost of that investment does not indicate whether the investment should take place. What matters is the relationship between the "social" rate of return and the opportunity cost of public investment (see Arrow (1966), Marglin (1963)). The social rate can, and often does, fall short of cash rates of return because of the difficulty of imposing user charges sufficiently high enough to recover costs. There are often income distributional reasons for providing public capital services which outweigh financial profit/loss considerations.

While the comprehensive government accounting framework is typically used to evaluate the intertemporal consistency of fiscal and financial policies, I believe the framework is also useful for assessing whether changes in government size are sustainable. Equation (3) indicates that the present value of government consumption is constrained by the public sector's net worth<sup>1</sup>. Existing studies on government size tend to focus on two measures of government size: absolute size given by levels of  $c_g$  or relative size given by the ratio  $\frac{c_g}{y}$ , where  $y$  is output. Concern is

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<sup>1</sup> There may be some objections to calling NW net worth since much of what NW consists of is subject to the government's choice. NW is thus not parametric to the government. Neither is private sector net worth entirely parametric to private agents since human wealth, for example, depends on individual employment choices. Whatever 'name' is attached to NW, the relationship in (3) does convey a sense of the intertemporal consistency required between spending and financing plans in the short and long runs.



often expressed about the growing size of the public sector if the sequence  $\{c_g(s)\}$  or  $\{\frac{c_g(s)}{y(s)}\}$ , for  $s \in [t, \infty)$ , appears to exhibit a persistently rising trend. I would argue that this is not a useful way to determine whether there is unbounded, unsustainable growth in government size. What matters is whether the present value of the stream of government consumption violates the net worth constraint. If increases in government size, as measured either by increasing levels of public consumption or by an increasing share of public consumption in GDP, result in a time path for public consumption which violates the net worth constraint, there will need to be revisions in fiscal policy since the existing fiscal spending and/or financing plans would be mutually incompatible.

Existing work has also placed much emphasis on the official liabilities of the government - particularly public debt. Focus should also be placed on public assets, their size and composition, and on net worth. This would provide a better perspective of the economic implications of the public debt, and of the implied future tax obligations facing private agents. Like public debt, 'net worth' alone is also not a reliable index of fiscal stance. An economic model must also be specified since in general there are no model-free measures of fiscal stance.

The results of a study by Boskin et al. (1987) should help us appreciate the importance of the net worth concept. The study finds that in the U.S., the market value of official government assets typically exceeds the market value of official government liabilities. In 1980 the net worth of the U.S. Federal government was \$1.1 trillion (in 1985 dollars) and was one-third of that in 1985 due to the growth in Federal liabilities since 1980. The net worth of U.S. State and Local governments was \$2.5 trillion (in

1985 dollars) in both 1980 and 1985. The U.S. public sector owns substantial amounts of land, mineral rights, real estate, equipment, gold and other financial assets. Income from the ownership of minerals is the second largest source of revenue to the U.S. Federal government, next to taxes.

### **III. 'Positive' Impacts of Government Spending**

An ongoing debate in the equilibrium approach to fiscal policy literature is whether the output effects of an increase in government purchases are greater when the spending change is temporary or when it is permanent. In some versions only temporary changes can affect output. Permanent changes may not even affect the short run dynamics or long run real equilibria. Whether the spending changes are permanent or temporary, the existence of any "Keynesian multiplier"<sup>1</sup> is denied. These versions assume that the stock of capital is fixed<sup>2</sup>. In other versions, in which capital accumulation is allowed, permanent government spending changes are more expansionary and can produce "multiplier" effects.

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<sup>1</sup> That is, an output change greater than or equal to the government spending change.

<sup>2</sup> Interestingly the normative part of the literature, namely the work on taxation-smoothing, also refers to models without capital. The presence of a predetermined stock of capital raises time-inconsistency problems due to the possible imposition of surprise capital levies by the fiscal authorities. One way around this, according to Lucas (1986), is to impose a legal constraint that outlaws capital levies. Time-inconsistency is then avoided if, in addition, surprise inflation is also outlawed by having the monetary authorities precommit to a specified path of nominal prices and if the government can manage the maturity structure of the debt it inherits in some "judicious" way.

Bailey (1971), Hall (1980), and Barro (1981)

The origins of the modern equilibrium approach to fiscal policy can be traced back to Bailey (1971). In this work, Bailey states the Ricardian equivalence proposition, shows that if government consumption substitutes perfectly for private consumption the Keynesian government multiplier vanishes, and derives rules for optimal government spending, namely that:

"... resources should be used in government activity until the services rendered by the last, or marginal, units of the resources are just as valuable as they would have been in private employment ...." (Bailey (1971), p. 187)

The optimal fiscal rule for public investment, for example, would be that both government and private investment have the same marginal productivity. Later authors have expanded upon Bailey (1971), introducing rational expectations, microfoundations, and the 'timing' of government spending.

Hall (1980) highlights the relationship between labour supply and interest rates in an intertemporal setting. If leisure is a normal good a higher interest rate will encourage agents to postpone leisure and increase labour supply. Transitory government spending increases are argued to produce greater output responses because the higher interest rate resulting from an increase in aggregate demand increases labour supply, as agents intertemporally substitute their work effort forward in time. Aggregate demand and interest rates are relatively unaffected when the government spending increases are permanent because agents expect current and future

taxes to be higher and accordingly make permanent adjustments in their consumption plans.

Barro (1981) finds similar results concerning temporary versus permanent changes in government spending. Unlike Bailey (1971), Barro allows government consumption to substitute imperfectly for private consumption:

$$\text{ie. } c^*(t) = c(t) + \theta g(t), \quad 0 < \theta < 1$$

where  $c^*$  represents effective private consumption<sup>1</sup>. Each unit of government purchases per capita,  $g$ , provides utility services that are equivalent to a fraction  $\theta$  of a unit of private consumption. Bailey had examined the  $\theta = 1$  case. When  $g$  increases, aggregate commodity demand rises as a fraction  $(1-\theta)$  of the change in  $g$ . Barro also allows for the possibility that  $g$  affects production but that the marginal benefits,  $MPG$ , do not exceed the social cost of providing an extra unit of  $g$ , which is unity - thus  $MPG \leq 1$ .

If government purchases increase temporarily (and unexpectedly) there is a shift in excess demand by  $(1 - \theta - MPG)$  which if positive causes interest rates, labour supply, and output to rise. If the increase in government purchases is permanent and unanticipated, the temporary effect just described is offset by the effects which transpire in the remaining time (to infinity)<sup>2</sup>. In the remaining time there is a loss in private permanent

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<sup>1</sup>  $c, c^*$  should be interpreted as per capita magnitudes since the model is cast in terms of a representative agent economy.

<sup>2</sup> i.e if  $t$  indexes time, and the current period is  $t=0$ , the effects of a permanent change can be thought of as combining the effects occurring between  $t=0$  and  $t=1$  and the effects transpiring from  $t=1$  to infinity.

wealth which works to lower aggregate demand. By the intertemporal government budget constraint (which is simply that the present value of lump-sum taxes equals the present value of government purchases), a permanent change in  $g$  raises the annuity value of tax liabilities on a one-to-one basis. Because of Ricardian equivalence, the rise in the present value of tax obligations lowers private permanent income on a one-to-one basis also. Permanent income is sufficiently lowered so as to crowd out enough private spending to accommodate the larger permanent share of government purchases in aggregate resources.

In a small open-economy setting, the interest rate is constant and the current account balance adjusts instead to relieve excess demand pressures arising from increases in government purchases. Ahmed (1986) finds that temporary increases in government purchases produce larger output responses and current account deficits (as the home economy borrows from abroad to finance expansions in domestic demand) than do permanent increases.

#### Aschauer-Greenwood (1985)

This paper adds microeconomic foundations to a two-period (infinite-horizon) model.

The intertemporal utility function is:

$$(4) \quad U = u(c_1^*) + v(l_1) + \beta [u(c_2^*) + v(l_2)]$$

where  $c_t^* = c_t + \theta c_{gt}$ ,  $c^*$  is again effective private consumption,  $\theta$  is again the index of substitutability,  $l$  is labour supplied ( $v'(\cdot) < 0$ ), and  $c_g$  is

government consumption. Subscripts denote time,  $t = 1, 2$ .  $\beta$  is the discount factor.

The production function is given by:

$$(5) \quad y_t = \delta_t + f(l_t, g_t^l) + h(i_t, g_t^i) + i_t,$$

where  $\delta$  is a time varying exogenous productivity parameter,  $y$  output,  $i$  private investment,  $g^i$  public investment, and  $g^l$  public services. This rather specialized production function assumes that  $g^l$  has no interaction with the marginal product of private investment and  $g^i$  has no interaction with the marginal product of labour. The authors impose  $\frac{\partial h}{\partial i} > \frac{\partial h}{\partial g^i}$  - ie. that the marginal product of government investment is less than the marginal product of private investment.

Market clearing in each period implies the following:

$$(6a) \quad c_1 + i + c_{g1} + g_1^l + g_1^i = \delta_1 + f(l_1, g_1^l)$$

$$(6b) \quad c_2 + c_{g2} + g_2^l = f(l_2, g_2^l) + h(i, g^i) + i + g^i$$

The intertemporal government budget constraint is:

$$(7) \quad g_1 + \tau_1 + \frac{g_2 + \tau_2}{1 + r} = 0,$$

where  $g_1 = c_{g1} + g_1^l + g_1^i$  and  $g_2 = c_{g2} + g_2^l - g^i$ , and  $\tau_1, \tau_2$  are lump-sum taxes.

Consider first a temporary rise in public services in the current period. Public services is the sum of  $c_g$  and  $g^1$ . Fewer resources are available for first period consumption according to 6(a). The agent smooths effective consumption by decreasing investment. The real rate of return rises and labour supply increases in the current period. Output therefore rises in the current period and falls in the future period because of the decreased first period investment - see the  $h( , )$  term in 6(b). Consider now an anticipated future increase in public services. In the first period, agents prepare for the expected excess demand for resources in period 2 by increasing investment in period 1.

A permanent increase in public consumption can be analyzed by combining these two experiments (ie. the temporary increase in public services in period 1 and the anticipated increase in public services in period 2). Labour supply in both periods will be higher since the higher  $g^1$  raises the marginal product of labour in both periods. The response of investment however is ambiguous. If the interest factor  $(1 + r)$  were equal to the time discount factor  $\beta$  there would be no effect on investment since there would be no inducement to save or dissave<sup>1</sup>. In this case the agent would distribute the burden of the government spending shock evenly over time. In general, investment or disinvestment occurs depending on the time profiles of the marginal propensities to consume and take leisure, as these determine how the agent distributes the burden of the spending shock over time.

Consider now a permanent increase in the level of public investment. As long as public and private investment are complementary - ie.

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<sup>1</sup> That is, the time preference rate would equal the interest rate so that private consumption (of a representative agent) would be steady over time.

$\frac{\partial^2 h}{\partial i \partial g^i} > 0$  - an increase in public investment will raise the marginal product of private investment and the real return. There will be an intertemporal reallocation of work effort to the current period which will further raise the marginal product of investment, and thus boost private investment. Thus output expands through factor supply channels.

In their open-economy extension the interest rate stays constant and net foreign lending and borrowing adjusts instead. The authors find that an unanticipated temporary increase in public services worsens the current trade balance, while an anticipated future increase in public services improves the current trade balance. An unanticipated permanent increase in public services combines these effects to produce no net change in the current trade balance. A permanent increase in public investment causes the first-period trade balance to worsen. The economy finances its public investment by borrowing from abroad against its increased future output.

### Baxter-King (1990)

Their paper overturns the earlier "equilibrium approach" results concerning the efficacy of temporary versus permanent government spending increases. The primary difference is that capital accumulation is permitted. While Aschauer-Greenwood allow for private investment, the production structure in their model was rather specialized: public investment did not affect the marginal product of labour, and flows of investment entered the production function rather than stocks of capital.



Baxter-King in contrast consider a neoclassical production function extended to allow for public capital:

$$(8) \quad Y(t) = F(K(t), K_g(t), N(t))$$

where  $Y$  is aggregate output,  $K$  the stock of private capital,  $K_g$  the stock of public capital, and  $N$  the quantity of labour input. Assume constant returns to scale in private capital and labour.

The representative agent's welfare is given by:

$$(9) \quad U = \sum_{t=1}^{\infty} \beta^{t-1} u(C(t), L(t), C_g(t))$$

where  $C$  is private consumption,  $C_g$  is public consumption,  $L$  is leisure (assume that  $N(t) + L(t) = 1$ ), and  $0 < \beta < 1$  is the time discount factor. Let  $I$  be private investment and  $I_g$  public investment. Assume that both public and private capital depreciate at the same rate:

$$(10a) \quad K(t+1) = (1 - \delta)K(t) + I(t)$$

$$(10b) \quad K_g(t+1) = (1 - \delta)K_g(t) + I_g(t)$$

The national resource constraint (NRC) and the government budget constraint (GBC) are respectively:

$$(11) \quad Y(t) = C(t) + I(t) + G(t)$$

$$(12) \tau(t)Y(t) = G(t) + TR(t)$$

where  $\tau$  is the income tax rate,  $G = G^B(t) + C_g(t) + I_g(t)$  where  $G^B$  is unproductive government spending (which can be thought of as basic administrative expenses), and  $TR$  transfer payments. The private budget constraint is given by:

$$(13) C(t) + I(t) = (1-\tau(t))Y(t) + TR(t)$$

The authors resort to numerical simulations in order to analyze the impacts of fiscal spending changes. Consider first an unanticipated permanent increase in  $G^B$ . The long run interest rate is pinned down by the time preference rate. Consequently there is no long run change in the private capital-labour ratio. However both the absolute levels of private capital and labour are higher in the long run, and thus long run output is higher. In their simulations the immediate impact of the spending increase is to raise the interest rate. This brings forth an increase in labour supply and a reduction in private consumption. The higher savings are channelled into increased capital accumulation which is encouraged by the higher marginal productivity of private capital resulting from an increased supply of labour. In their simulations the long run multiplier exceeds unity and is greater the more intertemporally substitutable leisure is. The multiplier is highest for an indivisible labour economy where labour supply is infinitely elastic.

A temporary increase in  $G^B$  on the other hand creates a decline in investment on impact and further declines over the duration of the

temporary  $G^B$  increase. As in Aschauer-Greenwood, investment falls because the government spending increase absorbs national resources and private agents prefer to keep their consumption profile smooth in the face of this shock. Since the spending increase is temporary the work effort that is induced (by the temporarily higher interest rate) is insufficient to offset the decline in capital accumulation. As the duration of the  $G^B$  increase is lengthened, the results approach the case where the  $G^B$  increase is permanent, as work effort rises for a longer period and raises the marginal productivity of private capital.

If either the temporary or permanent  $G^B$  increase is financed by distortionary income taxes, output declines. Higher taxes reduce work and investment incentives. As incomes fall, the tax base shrinks, necessitating further increases in tax rates (which increase more the more elastic labour supply is). The authors advocate taxation-smoothing in the case when the  $G^B$  increase is temporary. This would smooth tax distortions over time and minimize output losses.

The authors also find that a permanent increase in public investment is very stimulative. They identify two types of effects: (i) the direct effects on  $Y$  through the production function, holding  $K$  and  $N$  constant; (ii) the indirect effects on  $Y$  of public investment stimulating  $K$ ,  $N$  (ie. the "supply-side responses"). For example, if the output elasticity of public capital were 0.2, the direct multiplier  $\frac{\Delta Y}{\Delta I_g}$  would be 4 and the overall multiplier, taking into account the supply-side responses, would be 7.09. If the output elasticity of public capital were 0.4, the direct multiplier would be 8 and the overall multiplier 13.02.

Finally, in their analysis of public consumption, the authors do not take the standard route of defining effective consumption as  $C^* = C + \theta C_g$ . Rather they assume that  $C$  and  $C_g$  are complements with an elasticity of substitution of 0.5. An increase in  $C_g$  raises the marginal utility of private consumption. From the private 'static' first-order condition (namely that the marginal utility of consumption equal the marginal utility of leisure), we can see that this causes leisure to fall or labour supply to rise. The marginal product of capital rises and encourages capital accumulation. In the long run the multiplier  $\frac{\Delta Y}{\Delta C_g}$  is 1.5.

Another study by Aiyagari, Christiano, and Eichenbaum (1990) confirms that when capital accumulation is allowed, the output effects of a permanent increase in government purchases will exceed those of temporary increases, and there can be significant long run multipliers, even when, as in their model, government purchases do not enter the utility function or the production function. A novelty of their model is that government spending is driven by stochastic shocks - the temporary component of government spending is i.i.d. while the permanent component follows a Markov process.

#### Frenkel-Razin (1987)<sup>1</sup>

The authors focus on the open-economy dimensions of the temporary versus permanent government spending issue. They also provide an

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<sup>1</sup> This sub-section draws on Chapter 7 of Frenkel-Razin (1987)

analysis of the optimal size of government. Assume again that there is no investment.

Consider a two-period (infinite-horizon) model with the following intertemporal welfare function for a representative agent:

$$(14) \quad V(C_1, C_2, G_1, G_2) = \max_{\{C_1, C_2\}} U(C_1, C_2) + \beta U(G_1, G_2)$$

$$\text{subject to (14a) } C_1 + \alpha C_2 = W_1$$

where  $\alpha = \frac{1}{1+r_2}$ ,  $\beta$  is the discount factor, and  $r$  is the interest rate.

(Subscripts denote the time period). Initial wealth is given by:

$$(14b) \quad W_1 = (Y_1 - T_1) + \alpha(Y_2 - T_2)$$

where  $Y_1, Y_2$  are exogenous endowments,  $T_1, T_2$  are lump sum taxes. A necessary condition for maximizing the private agent's welfare is:

$$(14c) \quad \frac{U_c(C_1, G_1)}{U_c(C_2, G_2)} = \frac{\beta}{\alpha}$$

The intertemporal government budget constraint (GBC) is

$$(15) \quad G = G_1 + \alpha G_2 = T_1 + \alpha T_2 - (1+r_0)B_0$$

where  $(1+r_0)B_0$  represents the government's initial debt commitment,  $B_0$  being the past stock of debt. The private agent "sees through" the

intertemporal GBC and recognizes fully the precise dependence between government spending and taxes. The agent internalizes the activities of the government. The details of public finance (ie. composition between debt and taxes, and the timing of taxes) do not influence the agent's behaviour. This is reflected in the consolidated public and private intertemporal budget constraint:

$$(16) C_1 + \alpha C_2 = (Y_1 - G_1) + \alpha(Y_2 - G_2) - (1+r_0)B_0$$

The government influences private behaviour through two channels: (a) the resource withdrawal channel reflected in  $(Y_1 - G_1)$ ,  $(Y_2 - G_2)$ . Resources are absorbed that would otherwise have been available to the private sector; (b) the consumption-tilting channel which arises because the marginal rate of substitution of consumption between periods depends on levels of government spending.

If  $C$ ,  $G$  are complements (ie. the marginal utility of private consumption rises with higher levels of government spending), a temporary rise in current (period 1) government spending raises the marginal rate of substitution while an anticipated rise in future (period 2) government spending lowers it. The opposite results if  $C$ ,  $G$  are substitutes. The consumption tilting effects cancel out when government spending changes are permanent. There are also no tilting effects if  $C$ ,  $G$  are separable in the private utility function.

Consider a two-country world. The impact of fiscal spending on the world interest rate will be discussed. It is through the interest rate ('price

channel'<sup>1</sup>) that the effects of fiscal policy in one country are transmitted to the rest of the world. For simplicity assume that  $C$  and  $G$  are separable so that no consumption tilting effects occur.

Consider first a temporary expansion in government spending in the first period. As long as the private agent's marginal propensity to spend is less than one, the  $G_0$  increase will be larger than the decline in first period private consumption. On net there will be an excess demand for current period goods. This is eliminated by a rise in the relative price of present goods in terms of future goods - ie. a rise in the interest rate. The higher interest rate induces substitution away from current period private consumption towards future private consumption. The higher interest rate also crowds out foreign private consumption by lowering foreign wealth (that is, the discounted sum of foreign disposable incomes). Therefore a temporary increase in current government spending raises the world interest rate and crowds out both domestic and foreign first period private consumption.

Next an anticipated increase in future government spending induces an excess world demand for future goods. The relative price of current period goods falls - that is, the interest rate falls. This causes foreign wealth and private consumption to rise in the first period. A permanent expansion in government spending combines these two previous cases. The rate of interest rises or falls depending on whether the decrease in private sector demand for current goods is less or greater than the decline in the private sector demand for future goods, respectively.

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<sup>1</sup> As opposed to technological channels analyzed in Chapters 2 and 4.

We turn now to the issue of the optimal size of government. Define the following indirect social welfare function:

$$(17) \quad V = \max_{G_1, G_2, T_1, T_2} V(G_1, G_2, T_1, T_2)$$

subject to (15) the intertemporal government budget constraint. The necessary conditions are (14c) and:

$$(18) \quad \frac{U_G(C_1, G_1)}{U_C(C_2, G_2)} = \frac{U_G(C_1, G_1)}{U_C(C_2, G_2)} = 1$$

Condition (18) indicates that the marginal rate of substitution between C and G in each period should be equal, and equal to the marginal rate of transformation between C and G, which is unity (from (16)). From the necessary conditions and the consolidated intertemporal budget constraint (16), we can derive the following demand functions:

$$(19a) \quad C_t = C_t [ \alpha ; Y_1 + \alpha Y_2 - (1 + r_0)B_0 ]$$

$$(19b) \quad G_t = G_t [ \alpha ; Y_1 + \alpha Y_2 - (1 + r_0)B_0 ]$$

for  $t = 1, 2$ .

We turn now to the effects of supply shocks. Suppose there is a temporary supply shock in the first period (ie.  $\Delta Y_1 > 0$ ). Assuming C, G are normal,  $C_1, C_2, G_1, G_2$  all increase, but each increases by less than the increase in  $Y_1$ . Hence the current period trade balance improves. If the



supply shock is an anticipated future increase in  $Y_2$  with no change in  $Y_1$ , the current period trade balance worsens. If the supply shock is permanent, there are no changes in the trade balance. This analysis shows that when the government optimizes, the trade balance reflects the consumption smoothing motive of the private agent.

Furthermore, when the government optimizes, the correlation between  $G$  and output  $Y$  is a "consequence" of particular events and not, as in models with exogenous government behaviour, the effect of government spending on output. When the supply shocks are temporary, there are positive correlations between  $G$  and  $Y$ , but when the shocks are permanent, there are no correlations. If government spending were used to stabilize fluctuations in output, the contemporaneous correlations between  $Y$  and  $G$  would be negative. Thus inferences on the relationship between output and government spending hinge on the underlying assumptions about government behaviour.

I conclude this section with some observations about the papers reviewed. An important feature in these equilibrium approaches not sufficiently dealt with thus far is the time horizon of agents relevant to their optimization decisions. The horizon is assumed to be infinite. Society's time horizon and the private sector's horizon coincide. Agents with infinite horizons internalize the welfare of all future generations. In contrast, in overlapping generations (OLG) models with finite lifespans and without operative intergenerational gifts and bequests, the planning horizon of private agents coincides with their own lifespans. In such models the competitive intertemporal equilibrium need not coincide with the social optimum. Individual generations select choices - in Nash fashion - without

taking into account the interests of future generations. Forward markets are incomplete because of the constraint that private agents cannot make binding contracts with the unborn. Consequently there is scope for public intervention to improve social welfare.

In the equilibrium approach, the arguments for fiscal intervention are rarely, if ever, specified. Suppose there is an economy in which an exogenous level of government spending is financed by lump-sum taxes, and suppose that this spending yields no utility nor affects production. As long as the government sector is treated parametrically by the private sector, the decentralized economy will be Pareto efficient if consumers maximize intertemporal welfare, producers maximize profits, all agents are price-takers, markets are complete, and appropriate conditions on preferences and technology are satisfied. In this economy why should the optimal level of government spending be different from zero? A positive amount of spending serves only to lower the steady-state level of output, consumption, and welfare, while preserving all the private efficiency conditions.

If, however, government spending yields utility and/or affects production, variations in spending will affect equity and efficiency when the existing amount of spending is away from the optimum. The optimal government size in this instance should be non-zero. These issues have not been addressed in the equilibrium approach literature whose focus has been primarily to characterize the 'positive' impacts of changes in the level, composition, and timing of government spending. The impacts of these policy changes on welfare have been largely neglected. The literature does not address why government spending should be altered, what the optimal amount of spending is, and how the 'positive' and welfare impacts of

spending depend, if at all, on how close the economy initially is to providing the optimal amount of government spending. Moreover, if government spending were endogenous, it would be difficult to interpret the 'positive' effects that government spending changes have on the real economy because government spending would be reacting to circumstances in the real economy.

Another important feature of the equilibrium approaches is their maintained hypothesis of Ricardian equivalence. This feature serves to minimize the role of fiscal intervention, since government financing is rendered irrelevant. Private agents incorporate fully the intertemporal government budget constraint into their own intertemporal resource constraints. Hence their spending behaviour is constrained only by the permanent (consolidated) income of society. Buiter (1983a,b) points out that the conditions required for private agents to be constrained by comprehensive wealth or permanent income are quite stringent. In particular there must be perfect internal and external capital markets. Because of the illiquidity and nonmarketability of certain assets (for example pension rights, human capital, and expected future tax cuts), cash flow constraints, credit rationing, and lack of collateral, private agents are often incapable of intertemporally shifting resources so as to keep their spending in line with permanent income.

In a world characterized by imperfect capital markets, fiscal authorities could pursue financial policies so as to induce an evolution of wealth and income "that permit private agents and national economies ... to approximate the behaviour that would be adopted if either comprehensive wealth or permanent income were the only binding constraint on economic

behaviour"<sup>1</sup>. The government as 'Borrower of First Resort' could aim financial policies toward neutralizing constraints on private spending other than permanent income. In this instance, the use of government financial policies would better enable private agents to smooth their consumption over time.

In the intertemporal equilibrium approaches there is no scope for Pareto-improving government financial policies. Private disposable incomes in these models are already in line with permanent incomes and there are no other constraints such as illiquidity and nonmarketability of private assets. Moreover, private agents have access to capital markets on the same terms available to the government.

As pointed out earlier, in OLG economies, dynamic competitive equilibria may be inefficient and government financial policies will matter. Consequently, in OLG economies, optimal fiscal policy involves choosing both exhaustive government spending and financing plans optimally with respect to some appropriate welfare criterion. Of course in the presence of distortionary taxes, the intertemporal equilibrium approaches do allow for a passive form of efficient financing - namely taxation-smoothing and countercyclical deficits. However the role of deficits in these approaches strongly contrasts with the more active use of deficit finance under Keynesian stabilization schemes or under intergenerational schemes for redistributing resources across and between generations.

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<sup>1</sup> pp. 308-309, Buiter (1983a).

#### IV. 'Welfare' Impacts of Government Spending

In this section I explore the steady welfare effects of changes in public capital and public consumption in a simple OLG model, based on Diamond (1965). Closely related welfare studies, using an OLG framework, are Persson (1985) who analyzes the steady-state welfare effects of public debt and Lindbeck-Weibull (1988) who also examine the welfare effects of public spending. However, Lindbeck-Weibull work with a richer, more specialized model than the one presented here<sup>1</sup>. There is scarcely any mention of the welfare impacts of public spending in the equilibrium approach literature. The literature does not discuss whether welfare improves or worsens as a result of variations in government spending, thereby leaving aside the reasons for changing government spending plans. There is also very little 'normative' work on what the optimal fiscal spending should be. I will not delve into this here but will defer the topic to Chapters 2 and 3.

This section highlights the role dynamic efficiency plays in determining the steady-state welfare impacts of changes in public capital and public consumption. When the economy operates near or at the point where the "golden rule" condition is fulfilled, the welfare impacts are small. This confirms the point drawn in the last section that the welfare

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<sup>1</sup> In their model, individuals allocate their time between taxed-market work and tax-free home production. Individuals consume what they purchase in the market and what they produce at home. Home output is produced using as inputs labour and public services. Market output is produced using private capital and labour. Public investment augments the efficiency of private capital and labour inputs in the 'market' production function.

effects of public spending depend on how close the economy is initially to some benchmark optimum.

In this economy (both a small open-economy and a closed economy will be considered<sup>1</sup>), each individual lives for two periods, working in the first period and being retired in the second. At time  $t$ , there are  $N_t$  young agents and  $N_{t-1}$  old agents. The population growth rate is  $n$  (ie.  $N_t = (1+n)N_{t-1}$ ). Each young individual at time  $t$  supplies  $e_t$  units of labour. The aggregate quantity of labour at time  $t$  is  $L_t = N_t e_t$ .

-- Technology --

The aggregate production function is given by:

$$(20) \quad Y_t = F(K_t, Kg_t, L_t)$$

where  $Y$  is output,  $K$  is private capital, and  $Kg$  is public capital. I assume that the stocks of capital do not depreciate. I also assume that all factors are paid their marginal products:  $r_t = F_{K_t}$ ,  $w_t = F_{L_t}$ , and  $\rho_t = F_{Kg_t}$ , where  $r_t$  denotes the interest rate,  $w_t$  the wage rate,  $\rho_t$  the rental rate on public capital, and  $F_j$  the partial derivative of  $F$  with respect to  $j$ , where  $j = \{K_t, Kg_t, L_t\}$ . Thus,

$$(20a) \quad Y_t = r_t K_t + \rho_t Kg_t + w_t L_t$$

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<sup>1</sup> I will implicitly assume in the background that the world economy is characterized by perfect financial capital mobility and is a one-commodity world.

This could be rewritten in intensive form by dividing through by  $N_t$ :

$$(20b) \quad y_t = r_t k_t + \rho_t k g_t + w_t e_t$$

where  $y_t = \frac{Y_t}{N_t} = f(k_t, k g_t, e_t)$ ,  $k_t = \frac{K_t}{N_t}$ ,  $k g_t = \frac{K g_t}{N_t}$ , and  $r_t = f_{k_t}(k_t, k g_t, e_t)$ ,  $\rho_t = f_{k g_t}(k_t, k g_t, e_t)$ , and  $e_t w_t = f(k_t, k g_t, e_t) - f_{k_t} k_t - f_{k g_t} k g_t$ . Again the  $f_j$ 's represent the partial derivatives of  $f$  with respect to  $j$ , where  $j = \{k_t, k g_t\}$ .

-- Individual Behaviour --

The intertemporal welfare function for a typical individual born at time  $t$  is:

$$(21) \quad U_t = U(c^1_t, c^2_{t+1}, e_t, c g_t, c g_{t+1}) = u^1(c^1_t, c g_t, e_t) + \beta u^2(c^2_{t+1}, c g_{t+1})$$

which I assume is (locally) concave and increasing in all its arguments except labour  $e_t$ .  $\beta$  is the time discount factor, and  $c^i$  is private consumption, where the superscript  $i$  denotes the period of life,  $i = 1, 2$ . Note that the per capita supply of public consumption  $c g$  does not depend on the vintage of agents - at any moment in time the same per capita amount is available to all living generations.

The individual who is born at time  $t$  faces the following period-by-period budget constraints:

$$(22a) \ c^1_t + a_{t+1} = e_t w_t (1 - \tau^w_t) - \tau^1_t$$

$$(22b) \ c^2_{t+1} = (1 + r_{t+1}) a_{t+1} - \tau^2_{t+1}$$

$$(22c) \ a_{t+1} = k_{t+1} + b_{t+1} + z_{t+1}$$

where  $\tau^1, \tau^2$  represent lump-sum taxes imposed in the first and second periods respectively,  $\tau^w$  is the wage tax rate,  $a$  denotes the stock of assets,  $b$  government bonds, and  $z$  net foreign assets.

Note that in a closed economy  $z = 0$  for all time and  $r$  is endogenous. If the economy is small  $r$  is given exogenously and  $z$  is endogenous. If the economy is large, both  $r$  and  $z$  are endogenous.

Equations (22a) and (22b) can be collapsed into a lifetime budget constraint:

$$(23) \ c^1_t + \frac{c^2_{t+1}}{1 + r_{t+1}} = e_t w_t (1 - \tau^w_t) - \tau^1_t - \frac{\tau^2_{t+1}}{1 + r_{t+1}}$$

The LHS is the present discounted value of lifetime consumption and the RHS the present value of lifetime income net of tax obligations. Maximizing (21) subject to (23) gives the following standard first-order conditions:

$$(24a) \ \frac{u^2_c(.)}{u^1_c(.)} = \frac{1}{\beta(1+r_{t+1})}$$

$$(24b) \ - \frac{u^1_e(.)}{u^1_c(.)} = w_t(1 - \tau^w_t)$$



(where the subscripts denote the arguments with respect to which the partial derivatives of the utility functions  $u^1, u^2$  are taken). The first-order conditions indicate the margins at which the individual is indifferent between consuming in the first versus last period of life and indifferent between consuming goods or leisure in period 1.

From equations (24a), (24b), and (22a) we can derive the following implicit individual savings function:

$$(25) \quad s_t = s(e_t w_t(1 - \tau^w_t) - \tau^l_t, r_{t+1})$$

where  $0 < s_1 < 1$  and  $s_2 \leq 0$ ,  $s_1, s_2$  being the partial derivatives of  $s$  with respect to the first and second arguments respectively.

-- Aggregate Relations --

The following equations describe the aggregate economy (uppercase letters denote the aggregate magnitudes):

$$(26a) \quad C_t = N_t c^1_t + N_{t-1} c^2_t$$

$$(26b) \quad S_t = N_t s_t = A_{t+1} = K_{t+1} + B_{t+1} + K_{t+1}$$

$$(26c) \quad Cg_t = cg_t (N_t + N_{t-1})$$

$$(26d) \quad Z_{t+1} = (1 + r_t)Z_t + Y_t - (K_{t+1} - K_t) - (Kg_{t+1} - Kg_t) - C_t - Cg_t$$

$$(26e) B_{t+1} = (1 + r_t)B_t + (Kg_{t+1} - Kg_t) + Cg_t - T_t - \rho_t Kg_t$$

$$(26f) T_t = N_t \tau^1_t + N_{t-1} \tau^2_t + N_t e_t w_t \tau^w_t$$

Equation (26b) is the capital market equilibrium condition, (26d) is the national resource constraint, (26e) the government budget constraint, and (26f) tax revenues. The economy is a creditor (debtor) nation if  $Z > 0$  (if  $Z < 0$ ); the economy is closed if  $Z = 0$ .

In per capita aggregate terms (ie. by dividing through by  $N_t$ ) equations (26d) and (26e) are:

$$(26d)' (1 + n) z_{t+1} = (1 + r_t) z_t + y_t - (1+n) k_{t+1} + k_t - (1+n) kg_{t+1} + kg_t \\ - c^1_t - \frac{c^2_t}{1 + n} - c^g_t \left(1 + \frac{1}{1+n}\right)$$

$$(26e)' (1 + n) b_{t+1} = (1 + r_t) b_t + (1+n) kg_{t+1} - kg_t + c^g_t \left(1 + \frac{1}{1+n}\right) \\ - \rho_t kg_t - t_t$$

$$(26f)' t_t = \tau^1_t + \frac{\tau^2_t}{1 + n} + e_t w_t \tau^w_t$$

In steady-state the national resource constraint and government constraint become:

$$(27a) (n - r)z = y - nk - nkg - c^1 - \frac{c^2}{1+n} - c^g \left(\frac{2+n}{1+n}\right)$$

$$(27b) (n - r)b = (n-\rho) kg + c^g \left(\frac{2+n}{1+n}\right) - \tau^1 - \frac{\tau^2}{1 + n} + ew\tau^w$$

-- Stability Property of the Model --

There is one useful property of the model that will help determine some of the 'signs' in the comparative steady-state analyses that follow. Consider a closed-economy. Substituting the capital market equilibrium condition into the definition of the marginal productivity of capital gives:

$$r_{t+1} = f_{k_t} \left( \frac{s_t}{1+n} - b_{t+1}, k_{t+1}^g, e_t \right)$$

After inserting in the saving function, wage, and government budget constraint, we can totally differentiate the above and derive the following stability condition:

$$\left| \frac{r_{t+1}}{r_t} \right| < 1 \quad \text{if} \quad (1+n) + f_{kk} s_1 (k_t + b_t) - f_{kk} s_2 > 0$$

where  $s_1, s_2$  are the partial derivatives of the saving function with respect to the first and second arguments, and where

$$f_{kk} = \frac{\partial^2 f}{\partial k_t^2} \quad \text{- ie. the second partial derivative of } f \text{ with respect to } k_t$$

This condition ensures Walrasian stability. The capital market will be in equilibrium if young agents save sufficiently to hold existing assets. The young are the suppliers of savings and the entrepreneurs the demanders, and the interest rate is the price that clears the market. The interest rate rises if there is excess demand in the capital market.

-- Steady-State Welfare Analysis --

Throughout I focus on the steady-state lifetime welfare of a typical individual<sup>1</sup>. Substituting the first-order conditions (24a), (24b) back into the intertemporal utility function gives us the following indirect utility function<sup>2</sup>:

$$(28a) \quad V(w(1-\tau^w), r, \tau^1, \tau^2, c^g) = \max u^1(c^1, c^g, e) + \beta u^2(c^2, c^g)$$

subject to the steady-state intertemporal budget constraint:

$$(28b) \quad c^1 + \frac{c^2}{1+r} = ew(1-\tau^w) - \tau^1 - \frac{\tau^2}{1+r}$$

where<sup>3</sup>

$$(29a) \quad V_{w(1-\tau^w)} = u_c^1 e > 0 \quad (29b) \quad V_r = u_c^1 \frac{a}{1+r} \geq 0 \text{ as } a \geq 0,$$

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<sup>1</sup> Because we are working in steady-state, examining the lifetime allocation of a typical individual is akin to examining in a single time period the allocation between representative individuals of different ages.

<sup>2</sup> Time subscripts are omitted now since we are working in steady-state.

<sup>3</sup> The following partial derivatives of  $V$  are obtained by using the following formula:

$$V_i = \frac{\partial u^1}{\partial c^1} \frac{\partial c^1}{\partial i} + \frac{\partial u^1}{\partial e} \frac{\partial e}{\partial i} + \frac{\partial u^1}{\partial c^g} \frac{\partial c^g}{\partial i} + \beta \frac{\partial u^2}{\partial c^2} \frac{\partial c^2}{\partial i} + \beta \frac{\partial u^2}{\partial c^g} \frac{\partial c^g}{\partial i}$$

and equation (28b), where  $i = \{w(1-\tau^w), r, \tau^1, \tau^2, c^g\}$ .

$$(29c) \quad V_{\tau 1} = -u_c^1 < 0 \quad (29d) \quad V_{\tau 2} = -u_c^1 \frac{1}{1+r} < 0$$

$$(29e) \quad V_{cg} = \frac{\partial u^1}{\partial c^g} + \beta \frac{\partial u^2}{\partial c^g} > 0$$

$$\text{where } u_c^1 = \frac{\partial u^1}{\partial c^1}.$$

To analyze the impacts of changes in  $kg$ ,  $cg$ , I use the following formula:

$$(30) \quad \frac{dV}{dx} = V_{w(1-\tau^w)} \frac{dw(1-\tau^w)}{dx} + V_r \frac{dr}{dx} + \\ V_{\tau 1} \frac{d\tau^1}{dx} + V_{\tau 2} \frac{d\tau^2}{dx} + V_{cg} \frac{dcg}{dx}$$

where  $x = \{kg, cg\}$ . Note that I will be comparing across steady-states.

I will consider four cases, each one highlighting different sets of assumptions as to how government spending is financed and as to whether labour supply is elastic or inelastic. Both the closed and small open-economies are considered. For simplicity I assume  $\tau^2 = 0$  - ie. no lump sum taxes on the old. I initially assume that labour supply is inelastic (ie.  $e = 1$ ). I later allow for an elastic labour supply when I consider the distortionary financing of government spending (by wage taxation). Throughout I assume that the government is optimizing in the sense that the marginal products of public capital and private capital are equal. Hence  $f_k = f_{kg}$  or  $r = \rho$ .

### Case 1: Inelastic Labour Supply, Lump-Sum Tax Financing of Public Investment

Assume  $\tau^w = c_g = 0$ . The government budget constraint becomes  $\tau^1 = (n - r)(k_g - b)$ . Now that  $e=1$ , we have  $r=f_k(k, k_g)$  and  $w = f - f_k k - f_{kg} k_g$ . From the capital market equilibrium condition,

$$r = f_k \left( \frac{s(w - \tau^1, r)}{1+n} - b, k_g \right)$$

I substitute in expressions for  $w$ ,  $\tau^1$  and totally differentiate the condition to get:

$$\frac{dr}{dk_g} = \frac{s_1 f_{kk} (r - n) + (1 + n) f_{kkg}}{1 + n + f_{kk} s_1 (k + b) - f_{kk} s_2}$$

where  $f_{kk}$  is the own second partial derivative and  $f_{kkg}$  is the cross partial derivative of  $f$ .

The impact of a higher stock of public capital on the steady-state interest rate is ambiguous. The denominator is positive by the stability condition, but the numerator is ambiguous. If the economy is dynamically efficient ( $r > n$ ) and public capital does not affect the marginal productivity of private capital, an increase in steady-state  $k_g$  will cause the steady-state interest rate to decline. A larger stock of public capital is associated with lower steady-state taxes, so that net wage receipts and thus savings will be higher. However to the extent that public and private capital are complementary, a larger stock of public capital raises the marginal product

of private capital and increases the stock of private capital. In order to maintain  $r = f_k$ , a higher interest rate is required to induce savers to absorb the increased stock of private capital.

Next the net wage is given by  $\hat{w} = w - \tau^1$ , from which:

$$\frac{d\hat{w}}{dk^g} = - (k + b) \frac{dr}{dk^g} + (r - n)$$

which, if  $r > n$ , will be positive or negative depending on whether the steady-state interest rate will fall or rise.

Using (30), the steady-state change in welfare is thus

$$\frac{dV}{dk^g} = u_c^1 \left[ \frac{d\hat{w}}{dk^g} + \frac{a}{1+r} \frac{dr}{dk^g} \right] = u_c^1 (r - n) \left[ 1 - \frac{k+b}{1+r} \frac{dr}{dk^g} \right]$$

Assuming dynamic efficiency, we see that a higher stock of public capital will improve welfare if it succeeds in lowering the interest rate (and thereby raising net wages). If the economy is operating at the golden-rule point (ie.  $r = n$ ), there will be no change in welfare. This is in accord with the intuition that small changes near the optimum will not alter the value of the objective function.

In the open-economy the interest rate is exogenously given and  $\frac{dr}{dk^g} = 0$ . Thus the steady-state change in welfare is higher or lower depending on whether  $r$  exceeds or falls short of  $n$ .

### Case 2: Inelastic Labour Supply, Debt Financing of Public Investment

Here  $d\tau^1 = 0$  and  $db = dk^g$ . Again assume  $c_g = 0$ . From the capital market equilibrium condition given by:

$$r = f_k\left(\frac{s(f - r(k + k^g) + (r - n)(k^g - b), r)}{1+n} - b, k^g\right)$$

we can derive:

$$\frac{dr}{dk^g} = \frac{(1+n)(-f_{kk} + f_{kk}k^g)}{(1+n) + f_{kk} s_1 (k + b) - f_{kk} s_2} > 0$$

A larger stock of  $k^g$  increases the stock of private capital and thereby raises the steady-state interest rate because agents must be induced to absorb both the larger stock of private capital and government debt.

From  $\hat{w} = f - r(k + k^g) - \tau^1$ ,  $\frac{d\hat{w}}{dk^g} = -(k + k^g) \frac{dr}{dk^g} < 0$ . Thus using (30):

$$\frac{dV}{dk^g} = u_c^1 \frac{dr}{dk^g} \left[ \frac{(k+b)(1+n)}{1+r} - k - k^g \right]$$

which shows that the welfare impact is ambiguous because on the one hand the steady-state interest rate is higher and thus asset income is higher. On the other hand, the higher interest rate and stocks of capital reduce the labour share of factor payments. Steady-state welfare under dynamic efficiency will be improved only if  $\frac{b}{k^g} \gg \frac{1+r}{1+n} > 1$ . The intuition is that if the stock of public capital is initially greater than the stock of public debt,



the loss in wages will be more significant; if on the other hand  $b > kg$  initially, the gain in asset income of a higher interest rate will be more significant.

In a small open-economy there is no welfare change under a debt-financed increase in public capital. Since the interest rate does not change, the gross wage does not change; and since steady-state lump sum taxes do not change, the net wage does not change either.

### Case 3: Elastic Labour Supply, Wage-Tax Financing of Public Investment

The capital market equilibrium condition is:

$$r = f_k\left(\frac{s(ew(1 - \tau^w), r)}{1+n} - b, kg, e\right)$$

and the steady-state government budget constraint is (assuming no lump-sum taxes and public consumption)  $ew\tau^w = (r - n)(b - kg)$ . Thus, using the factor-price frontier,  $ew(1 - \tau^w) = f - r(k + kg) - (r - n)(b - kg)$ .

Labour supply  $e = e(w(1 - \tau^w), r)$  and  $de = e_1.dw(1 - \tau^w) + e_2.dr$ , where  $e_1, e_2$  are the partial derivatives of  $e$  with respect to its first and second arguments. Now,

$$\frac{dw(1 - \tau^w)}{dk_g} = \frac{1}{e(1 - \epsilon_L)} \left[ (r - n) - \frac{dr}{dk_g} (b + k + (1 - \tau^w)e_2w) \right]$$

where  $\varepsilon_L = \frac{e_1 w(1 - \tau^w)}{e}$  is the elasticity of labour supply with respect to the after tax wage rate. The impact of  $k^g$  on net wages depends on the response of the interest rate. If the interest rate rises, net wages are lowered by the fact that labour loses its factor share of income and by the labour force increase which works to depress wages, assuming  $e_2 > 0$ , ie. that labour responds positively to a higher interest rate. Holding  $r$  constant, we see that a larger stock of public capital works to boost net wages because a larger stock of  $k^g$  lowers the steady-state amount of wage taxes, provided  $r > n$ .

Next,

$$\frac{dr}{dk^g} = \frac{s_1 f_{kk} (r - n) + (1 + n) f_{kk} k^g}{1 + n + f_{kk} s_1 (k + b) - f_{kk} s_2}$$

is ambiguous for the same reasons given in Case 1 - note however the  $s$  function here is different from that in Case 1.

The steady-state welfare impact is thus:

$$\frac{dV}{dk^g} = u_c^1 \left[ \frac{(r - n)}{1 + \varepsilon_L} - \frac{dr}{dk^g} \left[ \frac{(1 - \tau^w) e_2 w}{1 + \varepsilon_L} + (k + b) \frac{(r - n) + (1 + n) \varepsilon_L}{(1 + r)(1 + \varepsilon_L)} \right] \right]$$

When  $\varepsilon_L$  approaches zero, then  $e_2 = 0$ , and the above result will be identical to that obtained in Case 1. A change in the steady-state interest rate impacts on welfare by affecting the after-tax factor share of labour income, represented by the term in brackets next to  $\frac{dr}{dk^g}$ . The after-tax wage income will be sensitive to the response of labour supply to changes

in the interest rate, as well as to the value of the tax rate  $\tau^w$  necessary to finance  $k\bar{g}$ , given an endogenous tax base which depends on labour supply. When  $\varepsilon_L$  is infinite, ie. labour supply is perfectly elastic, the effect on welfare through the after-tax wage term,  $w(1 - \tau^w)$ , disappears because labour is able to escape taxes fully. In this case the welfare impacts depend entirely on the interest rate response. That is,

$$\frac{dV}{dk^g} = u_c^1 \frac{dr}{dk^g} (k+b) \frac{(1+n)}{(1+r)} \quad \text{when } \varepsilon_L = \infty$$

The small open-economy impacts can be obtained by setting the interest rate channels to zero.

#### Case 4: Inelastic Labour Supply, Lump-Sum Tax Financing of Public Consumption

Assume that  $\tau^w = k\bar{g} = 0$ . Thus  $w = f - kr$ . The steady-state government budget constraint is  $\tau^1 = \frac{n+2}{n+1} c\bar{g} + (r - n)b$ . Lump-sum taxes fall on the young. Consequently the young bear the burden of financing public consumption which benefits both old and young. The capital market equilibrium condition is:

$$r = f_k \left( \frac{s(f - kr - \frac{n+2}{n+1} c\bar{g} - (r - n)b, r)}{1+n} - b \right)$$

from which we get:

$$\frac{dr}{dcg} = \frac{-f_{kk} \frac{n+2}{n+1} s_1}{1+n + f_{kk} s_1(b+k) - f_{kk} s_2} > 0$$

Thus a higher steady-state level of public consumption raises the steady-state real interest rate. Increased public consumption is associated with higher long run taxes, lower net wages, and thus lower savings. To re-equilibrate the capital market, the interest rate must be higher (ie. a higher  $r$  is needed because the higher  $\tau^1$  reduces the pool of savings relative to the demand for savings).

$$\text{From } \hat{w} = f - rk - \tau^1,$$

$$\frac{d\hat{w}}{dcg} = -(k+b) \frac{dr}{dcg} - \frac{n+2}{n+1} < 0$$

confirming that steady-state net wages are lower.

Putting these results into (30) gives:

$$\frac{dV}{dc^g} = -u_c^1 \left[ \frac{dr}{dk^g} (k+b)(r-n) + \frac{(n+2)}{(n+1)} \right] + V_c^g$$

If public consumption did not enter the private utility function (ie.  $V_c^g = 0$ ), welfare would be unambiguously lower. This serves to remind us that even if the effect of an increase in public consumption is to crowd out long run private consumption (as found to be the case in several of the papers reviewed in section III), it is not necessarily the case that private welfare is lowered unless  $V_c^g \leq 0$ .

## **V. Empirical Evidence**

A small number of empirical studies employing postwar U.S. time-series data have explored the relationship between public capital and private sector productivity<sup>1</sup>. Cross-country studies have tended to focus on GDP growth and government consumption<sup>2</sup>. Public R&D studies also shed light on the relationship between public investment and productivity<sup>3</sup>. Most of these studies build on the aggregate production function framework<sup>4</sup>.

Aschauer (1989a) obtains strong support for the view that public capital contributes to private sector output. This is especially the case with 'core' infrastructure (streets, highways, airports, mass transit, sewers, and water systems). Aschauer finds that the stock of non-military public capital is more important in determining productivity than is the flow of non-military public capital spending. This supports the view that the stock of public capital should enter into production functions rather than the flow of public investment. The stock of military capital is found to have no relation to productivity.

In another study Aschauer (1989b) looks at the relationship between private investment and public investment. On the one hand he finds that public investment crowds out private investment nearly one-to-one, given the rate of return to private capital, since higher public investment raises the national rate of capital accumulation above that chosen by private

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<sup>1</sup> see Holtz-Eakin (1988), Garcia-Mila and McGuire (1989), Aschauer (1989a,b), and Munnell (1990).

<sup>2</sup> see Kormendi-McGuire (1985), Grier-Tullock (1989), and Barro (1989b, 1990).

<sup>3</sup> see Levy-Terleckyj (1983), Griliches-Lichtenberg (1984), and Bartelsman (1990).

<sup>4</sup> For a duality approach see Shah (1988).

agents. On the other hand, increased public capital raises the rate of return to private capital and stimulates private investment. His finding is that the latter effect dominates the first, so that the net impact of an increase in public investment is to stimulate private investment.

Holtz-Eakin (1988) also finds that public capital is an important component of the aggregate private sector production function. The author also breaks down public capital by state/local and federal. State-local capital is by far the more significant. Munnell (1990) argues that much of the decline in published multifactor productivity figures could be the result of omitting public capital in the input calculations rather than the result of a decline in technological innovation. Garcia-Mila and McGuire (1989) study public capital at the state/local level and focus on public education and highway capital, both of which they find to be significant inputs.

Cross-country studies mix developed, middle-income, and developing countries in their sample. Kormendi-McGuire (1985) find a positive but weak relationship between the mean growth of the ratio of government consumption to GDP and the mean annual growth rate of aggregate real GDP. Grier-Tullock (1989) expand upon the Kormendi-McGuire study and find a significantly negative relationship between real GDP growth and the growth rate of the share of government consumption in GDP. In their framework they are implicitly assuming that the production function is a function of the "flow" of government consumption. Barro (1989b) finds a negative relationship between the growth rate of real per capita GDP and the ratio of government consumption to GDP and a positive but weakly significant relationship between the ratio of government investment to GDP and the growth rate of real per capita GDP.

In Barro's (1990) theoretical exposition there is a Laffer-type relationship between government investment and output growth. Initially public investment is expansionary but at some point the higher taxes associated with increased government size work to lower output growth owing to their disincentive effects on investment and saving. The possibility that governments might be optimizing complicates the empirical analysis to the extent that if countries were similar the data would cluster about the point where government size is optimal, and thus produce no correlation between government investment and output growth. On the other hand the cross-sectional variations in the data might be due to different economies reflecting different levels of public sector productivity. Barro thus argues that the positive but weakly significant influence of public investment on output growth may be consistent with the view that government investment is at its optimal level.

In the remaining part of this section I consider whether the empirical findings strongly support the appropriateness of the equilibrium approach to fiscal policy. My view is that the evidence has yet to confirm or deny the appropriateness of the approach.

Recall that the earlier versions of the literature argued that temporary expenditures were more expansionary than permanent ones. More recent versions have argued the opposite. This points to the need to distinguish business cycle from growth effects. It could be that temporary expenditures have the stronger cyclical output effects and permanent ones the stronger trend effects. It could also be that certain types of spending, such as public investment, have largely long-term supply effects and others, such as defense spending, have largely cyclical effects.

The empirical results surveyed in this section would not support the earlier version of the equilibrium approach since public investment is found to have important long run effects, assuming that public investment is more of a permanent type of government expenditure since it may require several periods of investment activity to complete a project and then maintain it.

Empirical support for the equilibrium approach was obtained by Barro (1981) and Ahmed (1986). These authors identified temporary expenditures largely with defense spending and permanent expenditures with non-defense spending, and found that temporary expenditures were more expansionary. However, their measure of non-defense spending included public investment and yet the output effects of this spending measure were weak. It is possible that if public investment alone were investigated the results might have been different and in line with the time-series studies discussed in this section.

Nonetheless I would argue that it is insufficient to claim support for the equilibrium approach based on discovering that public investment is productive. This is not the feature that separates the equilibrium approach from other approaches. A productive government sector can be introduced in almost any model, and the positive and welfare impacts of spending may very well be similar to those obtained in the equilibrium models. The equilibrium approach models are built on stronger assumptions, namely the maintained hypotheses that the private market is efficient, that there are no binding constraints on agents' behaviour other than permanent income, that agents can substitute leisure, consumption, and resources intertemporally without restriction, and that the economy exhibits Ricardian equivalence.



Empirical testing of these aspects should provide better criteria for judging the appropriateness of the equilibrium approach.

The empirical studies on public consumption are also not sufficient evidence in favour of the equilibrium approach. The equilibrium approach models typically assume that government behaviour is exogenous. If government behaviour is endogenous and public consumption yields utility to private agents, empirical studies may have captured a demand function for public consumption that depends on the state of economy. It is possible that the relationship picked up between public consumption and the real economy cannot discriminate between models that assume exogenous government behaviour and models that assume endogenous behaviour, thus providing no information one way or another on the appropriateness of the equilibrium approach.

## **VI. Conclusion**

In the past much of the literature on fiscal policy had focused on the financing aspects, such as taxation and deficit finance. The equilibrium approach has helped shift attention to the much neglected area of fiscal spending. The approach has stressed the importance of paying attention to details such as the composition and timing of fiscal spending. This influence is well reflected in the insightful theoretical and empirical works produced recently. It would be advantageous, however, to see more progress on normative issues, especially since the approach has been silent as to its version of why public policy should be used and the circumstances under which fiscal intervention is desirable. At the same time its de-

emphasis on issues that had mattered in the past appears in some sense extreme. For example, the approach downplays the possibility that the private market operates inefficiently, that effective demand failures and capital market imperfections exist, and that through government finance society can effect beneficial intertemporal redistributions of resources.

In the remaining chapters I aim to contribute to the ongoing literature on fiscal spending both by presenting alternatives to the equilibrium approach and by building upon some of the strengths of this literature. In the next chapter I study an intertemporal, optimizing open-economy model in which government financing matters. I further allow government behaviour to be endogenous. In Chapters 2 and 3 governments choose fiscal expenditures to maximize social welfare. I will be focused predominantly on 'permanent' fiscal changes. Fiscal spending will have effects not only through price variables but also through 'technological' channels. In Chapter 4 I will continue to explore the empirical relationship between productivity and public investment. I will extend the empirical analysis to a broader open-economy framework, one which will allow for an investigation of public goods spillovers.

**Chapter 2:**  
**Fiscal Externalities in the Open Economy**

## **I. Introduction**

Since the late 1970s industrial economies have experienced massive cutbacks in public investment expenditures, all during periods of slow productivity growth, real resource scarcities, and record high government budgetary deficits. Some of this retrenchment might have been necessary to improve public sector efficiency or to deal with revenue shortfalls. In any event serious concerns have been raised about the consequences for future economic growth and welfare, and for a country's ability to compete internationally.

In Chapter 4 I conduct an empirical study of public investment using OECD panel and individual country data. From an aggregate growth accounting framework, I find significant output elasticities for public infrastructural capital and public R&D capital. Moreover I find positive cross-country spillover effects associated with public R&D capital. Public investments are also found to stimulate private investment nationally and globally through spillovers. This gives some inkling of the costs, or forgone benefits, due to public investment reductions in the OECD - not to mention cutbacks in non-OECD economies.

This chapter turns to a theoretical investigation of public investment in an open-economy, overlapping-generations (OLG) framework<sup>1</sup>. The short run and long run impacts on private consumption, international

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<sup>1</sup> I have chosen an OLG framework to avoid Ricardian equivalence between debt issue and taxation, and to avoid the restriction found in representative agent models that require the steady-state interest rate to equal the subjective time preference rate, or else steady-state consumption will not exist. Unless time preference rates are endogenous, the restriction fixes the steady-state stock of capital and output exogenously.

competitiveness, and current account behaviour are evaluated. The framework adopted is along the lines of previous work by Alogoskoufis-van der Ploeg (1991), Buitier (1987), Frenkel-Razin (1987), Giovannini (1988), and Obstfeld (1989). There are three main points of departure. First, I focus on productive government spending. In previous work the attention is on taxation and budgetary deficits; government spending is assumed not to affect private production or marginal utility. In models where government spending is unproductive, a permanent increase in government spending is a resource withdrawal and is tantamount to a reduction in the lifetime wealth of private agents. Wealth is further reduced if increased external debt or crowding out of private capital results. In the model of this chapter, productive government spending boosts supply, stimulates private capital, raises lifetime human wealth and possibly financial wealth.

Secondly, previous work tends to stress "price" channels - such as the interest rate, terms of trade, real exchange rate - as the means by which fiscal policies in one country are transmitted to the rest of the world. Interdependence through prices generates "pecuniary externalities". This chapter, however, stresses "technological" channels: public capital of one country serves as an input not only in the country's own production function but also in the production functions of other countries. This type of direct interdependence generates "technological externalities" (assuming there is no international compensation mechanism to internalize the external effects of public capital). The distinction between these two types of externalities is elaborated in Scitovsky (1954), and the relevance for the open-economy discussed in Buitier-Kletzer (1990). International interdependence through prices is not by itself a source of market failure,

requiring international fiscal policy coordination. Indeed it reflects the market mechanism at work, provided every economic influence is transmitted through prices. Factor-use-related externalities of the sort I look at create direct dependence and do not operate through the market mechanism, and is therefore a source of divergence between private and social benefits. As Scitovsky (1954, p. 146) notes:

"What is puzzling ... is that interdependence through the market mechanism should be held to account for the failure of the market economy to lead to the socially desirable optimum, when equilibrium theory comes to the opposite conclusion and relies on market interdependence to bring about an optimum situation."

A third point of departure is that I allow for endogenous government behaviour. In particular, governments can choose a stream of public investment to maximize a social welfare function, of which there are two kinds. The social welfare function for a national planner is defined over the welfare of agents alive and to-be-born in a nation. The global planner's social welfare function is a weighted average of national social welfare functions. While the command optimum under each type of planner is time-consistent, the key difference is that the global planner takes into account the cross-country spillovers from public capital.

An earlier paper on public investment in the open-economy is presented in Devereux (1987a). The author develops a two-period model with no private capital (the national stock of capital could be thought of as being entirely government-owned) and no installation costs to capital formation. The paper presents an example of spillovers occurring through price variables. For instance, in a Nash equilibrium a creditor country overinvests (relative to an internationally coordinated investment level) and

a debtor country underinvests because an increase in public investment raises the world interest rate which benefits the creditor and hurts the debtor. A recent paper on "technological" spillovers is presented in Alogoskoufis-van der Ploeg (1991). The authors extend an endogenous growth model to a two-country, one-commodity setting. Endogenous growth is driven by externalities from private capital. It is assumed that the average stock of national private capital embodies 'knowledge'. International spillovers of knowledge are captured by entering each country's capital stock into each other's production function. Unfortunately I do not find empirical support for this type of production function. I fail to find the presence of external effects in the aggregate or aggregate per-capita stock of private capital. The authors also treat public expenditures as unproductive, and find that a higher global ratio of public consumption to GDP reduces world savings and growth.

The chapter is organized as follows: the next section describes the model. Section III discusses endogenous government behaviour (Appendix 2.1 explains how government objective functions are derived). Section IV analyzes the small open-economy, and Section V the two-country version. Both steady-state and dynamic analyses are provided (Appendix 2.2 provides background detail on the dynamic simulations). Section VI summarizes the main findings and suggests extensions for further research.

## **II. Model**

The following model describes a two-country world inhabited by utility-maximizing consumers and value-maximizing firms. Each country is specialized in the production of a tradable good, and agents desire the

consumption of both the domestic and foreign good. A perfectly integrated international financial market exists in which a global bond (denominated in terms of the domestic good) is traded. Labour and goods markets clear continuously.

Governments in each country engage in public capital formation which not only enhances own private-sector productivity but also spills over to enhance private-sector productivity abroad. Public capital formation is broadly defined here to cover not only physical capital (such as social infrastructure, telecommunications, transportation) but also intangible capital (such as public R&D capital). Governments can only tax and borrow from their own residents.

I will focus on developing the home country. The foreign economy is developed in an analogous manner and is summarized in Table 2.1. (Foreign variables are denoted by asterisks.)

### Production Sector:

The production function is assumed to be Cobb-Douglas over private inputs K capital, and L labour:

$$(1) \quad Y = A F(K, L) = A K^{\beta_K} L^{\beta_L}, \quad 0 < \beta_K, \beta_L < 1$$

I specify the total factor productivity term as:  $A = \pi G^{\beta_G} G^{*\beta_{G^*}}$  where  $\pi$  is a constant to which we assign all omitted variables, and where  $G$  and  $G^*$  are the stocks of domestic and foreign public capital respectively. They are assumed to be imperfectly substitutable and essential.

The production function can be rewritten as:



$$(1)' Y = F(G, G^*, K, L) = \pi G^{\beta_G} G^{*\beta_{G^*}} K^{\beta_K} L^{\beta_L}, \quad \pi > 0$$

It is still necessary to decide upon the "returns to scale". I examine three possibilities:

$$\beta_G + \beta_{G^*} + \beta_K + \beta_L = 1 \quad \text{CRTS across all inputs}$$

$$\beta_G + \beta_K + \beta_L = 1 \quad \text{CRTS across domestic inputs}$$

$$\beta_K + \beta_L = 1 \quad \text{CRTS across private inputs}^1,$$

where CRTS denotes "constant returns to scale".

For simplicity I assume that labour supplies are inelastic and normalized to unity ( $L = L^* = 1$ ). Firms are identical price-takers and face quadratic internal costs of adjustment. The firm's valuation functional is<sup>2</sup>:

$$(2) \max V(t) = \int_t^\infty \{ (1 - \tau_k)(F(\cdot) - w(s)L(s)) - J_p(s) \} e^{-\int_t^s r(u)du} ds$$

$$\text{where } J_p(s) = I_p(s) + \frac{\psi I_p(s)^2}{2 K(s)}$$

$$\text{and } \dot{K}(s) = I_p(s) - \delta K(s)$$

and where  $\delta$  is the geometric depreciation rate,  $\psi > 0$  the adjustment cost parameter, and  $\tau_k$  the capital taxation rate. The firm maximizes  $V(t)$  by

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<sup>1</sup> In my empirical study of OECD economies I tend to find support for CRTS across private  $K$  and  $L$ .

<sup>2</sup> To avoid cluttering up the notation I omit time subscripts in the production function  $F(\cdot)$  and in the marginal products, such as  $F_K(\cdot)$ ,  $F_G(\cdot)$ .

choosing  $L(s)$  and  $K(s)$  for all  $s \in [t, \infty)$ .  $F(\cdot)$  is the production function,  $r$  the domestic interest rate,  $w$  the wage rate, and  $J_p$  is the sum of gross private investment  $I_p$  and output used up in transforming goods into capital.  $K$  is thus the net of depreciation stock of private capital. Only home output is used for domestic capital formation.

The solutions to the firm's maximization are:

$$(3a) \quad w(s) = F(\cdot) - K(s) F_K(\cdot)$$

$$(3b) \quad q_p(s) = 1 + \psi\left(\frac{I_p(s)}{K(s)}\right) \equiv \frac{\partial J_p(s)}{\partial I_p(s)}$$

$$(3c) \quad r(s) = \frac{\dot{q}_p(s)}{q_p(s)} + \frac{(1 - \tau_k)F_K(\cdot) + \frac{\psi\left(\frac{I_p(s)}{K(s)}\right)^2}{2}}{q_p(s)} - \delta$$

for all  $s \in [t, \infty)$ .  $F_K$  is the marginal product of capital. Condition (3a) is the familiar marginal product of labour equated to the competitive wage. Condition (3b) is similar; the shadow price of capital,  $q_p$ , is equal to its replacement cost. Condition (3c) states that the value of the marginal product of capital (ie. the sum of the after-tax marginal product of capital in production and the marginal contribution of capital in raising the productivity of investment in installation) equals the cost of capital (ie. the sum of the interest rate and depreciation rate less capital gains).

Rewriting (3c) as:

$$(3c)' \quad \dot{q}_p(s) = (r(s) + \delta) q_p(s) - \left[ (1 - \tau_k)F_K(\cdot) + \frac{\psi\left(\frac{I_p(s)}{K(s)}\right)^2}{2} \right],$$

multiplying both sides by  $K(s)$  and solving forward (and imposing a no-bubble condition<sup>1</sup>) gives

$$(3d) \quad q_p(t)K(t) = \int_t^\infty e^{-\int_t^s r(u)du} [(1 - \tau_k)F_K K(s) - J_p(s)] ds.$$

Assume further that there are constant returns to scale across private inputs so that when  $K, L$  are paid their marginal products, aggregate output is fully exhausted - ie.  $F(.) = F_L.L + F_K.K$ . Substituting this relationship into (3d) gives  $q_p(t)K(t) = V(t)$ , or

$$(3d)' \quad q_p(t) = \frac{V(t)}{K(t)}$$

That is, the value of the private capital stock equals the value of firms, which in turn equals the present discounted sum of the future marginal value products of private capital.

### Consumer Sector:

The economy is populated by overlapping generations of agents whose lifespans are uncertain<sup>2</sup>. The birth and death rates are both given by  $\lambda$  as is the size of cohorts born at every instant. This suitable normalization yields a population size of unity at all times:

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<sup>1</sup>

ie.  $\lim_{s \Rightarrow \infty} K(s)q_p(s) e^{-\int_t^s r(u)du} = 0$

<sup>2</sup> The approach is based on Blanchard (1985).

$$\text{ie. } \int_{-\infty}^t \lambda e^{-\lambda(t-s)} ds = 1$$

All agents earn the same wage rate and pay the same lump sum taxes. The rest of this section is divided into two parts: the first part will describe an individual agent's consumption and savings decisions; the second part will aggregate across individuals.

### Individual Consumer Behavior

For each variable  $x(t,v)$ ,  $t$  indexes time and  $v$  the vintage of the agent - ie. the time of birth ( $v \leq t$ ). The individual agent allocates financial wealth among three perfectly substitutable assets - government bonds, international bonds, and claims on the stock of private home country capital (equities) - each yielding a rate of return  $r(t)$ . The government and international bonds are of the fixed market value and variable interest rate type while equities have variable market value,  $q_p$ , as derived earlier.

The individual derives utility homothetically from the domestic and foreign good, faces a constant probability of death, and has rational, single-valued expectations. The individual's intertemporal objective functional is thus:

$$(4) \quad U(t,v) = \int_t^{\infty} u(\tilde{c}(s,v)) e^{-(\rho + \lambda)(s-t)} ds$$

where  $\rho$  is the time preference rate and

$$(4a) \quad u(\tilde{c}(s,v)) = \frac{\tilde{c}(s,v)^{1-\sigma}}{1-\sigma} \quad \text{for } \sigma \neq 1$$

$$(4b) \quad u(\tilde{c}(s,v)) = \log(\tilde{c}(s,v)) \quad \text{for } \sigma = 1$$

$$(4c) \quad \tilde{c}(s,v) = c_H(s,v)^\alpha c_F(s,v)^{1-\alpha}$$

$c_H$  is the consumption of domestic output

$c_F$  is the consumption of foreign output

where  $\frac{1}{\sigma}$  is the intertemporal elasticity of substitution and  $\alpha$  is the share of the consumer's expenditure on domestic output.

The individual maximizes (4) subject to the following flow budget constraint for each time  $s \in [t, \infty)$  :

$$(5) \quad \dot{a}(s,v) = (r(s) + \lambda) a(s,v) + w(s,v) - \tau(s,v) - (1 + \tau_c) c(s,v)$$

where

$$(5a) \quad a(s,v) = b(s,v) + z(s,v) + q_p(s)k(s,v)$$

$$(5b) \quad c(s,v) = c_H(s,v) + p(s) c_F(s,v)$$

$$(5c) \quad c_H(s,v) = \alpha c(s,v)$$

$$(5d) \quad p(s)c_F(s,v) = (1-\alpha) c(s,v)$$

(5c) and (5d) are obtained by solving the intratemporal problem of maximizing (4c) subject to (5b).  $c(s,v)$  represents total consumption expenditures,  $p$  the relative price of the foreign good in terms of the domestic good,  $\tau$  lump-sum taxes (age-independent),  $\tau_c$  the consumption tax rate,  $b$  holdings of government bonds, and  $z$  holdings of the internationally traded bond. The term  $\lambda a(s,v)$  reflects the presence of a

perfectly competitive annuity market. Agents are assumed not to leave bequests. Instead each agent contracts with an annuity company to transfer all her asset holdings  $a(s,v)$  to the company upon accidental death, and in return to earn interest on her asset holdings while alive - say  $i(s)a(s,v)$  where  $i(s)$  is the rate of interest. Since  $\lambda$  people die each instant, the annuity firm receives  $\lambda a(s,v)$  and pays out  $i(s)a(s,v)$  each instant. Perfect competition in the annuity market leads to zero profits so that  $\lambda = i(s)$ .

Maximizing (4) subject to (5), after substituting (5c,d) into (4c), yields:

$$(6a) \quad \frac{\dot{c}(s,v)}{c(s,v)} = \frac{1}{\sigma} \left( r(s) - \rho + (1 - \alpha)(1 - \sigma) \frac{\dot{p}(s)}{p(s)} \right) \text{ for } s \in [t, \infty)$$

which describes the optimal evolution of consumption for the individual agent<sup>1</sup>. The term inside the brackets represents the rate of interest in terms of the domestic consumption basket. A rising  $\frac{\dot{p}(s)}{p(s)}$  represents a declining terms of trade. The reaction of the individual to it depends on the intertemporal elasticity parameter times the share of imported goods in her consumption; if for example,  $\sigma > 1$ , the adverse wealth effect dominates and the consumer consumes less relative to the future. If  $\sigma < 1$  the substitution effect dominates and the consumer consumes more relative to the future in anticipation of higher future prices.

Rolling the budget constraint forward<sup>2</sup> and substituting (6a) into it gives:

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<sup>1</sup> The consumption tax rate drops out since it is assumed to be a time-invariant exogenous policy instrument. As a result no intertemporal distortion is introduced in the time profile of the individual's consumption expenditures.

<sup>2</sup> Imposing also the transversality condition:

$$(6b) \quad c(t,v) = \theta(t) [a(t,v) + h(t,v)]$$

where

$$(6c) \quad \theta(t) = \left\{ \int_t^\infty e^{-\int_t^s \frac{1}{\sigma} \left( (1-\sigma)(r(u) + (\alpha-1)\frac{\dot{p}(s)}{p(s)} - \rho - \lambda\sigma \right) du} ds \right\}^{-1}$$

$$(6d) \quad h(t,v) = \int_t^\infty e^{-\int_t^s (r(u) + \lambda) du} [w(s,v) - \tau(s,v)] ds$$

Consumption is thus a linear function of the present value of human  $h(t,v)$  and non-human wealth  $a(t,v)$ . The discount factor for human wealth effectively incorporates the probability of death.  $\theta(t)$  is the marginal propensity to consume out of total wealth; note that  $\frac{\partial \theta(t)}{\partial r(t)} \geq 0$  as  $\frac{1}{\sigma} \leq 1$  and vice versa.

### Aggregate Consumer Behavior

For each individual variable  $x(t,v)$ , the aggregate counterpart  $x(t)$  can be obtained by aggregating across all cohorts alive:

$$x(t) = \int_{-\infty}^t x(t,v) \lambda e^{-\lambda(t-v)} dv$$

Applying this procedure provides the following:

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$$\lim_{s \Rightarrow \infty} a(s,v) e^{-\int_t^s (r(u) + \lambda) du} = 0$$

$$(7a) \quad (1 + \tau_c) c(t) = \theta(t) [a(t) + h(t)]$$

$$(7b) \quad c(t) = c_H(t) + p(t) c_F(t) \quad \text{where} \quad c_H(t) = \alpha c(t) \quad \text{and} \\ p(t) c_F(t) = (1 - \alpha)c(t)$$

$$(7c) \quad a(t) = b(t) + z(t) + q_p(t)k(t)$$

$$(7d) \quad h(t) = \int_t^\infty e^{-\int_t^s (\tau(u) + \lambda) du} [w(s) - \tau(s)] ds$$

$$(7e) \quad \theta(t) = \left\{ \int_t^\infty e^{-\int_t^s \frac{1}{\sigma} \left( (1-\sigma)(r(u) + (\alpha-1)\frac{\dot{p}(s)}{p(s)} - \rho - \lambda\sigma \right) du} ds \right\}^{-1}$$

The dynamical counterparts are:

$$(8a) \quad (1 + \tau_c) \dot{c}(t) = \frac{1}{\sigma} \left( r(t) - \rho + (1 - \alpha)(1 - \sigma) \frac{\dot{p}(t)}{p(t)} \right) (1 + \tau_c) c(t) - \lambda \theta(t) a(t)$$

$$(8b) \quad \dot{\theta}(t) = \theta(t)^2 + \theta(t) \left( \frac{1}{\sigma} \{ (1 - \sigma)[r(t) + (1 - \alpha) \frac{\dot{p}(t)}{p(t)}] - \rho - \sigma\lambda \} \right)$$

$$(8c) \quad \dot{h}(t) = (r(t) + \lambda) h(t) + \tau(t) - w(t)$$

$$(8d) \quad \dot{a}(t) = r(t) a(t) + w(t) - \tau(t) - (1 + \tau_c) c(t)$$

In deriving  $h(t)$  it is assumed that wages and lump-sum taxes are vintage independent. In deriving  $a(t)$  it is assumed that newly born agents possess no financial wealth, ie.  $a(t, t) = 0$ . In (8a) it is seen that if  $\lambda = 0$  the standard Euler equation for an infinitely-lived representative agent is obtained. Otherwise aggregate consumption will differ from that in a representative agent economy by  $\lambda\theta(t)a(t)$  - ie. since the newly born agents



have no financial wealth, one must deduct the marginal propensity to consume times the level of financial wealth ( $\theta(t)a(t)$ ) from each of the  $\lambda$  new agents. In (8d) note that the  $\lambda a(t)$  interest income term cancels in the aggregate. The equal birth and death rate of agents simply results in  $\lambda a(t)$  being transferred from the deceased population to the living.

### Government Sector

In this subsection I outline the basic government budget constraint. The next section derives behavioral specifications for the government.

Government spending here includes public investment  $i_g$  and unproductive government consumption expenditures  $c_g$  such as those purchases necessary to maintain the system of government but which have otherwise no direct effect on private sector production or utility.<sup>1</sup> Government revenues come from lump-sum taxes, capital income taxes, and consumption taxes imposed on domestic agents. Under certain circumstances (to be discussed in the next section) the government can also obtain revenues by charging a user fee for its stock of capital  $G$  or allow private firms to rent  $G$ . Finally the government can also borrow from the domestic private sector. Let  $b(t)$  denote the stock of government debt. The government budget constraint is thus:

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<sup>1</sup> In order to isolate the impacts of "productive" public investment, I abstract from other public goods spending that may impinge on private sector productivity or utility directly. In a separate chapter I consider the optimal public provision of  $c_g$  when  $c_g$  enters household utility. Here  $c_g$  is treated exogenously. By treating it this way, I can better differentiate the impacts of  $G$  versus  $c_g$ . The way I treat  $c_g$  is the way government spending is treated in much of the existing literature.

$$(9) \dot{b}(t) = r(t) b(t) + J_g(t) + c_g(t) - \tau(t) - \tau_c c(t) - \tau_k F_K K(t) - \xi(t) G(t)$$

where

$$J_g(t) = I_g(t) + \frac{\psi I_g(s)^2}{2 G(s)}$$

$$\text{and } \dot{G}(s) = I_g(s) - \delta G(s)$$

I assume that there are also adjustment costs to installing public capital. For simplicity I assume that the adjustment cost parameter and depreciation rate are identical to those for private capital. Again only home output can be used to augment the stock of domestic public capital<sup>1</sup>. The user fee for public capital is given by  $\xi(t)$ .

Solving (9) forward yields the intertemporal government budget constraint<sup>2</sup>:

$$\begin{aligned} (9a) \quad & \int_t^\infty e^{-\int_t^s r(u) du} [\tau(s) + \tau_c c(s) + \tau_k F_K K(s) + \xi(s) G(s)] ds \\ & = \int_t^\infty e^{-\int_t^s r(u) du} [J_g(s) + c_g(s)] ds + b(t) \end{aligned}$$

The LHS is the present discounted value of future revenues; the RHS is the present value of future spending plans plus initial debt. In order to help

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<sup>1</sup> Similarly, only foreign output can be used to augment the stock of foreign public and private capital.

<sup>2</sup> After imposing the transversality condition:

$$\lim_{s \Rightarrow \infty} b(s) e^{-\int_t^s r(u) du} = 0$$

ensure long run public sector solvency I will be adopting the following feedback rule for lump-sum taxation:

$$(9b) \quad \tau(t) = \tau_0 + \eta \dot{b}(t) \quad \eta < -1$$

where  $\tau_0$  is the exogenous component of lump sum taxes.

### Linkages with the Foreign Sector

There is an internationally traded bond denominated in terms of domestic output, denoted by  $z$ , that is held by wealth holders in both countries. If  $z > 0$  domestic agents are net lenders to the rest of the world, and net borrowers if  $z < 0$ . On net, the global capital market clears:

$$(10a) \quad z(t) + p(t)z^*(t) = 0$$

$$(10b) \quad r(t) = r^*(t) + \frac{\dot{p}(t)}{p(t)}$$

where asterisks denote foreign country variables. (10b) is the uncovered real interest parity condition.

The home country's trade balance is  $TB(t) = c_H^*(t) - p(t)c_F(t) = \alpha^*p(t)c^*(t) - (1 - \alpha)c(t)$ , where  $\alpha^*$  is the share of foreign consumption expenditures on the domestically produced good. The home balance of payments identity is:

$$(11) \quad \dot{z}(t) = r(t) z(t) + TB(t)$$

The LHS is minus the capital account balance; the RHS is the current account balance (sum of the service account balance  $r(t)z(t)$  and the trade balance). The demand for domestic output  $y(t)$  comes from domestic and foreign consumers and domestic investors:

$$y(t) = c_H(t) + c_H^*(t) + c_g(t) + J_p(t) + J_g(t)$$

$$\text{or } y(t) = \alpha c(t) + \alpha^* p(t) c^*(t) + c_g(t) + J_p(t) + J_g(t)$$

Substituting this into (11), using the TB(t) definition, provides another way to express the home country's balance of payments identity:

$$(11)' \quad \dot{z}(t) = r(t) z(t) + y(t) - c(t) - c_g(t) - J_p(t) - J_g(t)$$

I also note for future reference the foreign economy's balance of payments identity and goods market-clearing condition:

$$(11)'' \quad \dot{z}^*(t) = r^*(t) z^*(t) + y^*(t) - c^*(t) - c_g^*(t) - J_p^*(t) + J_g^*(t)$$

$$\text{where } y^*(t) = c_F(t) + c_F^*(t) + c_g^*(t) + J_p^*(t) + J_g^*(t)$$

$$\text{or } y^*(t) = (1 - \alpha) \frac{c(t)}{p(t)} + (1 - \alpha^*) c^*(t) + c_g^*(t) + J_p^*(t) + J_g^*(t)$$

### **III. Endogenous Government Behavior**

In the previous section public investment ( $i_g$ ) and consumption ( $c_g$ ) were exogenous. Public capital formation will now be chosen optimally. There is a shadow price,  $q_g$ , associated with public capital which equals the present discounted value of its future marginal products. I will first characterize the optimum conditions and discuss alternative ways to charge for the services of public capital. There are two types of planners: national and global. The national planner chooses a public investment strategy to maximize national welfare while the global planner chooses public investments in each country to maximize world welfare.

I will continue to leave  $c_g$  exogenous. It seems to be a good benchmark for comparing the open-economy effects of productive versus unproductive government expenditure. As indicated earlier, the treatment of government spending in much of the previous literature is similar to the way  $c_g$  is treated here. I will also abstract from issues of optimal taxation. Government behavior is optimal as far as its choice of investment is concerned. However it is not a fully optimizing government because its choice of financing mix need not be optimal. If the government were fully optimizing and no distortionary taxes were needed, the planner would choose public capital formation optimally and carry out a system of intergenerational and intragenerational redistributions of lump-sum taxes (age-dependent, or age-independent with debt policy) such that at the margin the planner had no further incentive to shift one unit of resource between periods and/or between generations alive contemporaneously. That is, if fiscal policy were fully optimal (and instruments fully available) there should be no need to alter the fiscal policies set, except in

circumstances of unexpected shocks, since the fiscal policies in place would be Pareto-optimal intergenerationally and intragenerationally. For this reason, I assume that governments are not fully optimizing over all their instruments (in particular, those related to financing) so that exogenous variations in fiscal instruments such as lump-sum taxes or distortionary taxes can be studied<sup>1</sup>.

I assume governments are non-paternalistic, utilitarian planners. Appendix 2.1 outlines how individual welfare functions in an OLG model can be aggregated. The home-country national planner maximizes (with respect to choices of  $K, \dot{K}, G, \dot{G}, z, \dot{z}$ , and  $c$ ):

$$(12) \quad SW_t = \int_t^{\infty} u(c(s)) e^{-\gamma(s-t)} ds$$

(where  $\gamma$  is the planner's time preference rate) subject to the national resource constraint:

$$(11)' \quad \dot{z}(s) = r(s) z(s) + y(s) - c(s) - c_g(s) - J_p(s) - J_g(s), \text{ for all } s \in [t, \infty).$$

Assume there is no need for distortionary taxation - ie.  $\tau_K = \tau_c = 0$  for all  $s$ . The necessary conditions are for  $s \in [t, \infty)$ :

$$(13a) \quad \frac{\dot{c}(s)}{c(s)} = \frac{1}{\sigma} [r(s) + (\alpha-1)(1-\sigma) \frac{\dot{p}(s)}{p(s)} - \gamma]$$

---

<sup>1</sup> It would be desirable to introduce different types of uncertainty and observe how a fully optimizing planner responds, and examine what kinds of optimal policy rules emerge in the presence of uncertainty.

$$(13b) \quad \frac{\dot{q}_p(s)}{q_p(s)} + \frac{F_K + \frac{\psi \left( \frac{i_p(s)}{K(s)} \right)^2}{2}}{q_p(s)} - \delta = \frac{\dot{q}_g(s)}{q_g(s)} + \frac{F_G + \frac{\psi \left( \frac{i_g(s)}{G(s)} \right)^2}{2}}{q_g(s)} - \delta = r(s)$$

where

$$(13c) \quad q_p(s) = 1 + \psi \left( \frac{i_p(s)}{K(s)} \right)$$

$$(13d) \quad q_g(s) = 1 + \psi \left( \frac{i_g(s)}{G(s)} \right)$$

In steady state,  $q_p = q_g = 1 + \psi\delta$ , which is not equal to one because there is a constant amount of steady state investment (namely  $i_g = \delta G$ ,  $i_p = \delta K$ ). The two  $q$ 's are also equal in steady-state because it costs the same to replace the two types of capital<sup>1</sup>. The national planner chooses public capital in such a way that the returns (capital gains plus the marginal productivity terms less depreciation) to the two types of capital are equal, and equated to the interest rate (ie. intertemporal terms of trade). Aggregate consumption evolves according to the discrepancy between the real interest rate (in terms of the domestic consumption basket) and the planner's time preference rate. The long run interest rate equals the planner's time preference rate.

The global planner maximizes (with respect to  $K, \dot{K}, G, \dot{G}, K^*, \dot{K}^*, G^*, \dot{G}^*, c$ , and  $c^*$ ):

$$(14) \quad GW_t = \int_t^\infty [u(c(s)) + u^*(c^*(s))] e^{-\gamma'(s-t)} ds$$

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<sup>1</sup> In a linearized model where it is only valid to study small changes around the neighbourhood of steady-state, the two  $q$ 's tend to be equal even out of steady-state.

subject to the world resource constraint:

$$(14a) \quad y(s) + p(s)y^*(s) = c(s) + p(s)c^*(s) + c_g(s) + p(s)c_g^*(s) \\ + J_p(s) + p(s)J_p^*(s) + J_g(s) + p(s)J_g^*(s)$$

for  $s \in [t, \infty)$ .

The global planner is assumed to weight home and foreign welfare equally and apply the same time preference rate  $\gamma'$  to both countries. Let  $\mu(s)$  be the shadow value of the world resource constraint. Necessary conditions for an optimum are:

$$(15a) \quad \frac{\dot{\mu}(s)}{\mu(s)} = (\alpha-1)(1-\sigma)\frac{\dot{p}(s)}{p(s)} - \sigma\frac{\dot{c}(s)}{c(s)} = (\alpha^*(1-\sigma^*)-1)\frac{\dot{p}(s)}{p(s)} - \sigma^*\frac{\dot{c}^*(s)}{c^*(s)}$$

$$(15b) \quad -\frac{\dot{\mu}(s)}{\mu(s)} = \frac{\dot{q}_g(s)}{q_g(s)} + \frac{F_G + p(s)F_G^* + \frac{\psi\left(\frac{i_g(s)}{G(s)}\right)^2}{2}}{q_g(s)} - \gamma' - \delta$$

$$(15c) \quad -\frac{\dot{\mu}(s)}{\mu(s)} = \frac{\dot{q}_p(s)}{q_p(s)} + \frac{F_K + \frac{\psi\left(\frac{i_p(s)}{K(s)}\right)^2}{2}}{q_p(s)} - \gamma' - \delta$$

$$(15d) \quad -\frac{\dot{\mu}(s)}{\mu(s)} - \frac{\dot{p}(s)}{p(s)} = \frac{\dot{q}_g^*(s)}{q_g^*(s)} + \frac{F_{G^*} + p(s)F_{G^*}^* + \frac{\psi^*\left(\frac{i_g^*(s)}{G^*(s)}\right)^2}{2}}{q_g^*(s)} - \gamma' - \delta^*$$

$$(15e) \quad -\frac{\dot{\mu}(s)}{\mu(s)} - \frac{\dot{p}(s)}{p(s)} = \frac{\dot{q}_p^*(s)}{q_p^*(s)} + \frac{F_{K^*}^* + \frac{\psi^*\left(\frac{i_p^*(s)}{K^*(s)}\right)^2}{2}}{q_p^*(s)} - \gamma' - \delta^*$$



where  $q_p, q_g$  are as defined before. Analogously,

$$(15f) \quad q_p^*(s) = 1 + \psi^* \left( \frac{i_p^*(s)}{K^*(s)} \right)$$

$$(15g) \quad q_g^*(s) = 1 + \psi^* \left( \frac{i_g^*(s)}{G^*(s)} \right)$$

The interpretation of (15a) is that  $-\frac{\dot{\mu}}{\mu}$  plus the planner's time preference rate represent the return to consumption<sup>1</sup>. The returns to each of the four stocks of capital are equated to this return to consumption (denominated in terms of home country output). This also implies that the returns (capital gains plus the marginal productivity terms less depreciation) from each type of capital are equated.

To summarize, (13a-b) constitute consumption and production efficiency conditions for a national planner and (15a-e) the same for a global planner. A key difference is that the global planner takes into account the contribution each country's public capital stock makes to the other's production possibilities. These conditions turn public capital formation into an endogenous "policy rule".

There are special cases of (15a) that are worth noting - namely when any one of the following three holds: (i)  $p = 1$ ; (ii)  $\sigma = 1$ ; (iii)  $\alpha = \alpha^*$ . Each leads to:

$$c = pc^*$$

This simplifies the global model as the global planner treats the two countries as essentially two identical economies, so that  $K=pK^*$ ,  $G=pG^*$ ,

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<sup>1</sup> In a decentralized economy this return equals the competitive interest rate.

and  $y=py^*$  and so forth, where  $p$  is the slope of the global production possibilities frontier. The case  $p=1$  can be justified if the production functions are identical and global opportunity costs are constant (ie. equal ratios of marginal products among the factors of production in each country);  $\sigma=1$  if the intertemporal substitution and wealth effects on consumption of intertemporal price changes cancel, and  $\sigma=1$  causes the evolution of consumption to be independent of intratemporal price changes as well; finally,  $\alpha=\alpha^*$  implies basically a one commodity world - domestic agents have the same propensity to consume home output as foreign agents have to consume home output.

I turn now to issues related to the valuation of public capital. I consider the national and global planners in turn.

### National Planner

First, from (13b):

$$(13b)' \quad \dot{q}_g(t) = (r(s) + \delta) q_g(t) - (F_G + \frac{\psi \left( \frac{i_g(t)}{G(t)} \right)^2}{2})$$

$$\text{or (16)} \quad q_g(t)G(t) = \int_t^\infty e^{-\int_t^s r(u)du} [F_G G(s) - J_g(s)] ds$$

As long as there are constant returns across private inputs - ie.  $F = F_K K + F_L L$  - it is not possible to impute a convenient market share of output to government capital. Neither can it be assumed that  $F_G = 0$  since that would contradict the assumption that public capital is productive. In the case where public capital cannot be priced or valued in the market I assume that it is provided "free" to the private sector despite a non-zero

FG. The shadow price  $q_g$  remains unobservable (known only to the planner) as there is no market counterpart to it as there is for  $q_p$ .

If however there are less than constant returns across private inputs, total national output will not be fully exhausted. There are two possible ways the government can exact its share of aggregate output: it can charge a user fee for the services of public capital or lend its stock to private firms. This last scheme is analogous to the way agents in the model who own the stock of private capital lend their capital ( $q_p K$ ) to private firms and earn a market return on it ( $r q_p K$ ). I will indicate under what conditions the two schemes are equivalent.

To illustrate I rewrite the flow government budget constraint as:

$$(17) \dot{a}_g(t) = r(t)a_g(t) + \tau(t) - c_g(t)$$

where

$$(17a) a_g(t) = q_g(t)G(t) - b(t)$$

$a_g(t)$  represents the explicit net worth of the government (assets less liabilities)<sup>1</sup>. The LHS of (17) represents a net debit on the government's capital account balance while the RHS represents a net credit on the government's current account balance.

#### (i) User Fee Approach

Substituting (17a) and the time-differentiation of (16) into (17) gives:

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<sup>1</sup> ie. not including the implicit assets represented by the present value of future taxes.

$$(18) \dot{b}(t) = r(t)b(t) + J_g(t) + c_g(t) - \tau(t) - F_G G(t)$$

From (18) an appropriate user fee for government capital would be  $\xi(t) = F_G$ . Then (18) can be rewritten as:

$$(18)' \dot{b}(t) = r(t)b(t) + J_g(t) + c_g(t) - \tau(t) - \xi(t)G(t)$$

The value-maximizing firm's objective functional should also be changed to:

$$(19) V(t) = \int_t^\infty e^{-\int_t^s r(u)du} [F(.) - w(s) - J_p(s) - \xi(s)G(s)]ds$$

but the first-order conditions would be as before (since firms treat the stock of  $G$  as exogenously given).

The optimizing government would charge users the marginal product of its capital. Any other government might charge an  $\xi(t)$  differing from the 'true'  $F_G$  or keep  $\xi(t)$  fixed at some  $\xi_0$ . However, unless  $\xi(t) = F_G$ ,  $q_p K$  will not equal the value of firms  $V$ . Recall from

$$(3d) q_p(t)K(t) = \int_t^\infty e^{-\int_t^s r(u)du} [F_K K(s) - J_p(s)]ds$$

that the RHS equals  $V$  only if  $F = F_K K + F_L L + F_G G = F_K K + wL + \xi G$ , where  $w = F_L$ . In the event that  $F > F_K K + F_L L + F_G G$ , say due to the contribution of foreign public capital (ie.  $F = F_K K + F_L L + F_G G + F_G^* G^*$ ), the home government is assumed to overcharge; ie. it charges  $\xi(t) = F_G + F_G^* \left(\frac{G^*}{G}\right) > F_G$ . Conversely if  $F < F_K K + F_L L + F_G G$ , the government undercharges for the use of its capital.

## (ii) Rental Approach

The government under this approach lends its stock of capital to the private sector. The value functional is rewritten as:

$$(20) \quad V(t) = \int_t^\infty e^{-\int_t^s r(u)du} [F(.) - w(s) - J_p(s) - J_g(s)] ds$$

Firms now choose  $G$  and  $K$  to maximize value. A price  $q_p$  is paid to the owners of  $K$ , namely households, and  $q_g$  to the owner of  $G$ , the government. By the arbitrage condition (13b),  $K$  and  $G$  both earn the same market rate of return. If  $F = F_K K + F_L L + F_G G$ , then (3d) and (16) can be substituted into (20) to give:

$$(20)' \quad V = V_K + V_G, \text{ where } V_K = q_p K \text{ and } V_G = q_g G$$

that is, the value of a firm is divided between the private owners and the public sector.

It is possible (but not for certain) that  $q_p = q_g$  even out of steady-state since both  $G$  and  $K$  come from the same good, face the same installation costs and depreciation rates. In addition firms may develop criteria which lead them to select  $G$ ,  $K$  optimally only if these prices are equal. Again if  $F > F_K K + F_L L + F_G G$ , the gap being attributable to  $F_G^* G^*$ , the price  $q_g$  will be greater than what (16) dictates. Firms are assumed to perceive the marginal product of domestic capital to be  $F_G'$  where  $F_G' G = [F - F_K K - F_L L]$ , the residual from total product after deducting private factor shares. The new price of public capital becomes  $q_g' = q_g(1+v)$  where  $v$ , a markup, is obtained from  $F_G' G = F_G G(1+v)$  and

$F_G$  is the true marginal product<sup>1</sup>. Similarly if  $F < F_K K + F_L L + F_G G$ , the market  $q_g$  will be underpriced and  $v$  will be a markdown..

Note that the user fee and rental approaches are equivalent when

$$\xi(t) = F_G = \frac{r(t)q_g(t)G(t) - (q_g(t)\dot{G}(t)) + J_g}{G(t)}.$$

The  $F_G$  expression on the right is just the time-differentiation of (16). The numerator is the rental earned less capital gains plus gross investment (including installation). Substituting the above into (18)', the government budget constraint under the user fee approach, gives (17), the net worth identity for the government. In other words the government can charge a user fee equal to what it would have earned had it lent its capital to the private sector. The value of a firm is  $V_K = q_p K$ . Through appropriate fees, the government can siphon its 'implicit' market share of the value of firms (ie.  $V_G = V - V_K$ ).

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<sup>1</sup> The  $v$  comes from revising (16) to

$$(16)' \quad q_g(t)'G(t) = \int_t^\infty e^{-\int_t^s r(u)du} [F_G' - J_g(s)]ds$$

which gives:

$$\frac{q_g(t)'G(t)}{1+v} = \int_t^\infty e^{-\int_t^s r(u)du} [F_G - \frac{J_g(s)}{1+v}]ds$$

Comparing the LHS of the above to the LHS of the original (16), we find the new price to be  $q_g' = q_g(1+v)$ . Moreover  $J_g'(s) = [J_g(s)/(1+v)]$ ; that is, it is necessary to revise  $J_g$  as well because by thinking that domestic public capital actually accounts for a fraction  $(F_G G + F_G^* G^*)/F$  of national output will overstate the actual size of the stock of  $G$ . Thus firms value public capital according to (16)' above, not (16), and by comparing (16) and (16)' we can determine the extent to which  $q_g'$  and  $J_g'$  differ from the 'true'  $q_g$  and  $J_g$ .

Finally note that in steady-state the stock of government debt and the stock of public capital can be positively or negatively associated. This can be seen from the government budget constraint (18), or (17) using (13b)'. Setting time-derivatives to zero:

$$0 = rb + \delta(1 + \frac{\psi\delta}{2})G + c_g - \tau - F_G G$$

which states that if  $F_G > \delta(1 + \frac{\psi\delta}{2})$  long run  $G$  and  $b$  will be positively associated. The interpretation is that the marginal product of government capital exceeds the steady-state expenditures needed to maintain a unit of a constant stock of public capital. In steady-state the depreciation component acts much like a government expenditure item, like  $c_g$ , and has a negative association with the long run stock of government debt. Government revenues (such as government earnings from public capital services) have a positive association with the long run stock of  $b$ . Thus if  $F_G > \delta(1 + \frac{\psi\delta}{2})$ , the revenue effect dominates in the long run.

### Global Planner

Since the user and rental schemes can be equivalent, I discuss only the rental approach under global planning.

Assume constant returns across all inputs within each country (ie.  $F = F_K K + F_L L + F_G G + F_{G^*} G^*$  and  $F^* = F_{K^*} K^* + F_{L^*} L^* + F_{G^*} G^* + F_{G^*} G$ ). The following results depend on this assumption. Using (15b,d) and  $-\frac{\dot{\mu}}{\mu} = r - \gamma'$  gives:

$$(21a) \quad q_g(t)G(t) = \int_t^\infty e^{-\int_t^s r(u)du} [(F_G + p(s)F_G^*)G(s) - J_g(s)]ds$$

$$(21b) \quad q_g^*(t)G^*(t) = \int_t^\infty e^{-\int_t^s r^*(u)du} \left[ \left( \frac{F_G^*}{p(s)} + F_{G^*}^* \right) G^*(s) - J_g^*(s) \right] ds$$

Domestic and foreign firms maximize respectively,

$$(21c) \quad V(t) = \int_t^\infty e^{-\int_t^s r(u)du} [F(.) - w(s) - J_p(s) - J_g(s)]ds$$

$$(21d) \quad V^*(t) = \int_t^\infty e^{-\int_t^s r^*(u)du} [F^*(.) - w^*(s) - J_p^*(s) - J_g^*(s)]ds$$

Substituting (21a,b) into (21c,d) and using (15c,e) yields:

$$(22a) \quad q_g(t)G(t) + q_p(t)K(t) = V(t) - \Omega_{G^*} + \Omega_G$$

$$\text{where } \Omega_{G^*} = \int_t^\infty e^{-\int_t^s r(u)du} [F_{G^*} G^*(s)]ds \text{ and } \Omega_G = \int_t^\infty e^{-\int_t^s r(u)du} [p(s)F_G^* G(s)]ds$$

$$(22b) \quad q_g^*(t)G^*(t) + q_p^*(t)K^*(t) = V^*(t) - \Omega_{G^*}^* + \Omega_G^*$$

$$\text{where } \Omega_G^* = \int_t^\infty e^{-\int_t^s r^*(u)du} [F_G^* G(s)]ds \text{ and } \Omega_{G^*}^* = \int_t^\infty e^{-\int_t^s r^*(u)du} \left[ \frac{F_{G^*}^*}{p(s)} G^*(s) \right] ds$$

$\Omega_{G^*}$ ,  $\Omega_G^*$  are the present discounted stream of payments made from one country to the other for the services of (or positive externality generated by) the other's stock of public capital, and  $\Omega_G$ ,  $\Omega_{G^*}^*$  represent the



corresponding receipts. In other words the global planner sets up a system of international side-payments to create a market for the value of external effects generated by  $G, G^*$ .

Note that when  $p(t)$  is constant<sup>1</sup> so that  $r(t) = r^*(t)$ ,

$$V + pV^* = q_p K + q_g G + p [q_p^* K^* + q_g^* G^*]$$

that is, the total ownership claims to all of the world's stocks of productive capital are equal to the total world value of firms.

#### IV. Small Open-Economy Model

This section examines the impacts of public investment and public consumption for a small-open economy. Certain properties of this version should aid in understanding the full two-country general-equilibrium model. In this simpler version, there are no distortionary taxes, public capital is provided "free", and  $\sigma=1$ . The interest rate and terms of trade are treated as given. Time subscripts are now omitted except where ambiguities may arise. I first consider an 'exogenous' government and then turn to a national planner.

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<sup>1</sup> A global planner who weights the welfare of the two countries equally or symmetrically may desire a constant terms of trade, for otherwise one country's purchasing power would rise at the expense of the other's. Such a planner would take actions to offset declining or appreciating terms of trade by judiciously shifting resources between countries. The constancy of  $p$  under global planning was not proven formally, but we did show that  $p$  equal to 1 or to any other constant was feasible under certain conditions.

## Exogenous Government

Substituting the definition of wealth,  $a$ , into the equations of motion for aggregate wealth (8d) and consumption (8a), I arrive at the following four state equations to describe the small open-economy:

$$(23a) \dot{c} = (r - \rho)c - \lambda\theta [z + b + q_p K]$$

$$(23b) \dot{z} = r z + y - c - c_g - J_p - J_g$$

$$(23c) i_p = \frac{1}{\psi}(q_p - 1)K$$

$$(23d) \dot{q}_p = (r + \delta)q_p - F_K - \frac{\psi}{2}\left(\frac{i_p}{K}\right)^2$$

Linearizing them around steady state gives:

$$(24) \quad \begin{bmatrix} \dot{c} \\ \dot{z} \\ \dot{K} \\ \dot{q}_p \end{bmatrix} = \begin{bmatrix} (r-\rho) & -\lambda\theta & 0 & -\lambda\theta(K + \frac{q_p r}{F_{KK}}) \\ -1 & r & F_K - \delta(1 + \frac{\delta\psi}{2}) & -\frac{q_p K}{\psi} \\ 0 & 0 & 0 & \frac{K}{\psi} \\ 0 & 0 & -F_{KK} & r \end{bmatrix} \begin{bmatrix} dc \\ dz \\ dK \\ dq_p \end{bmatrix}$$

$$+ \begin{bmatrix} -\lambda\theta & 0 & 0 & \lambda\theta \frac{q_p F_{KG}}{F_{KK}} & \lambda\theta \frac{q_p F_{KG^*}}{F_{KK}} \\ 0 & -1 & -(1+\psi\delta) & F_G + \frac{\delta^2 \psi}{2} & F_{G^*} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -F_{KG} & -F_{KG^*} \end{bmatrix} \begin{bmatrix} db \\ dc_g \\ di_g \\ dG \\ dG^* \end{bmatrix}$$

For each variable  $x$  above, let  $dx = x - \bar{x}$ , where  $\bar{x}$  is the steady-state value of  $x$ .  $c, q_p$  are non-predetermined and  $K, z$  are predetermined. The roots are  $\kappa_1 = r + \lambda > 0$ ,  $\kappa_2 = r - \theta < 0$ ,

$$\kappa_3 = \frac{r + \sqrt{r^2 + \frac{4K}{\psi}(-F_{KK})}}{2} > 0, \kappa_4 = \frac{r - \sqrt{r^2 + \frac{4K}{\psi}(-F_{KK})}}{2} < 0.$$

Thus the above system is saddlepoint stable<sup>1</sup>.

The system above is also block-recursive. The dynamics of the  $q_p, K$  subsystem feed into the dynamics of the  $c, z$  subsystem. Both subsystems will be affected by domestic and foreign public investment shocks. These shocks work their way directly to the  $c, z$  subsystem and indirectly via the  $q_p, K$  subsystem. Some steady-state impacts of balanced-budget public consumption shocks are:

$$\left. \frac{dc}{dc_g} \right|_{\Delta c_g = \Delta \tau} = -\frac{\lambda\theta}{\Omega} < 0$$

<sup>1</sup>  $\kappa_4$  is negative so long as  $F_{KK} < 0$ .  $\kappa_2$  is negative because if there are many economies in the world lending to and borrowing from one another, world equilibrium requires that the steady-state interest rate lie between  $\delta$  and  $\theta = \delta + \lambda$ .

$$\left. \frac{dz}{dc_g} \right|_{\Delta c_g = \Delta \tau} = - \frac{(r-\rho)}{\Omega} \quad \text{depends on } \text{sgn}(r-\rho)$$

where  $\Omega = (r + \lambda)(\theta - r) > 0$  is a condition needed to ensure that the marginal propensity to consume out of financial wealth is positive.

A long run balanced budget increase in public consumption crowds out long run private consumption<sup>1</sup> - see Figure 2.1. It is assumed that the  $c_g$  increase is unanticipated and permanent. Point  $E_0$  denotes an initial equilibrium,  $E_1$  a final equilibrium, and  $E_2$  an intermediate point, if any. The effect on  $z$  depends on the sign of  $r - \rho$ . In steady-state the stock of wealth  $a (= b + z + q_p K)$  is zero when  $r = \rho$ , negative when  $r < \rho$ , and positive when  $r > \rho$ <sup>2</sup>. When  $r < \rho$ , the  $c_g$  shock lowers  $c$  in the long run but by less than the rise in  $c_g$ . Thus long run absorption is higher. Across steady states this can only occur if  $z$  is higher (that is, a country runs a steady-state trade deficit only if its net stock of foreign assets is positive). To lead to a long run situation in which  $z > 0$ , home private consumption falls on impact by more than the increase in public consumption; that is, private consumption overshoots its long run equilibrium level. Then, along the transition path to the long run, the economy runs current account surpluses. However when  $r > \rho$ , the decrease in long run  $c$  is greater than

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<sup>1</sup> To evaluate the impact when  $\lambda = 0$  it is also necessary to set  $r = \rho$ , a steady state requirement in representative agent models. Then  $\theta = \lambda + \rho$ . Upon substituting this into  $\Omega$ , it will be found that public consumption crowds out private consumption one-for-one in the long run.

<sup>2</sup> This can be seen by setting  $\dot{c} = \dot{a} = 0$  and solving simultaneously to get:

$$a = \frac{(r - \rho)(F_L - \tau)}{\Omega}.$$

the increase in long run  $c_g$ ; that is, long run absorption decreases. Thus along the transition path to the long run the economy runs current account deficits. On impact, therefore,  $c$  undershoots its long run equilibrium level. When  $r = \rho$  public consumption crowds out private consumption one-for-one in the long run, and there are no steady-state or transitional changes in  $z$ .

I turn now to long run public capital shocks. Again I assume balanced-budget financing<sup>1</sup> and that shocks are permanent and unanticipated. Some steady-state impacts are:

$$\left. \frac{dc}{dG} \right|_{\Delta b = 0} = \frac{\lambda \theta [\delta(1 + \frac{\psi \delta}{2})(\frac{F_{KG}}{-F_{KK}} - 1) + F_G]}{\Omega} > 0 \quad \text{provided } \delta \text{ is not "too" large}$$

$$\left. \frac{dz}{dG} \right|_{\Delta b = 0} = \frac{(r - \rho)[F_G + \frac{F_K F_{KG}}{-F_{KK}} - \delta(1 + \psi \delta)] - \lambda \theta q_p \frac{F_{KG}}{-F_{KK}}}{\Omega}$$

In what follows I assume:  $[F_G + \frac{F_K F_{KG}}{-F_{KK}} - \delta(1 + \psi \delta)] > 0$ . That is, a higher steady-state  $G$  stock raises output (directly, and indirectly by stimulating an increase in  $K$ ) sufficiently to keep itself maintained against depreciation. If  $F_{KG} < -F_{KK}$ , as long as the depreciation rate is not too large, a balanced-budget increase in the long run stock of public capital will increase long run private consumption. The positive effect comes from the impact public capital has on long run output and on private sector wealth through an increase in the stock of  $K$ . If steady-state depreciation is

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<sup>1</sup> In this case the required change in long run taxes is  $\Delta \tau = \delta(1 + \frac{\psi \delta}{2}) \Delta G$ .

high, it is burdensome to maintain the stock of  $K$  and  $G$  in the long run because a larger share of resources will be absorbed by depreciation rather than by private consumption. The long run impact on  $z$  depends on whether the output effect is stronger or the consumption effect (given by the last term on the numerator of  $\frac{dz}{dG}$  above). Whether the output effect will outweigh the latter depends on the sign of  $(r - \rho)$ . It is necessary but not sufficient that  $r > \rho$  in order for long run  $z$  to increase - that is, the economy must have a high saving propensity.

The transitional dynamics are depicted in Figure 2.2. Both loci are shifted to the left. When  $r \leq \rho$ , the economy has a low saving propensity - ie.  $a \leq 0$  - and long run  $z$  is lower when the long run stock of public capital  $G$  is higher. Recall that  $a = b + q_p K + z$ .  $K$  is higher and, by definition,  $b$  is constant because of the balanced budget assumption. This leaves  $z$  to be negative in order that  $a \leq 0$ . Because agents desire to hold non-positive wealth and are induced to hold more  $K$  in their portfolio,  $z$  is crowded out. To reach a situation in which  $z$  is lower, the economy must run current account deficits along the transition path; this causes private consumption on impact to overshoot its long run level. When  $r > \rho$  there are two possibilities, both associated with a positive long run stock of wealth,  $a$ . The first is a repeat of the process just described - long run  $a > 0$  even if  $z$  falls, provided  $K$  increases more. The second possibility is that long run  $z$  increases. Agents desire such large positive holdings of wealth that the increase in  $K$  is insufficient;  $z$  must be higher also. If this is the case, consumption on impact undershoots its long run level and the economy runs current account surpluses along the transition path.

Thus in contrast to balanced-budget unproductive  $c_g$ , productive balanced-budget public spending has a tendency to raise long run private consumption for a small open-economy. Unless the economy has a high enough saving propensity, however, in the long run the economy could become a debtor because of the strong stimulus to short run consumption. It should be noted that these results depend on depreciation being not too large, for if the steady-state maintenance expenses were so large as to absorb greater steady-state resources, long run private consumption could be lower. This would be depicted by the  $\dot{c} = 0$  locus shifting very far to the left. In this case the private  $K$  increase would be such that its contribution to output would fall short of the increase in the cost of maintaining  $K$  in steady-state<sup>1</sup>.

### National Planner

In this economy  $G$  is endogenous and a price  $q_g$  associated with it is introduced. The model is still block recursive - that is, the dynamics of the  $(q_g, q_p, G, K)$  subsystem still feed into the  $(c, z)$  subsystem. The new subsystem is:

$$(25a) \quad \dot{q}_p = (r + \delta)q_p - F_K - \frac{\Psi}{2} \left( \frac{i_p}{K} \right)^2$$

$$(25b) \quad \dot{q}_g = (r + \delta)q_g - F_G - \frac{\Psi}{2} \left( \frac{i_g}{G} \right)^2$$

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<sup>1</sup> I have ruled out that this is the case with a higher stock of  $G$  by the condition:

$$[F_G + \frac{F_K F_{KG}}{-F_{KK}} - \delta(1 + \psi\delta)] > 0$$

$$(25c) \dot{K} = \frac{K}{\psi}(q_p - 1) - \delta K$$

$$(25d) \dot{G} = \frac{G}{\psi}(q_g - 1) - \delta G$$

The linearized model (around steady-state) is:

$$(26) \begin{bmatrix} \dot{q}_p \\ \dot{q}_g \\ \dot{K} \\ \dot{G} \end{bmatrix} = \begin{bmatrix} r & 0 & -F_{KK} & -F_{KG} \\ 0 & r & -F_{GK} & -F_{GG} \\ \frac{K}{\psi} & 0 & 0 & 0 \\ 0 & \frac{G}{\psi} & 0 & 0 \end{bmatrix} \begin{bmatrix} dq_p \\ dq_g \\ dK \\ dG \end{bmatrix} + \begin{bmatrix} -F_{KG}^* \\ -F_{GG}^* \\ 0 \\ 0 \end{bmatrix} dG^*$$

Again for each variable  $x$  above,  $dx = x - \bar{x}$ , where  $\bar{x}$  is the steady-state value of  $x$ . The foreign stock of public capital  $G^*$  is treated as given.  $K$ ,  $G$  are predetermined and their prices are non-predetermined. The roots of the above subsystem are:

$$\kappa_1 = \frac{r + \sqrt{r^2 - 2e_1 - 2e_2}}{2} > 0, \quad \kappa_2 = \frac{r + \sqrt{r^2 - 2e_1 + 2e_2}}{2} > 0$$

$$\kappa_3 = \frac{r + \sqrt{r^2 - 2e_1 - 2e_2}}{2} < 0, \quad \kappa_4 = \frac{r + \sqrt{r^2 - 2e_1 + 2e_2}}{2} < 0$$

where

$$e_1 = \left( \frac{G}{\psi} F_{GG} + \frac{K}{\psi} F_{KK} \right) < 0, \quad e_2 = \sqrt{\left( \frac{G}{\psi} F_{GG} - \frac{K}{\psi} F_{KK} \right)^2 + \frac{4KG}{\psi^2} F_{KG}^2} > 0.$$



Saddlepoint stability (two positive and two negative roots) will be ensured if  $\sqrt{r^2 - 2e_1 \pm 2e_2} > r > 0$  which holds if  $e_1 + e_2 < 0$ , which in turn holds if  $F_{GG}F_{KK} - F_{KG}^2 > 0$ ; hence all that is required for saddlepoint stability is the requirement that the production function be concave (locally)<sup>1</sup>.

Within this subsystem the time paths of  $K$  and  $G$  are correlated in response to exogenous shocks. Figure 2.3 depicts the response to a permanent increase in the long run stock of foreign public capital. This is very much like an exogenous productivity shock. If the shock is unanticipated, the transitional path is given by  $E_0$ - $E_2$ - $E_1$ ; if anticipated, the path is  $E_0$ - $E_{01}$ - $E_{02}$ - $E_1$ . Thus locally, if public capital is optimally chosen, we should see the prices of public and private capital moving in concert - ie. their rates of change covarying positively. The valuation of public capital using private market prices may not be a bad approximation in the neighborhood of steady-state, although the subsequent price movements will depend on the size of the shifts in the  $\dot{q}_p = 0$  and  $\dot{q}_g = 0$  loci, which in turn will hinge on how close the marginal products  $F_K$  and  $F_G$  are to each other on impact. In initial steady-state,  $F_K$  and  $F_G$  are equal.

## **V. Two-Country Model**

This section investigates the effects of public spending and spillovers in a full general equilibrium open-economy setting. The two-country

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<sup>1</sup> The extension of the second-derivative test to functions of three or more variables is just the condition that the principal minors alternate in sign, beginning with negative own-second partials.

setting appears to be a useful vehicle for highlighting the role played by cross-country technological interdependence<sup>1</sup>. The consequences and repercussions for domestic and foreign national consumption, capital accumulation, current account, and the terms of trade are simultaneously derived in this setting. Table 2.1 summarizes the two-country model. This section contains two parts: (A) Steady-state properties of the model; (B) Transitional dynamics of the model.

### (A) Steady-State

In the long run all state variables are stationary and exogenous variables constant. The long run growth rate is zero. The key long-run equilibrium conditions for the case of exogenous government are, assuming no user fee or rental charges:

$$(27i) \quad r = r^*$$

$$(27ii) \quad q_p = 1 + \psi\delta$$

$$(27iii) \quad q_p^* = 1 + \psi^*\delta^*$$

$$(27iv) \quad J_g = \delta(1 + \frac{\psi\delta}{2})G$$

$$(27v) \quad J_g^* = \delta^*(1 + \frac{\psi^*\delta^*}{2})G^*$$

$$(27vi) \quad J_p = \delta(1 + \frac{\psi\delta}{2})K$$

$$(27vii) \quad J_p^* = \delta^*(1 + \frac{\psi^*\delta^*}{2})K^*$$

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<sup>1</sup> It would be a useful future exercise to increase the number of countries. This would allow for various kinds of asymmetric cases, such as one where a third country creates no spillovers but enjoys "spillins", or where a single-country provides all the spillovers, such as defense protection and R&D investment, but receives no spillins.

$$(27viii) (1-\tau_k)F_K = r q_p + \delta(1+\frac{\psi\delta}{2}) \quad (27ix) (1-\tau_k^*)F_K^* = r q_p^* + \delta^*(1+\frac{\psi^*\delta^*}{2})$$

$$(27x) (1 + \tau_c)c = \theta[a + h] \quad (27xi) (1 + \tau_c^*)c^* = \theta^*[a^* + h^*]$$

$$(27xii) (1 + \tau_c)(r-\rho)c = \lambda\theta a \quad (27xiii) (1 + \tau_c^*)(r-\rho^*)c^* = \lambda^*\theta^*a^*$$

$$(27xiv) h = \frac{F_L - \tau}{r + \lambda} \quad (27xv) h^* = \frac{F_L^* - \tau^*}{r + \lambda^*}$$

$$(27xvi) \theta = r + \lambda - \frac{r-\rho}{\sigma} \quad (27xvii) \theta^* = r + \lambda^* - \frac{r-\rho^*}{\sigma^*}$$

$$(27xviii) \tau = rb + J_g + c_g - \tau_c c - \tau_k F_K K$$

$$(27ixx) \tau^* = rb^* + J_g^* + c_g^* - \tau_c^* c^* - \tau_k^* F_K^* K^*$$

$$(27xx) rz = (1-\alpha)c - \alpha^* p c^* = - rz^* p$$

$$(27xxi) y = \alpha c + \alpha^* p c^* + c_g + J_g + J_p$$

$$(27xxii) y^* = (1-\alpha)\frac{c}{p} + (1-\alpha^*)c^* + c_g^* + J_p^* + J_g^*$$

The steady-state properties of the model will be analyzed from three angles. First, I will look at the comparative steady-state impacts of exogenous variables on international prices - namely the world interest rate

$r$  and terms of trade  $p$ . I then turn to the impacts on production  $y$  and  $y^*$ , and finally to consumption  $c$  and  $c^*$ .

- International Prices -

For simplicity assume  $\sigma = \sigma^* = 1$ . Using (27x,xi,xii,xiii):

$$(28a) \quad c = \frac{-\lambda\theta(F_L - \tau)}{(r(r-\rho)-\lambda\theta)(1 + \tau_c)} \quad , \quad (28b) \quad c^* = \frac{-\lambda^*\theta^*(F^*_L - \tau^*)}{(r(r-\rho^*)-\lambda^*\theta^*)(1 + \tau_c^*)}$$

Substituting these into (27xxi,xxii) and totally differentiating gives:

(29)

$$\begin{bmatrix} x_{01} & x_{10} \\ x_{02} & x_{20} \end{bmatrix} \begin{bmatrix} dp \\ dr \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} & x_{17} & x_{18} & x_{19} & 0 \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} & x_{27} & x_{28} & 0 & x_{29} \end{bmatrix} \begin{bmatrix} dG \\ dG^* \\ d\tau \\ d\tau^* \\ d\tau_c \\ d\tau_c^* \\ d\tau_K \\ d\tau_K^* \\ dc_g \\ dc_g^* \end{bmatrix}$$

where

$$x_{01} = \alpha^*c > 0$$

$$x_{10} = \alpha j_3(2r-\rho)(1+\tau_\rho) + \alpha^* p j_4(2r-\rho^*)(1+\tau_{\rho^*}) \\ + \frac{q_p(F_K - \alpha j_1 F_{LK} - \delta(1+\frac{\psi\delta}{2}))}{-F_{KK}(1-\tau_K)} - \frac{q_p \alpha^* p j_2 F_{L^*K^*}^*}{-F_{K^*K^*}^*(1-\tau_{K^*}^*)}$$

$$x_{02} = \frac{-(1-\alpha)c}{p^2} < 0$$

$$x_{20} = (1-\alpha^*)j_4(2r-\rho^*)(1+\tau_{\rho^*}) + \frac{(1-\alpha)}{p}j_3(2r-\rho)(1+\tau_\rho) \\ + \frac{q_p^*(F_{K^*}^* - (1-\alpha^*)j_2 F_{L^*K^*}^* - \delta^*(1+\frac{\psi^*\delta^*}{2}))}{-F_{K^*K^*}^*(1-\tau_{K^*}^*)} - \frac{q_p^*(1-\alpha)j_1 F_{LK}}{-p F_{KK}(1-\tau_K)}$$

$$x_{11} = F_G - \alpha j_1 F_{LG} - \alpha^* p j_2 F_{L^*G}^* - \delta(1+\frac{\psi\delta}{2}) + \frac{F_{KG}}{-F_{KK}}(F_K - \alpha j_1 F_{LK} - \delta(1+\frac{\psi\delta}{2})) \\ - \frac{F_{K^*G}^*}{-F_{K^*K^*}^*} \alpha^* p j_2 F_{L^*K^*}^*$$

$$x_{12} = F_{G^*} - \alpha j_1 F_{LG^*} - \alpha^* p j_2 F_{L^*G^*}^* + \frac{F_{KG^*}}{-F_{KK}}(F_K - \alpha j_1 F_{LK} - \delta(1+\frac{\psi\delta}{2})) \\ - \frac{F_{K^*G^*}^*}{-F_{K^*K^*}^*} \alpha^* p j_2 F_{L^*K^*}^*$$

$$x_{13} = \alpha j_1 > 0$$

$$x_{14} = \alpha^* p j_2 > 0$$

$$x_{15} = \alpha j_3 (r(r-\rho) - \lambda\theta) > 0$$

$$x_{16} = \alpha^* p j_4 (r(r-\rho^*) - \lambda^*\theta^*) > 0$$

$$x_{17} = \frac{-F_K}{-F_{KK}(1-\tau_K)}(F_K - \alpha j_1 F_{LK} - \delta(1 + \frac{\psi\delta}{2}))$$

$$x_{18} = \frac{F^*_{K^*}}{-F^*_{K^*K^*}(1-\tau_{K^*})}\alpha^* p j_2 F^*_{L^*K^*} > 0$$

$$x_{19} = -1 < 0$$

$$x_{21} = F^*_G - \frac{(1-\alpha)}{p} j_1 F_{LG} - (1-\alpha^*) j_2 F^*_{L^*G} + \frac{F^*_{K^*G}}{-F^*_{K^*K^*}}(F^*_{K^*} - (1-\alpha^*) j_2 F^*_{L^*K^*} - \delta^*(1 + \frac{\psi^*\delta^*}{2})) - \frac{F_{KG}(1-\alpha)}{-F_{KK}} j_1 F_{LK}$$

$$x_{22} = F^*_{G^*} - \frac{(1-\alpha)}{p} j_1 F_{LG^*} - (1-\alpha^*) j_2 F^*_{L^*G^*} - \delta^*(1 + \frac{\psi^*\delta^*}{2}) + \frac{F^*_{K^*G^*}}{-F^*_{K^*K^*}}(F^*_{K^*} - (1-\alpha^*) j_2 F^*_{L^*K^*} - \delta^*(1 + \frac{\psi^*\delta^*}{2})) - \frac{F_{KG^*}(1-\alpha)}{-F_{KK}} j_1 F_{LK}$$

$$x_{23} = \frac{(1-\alpha)}{p} j_1 > 0$$

$$x_{24} = (1-\alpha^*) j_2 > 0$$

$$x_{25} = \frac{(1-\alpha)}{p} j_3 \lambda \theta > 0$$

$$x_{26} = (1-\alpha^*) j_4 \lambda^* \theta^* > 0$$

$$x_{27} = \frac{F_K}{-F_{KK}(1-\tau_K)} \frac{(1-\alpha)}{p} j_1 F_{LK} > 0$$

$$x_{28} = \frac{-F^*_{K^*}}{-F^*_{K^*K^*}(1-\tau_{K^*})}(F^*_{K^*} - (1-\alpha^*) j_2 F^*_{L^*K^*} - \delta^*(1 + \frac{\psi^*\delta^*}{2}))$$

$$x_{29} = -1 < 0$$

where  $\Delta = x_{01} x_{20} - x_{10} x_{02}$ , and

$$j_1 = \frac{-\lambda\theta}{(r(r-\rho) - \lambda\theta)(1+\tau_c)} > 0, \quad j_2 = \frac{-\lambda^*\theta^*}{(r(r-\rho^*) - \lambda^*\theta^*)(1+\tau_c^*)} > 0,$$

$$j_3 = \frac{(F_L - \tau)j_1^2}{\lambda\theta} > 0, \quad j_4 = \frac{(F_L^* - \tau^*)j_2^2}{\lambda^*\theta^*} > 0,$$

under the assumptions that  $\lambda\theta > r(r - \rho)$  and  $\lambda^*\theta^* > r(r - \rho^*)$ . These conditions ensure that the marginal propensity to consume out of financial wealth is positive - see (27xxii,xxiii). From (28a,b) it can be seen that  $j_1$ ,  $j_2$  are the marginal propensities to consume out of labour income less taxes<sup>1</sup>.

Whether  $\Delta$  is positive or negative depends on whether  $x_{10}$ ,  $x_{20}$  are both positive or both negative. They will both be positive if a higher world interest rate results in an excess demand for both the domestic and foreign good. A higher world interest rate lowers the steady-state stocks of capital and output, lowers steady-state wages, but raises steady-state asset income; the assumption here is that on net there is excess demand.

With  $\Delta > 0$ , the following steady-state impacts can be derived (evaluated at  $\tau_c = \tau_c^* = \tau_K = \tau_K^* = 0$ ):

$$\text{sgn } \frac{dr}{d\tau} = \text{sgn } (x_{01}x_{23} - x_{13}x_{02}) > 0$$

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<sup>1</sup> Definitions  $j_3$ ,  $j_4$  emerge when differentiating consumption with respect to the consumption tax rate and interest rate.

$$\text{sgn } \frac{dp}{d\tau} = \text{sgn } (x_{13}x_{20} - x_{23}x_{10}) ?$$

Figure 2.4a depicts the impact of an increase in the home country's steady-state lump-sum taxes<sup>1</sup>.  $yy$  is the goods market equilibrium locus for the home good and is downward-sloping because as  $r$  rises an excess demand for home output is assumed to result, requiring the relative price of the foreign good,  $p$ , to fall (or the relative price of the home good to rise). Similarly  $y^*y^*$  is the goods market equilibrium locus for the foreign good and is upward-sloping because a higher  $r$  requires a higher  $p$  (or higher relative price for the foreign good) to eliminate an incipient excess demand for the foreign good. An increase in  $\tau$  shifts both loci upward (since  $x_{23}, x_{13} > 0$ ). The initial equilibrium is given by  $E_0$  and final by  $E_1$ . A higher steady-state level of  $\tau$  is associated with a higher stock of government debt,  $b$ , which is net wealth to domestic consumers. Domestic consumers thus increase their steady-state consumption of both goods. The net effect on the terms of trade of an increase in the consumption of both goods is ambiguous.

The steady-state impacts of other taxes are:

$$\text{sgn } \frac{dr}{d\tau_k} = \text{sgn } (x_{01}x_{27} - x_{17}x_{02}) ?$$

$$\text{sgn } \frac{dp}{d\tau_k} = \text{sgn } (x_{17}x_{20} - x_{27}x_{10}) < 0$$

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<sup>1</sup> Recall that there are two components to lump-sum taxes: an exogenous component and an endogenous component. The steady-state derivatives are taken with respect to the exogenous component.



$$\text{sgn } \frac{dr}{d\tau_c} = \text{sgn } (x_{01}x_{25} - x_{15}x_{02}) > 0$$

$$\text{sgn } \frac{dp}{d\tau_c} = \text{sgn } (x_{15}x_{20} - x_{25}x_{10}) ?$$

The impact of higher consumption taxes mimics that of higher lump-sum taxes; that is, a larger stock of government debt is associated with higher consumption tax revenue in the long run. If  $x_{17}$  is sufficiently negative it is possible that the long run interest rate can fall in response to a higher capital income tax rate.  $x_{17} < 0$  requires  $F_K - \alpha j_1 F_{LK} - \delta(1 + \frac{\psi\delta}{2}) > 0$ . That is, a rise in  $\tau_k$  must lower supply (by lowering the stock of capital  $K$  - the first term) by more than it lowers domestic absorption (ie. the sum of the second term - which represents the change in consumption due to a change in labour income - and the third term - which represents steady-state depreciation per unit of capital  $K$ ). The steady-state interest rate must then fall to eliminate an incipient excess demand. This case is depicted in Figure 2.4b.

I turn now to some expenditure impacts. I assume that increases in long run spending are all deficit-financed:

$$\text{sgn } \frac{dr}{dc_g} = \text{sgn } (-x_{19} x_{02}) < 0$$

$$\text{sgn } \frac{dp}{dc_g} = \text{sgn } (x_{19} x_{20}) < 0$$

$$\text{sgn } \frac{dr}{dG} = \text{sgn } (x_{01} x_{21} - x_{11} x_{02}) < 0$$

$$\text{sgn } \frac{dp}{dG} = \text{sgn } (x_{20} x_{11} - x_{21} x_{10}) ?$$

The impact of  $c_g$  is shown in Figure 2.4c. Higher public consumption is associated with a lower long run interest rate. This results because across steady-states, higher  $c_g$  is associated with a lower stock of public debt. An increased demand for domestic goods by the government appreciates the terms of trade. In Buiter (1987) the impact of  $c_g$  on  $p$  is ambiguous. The difference in result is due to the fact that the domestic government in his model purchases both domestic and foreign goods; in this model,  $c_g$  falls exclusively on domestic output.

The impact of a higher long run  $G$  depends on the signs of  $x_{21}$ ,  $x_{11}$ . If these are sufficiently negative, long run global crowding in can occur - as depicted in Figure 2.4d. Their signs are negative if a higher stock of public capital and higher induced stocks of private capital increase long run production but increase steady-state absorption even more so as to produce an incipient excess demand. The steady-state world interest rate must then fall. From  $x_{21}$ ,  $x_{11}$ , the following conditions help improve this possibility: (i) large cross-country spillover effects - particularly on raising the wages of the other country - along with significant enough import propensities ( $\alpha^*$ ,  $1-\alpha$ ); (ii) significant enough depreciation rates so that in steady-state a significant enough share of resources is absorbed into maintaining the steady stocks of capital - but not too much as to crowd out steady-state consumption<sup>1</sup>; (iii) high labour shares of output ( $F_L$ ,  $F_L^*$ ) which help magnify the impact of increased public capital on human wealth; (iv) large

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<sup>1</sup> For normative reasons only. As a positive feature, higher steady-state depreciation contributes to creating incipient excess demand.

$j_1, j_2$ , ie. the marginal propensities to consume out of after-tax labour income.

In the steady-state analyses so far, the impacts on production and consumption are implicit, the focus being on the effects on world market clearing prices. In the next couple of subsections I isolate the impacts on production and consumption, conditional on world prices not having changed. To derive the total effect on production or consumption, one must therefore evaluate the direct and indirect effects of a change in some exogenous variable  $X$  as:

$$\frac{\Delta y}{\Delta X} = \left. \frac{\partial y}{\partial X} \right|_{\text{direct}} + \left. \frac{\partial y}{\partial r} \frac{\partial r}{\partial X} \right|_{\text{indirect}} + \left. \frac{\partial y}{\partial p} \frac{\partial p}{\partial X} \right|_{\text{indirect}}$$

$$\frac{\Delta c}{\Delta X} = \left. \frac{\partial c}{\partial X} \right|_{\text{direct}} + \left. \frac{\partial c}{\partial r} \frac{\partial r}{\partial X} \right|_{\text{indirect}} + \left. \frac{\partial c}{\partial p} \frac{\partial p}{\partial X} \right|_{\text{indirect}}$$

The next two subsections will study some of these "direct" impacts.

#### - Production -

Recall that when  $\dot{q}_p=0$ ,  $(1-\tau_k)F_K = r q_p + \delta(1 + \frac{\psi \delta}{2})$ . Totally differentiating this and substituting the result into the totally differentiated production function  $dY = F_K dK + F_G dG + F_{G^*} dG^*$  (assuming  $dL=0$  since labour is fixed in supply), I obtain the steady-state output effects of public

capital, the capital income tax rate, and the world interest rate (again evaluated at  $\tau_k = \tau_k^* = \tau_c = \tau_c^* = 0$ )<sup>1, 2</sup>:

$$(30) \quad dY = \beta_G dG + \beta_{G^*} dG^* + \beta_{\tau_K} d\tau_K + \beta_r dr$$

$$\text{where } \beta_G = F_G - \frac{F_K}{F_{KK}} F_{KG} > 0, \quad \beta_{G^*} = F_{G^*} - \frac{F_K}{F_{KK}} F_{KG^*} > 0$$

$$\beta_{\tau_K} = \frac{F_K^2}{F_{KK}} < 0, \quad \beta_r = q_p \frac{F_K}{F_{KK}} < 0$$

Alternatively under national planning,  $G$  is endogenous. From  $\dot{q}_g = 0$ ,  $F_G = r q_g + \delta(1 + \frac{\psi\delta}{2})$ . Totally differentiating this and substituting the result into (30), we get:

$$(31) \quad dY = \beta_{G^*} dG^* + \beta_{\tau_K} d\tau_K + \beta_r dr$$

where

$$\beta_{G^*} = F_{G^*} + \frac{F_K}{F_{KK} F_{GG} - F_{KG}^2} [F_{GG^*} F_{KG} + F_{KG^*} F_{KG} - F_{KK} F_{GG^*} - F_{GG} F_{KG^*}] > 0$$

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<sup>1</sup> It is assumed that all cross partials - eg.  $F_{KG}$ , ... - are positive (ie. all inputs are complements) and that principal minors alternate in sign beginning with negative own second partials - eg.  $F_{KK} < 0$ ,  $F_{GG} < 0$ , and so forth.

<sup>2</sup> This type of production function is estimated - in both levels and first differences - in my empirical work with the exception that labour hours is not constrained to be fixed. I find  $\beta_G, \beta_{G^*} > 0$  (corrected for capacity utilization) but tend to find  $\beta_{\tau_K}$  to be insignificant and sometimes positive. The latter finding could indicate the presence of a public policy feedback rule relating capital income tax rates to changes in output growth and/or cycles. It would therefore be desirable to derive and analyze decision rules for taxes also.

$$\beta_{\tau_K} = \frac{-F_K^2(F_{GK}-F_{GG})}{F_{KK}F_{GG}-F_{KG}^2} < 0, \quad \beta_r = q_p \frac{F_K(F_{KK}+F_{GG}-2F_{KG})}{F_{KK}F_{GG}-F_{KG}^2} < 0$$

Under global planning, all capital stocks  $G$ ,  $G^*$ ,  $K$ ,  $K^*$  are endogenous. Setting  $\dot{q}_G=0$ ,  $\dot{q}_G^*=0$ ,  $\dot{q}_K=0$ ,  $\dot{q}_K^*=0$ , we obtain the following steady-state marginal productivity conditions:  $F_K = F_G + pF_G^* = F_{G^*}^* + \frac{F_{G^*}^*}{p} = F_{K^*}^*$ . After totally differentiating them and substituting the results into the global production function, we arrive at:

$$(32) \quad d(Y + pY^*) = \beta_p dp + \beta_r dr + \beta_{\tau_K} d\tau_K + \beta_{\tau_{K^*}} d\tau_{K^*}.$$

The coefficient details are omitted. I find that as long as the production functions  $F$  and  $F^*$  are each (locally) concave,  $\beta_r < 0$ ,  $\beta_{\tau_K} < 0$ , and  $\beta_{\tau_{K^*}} < 0$ . The value of  $\beta_p$  is  $\lesseqgtr 0$  according to whether:

$$\frac{pF_G^*}{-(F_{GG} + pF_{GG}^*)} < \frac{F_{G^*}^*}{-(F_{G^*G^*} + pF_{G^*G^*}^*)}$$

A rise in steady-state  $p$  has ambiguous effects on the steady state value of global output ( $Y + pY^*$ ). The supplies of  $Y^*$  and  $Y$  may change in response to the higher relative price of the foreign good. In particular if  $p$  is higher, the restoration of steady-state marginal productivity conditions (ie.  $\dot{q}_G = \dot{q}_K = \dot{q}_G^* = \dot{q}_K^* = 0$ ) requires that  $G$  rise and  $G^*$  fall<sup>1</sup>, the

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<sup>1</sup> This requires that the sum of pairs of own second partials (ie.  $F_{KK} F_{GG} + F_{GG} F_{G^*G^*} + F_{KK} F_{G^*G^*} + \dots$ ) be greater in absolute value than the sum of pairs of cross partials (ie.  $F_{KG} F_{KG^*} + F_{G^*K} F_{G^*G} + F_{GG^*} F_{GK} + \dots$ ). The reason is that when  $p$  is higher, to maintain  $\dot{q}_G=0 = \dot{q}_K$ , both  $G$  and  $G^*$  must rise (so as to reduce  $F_G$  and  $F_{G^*}^*$ ); on the other hand to maintain  $\dot{q}_G^*=0 =$

intuition being that when  $p$  is higher, the global planner finds it cheaper to use domestic output in production. The above expression states that whether the value of global output rises hinges on the positive output effect of a higher  $G$  on the production of  $Y^*$  being greater than the adverse effect of a lower  $G^*$  on the production of  $Y$ . On the other hand if  $p$  falls, the value of global output rises if the increase in  $G^*$  raises  $Y$  more than the decrease in  $G$  reduces  $Y^*$ . If production functions are identical and symmetric ( $G^*$  enters  $F(\cdot)$  in the same way  $G$  enters  $F^*(\cdot)$ ),  $\beta_p$  is zero.

In both the national and global planner cases, the consumption tax rate does not appear in the reduced-form production functions. This is not to say that consumption taxes have no effect on supply; their effects enter by influencing the world interest rate or terms of trade.

### - Consumption -

In this subsection I limit the analysis to exogenous governments only (the main changes to be made when considering endogenous governments are to incorporate the steady-state marginal productivity conditions discussed above). In order to analyze steady-state consumption and external asset/debt holdings, I work with the following three equations :

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$\dot{q}_p^*$ , both  $G$  and  $G^*$  must fall (so as to raise  $F_G^*$  and  $F_{G^*}^*$ ). The restriction just given is that in the first set of conditions  $\dot{q}_g=0=\dot{q}_p$ , a  $G$  increase dominates a  $G^*$  decrease, while in the second set of conditions  $\dot{q}_g^*=0=\dot{q}_p^*$ , a  $G^*$  decrease dominates a  $G$  increase.

$$(33a) (1 + \tau_c)c = \theta[a + h]$$

$$(33b) (1 + \tau_{c^*})c^* = \theta^*[a^* + h^*]$$

$$(33c) rz = (1 - \alpha)c - \alpha^*pc^*$$

Equations (33a, b) are the home and foreign consumption functions. The LHS of (33c) is the steady-state home country service account balance and the RHS the steady-state trade balance. Totally differentiating (33a-33c), evaluated at zero distortionary taxes, I arrive at:

(34)

$$\begin{bmatrix} c_{11} & 0 & c_{12} \\ 0 & c_{21} & c_{22} \\ -z_2 & -z_3 & 1 \end{bmatrix} \begin{bmatrix} dc \\ dc^* \\ dz \end{bmatrix} = \begin{bmatrix} dG \\ dG^* \\ dc_g \\ dc_g^* \\ dr \\ dp \\ d\tau \\ d\tau^* \\ d\tau_K \\ d\tau_K^* \\ d\tau_c \\ d\tau_c^* \end{bmatrix}$$

$$\begin{bmatrix} c_{13} & c_{14} & c_{15} & 0 & c_{16} & 0 & c_{17} & 0 & c_{18} & 0 & c_{19} & 0 \\ c_{24} & c_{23} & 0 & c_{25} & c_{26} & c_{30} & 0 & c_{27} & 0 & c_{28} & 0 & c_{29} \\ 0 & 0 & 0 & 0 & z_1 & z_4 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where

$$c_{11} = \frac{1}{\theta} > 0, \quad c_{12} = -1 < 0,$$

$$c_{13} = q_p \frac{F_{KG}}{-F_{KK}} - \frac{\delta(1+\psi\delta)}{r} + \frac{1}{r+\lambda} [F_{LG} + \frac{F_{LK}F_{KG}}{-F_{KK}}]$$

$$c_{14} = q_p \frac{F_{KG}^*}{-F_{KK}^*} + \frac{1}{r+\lambda} [F_{LG}^* + \frac{F_{LK}F_{KG}^*}{-F_{KK}^*}] > 0$$

$$c_{15} = -\frac{1}{r} < 0$$

$$c_{16} = \frac{-q_p^2}{-F_{KK}} - \frac{b}{r} - \frac{1}{r+\lambda} [h + \frac{F_{LK}q_p}{-F_{KK}}] < 0$$

$$c_{17} = \frac{1}{r} - \frac{1}{r+\lambda} > 0$$

$$c_{18} = \frac{-q_p F_K}{-F_{KK}} + \frac{F_{KK}}{r} - \frac{1}{r+\lambda} [\frac{F_{LK}F_K}{-F_{KK}}]$$

$$c_{19} = \frac{c}{r} - \frac{c}{\theta} > 0$$

$$c_{21} = \frac{1}{\theta^*} > 0, \quad c_{22} = \frac{1}{p} > 0,$$

$$c_{23} = q_p^* \frac{F^*_{K^*G^*}}{-F^*_{K^*K^*}} - \frac{\delta^*(1+\psi^*\delta^*)}{r} + \frac{1}{r+\lambda^*} [F^*_{L^*G^*} + \frac{F^*_{L^*K^*}F^*_{K^*G^*}}{-F^*_{K^*K^*}}]$$

$$c_{24} = q_p^* \frac{F^*_{K^*G}}{-F^*_{K^*K^*}} + \frac{1}{r+\lambda^*} [F^*_{L^*G} + \frac{F^*_{L^*K^*}F^*_{K^*G}}{-F^*_{K^*K^*}}] > 0$$



$$c_{25} = -\frac{1}{r} < 0$$

$$c_{26} = \frac{-q_p^{*2}}{-F^*_{K^*K^*}} - \frac{b^*}{r} - \frac{1}{r+\lambda^*} \left[ h^* + \frac{F^*_{L^*K^*} q_p^*}{-F^*_{K^*K^*}} \right] < 0$$

$$c_{27} = \frac{1}{r} - \frac{1}{r+\lambda^*} > 0$$

$$c_{28} = \frac{-q_p^* F^*_{K^*}}{-F^*_{K^*K^*}} + \frac{F^*_{K^*K^*}}{r} - \frac{1}{r+\lambda^*} \left[ \frac{F^*_{L^*K^*} F^*_{K^*}}{-F^*_{K^*K^*}} \right]$$

$$c_{29} = \frac{c^*}{r} - \frac{c^*}{\theta^*} > 0, \quad c_{30} = \frac{z}{p^2}$$

$$z_1 = -\frac{z}{r}, \quad z_2 = \frac{(1-\alpha)}{r} > 0, \quad z_3 = -\frac{\alpha^* p}{r} < 0, \quad z_4 = -\frac{\alpha^* c^*}{r} < 0.$$

As long as  $\lambda > 0$ , government debt is viewed as net wealth and  $c_{17}$ ,  $c_{27}$  are positive; that is, in  $c_{17}$ ,  $c_{27}$  the first terms reflect the fact that a higher long run  $\tau$  is associated with a larger long run stock of debt  $b$ , and the second terms indicate that future taxes on human wealth are discounted more heavily. Again I assume  $\theta, \theta^*$  exceed  $r$  for stability reasons, and hence  $c_{19}$ ,  $c_{29}$  are positive. As already noted, higher consumption taxes across steady-states are associated with a larger stock of government debt and greater private consumption. On the other hand, the impacts of capital income taxes are ambiguous because while higher capital income taxes are also associated with a larger steady-state stock of government debt, they lower the steady-state stock of private capital and reduce private wealth.

If depreciation is small enough,  $c_{13}, c_{23} > 0$ , so that  $G$  and  $G^*$  are associated with greater long run consumption. Once again the steady-state absorption share of maintaining public capital must not be "too" large.  $c_{14}, c_{24}$  are unambiguously positive because they are the spillover terms: agents can enjoy greater consumption allowed by an increase in spillover capital without having to finance it through higher long run taxes or maintain it against depreciation.  $c_{16}, c_{26}$  are negative because a higher interest rate works to depress human wealth (via heavier discounting) and non-human wealth by reducing the stock of private capital and by reducing the present value of government bonds.

The determinant of the LHS matrix in (34) (evaluated at  $\tau_c = \tau_c^* = 0$ ) is  $\lesseqgtr 0$  according to whether:

$$(34)' \quad (1-\alpha)\theta + \alpha^*\theta^* \gtrless r$$

(34)' has a 'transfer criterion' interpretation: the LHS is the sum of the marginal propensities to import by both countries; the RHS is the amount of wealth transferred between countries per external claim or liability. To see this, recall that in steady-state the capital account balance of the home country is zero:

$$\text{ie. } rz = (1-\alpha)c - \alpha^*pc^*.$$

In steady-state, a net creditor country ( $z > 0$ ) runs a steady-state trade deficit while a net debtor ( $z < 0$ ) runs a steady-state trade surplus. Differentiating the above with respect to  $z$  results in:

$$r + z \frac{dr}{dz} = (1-\alpha) \frac{dc}{dz} - \alpha^*p \frac{dc^*}{dz} - \alpha^*c \frac{dp}{dz}$$

$$\text{or (34)''} \quad r = (1-\alpha)\theta + \alpha^*\theta^* - z\frac{dr}{dz} - \alpha^*c\frac{dp}{dz}$$

where I made use of the consumption functions  $c=\theta[z + q_pK + b + h]$ ,  $c^*=\theta^*[z^* + q_p^*K^* + b^* + h^*]$ , and  $z^* = -\frac{z}{p}$ . From (34)'', it can be seen that (34)' reflects the effect of an increase in  $\Delta z$  on world import demands absent changes in world prices ( $r, p$ ). Thus (34)' will signal the changes needed in  $r$  and  $p$ .

To illustrate the use of (34)', I abstract from price changes. If  $(1-\alpha)\theta + \alpha^*\theta^* > r$ , an increase in wealth of  $\Delta z$  for the creditor country will produce a more than proportionate increase in the creditor country's steady-state trade deficit. That is, the creditor country raises its import demand while the debtor country reduces its import demand (or demand for the creditor's exports) in response to a loss in wealth  $-\Delta z^*$ . The transfer  $\Delta z$  therefore results in a higher demand for the debtor country's output, working to increase the debtor's terms of trade. Similarly, if  $(1-\alpha)\theta + \alpha^*\theta^* < r$ , a transfer of wealth to the creditor country will produce a less than proportionate expansion in the creditor's trade deficit, resulting in a relatively greater increase in the demand for the creditor country's own good, thus working to increase the creditor's terms of trade.

In summary, the interpretation of (34)' is that if  $(1-\alpha)\theta + \alpha^*\theta^* > r$  (labelled case 1), there is an incipient increase in the demand for the output of the 'transferer' country; if  $(1-\alpha)\theta + \alpha^*\theta^* < r$  (labelled case 2), there is an incipient increase in the demand for the output of the 'transferee' country.

Returning to the system in (34), I substitute the third equation (for  $dz$ ) into the other two equations. This yields two consumption equilibrium conditions. Figures 2.5a, 2.5b depict two possible configurations. In each, the vertical axis measures total home consumption expenditures, the horizontal total foreign consumption expenditures. The  $c_0$  locus is downward sloping because an increase in  $c^*$  requires, across steady-states, a larger stock of net external claims  $z^*$  (or lower  $z$ ) which reduces domestic nonhuman wealth and thus reduces  $c$ . The  $c_0^*$  locus is likewise also downward-sloping. Note that it is possible to ascertain the value of  $z$  from the diagrams. There is a locus of points  $c, c^*$  such that  $z = 0$ :

$$\text{ie. } 0 = rz = (1-\alpha)c - \alpha^*pc^*$$

$$\text{or } \frac{c}{c^*} = \frac{\alpha^*p}{1-\alpha}.$$

This locus is depicted in each figure; it need not of course coincide with the 45° line. Moreover the  $c_0, c_0^*$  need not have intersected at a point where  $z = 0$ . I have chosen  $z = 0$  for expositional reasons. North of the  $z = 0$  locus is the region where  $z > 0$  - that is, for each  $c^*$ , higher steady-state  $c$  is possible only if steady-state  $z$  is higher. South of  $z = 0$  is the region where  $z < 0$ .

Figure 2.5a illustrates case 1, where the home country's consumption equilibrium locus,  $c_0$ , is steeper than the foreign country's consumption equilibrium locus,  $c_0^*$ . Figure 2.5b illustrates case 2, where the slopes are reversed<sup>1,2</sup>. The slope configurations imply that under case 1 there is a

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<sup>1</sup> The slopes are:

higher demand for the debtor country's output when the creditor country accumulates more net external assets, and that under case 2 there is a higher demand for the creditor country's own output when the creditor country accumulates more net external assets. For example under case 1, when  $z > 0$ , the home country's steady-state trade deficit along the  $c_0$  locus is larger than the foreign country's steady-state trade surplus along the  $c_0^*$  locus. Thus disequilibria in this region are characterized by higher net import demands<sup>1</sup>, consistent with the transfer criterion  $(1-\alpha)\theta + \alpha^*\theta^* > r$ . Moreover, as  $z$  increases in this region, the home country's steady-state trade deficit along  $c_0$  and the foreign country's steady-state trade surplus along  $c_0^*$  widen. But since the  $c_0$  locus is steeper than the  $c_0^*$  locus, the consequent increase in the home country's import demand is more pronounced than the consequent increase in the foreign country's export supply, and there is an incipient excess demand for foreign output. It is in this sense that when net external assets are accumulated by a creditor, a net increase in the demand for the debtor's output results. In contrast, under case 2, when  $z > 0$ , the situation is reversed: the home country's steady-

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$$\left. \frac{dc}{dc^*} \right|_{c_0 \text{ locus}} = \frac{-\frac{\theta\alpha^*p}{r}}{\left[1 - \frac{\theta(1-\alpha)}{r}\right]}, \quad \left. \frac{dc}{dc^*} \right|_{c_0^* \text{ locus}} = \frac{\left[1 - \frac{\theta^*\alpha^*}{r}\right]}{-\frac{\theta^*(1-\alpha)}{rp}}$$

<sup>2</sup>

I restrict  $(1 - \frac{\theta^*\alpha^*}{r})$ ,  $(1 - \frac{\theta(1-\alpha)}{r})$  to be positive. These prevent the marginal propensity to import out of service income from exceeding 100%, as this would not be sustainable in steady state. Moreover, the  $c_0$ ,  $c_0^*$  loci would not be downward sloping, and the marginal propensity to consume from other income would be negative.

<sup>1</sup> ie. the sum of home and foreign import demands is greater than the sum of home and foreign export supplies.

state trade deficit along the  $c_0$  locus is smaller than the foreign country's steady-state trade surplus along the  $c_0^*$  locus. Disequilibria in this region involve lower net import demands<sup>1</sup>, and as net external assets are accumulated by a creditor, a relatively greater increase in demand for the creditor country's own output results.

I turn now to some comparative steady-state impacts. I examine the impacts of deficit-financed changes in long run public consumption and public investment on long run consumption and external assets, conditional on  $r$  and  $p$  remaining unchanged. I then turn to the impacts that changes in the world interest rate and terms of trade have on long run consumption and external assets. The initial steady-state  $z$  is assumed to be zero.

The impacts of public consumption are in Figures 2.6a and 2.6b. With  $c_{15}$  negative, the home consumption equilibrium locus shifts down and to the left: *ceteris paribus*, an increase in home public consumption takes resources away from private consumption<sup>2</sup>. Under case 1, however, the unilateral increase in home public consumption can 'crowd' in own private consumption, at the expense of foreign private consumption, while under case 2 the same policy can increase foreign private consumption at the expense of home private consumption. The intuition is that the public consumption shock transfers wealth to the foreign economy (as domestic absorption is higher), but under case 1, the transfer significantly raises the foreign economy's import demand for domestic output while the loss of home wealth reduces the home country's demand for foreign relative to

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<sup>1</sup> ie. the sum of home and foreign import demands is less than the sum of home and foreign export supplies.

<sup>2</sup> Note that if public consumption were balanced-budget (lump-sum tax) financed instead, the  $c_0$  locus would still shift to the left on net (ie. the sum of  $c_{15}$  and  $c_{17}$  is still negative).

domestic output; thus across steady states, the actual equilibrium occurs in the region  $z > 0$ . The home economy actually ends up being the net creditor and gets to enjoy steady-state trade deficits. Under case 2, the 'transfer' to the foreign economy results instead in an overall net increase in the demand for foreign output; the foreign economy is thus the long run creditor.

Figures 2.6c, 2.6d turn to the impacts of a higher long run stock of domestic public capital. This time regardless of the direction of the transfer criterion, long run private consumption of both countries can rise. The final impact on long run net external assets is ambiguous, though the diagrams have been drawn so that the new equilibria remain on the  $z=0$  locus. It is also possible that if the size of the shifts is asymmetric, one country's long run consumption can increase at the expense of the other's. That both the  $c_0, c_0^*$  loci shift out is attributable to the presence of spillovers. In the absence of spillovers an increase in long run  $G$  leads, under case 1, to an equilibrium point like  $E_1$  in Figure 2.6a. Again because of the transfer effect the increase in home country wealth (from increased production and income) results in an overall increase in demand for foreign output and an equilibrium where the home economy ends up being a steady-state debtor. The spillover effect thus works to boost foreign consumption so that the world equilibrium may occur at a point like  $E_2$ .

The long run impacts of a higher world interest rate on home and foreign consumption - evaluated at  $z = 0$  - can be obtained from (34):

$$(35a) \quad \frac{dc}{dr} = M^{-1} \left[ c_{16} \left( \frac{1}{\theta^*} - \frac{\alpha^*}{r} \right) - c_{26} \frac{\alpha^* p}{r} \right]$$

$$(35b) \quad \frac{dc^*}{dr} = M^{-1} [ c_{26} \left( \frac{1}{\theta} - \frac{1-\alpha}{r} \right) - c_{16} \frac{1-\alpha}{pr} ]$$

where

$$c_{26} = \frac{\partial(b^* + q_p^* K^* + h^*)}{\partial r} < 0, \quad c_{16} = \frac{\partial(b + q_p K + h)}{\partial r} < 0$$

and  $M$  is the determinant of (34), and  $\text{sgn } M \geq 0$  as  $r \geq (1-\alpha)\theta + \alpha^*\theta^*$ .

Now, if  $c_{26} = c_{16}$ , both  $\frac{dc}{dr} < 0$ ,  $\frac{dc^*}{dr} < 0$ . This can be shown as follows. First,  $c_{26} = c_{16}$  if the initial stocks of comprehensive wealth in both countries are equal - ie.  $a + h = a^* + h^*$ . Secondly, since  $z = 0$ , there is initially balanced trade:  $rz = 0 = (1-\alpha)c - \alpha^*pc^*$ . These two points imply that  $(1-\alpha)\theta = \alpha^*p\theta^*$ . Substituting this into (35a) and (35b) will show that the numerator (excluding  $c_{26}$  or  $c_{16}$ ) has the same sign as the denominator. Since  $c_{26}, c_{16}$  are negative, this produces the desired result.

Instead if (i)  $|c_{26}| \ll |c_{16}|$

$$(a) \quad \frac{dc}{dr} > 0, \quad \frac{dc^*}{dr} < 0 \quad \text{when } r < (1-\alpha)\theta + \alpha^*\theta^*$$

$$(b) \quad \frac{dc}{dr} < 0, \quad \frac{dc^*}{dr} > 0 \quad \text{when } r > (1-\alpha)\theta + \alpha^*\theta^*$$

and if (ii)  $|c_{26}| \gg |c_{16}|$



$$(a) \frac{dc}{dr} < 0, \frac{dc^*}{dr} > 0 \quad \text{when } r < (1-\alpha)\theta + \alpha^*\theta^*$$

$$(b) \frac{dc}{dr} > 0, \frac{dc^*}{dr} < 0 \quad \text{when } r > (1-\alpha)\theta + \alpha^*\theta^*$$

All of these possibilities are shown in Figures 2.7a, 2.7b. The final impacts depend on the values of  $c_{16}$ ,  $c_{26}$ . These coefficients represent the effects of a change in the world interest rate on non-human wealth (excluding  $z$  and  $z^*$ ) and human wealth in the home and foreign country respectively. Typically if a higher long run world interest rate depresses home and foreign wealth evenly, both countries will be consuming less in the long run, regardless of the direction of the transfer criterion. However if the higher long run world interest rate has an uneven impact on home and foreign wealth, there may be a possibility that one country's consumption rises at the expense of the other's; these are shown by points 1 and 3 in Figures 2.7a, 2.7b. Which country it is that gets to have the higher long run consumption level depends on the transfer criterion and on whose wealth it is that gets depressed more. If it is the home country whose wealth is depressed more  $|c_{26}| < |c_{16}|$ , the foreign country is relatively wealthier, and under case 1 of the transfer criterion, there will be a relatively greater increase in demand for the home country's output so that in the long run the home country will be a creditor and enjoy higher long run consumption. Instead under case 2 of the transfer criterion, there will be a relatively greater increase in demand for foreign country output, and an opposite long run scenario will emerge.

Finally the long run impact of a change in the terms of trade (evaluated at  $z=0$ ) is:

$$(36) \quad \frac{dc}{dp} = M^{-1} [ -c_{12} z_4 (c_{21} + z_3 c_{22}) + c_{12} c_{22} z_3 z_4 ]$$

where  $M$  is the determinant of (34), and the numerator is negative.

Thus  $\frac{dc}{dp} > 0$ , if  $M < 0$  (ie.  $r < (1-\alpha)\theta + \alpha^*\theta^*$ ) and  $\frac{dc}{dp} < 0$ , if  $M > 0$ , as shown in Figures 2.7c, 2.7d. The intuition is that an improvement in the terms of trade for the foreign country (ie. a  $p$  increase) makes the foreign country wealthier, and as before, if case 1 holds, will result in a net increase in the demand for home country output and in the home country becoming a creditor in the long run. If case 2 holds, the opposite outcome will result. Note that the  $z = 0$  locus rotates to the left since  $p$  is higher.

### B. Dynamic Simulations

The simulation experiments presented here are of three types: (a) public spending shocks (namely, public investment ( $i_g$ ) and public consumption ( $c_g$ ) shocks); (b) intertemporal redistributions of lump-sum taxes; (c) revenue-neutral distortionary tax reforms.

For (a) I have examined both tax and bond-financed increases in public spending; permanent and temporary; anticipated and unanticipated; unilateral and globally coordinated. For (b) I have considered current tax cuts followed by future tax increases; current tax increases followed by future tax cuts; unilateral and globally coordinated intertemporal

redistributions. For (c) I have examined revenue-neutral switches from capital income taxation to consumption taxation; from consumption taxation to capital income taxation; unilateral and globally coordinated tax reforms.

For each experiment I have investigated alternative specifications and assumptions, of which there are several to consider: (i) type of government ('exogenous', national planner, global planner); (ii) nature of returns to scale (constant over all inputs, domestic inputs, or private inputs) and associated issue of whether public capital is charged on a user fee basis, rented, or provided for free; (iii) whether an externality is present or absent, and if present, whether the externality is symmetric (countries find own public capital and spillover capital equally productive) or asymmetric (countries find own public capital more productive for own country production); (iv) parameter values and initial conditions - for example, the value of the intertemporal elasticity of substitution ( $1/\sigma$ ), the rates of time preference, the transfer criterion condition, and initial debtor/creditor status (ie. value of initial  $z$ ).

The experiments and specification tests were all conducted and I shall report on a select few, along with comments as to how the results would be affected, if at all, by changes in specification. Appendix 2.2 has the necessary simulation details. The nonlinear model given in Table 2.1 is linearized around a steady-state consistent with the assumed parameter values and initial conditions. For each policy experiment, the initial and final steady-states will be compared, and the time paths between the two steady-states will be plotted.

The qualitative results do not depend on the initial value of  $z$ . The nature of the time paths and the ratios of final to initial steady-state values

are similar whether  $z = 0$  or nonzero. For this reason I assume initial  $z = 0$ . I also assume initial  $p = 1$ . Furthermore, I report the results of only case 1 of the transfer criterion (ie.  $(1-\alpha)\theta + \alpha^*\theta^* > r$ , where initial  $r = 0.05$ ). The simulation results are consistent with the predictions of whichever configuration of the transfer criterion is assumed. I have also experimented with  $\sigma = \sigma^* = 1.5$  and  $\sigma = \sigma^* = 0.5$  and have found movements in  $\theta, \theta^*$  to be quite negligible<sup>1</sup>. I therefore assume  $\sigma = \sigma^* = 1$  which implies that  $\theta, \theta^*$  are constant (independent of interest rates).

The output elasticities for the production function range as follows: under CRTS across all inputs,  $\beta_G = 0.1, \beta_{G^*} = 0.1, \beta_K = 0.2, \beta_L = 0.6$ ; under CRTS across domestic inputs,  $\beta_G = 0.1, \beta_{G^*} = 0.1, \beta_K = 0.3, \beta_L = 0.6$ ; under CRTS across private inputs,  $\beta_G = 0.1, \beta_{G^*} = 0.1, \beta_K = 0.35, \beta_L = 0.65$ ; and likewise for the foreign production function.

#### (a) Public Spending Shocks

##### (i) Permanent, Unanticipated Balanced Budget Increase in $ig$ .

Consider a 25% increase in domestic long run public investment above its initial steady-state level, financed by higher long run lump sum taxes. Assume there are CRTS across all private inputs and no distorting taxes. In the long run, both home and foreign private capital stocks are higher (up 7.22% from its initial steady-state level), while only the home country's stock of public capital is 25% higher. Long run production is increased by 7.21% in both countries. Long run private consumption is higher in both countries, though relatively higher in the foreign country.

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<sup>1</sup> This may be different in non-linear simulations.

This last outcome has to do with the foreign economy not sharing any burden, in the long run or short run, of financing the additional creation of  $G$ . With the lump-sum tax feedback rule in place, home taxes initially rise on impact above their (higher) long run levels. Domestic human wealth falls on impact - hence the short run decrease in home consumption. Gradually both  $h$  and  $c$  rise as the effects of a higher  $G$  affect production and labour income. Foreign wealth rises on impact in response to news of a higher future stream of income.

The long run stock of domestic government debt is lower, since the long run stock of  $G$  is higher. There is some movement in the foreign stock of government debt  $b^*$  and foreign taxes  $\tau^*$  along the transition; this has to do with the interest rate being endogenous. As it changes so too do government interest payments. In the long run the domestic terms of trade appreciates, since there is a relatively higher long run demand for domestic output. The reason for this is that while consumption and private investment in both countries are higher, only domestic public investment is higher, so that steady-state domestic absorption ends up being relatively higher than that abroad. In the short run also, domestic absorption is higher because of greater domestic investment at home, causing the home economy to run current account deficits along the transition. In the long run, the foreign economy is the net creditor and has the relatively higher steady-state level of private consumption. The time paths for this experiment are given in Figure 2.8.

I now comment on the results of other specifications. When the returns to scale are changed there are no qualitative changes in the time paths, but there are quantitative changes in the final steady-state values. The general rule has been that the larger the aggregate returns to scale, the

larger the final impacts on output. For example when CRTS prevails across domestic inputs, long run foreign and domestic stocks of private capital increase by 6.17% of their initial steady-state levels and long run output in each country rises by 6.54%. These increases are less than the increases under CRTS across private inputs. When CRTS prevails across all inputs, long run private capital stocks rise by 3.67% and long run output in each country rises by 4%. One rather noticeable outcome in this last case is that home consumption actually falls in the long run. That is, with CRTS across all inputs, long run output does not rise sufficiently to raise long run labour income to offset the higher long run financing burden of increased public capital. (The foreign economy does, however, consume more in the long run, and is a free-rider). There must consequently be sufficient scale economies at the aggregate level to make public investment worthwhile for the home economy.

When governments either lend their stock of capital or charge a user fee for capital services, the long run stock of government debt increases in both countries. The reason is that as  $G$  increases, the resulting output increases induce fiscal revenues. In the long run higher government revenues are associated with a larger stock of government debt.

Returning to the case where CRTS prevails across private inputs, I illustrate the impacts of the same increase in  $i_g$  when no spillovers exist. Figure 2.9 plots the time paths. Essentially only the home country stocks of capital and output increase. In the long run the home economy's terms of trade deteriorates as there is a relatively greater increase in the supply of home output. The home economy's long run consumption is higher and is supported by a higher steady-state stock of external assets. However, along the transition to long run the home economy runs current account

deficits as absorption is higher (due to the home investment boom). This and the rising foreign terms of trade enable the foreign country to enjoy higher consumption temporarily. On impact domestic consumption and human wealth do not decline despite the higher short run taxes; home production and labour income increase more when there are no spillovers since global savings do not flow to the foreign country to help finance private capital accumulation there. As a result domestic  $K$  and  $y$  both increase by 10% of their initial steady-state levels.

A case intermediate to this one of no spillovers and spillovers with symmetric public capital output elasticities is the case of asymmetric output elasticities. Here 'own' country public capital stocks are more productive than the spillover public capital stocks. For example,  $\beta_{G^*}=0.05$ ,  $\beta_G=0.1$  and  $\beta^*_{G^*}=0.1$ ,  $\beta^*_G=0.05$ . The time paths are qualitatively similar (given the same  $i_g$  shock), but home private capital, output, and consumption all rise faster and reach higher long run levels than their foreign counterparts. The long run stock of net external assets  $z$  is negative, and the home country's terms of trade deteriorates. The latter occurs because the relative supply of home output is larger in the long run, and  $z < 0$  because, despite  $c > c^*$ , the  $p$  increase is relatively larger (ie.  $rz = (1-\alpha)c - \alpha^*pc^* < 0$ ).

(ii) Temporary, Immediately Implemented Balanced Budget Increase in  $i_g$ .

Figure 2.10 assumes the same experimental conditions underlying Figure 2.8 (ie. CRTS across all private inputs, and spillovers with 'symmetric' output elasticities of  $G$ ,  $G^*$ ). Time  $t_0$  is the date of implementation, and  $t_1$  is a future date at which the policy is reversed.

In the long run all variables return to their initial steady-state levels. There are no permanent effects. Capital accumulation is followed by decumulation, current account deficits (in the home country) followed by current account surpluses. The home terms of trade first increases on impact in response to a higher demand for domestic output, then depreciates between  $t_0$ ,  $t_1$  as foreign absorption is higher along this transition phase (primarily because  $c^* > c$  during this period). At  $t_1$  there is a discrete cut in public investment and a decline in demand for domestic output, causing the home terms of trade to decrease on impact at  $t_1$ . Thereafter the home terms of trade appreciates as domestic absorption is higher during the rest of the transition. The prices of capital  $q_p$ ,  $q_p^*$  take into account the future policy reversal, jumping only at the date of announcement, and by the time the policy reversal takes place at  $t_1$ , the prices of capital overshoot their long run equilibrium levels so as to bring about the expectation that they will be rising towards their long run levels.

(iii) Unanticipated, Permanent Deficit-Financed Increase in  $i_g$

The same increase in long run domestic public investment is financed by bonds. The time paths (see Figure 2.11) are strikingly similar to those in the balanced budget case. The reason is that the lump sum tax feedback rule automatically raises current lump sum taxes on impact, just as if the public investment were financed by taxes. The key difference is that the home country's long run lump sum tax burden is unchanged. As a result there is no gap between foreign and home consumption in the long run.

Moreover the final steady-state magnitudes are larger than they were under balanced-budgets: the private stocks of capital are 8% above their initial steady-state levels and supplies of output 7.5% above their initial



steady-state levels. The home economy is also a net creditor in the long run despite  $c = c^*$ . This is attributable to the long run terms of trade improvement for the home country; given a higher steady-state stock of domestic public capital  $G$ , unchanged  $G^*$ , and  $K = K^*$  and  $y = y^*$ , domestic absorption is on net higher than foreign absorption in the long run. Consequently there is a relatively higher demand for domestic output in the long run<sup>1</sup>.

A higher stock of government capital is associated with a smaller stock of government debt. But note that the two countries have the same long run stock of financial wealth and private capital; this implies that, for the home country, net external debt and government debt have moved in the same direction. In particular,  $z$  increased (or  $-z$  decreased) and  $b$  decreased. Thus a long-run deficit financed increase in public capital (which generates spillovers) works to eliminate the twin "debts".

#### (iv) A Coordinated Balanced-Budget Increase in $i_g, i_g^*$

In this experiment long run public investments in both countries are increased by 25% of their initial steady-state levels. The result of this simultaneous expansion is to magnify the effects that were generated by a unilateral expansion and to create offsetting changes in the terms of trade and net external asset positions. The stock of private capital in each country increases by 14.4% and output in each country by 14.38% of their respective initial steady-state levels. Global absorption is higher in the long run, necessitating a decline in long run world interest rates.

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<sup>1</sup> Note that it would be harder to make these inferences if domestic capital accumulation involved foreign country output as well as home output.

(v) Public Consumption Shocks  $c_g$ :

Domestic long run public consumption is increased by 25% of its initial steady-state level. The balanced-budget case is shown in Figure 2.12 and deficit-financed case in Figure 2.13. The primary purpose of this set of experiments is to contrast unproductive public spending with productive. The important contrast is that  $c_g$  works through "pecuniary" channels of transmission. Under a balanced-budget expansion, interest rates rise, private stocks of capital decumulate, and human wealth and private consumption decline. Foreign consumption however does rise in the long run. This is attributable to the home economy increasing its absorption and running current account deficits along the transition, enabling the foreign country to be a creditor in the long run. This is also mirrored in long run foreign financial wealth being larger than domestic financial wealth. In the long run the home country's terms of trade appreciates as the higher long run  $c_g$  makes domestic absorption relatively higher than that abroad. This depends on the home country's public consumption purchases falling on domestic output only.

Under a deficit financed expansion in home public consumption, the lump sum tax feedback rule has the effect of raising taxes in the short run and generating government surpluses along the transition. The long run stock of domestic government debt is lower and thus long run interest rates are lower. This results also in a very modest (1%) increase in long run private capital in each country. Total financial wealth falls for both countries, however. In the home country this arises because government bond holdings are lower; in the foreign country this arises because its external asset holdings are lower. Thus both home and foreign long run private consumption decline.

The reason long run  $z$  rises is that, along the transition, the home consumption path lies below the foreign consumption path, since the home country is undergoing a period of higher taxes, and foreigners are benefitting from higher financial wealth (due to higher  $q_p^*K^*$ ) and human wealth (due to higher expected labour income and lower interest rates). The home country thus runs current account surpluses. In the long run, since domestic and foreign financial wealth holdings are equal, and since the return of domestic  $\tau$  to  $\tau_0$  induces human wealth stocks to converge, long run private consumption in each country is the same. Since home  $c_g$  is higher, home steady-state absorption is higher, and thus the home country's terms of trade is also higher in the long run.

#### (b) Intertemporal Redistributions of Lump Sum Taxes

In the remaining group of experiments I will focus attention on endogenous government behaviour. Public investments,  $i_g$  and  $i_g^*$ , are now endogenous. To alter the stocks of public sector capital, governments will have to turn to other 'exogenous' policy instruments. They are: (1) lump sum taxes; (2) distortionary taxes (consumption taxes or capital income taxes). This section focuses on (1) and the next subsection (c) will deal with (2). Public consumption in each country is held fixed throughout. I also assume CRTS across private inputs and that spillovers are present.

##### (i) Unanticipated, Permanent Increase in Long Run Taxes $\tau_0$ .

Suppose the home country unilaterally increases  $\tau_0$  by 25% of its previous steady-state level. In order for long run future taxes to be higher

there must be transitorily lower short run taxes and a rise in budgetary deficits which translate into a higher stock of long run government debt, the higher long run interest payments on which need to be financed by higher steady-state taxes. The lump sum taxation feedback rule governs this intertemporal redistributive shift of the burden of taxation to the future.

This experiment was performed on all three types of governments: exogenous (for comparison), national planner, and global planner. The time paths of the variables under each type of government were qualitatively similar, though quantitatively, the final steady-state magnitudes tended to differ, which I will later discuss. A representative example for the case of a national planner is shown in Figure 2.14<sup>1</sup>.

In the long run the stock of home country government debt is permanently higher, leading to higher long run world interest rates. Private capital stocks are therefore displaced in the long run, as are public capital stocks (not drawn). Public investments and capital stocks react endogenously and decline in order to balance the marginal productivities

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<sup>1</sup> In this instance there were no qualitative differences in results over how governments provided public capital - ie. freely or on a rental/user fee basis. The long run association between the stock of public capital  $G$  and government debt  $b$  is the same whether the government charges for the use of public capital or not. This is merely due to the parameter values assumed. Recall from section III that a long run positive association between  $G$  and  $b$  emerges if  $F_G > \delta(1 + \frac{\psi\delta}{2})$  - that is, if marginal government revenues from public capital exceed steady-state expenditures needed to maintain public capital. Under our parameter values  $\delta(1 + \frac{\psi\delta}{2}) = 0.11 > F_G (= 0.1 \frac{Y}{G})$ , implying a long run negative relationship between  $b$  and  $G$  regardless of whether governments charge  $\xi = F_G$  or zero.

between private and public capital. Production in both countries also declines. During the transition to long run, the home country's current account goes into deficit as consumers increase their spending in response to lower short run taxes. Foreign consumption also declines since the value of capital ( $q_p * K^*$ ) falls, causing a drop in foreign financial wealth. The private consumption paths are "twisted" eventually as higher domestic taxes take effect and as foreigners gradually gain back wealth from their accumulation of external assets.

(ii) Unanticipated Coordinated Long Run Tax Cuts in  $\tau_0, \tau_0^*$ .

The previous policy led to a long run decline in the stocks of public capital in both countries and to a worsening of the net external asset position of the country which unilaterally pursued the policy. Suppose instead the national planners coordinated to do the reverse policy: namely decrease long run lump sum taxes in each country by 25% of their initial steady-state levels. The time paths are given in Figure 2.15.

To achieve lower long run taxes, both economies must undergo a transitional period of budgetary surpluses. In the long run public debt stocks decline as do world interest rates. Both public capital and private capital increase in the long run. There are no changes in the terms of trade or in current account balances, owing to offsetting pressures. Consumption falls on impact as human wealth is lower (since short run taxes are raised). In the long run taxes decline, human wealth recovers, output increases, and long run consumption increases. If the home country were to implement this long run tax cut policy unilaterally, only its long run private consumption would be higher; long run foreign private consumption would

be crowded out as the home country would gain external assets in the long run.

I now discuss the sensitivity of the results to the type of government assumed. There are no qualitative differences but there are quantitative. Typically if a fiscal policy change is expansionary (such as a long run tax cut), the global planner economy produces the greatest increases in long run output and consumption, secondly the national planner economy, and thirdly the exogenous government economy. Conversely if a contractionary fiscal policy is pursued (such as an increase in long run taxes), the global planner economy produces the greatest contraction in long run output and decline in consumption, secondly the national planner economy, and thirdly, the exogenous government economy. Another interpretation of this is that to achieve a given amount or degree of "contraction", the global planner needs only to raise a smaller amount of long run taxes than does a national planner, who in turn needs only to raise a smaller amount of long run taxes than does an exogenously behaving government. To achieve a given amount or degree of expansion, the global planner needs only to reduce a smaller amount of long run taxes than does a national planner, who in turn needs only to reduce a smaller amount of long run taxes than does an exogenously behaving government.

The intuition is that in the exogenous government economy, public investment does not react to changes in the world interest rate or to differences in the marginal productivities of public and private capital. It is the economy that is the least responsive to "fundamentals". The global planner economy is the most responsive; resources are shifted constantly and thoroughly so as to preserve global efficiency conditions at all times.

To provide an idea of how expansionary or contractionary the same policy changes can be under different types of government, I report the relative magnitudes of the final steady-state impacts in Tables 2.2 and 2.3. Table 2.2 refers to the experiment described earlier where the home country unilaterally raises long run taxes. Each entry is the ratio of the final steady-state value of a variable in question to its initial steady-state value. Table 2.3 refers to the experiment where both countries simultaneously lower long run taxes. In Table 2.2, the global planner creates the most contraction for the same lump-sum tax increase. Should the global planner want to achieve the same contraction shown in the exogenous government case, it could judiciously choose a lower long run tax increase. In Table 2.3, the global planner produces the most expansion for a given policy change compared to the other types of government.

### (iii) Pursuit of 'Asymmetric' Redistribution of Lump Sum Taxes

In this experiment the home country raises long run taxes while the foreign country reduces long run taxes by the same amount. Figure 2.16 depicts the time paths. This fiscal policy programme neutralizes each country's effect on interest rates. Thus the prices of capital, stocks of capital, and output levels do not move. Movements only occur in consumption related variables. In the short run, domestic human wealth rises (since short run domestic taxes are lower), causing home consumption to increase and home current account deficits to be triggered, while the opposite takes place in the foreign country. In the long run the foreign economy is a net creditor and gets to enjoy greater long run consumption.

### (c) Revenue-Neutral Tax Reforms

The experiments here are designed to see how endogenous public capital formation responds to variations in distortionary taxes. In particular I look at revenue neutral switches from capital income taxation to consumption taxation, and vice versa. The policy change is unanticipated and revenue-neutral with reference to the initial steady-state; as time progresses, tax bases will change, and the amount of revenue raised will be endogenously determined<sup>1</sup>. The production functions are CRTS across private inputs.

#### (i) Unilateral Switch from Capital Income Taxation to Consumption Taxation

The home country is assumed to initiate the reform unilaterally. The consequences are depicted in Figure 2.17. The switch stimulates private investment and public investment. Note that capital accumulation also takes place in the foreign economy. This occurs because of 'spillovers'. A higher stock of domestic public capital increases foreign productivity, as  $G$  is an input in the foreign production function. This causes  $q_p^*$  to increase and stimulate foreign private and public investment. Foreign capital and output growth are not as significant as the home economy's since foreign capital income taxes are still in place.

During the transition, the home country experiences budgetary surpluses as the long run stock of government debt is lower, an outcome

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<sup>1</sup> An alternative experiment would be to fix government debt for all time and let the distortionary tax instrument that is not reduced or eliminated by tax reform to be an endogenous variable.



that is consistent with a higher long run stock of public capital. In the foreign country the increase in  $G^*$  is modest but the increase in  $c^*$  and  $K^*$  induces higher foreign government (distortionary tax) revenues so that long run foreign government debt is higher.

During the transition the home economy also runs current account deficits, as domestic absorption is higher (particularly because of the investment boom at home). In the long run  $z < 0$ , and because there is a relatively greater supply of home goods in the long run, the home country's terms of trade declines in the long run. Had there been no spillovers the home country's terms of trade and external asset position would have deteriorated more. The spillover channel allows foreign absorption to rise somewhat along the transition and cushion the adverse impacts on the home country's terms of trade and balance of payments.

#### (ii) Unilateral Switch from Consumption Taxation to Capital Income Taxation

Again the home country unilaterally undertakes the reform. Figure 2.18 depicts the time paths. This time public and private capital stocks decumulate. The home country temporarily enjoys a 'consumption boom' because of the consumption tax cut. Again because of the presence of spillovers, the stocks of foreign public and private capital also decline, but not by as much. Thus along the transition, the home country's decline in absorption is greater than that of the foreign country. Consequently the home country runs current account surpluses and becomes a long run creditor; its terms of trade appreciates in the long run because of the relatively shorter supply of domestic output in the long run.

Domestic government debt is higher in the long run since the stock of domestic public capital is lower. The stock of foreign government debt is lower since foreign government revenues (from distortionary taxes) are lower, owing to the decreases in  $c^*$  and  $K^*$ .

### (iii) Globally Coordinated Tax Reforms

Under globally coordinated tax reforms the impacts on net external assets and terms of trade cancel. Figures 2.19, 2.20 show the global planner's coordinated version of the revenue-neutral tax reforms. The time paths of all the variables are identical for each country and the variables mirror the solid curves shown in the 'unilateral' reform cases (ie. Figures 2.17, 2.18). The impact effects, however, are larger under coordination. For example, the decline in world interest rates following a global switch to consumption taxation is greater than the decline following a unilateral switch. Similarly the rise in world interest rates following a global switch to capital income taxation is greater than the rise following a unilateral switch.

## VI. Conclusion

This chapter has studied public investment and technological externalities arising from public sector capital in the context of an open-economy. Both the small open-economy and two-country models were used for illustration. Government investment decisions were either specified exogenously or derived optimally within a utilitarian social planner framework. A government could either provide the services of

public capital for free or, if private factors of production do not fully exhaust national output, charge an efficient user fee or lend its capital to the private sector. A global planner could devise a system of international side payments which internalizes the cross-country spillovers from public capital.

Incorporating public investment has revealed a number of contrasts with 'unproductive' government spending. In a small open-economy a public consumption shock lowers long run private consumption and has an uncertain effect on external assets. A public investment shock, however, increases long run private consumption, although the effect on external assets is still ambiguous. Public investment works to augment supply, incomes, and lifetime personal wealth.

In a two-country model an increase in home country public consumption, however financed, lowers long run home private consumption; if tax-financed, long run foreign private consumption can rise since this policy turns the domestic economy into a long run debtor. On the other hand, an increase in the long run stock of home country public capital raises private consumption and crowds in private capital in both countries (provided spillovers are present). Long run foreign private consumption, however, tends to be relatively higher since foreigners do not bear any of the burden of financing the increased stock of an essentially global public good. When no spillovers exist, the benefits accrue solely to the home country. In fact the gains are much larger to the home country because the supply of global savings do not flow to the foreign country to help finance capital accumulation there as would otherwise occur if spillovers exist to stimulate foreign capital formation. An important qualification to the public capital expansion results is that steady-state

depreciation expenses must not be so large as to absorb a large share of steady-state resources. An increase in long run output is insufficient to make an investment in the long run worthwhile if the rise in steady-state expenses necessary to maintain the larger stocks of capital should leave less available for private consumption.

When public investment is endogenous, optimizing governments influence the growth of public capital through changes in tax policy. For example, globally coordinated decreases in long run lump sum taxes stimulate public and private capital accumulation worldwide, and raise long run private consumption in both countries. If only distortionary taxes exist, either a unilaterally implemented or globally coordinated conversion to consumption taxation crowds in public and private capital, and raises long run private consumption in both countries.

In this chapter I have tended to focus on private consumption more than on international competitiveness and external balances. While the consequences for the current account and terms of trade are vital, they are somewhat incidental to what happens to consumption in this model. For instance, it is possible for a country to be a long run debtor, experience a long run terms of trade loss, and still have a higher long run steady-state level of consumption than that of another country<sup>1</sup>. This occurs if the country's long run output supply is relatively larger and the country must undergo current account deficits (increased absorption) along the transition to achieve that state. The point is that while developments pertaining to the

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<sup>1</sup> For a steady-state debtor,  $rz = (1-\alpha)c - p\alpha^*c^* < 0$  and  $z < 0$ , or  $(1-\alpha)c < p\alpha^*c^*$ . If  $\alpha^*p > (1-\alpha)$ , it is easy to find values for  $c, c^*$  such that  $c > c^*$ ; for example, let  $\alpha^*=(1-\alpha)$ ,  $p=1.5$ ,  $c=100$  and  $c^*=90$ .

current account and terms of trade do often determine who enjoys the higher long run consumption, they do not always. For this reason I have viewed them as "means, not ends".

Several extensions come to mind. First, distortionary taxes have been treated exogenously. The social planner framework could be modified to allow governments to choose tax rates, as well as spending, optimally. A second extension is to consider optimal fiscal policy design under 'uncertainty'. A third is to add more countries. This would allow certain asymmetric features to be modelled, such as a case where some countries generate spillovers but enjoy no spillins, and vice versa. How these several countries might share the burden of financing global public goods should also be raised, and related to international tax harmonization. The local public finance literature on 'tax exporting' and public goods spillovers across communities may provide valuable insights at the international level. A fourth extension is to bring back "money", nominal rigidities, and nominal exchange rates, and reexamine the short run, if not long run, properties of the model. A fifth is to carry out a strategic differential game analysis using the endogenous government framework developed here. Learning behaviour, information sets, and game-equilibrium concepts, remain to be specified. Finally, the paper worked with linearized versions of the non-linear model; a next step would be to try some nonlinear approaches.

## **Appendix 2.1: Aggregation of Individual Welfare Functions**

Each country's government is assumed to be a utilitarian planner. The objective functionals for the domestic and foreign national planners are, respectively:

$$(A1) \quad SW_t = \int_t^{\infty} U(c(s)) e^{-\gamma(s-t)} ds, \quad (A1)' \quad SW_t^* = \int_t^{\infty} U^*(c^*(s)) e^{-\gamma^*(s-t)} ds$$

where  $\gamma, \gamma^*$  are the national planners' time preference rates,  $c(s)$  and  $c^*(s)$  aggregate consumption, and  $U(\cdot)$  and  $U^*(\cdot)$  aggregate (instantaneous) welfare<sup>1</sup>.

The global planner maximizes:

$$(A2) \quad GW_t = \omega SW_t + (1-\omega) SW_t^*$$

where  $\omega, (1-\omega)$  are the weights the planner attaches to the two economies. In the text I assume that the global planner weights the two countries equally ( $\omega = 0.5$ ) and discounts the welfare of future generations in the two countries at the same rate ( $\gamma' = \gamma = \gamma^*$ ).

This appendix shows how the welfare of agents alive and to-be-born within an economy can be aggregated and expressed by functions such as

---

<sup>1</sup> Aggregate and per capita aggregate magnitudes coincide here since aggregate population has been normalized to unity.

$U(.)$  and  $U^*(.)$  above. The derivation works through a sequence of indirect utility functions:

### Home Economy

$$(A3) \quad SW_t = \int_t^\infty \left\{ \int_v^\infty u(c(s,v)) e^{-(\rho + \lambda)(s-v)} ds \right\} e^{-\gamma(v-t)} dv \\ + \int_{-\infty}^t \left\{ \int_t^\infty u(c(s,v)) e^{-(\rho + \lambda)(s-v)} ds \right\} e^{-\gamma(v-t)} dv$$

Aggregate social welfare consists of two parts: the first part represents the discounted welfare of agents to be born and the second part represents the discounted welfare of agents already alive. In each part, the inner integral sums welfare over time for each agent of vintage  $v$  and the outer integral sums across vintages. Note that welfare for each agent is discounted back to the date of birth and not to calendar time. This is necessary to preserve time-consistency. Otherwise, as time progresses, those previously newly born will be amongst those already alive and will be treated differently from those who become currently newly born<sup>1</sup>. The private agent's time preference rate  $\rho$  need not be the same as the planner's  $\gamma$ .

Change the order of integration:

---

<sup>1</sup> see also Buiter-Kletzer (1990) and Calvo-Obstfeld (1988).

$$(A3)' \quad SW_t = \int_t^\infty \left\{ \int_{-\infty}^s u(c(s,v)) e^{-(\rho + \lambda)(s-v)} dv \right\} e^{-\gamma(s-t)} ds$$

and define:

$$(A4) \quad U(c(s)) = \max \int_{-\infty}^s u(c(s,v)) e^{-(\rho + \lambda)(s-v)} e^{-\gamma(s-v)} dv$$

$$\text{s.t.} \quad (A4a) \quad c(s) = \int_{-\infty}^s c(s,v) \lambda e^{-\lambda(s-v)} dv$$

$$\text{where} \quad (A4b) \quad c(s,v) = c_H(s,v) + p(s) c_F(s,v)$$

To illustrate, I assume a constant relative risk aversion (CRRA) utility function as in the main text:

$$(A5) \quad u(c(s,v)) = \max u[c_H(s,v), c_F(s,v)] = \frac{[c_H(s,v)^\alpha c_F(s,v)^{1-\alpha}]^{1-\sigma}}{1-\sigma}$$

subject to (A4b)

for each individual  $v$ . Solving problem (A5) yields:

$$u(c(s,v)) = \Gamma p(s)^{(1-\sigma)(\alpha-1)} \frac{c(s,v)^{1-\sigma}}{1-\sigma} \quad \text{where} \quad \Gamma = [\alpha^\alpha (1-\alpha)^{1-\alpha}]^{1-\sigma}$$

Substituting this into (A4) gives:



$$(A4)' \quad U(c(s)) = \max \int_{-\infty}^s \Gamma p(s)^{(1-\sigma)(\alpha-1)} \frac{c(s,v)^{1-\sigma}}{1-\sigma} e^{-(\rho+\lambda)(s-v)} e^{-\gamma(s-v)} dv$$

$$\text{s.t.} \quad (A4a)$$

The solution to problem (A4)' is:

$$(A6) \quad \frac{\frac{\partial c(s,v)}{\partial v}}{c(s,v)} = \frac{\rho - \gamma}{\sigma} \quad \text{for } v \in [s, \infty)$$

or

$$(A6)' \quad c(s,v) = c(s,s) e^{\left(\frac{\rho - \gamma}{\sigma}\right)(v-s)}$$

Substituting (A6)' into (A4a) gives:

$$c(s,s) = \left(\frac{\sigma\lambda + \rho - \gamma}{\sigma\lambda}\right) c(s)$$

and substituting this back into (A6)' gives:

$$(A7) \quad c(s,v) = \left(\frac{\sigma\lambda + \rho - \gamma}{\sigma\lambda}\right) e^{\left(\frac{\rho - \gamma}{\sigma}\right)(v-s)} c(s)$$

for which it is required that  $\sigma\lambda > \gamma - \rho$ , ie. that the discrepancy between the planner's time preference rate and the individual's rate not be "too large".

Substituting (A7) into (A4)' gives:

$$(A8) \quad U(c(s)) = \Phi \frac{c(s)^{1-\sigma}}{1-\sigma} p(s)^{(1-\sigma)(\alpha-1)}$$

$$\text{where } \Phi = \Gamma \left( \frac{\sigma}{\lambda\sigma + \rho - \gamma} \right)^{\sigma} \lambda^{\sigma-1} \text{ and } \Gamma = [\alpha^{\alpha} 1-\alpha^{1-\alpha}]^{1-\sigma}$$

Finally substituting (A8) into (A1) will give us the home country's national welfare functional, as used in the main text. The condition that  $(\sigma\lambda > \gamma - \rho)$  ensures that welfare is positive and bounded.  $\Phi$  will be referred to as the "taste" shifter.

### Foreign Economy

The derivations for aggregating foreign individual utilities are similar. I report the final outcome:

$$(A9) \quad U^*(c^*(s)) = \Phi^* \frac{c^*(s)^{1-\sigma^*}}{1-\sigma^*} p(s)^{(1-\sigma^*)\alpha^*}$$

$$\text{where } \Phi^* = \Gamma^* \left( \frac{\sigma^*}{\lambda^*\sigma^* + \rho^* - \gamma^*} \right)^{\sigma^*} \lambda^{*\sigma^*-1} \text{ and } \Gamma^* = [\alpha^{*\alpha^*} 1-\alpha^{*1-\alpha^*}]^{1-\sigma^*}$$

Substituting (A9) into (A1)' will give us the foreign country's national welfare functional.

## Appendix 2.2: Simulation Model

Let  $x^p$  be the predetermined state variables

$x^n$  the non-predetermined state variables

$v$  the output variables

$u$  the exogenous variables.

The state-space form of the model is:

$$\begin{aligned}\dot{x} &= Ax + Bu \\ v &= Cx + Du\end{aligned}$$

where  $A, B, C, D$  are matrices of coefficients and

$$x = \begin{bmatrix} x^p \\ x^n \end{bmatrix}$$

### Benchmark Model ( $\sigma = \sigma^* = 1$ ; Exogenous Government)

Predetermined state variables,  $x^p$ :  $K, K^*, G, G^*, z, b, b^*$

Non-predetermined state variables,  $x^n$ :  $q_p, q_p^*, h, h^*$

Output Variables,  $v$ :  $y, y^*, c, c^*, a, a^*, \tau, \tau^*, \dot{p}, p, r, r^*$

Exogenous Variables,  $u$ :  $\tau_0, \tau_0^*, i_g, i_g^*, c_g, c_g^*, \tau_k, \tau_k^*, \tau_c, \tau_c^*$ .

Notation: (\* denotes foreign country variables)

|                    |                                                                                                                  |                    |                           |
|--------------------|------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------|
| $K, K^*$           | private capital stocks                                                                                           | $G, G^*$           | public capital stocks     |
| $h, h^*$           | human wealth                                                                                                     | $a, a^*$           | financial wealth          |
| $y, y^*$           | output                                                                                                           | $c, c^*$           | private consumption       |
| $b, b^*$           | government debt stocks                                                                                           | $\tau, \tau^*$     | endogenous lump-sum taxes |
| $r, r^*$           | interest rates                                                                                                   | $\tau_0, \tau_0^*$ | exogenous lump-sum taxes  |
| $i_g, i_g^*$       | public investment                                                                                                | $c_g, c_g^*$       | public consumption        |
| $\tau_k, \tau_k^*$ | capital income tax rates                                                                                         | $\tau_c, \tau_c^*$ | consumption tax rates     |
| $z$                | net stock of foreign assets (denominated in home output)                                                         |                    |                           |
| $q_p, q_p^*$       | ratio of market value of equity capital to replacement cost                                                      |                    |                           |
| $p$                | relative price of foreign output in terms of home output<br>(or reciprocal of the home country's terms of trade) |                    |                           |

State Equations:

$$\dot{K} = \frac{1}{\psi}(q_p - 1)K - \delta K$$

$$\dot{K}^* = \frac{1}{\psi^*}(q_p^* - 1)K^* - \delta^* K^*$$

$$\dot{G} = i_g - \delta G$$

$$\dot{G}^* = i_g^* - \delta^* G^*$$

$$\dot{z} = rz + \alpha^* pc^* - (1-\alpha)c$$

$$\dot{b} = rb + i_g + \frac{1}{\psi} \left( \frac{i_g^2}{G} \right) + c_g - \tau - \tau_c - \tau_k \beta_k y - \xi G$$

$$\dot{b}^* = r^* b^* + i_g^* + \frac{1}{\psi^*} \left( \frac{i_g^{*2}}{G^*} \right) + c_g^* - \tau^* - \tau_c^* c^* - \tau_k^* \beta_k^* y^* - \xi^* G^*$$

$$\dot{q}_p = (r+\delta)q_p - \left[ (1-\tau_K) \frac{\beta_K y}{K} + \frac{\psi}{2} \left( \frac{\dot{K} + \delta K}{K} \right)^2 \right]$$

$$\dot{q}_p^* = (r^* + \delta^*)q_p^* - \left[ (1-\tau_K^*) \frac{\beta_K^* y^*}{K^*} + \frac{\psi^*}{2} \left( \frac{\dot{K}^* + \delta^* K^*}{K^*} \right)^2 \right]$$

$$\dot{h} = (r+\lambda)h + \tau - \beta_L y$$

$$\dot{h}^* = (r^* + \lambda^*)h^* + \tau^* - \beta_L^* y^*$$

Output Equations:

$$y = \pi G^{\beta_G^*} G^{\beta_G} K^{\beta_K} L^{\beta_L}$$

$$y^* = \pi^* G^{\beta_G^*} G^{*\beta_G^*} K^{*\beta_K^*} L^{*\beta_L^*}$$

$$c = \frac{1}{(1 + \tau_c)} \theta (a + h)$$

$$c^* = \frac{1}{(1 + \tau_c^*)} \theta^* (a^* + h^*)$$

$$a = b + z + q_p K$$

$$a^* = b^* - \frac{z}{p} + q_p^* K^*$$

$$\tau = \tau_0 + \eta \dot{b}$$

$$\tau^* = \tau_0^* + \eta^* \dot{b}^*$$

$$\frac{\dot{p}}{p} = r - r^*$$

$$y = \alpha^* p c^* + \alpha c + c_g + i_g + \frac{\psi i_g^2}{2 G} + (\dot{K} + \delta K) + \frac{\psi (\dot{K} + \delta K)^2}{2 K}$$

$$\dot{y} = \frac{d}{dt} \left[ \alpha^* p c^* + \alpha c + c_g + i_g + \frac{\psi i_g^2}{2 G} + (\dot{K} + \delta K) + \frac{\psi (\dot{K} + \delta K)^2}{2 K} \right]$$

$$\dot{y}^* = \frac{d}{dt} \left[ (1 - \alpha^*) c^* + \frac{(1 - \alpha)}{p} c + c_g^* + i_g^* + \frac{\psi^* i_g^{*2}}{2 G^*} + (\dot{K}^* + \delta^* K^*) + \frac{\psi^* (\dot{K}^* + \delta^* K^*)^2}{2 K^*} \right]$$

The last two equations are derived by time differentiating the two goods market clearing equilibrium conditions in order to obtain additional equations containing the domestic and foreign interest rates. The interest rates will appear after the appropriate state equations (such as the  $\dot{c}$ ,  $\dot{c}^*$  equations) are substituted into those equations.

To simulate this model I linearize the model around an initial stationary-state:

An example:

Parameters:  $\beta_G = \beta_G^* = \beta^* G = \beta^* G^* = 0.1$ ;  $\beta_K = \beta^* K = 0.35$ ,  $\beta_L = \beta^* L = 0.65$ ,  
 $\alpha^* = 1 - \alpha = 0.4$ ,  $\alpha = 0.6 = 1 - \alpha^*$ ,  $\psi = \psi^* = 2$ ,  $\delta = \delta^* = 0.1$ ,  
 $\eta = \eta^* = -2$ ,  $\rho = \rho^* = 0.043$ ,  $\lambda = \lambda^* = 0.03$ ,  $\xi = \xi^* = 0$ ,  $\pi = \pi^* = 8.8$

Exogenous Values:  $\tau_0 = \tau_0^* = 3.114$ ,  $\tau_k = \tau_k^* = \tau_c = \tau_c^* = 0.2$ ,  
 $i_g = i_g^* = 8.57$ ,  $c_g = c_g^* = 19.88$

These generate:

Steady-State Values:  $r = r^* = 0.05$ ,  $p = 1$ ,  $z = 0$ ,  $q_p = q_p^* = 1.2$ ,  $K = K^* = 240$ ,  
 $G = G^* = 85.7$ ,  $y = y^* = 145.7$ ,  $c = c^* = 90$ ,  $\theta = \theta^* = 0.073$ ,  
 $a = a^* = 328$ ,  $h = h^* = 1145$ ,  $b = b^* = 40$ ,  $\tau = \tau^* = 3.114$

Eigenvalues: -0.1045, -0.101, -0.1, -0.1, -0.055, -0.042, -0.025, 0.061,  
0.071, 0.18, 0.185.

Associated Settling Times (to reach within 1% of steady-state), respectively: 44.05, 45.64, 46.05, 46.05, 83.5, 110.04, 183.9, -75.4, -64.7, -25.5, -24.8<sup>1</sup>.

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<sup>1</sup> The eigenvalue with the smallest modulus is associated with net external assets,  $z$ . As a result, the current account dynamics in the system are most sluggish. It takes  $z$  183.9 periods to settle down within 1% of steady-state.



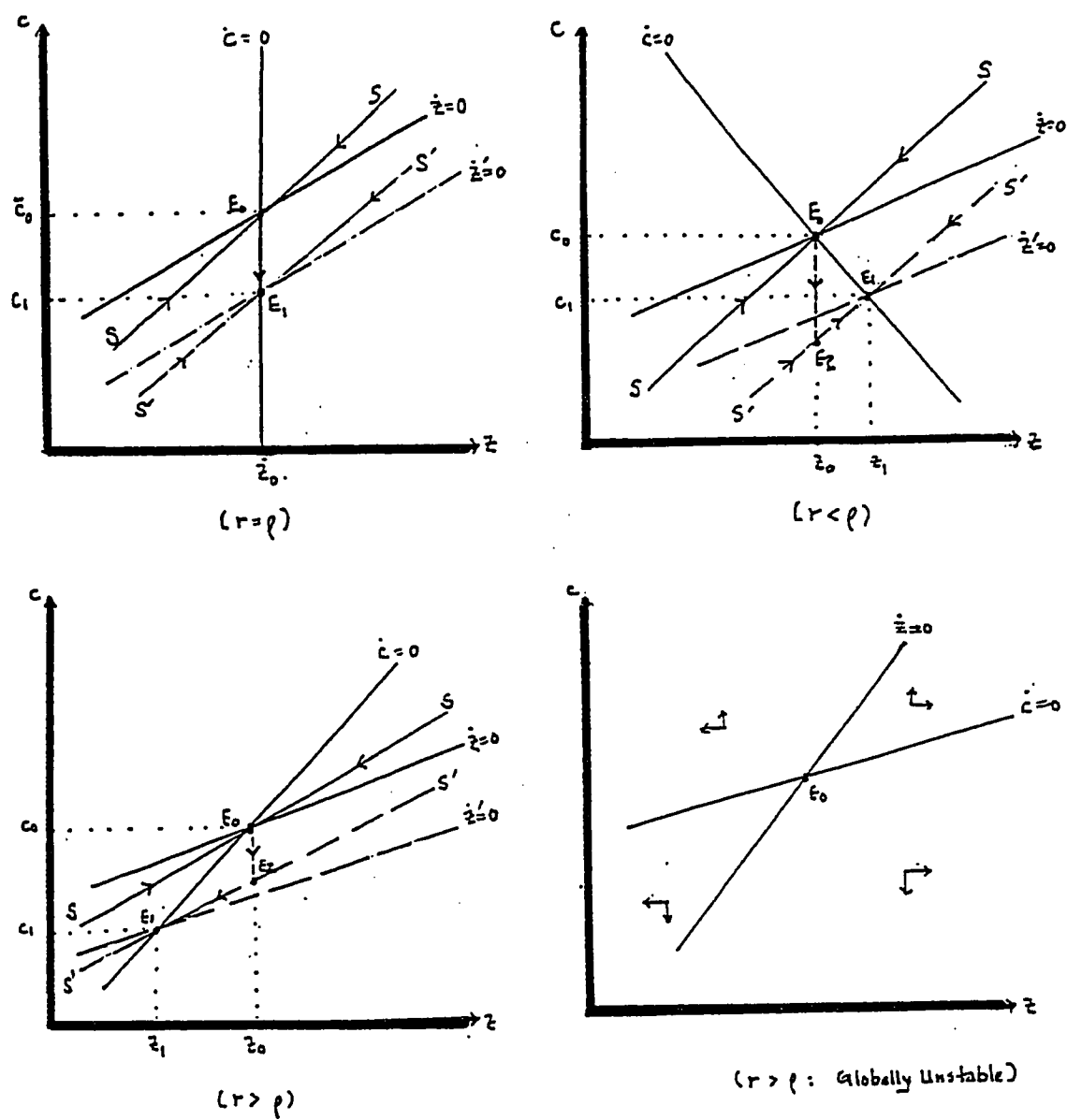


Figure 2.1: Small Open-Economy Public Consumption Shock

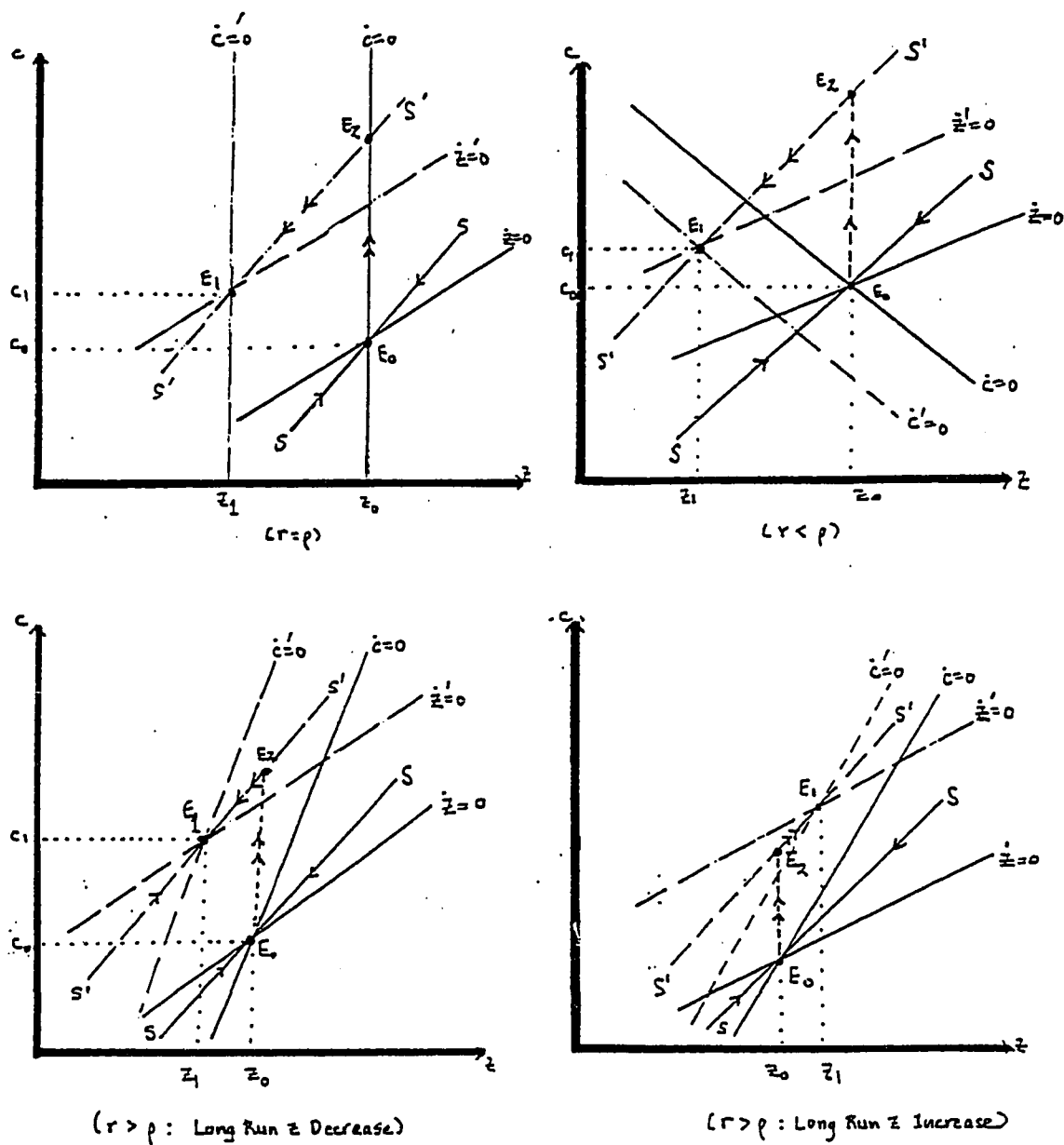


Figure 2.2: Small Open-Economy Public Investment Shock

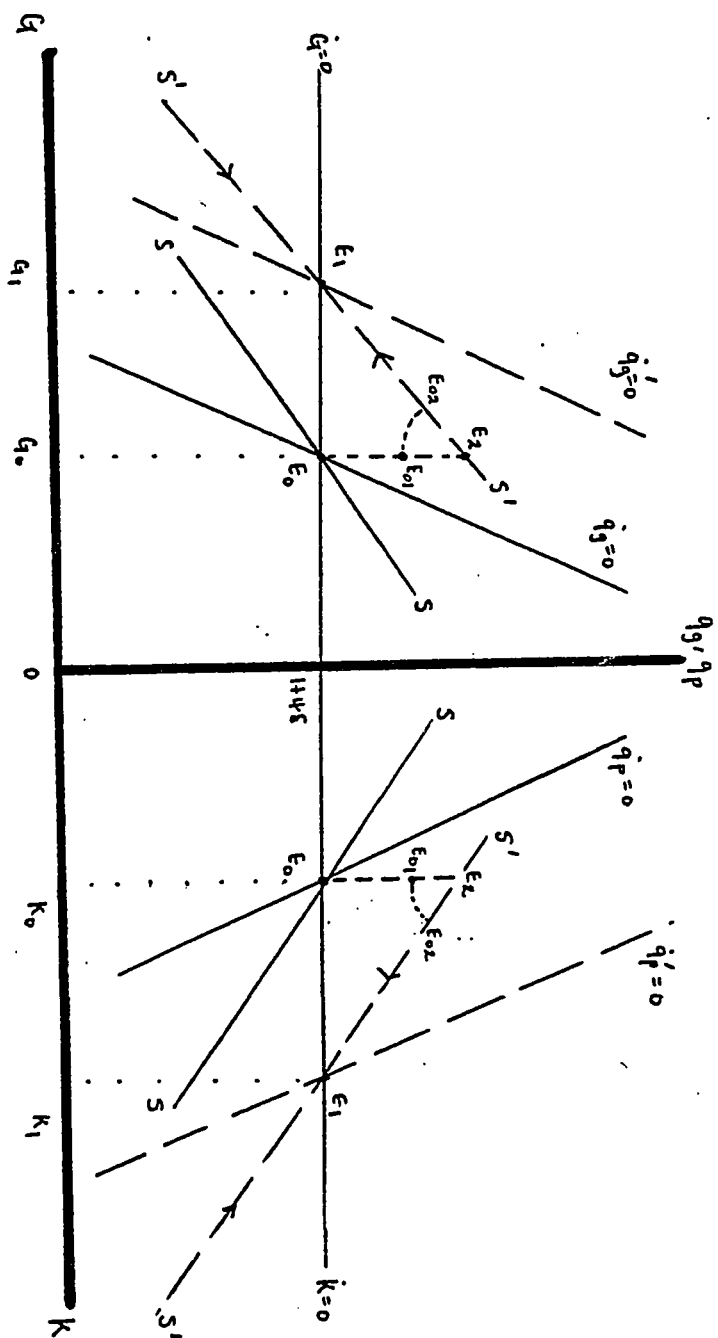


Figure 2.3: Permanent Increase in Foreign Public Investment

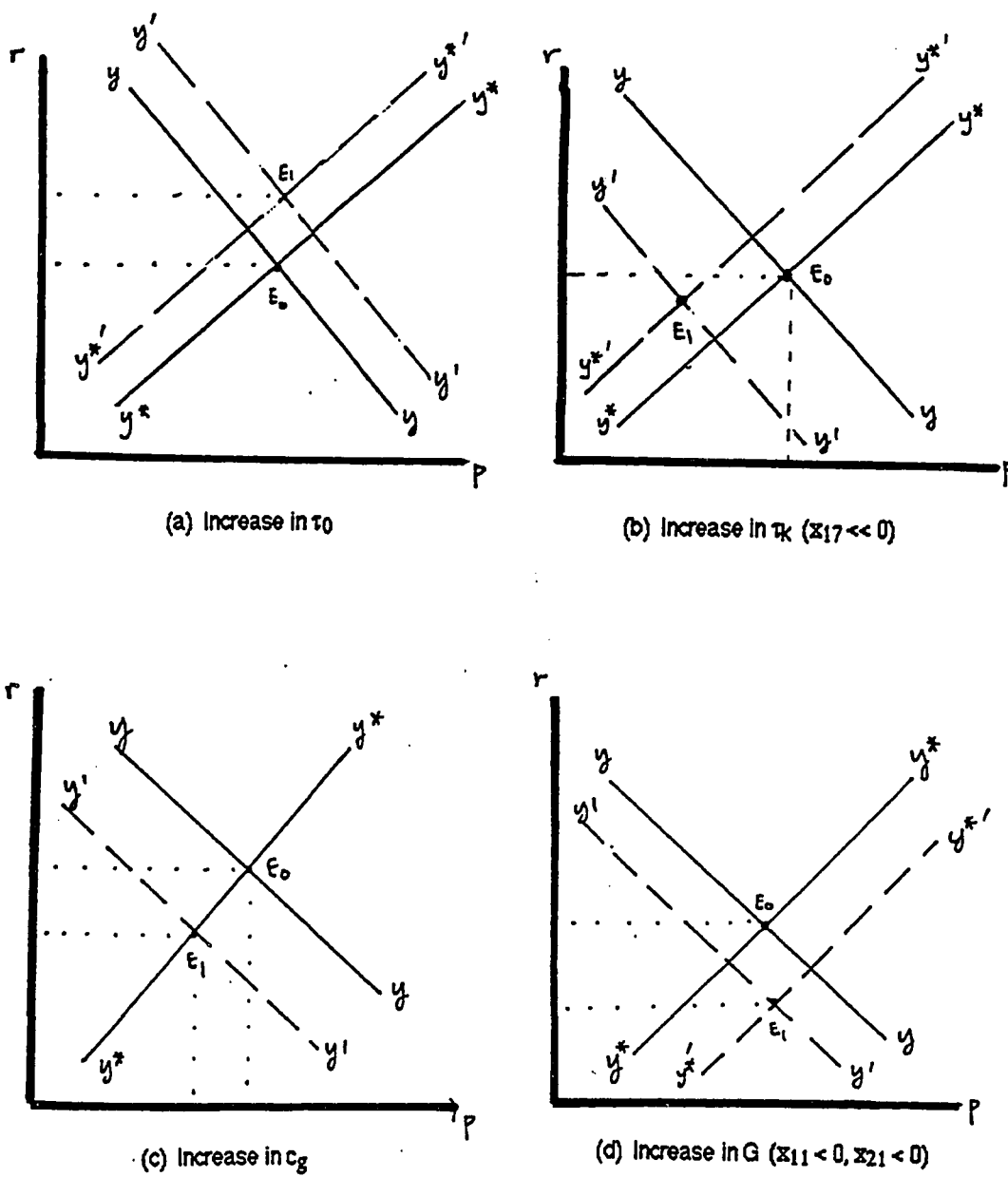


Figure 2.4: Steady-State Impacts on International Prices

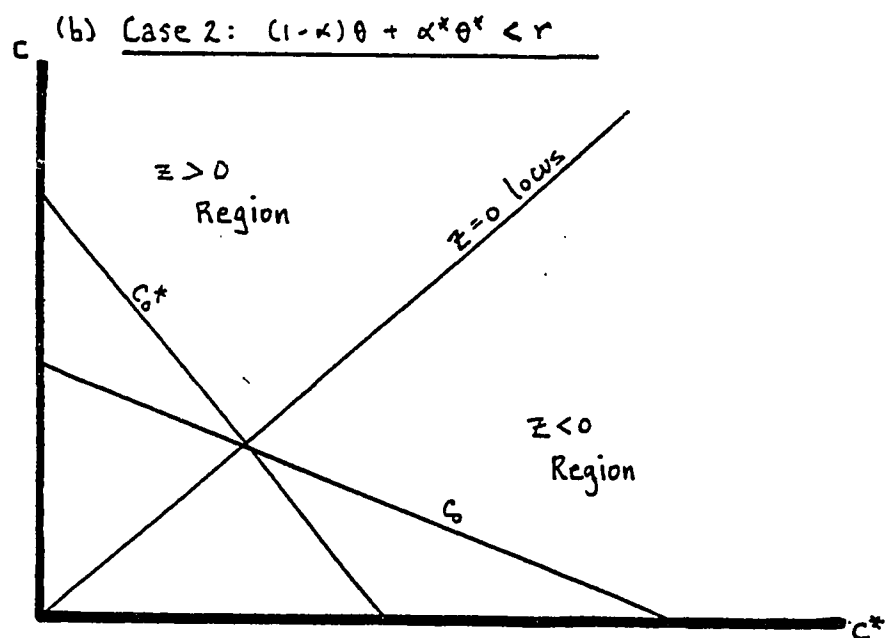
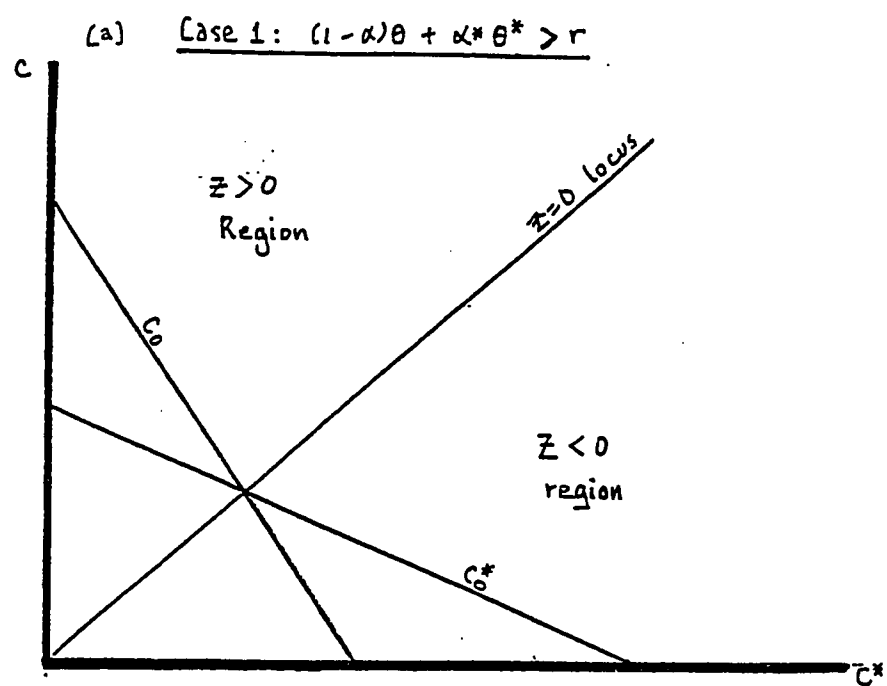
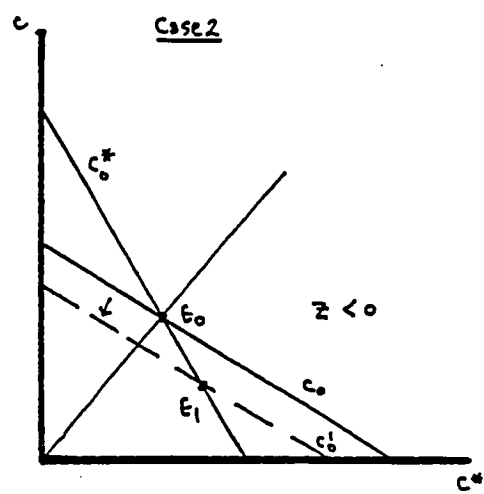
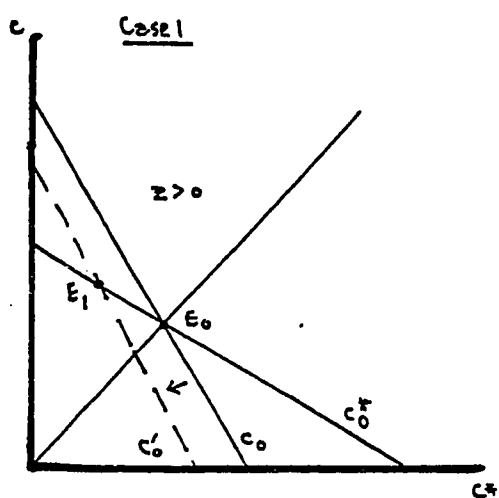
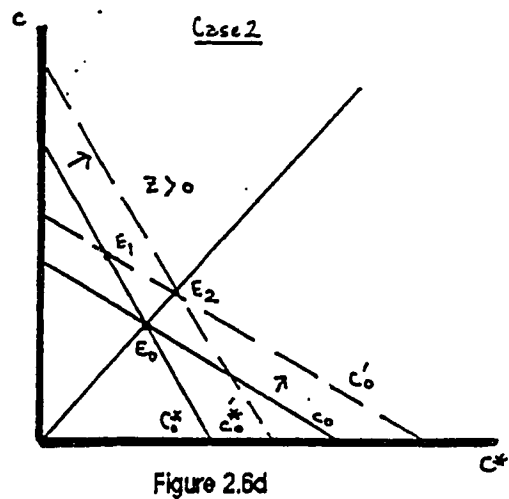
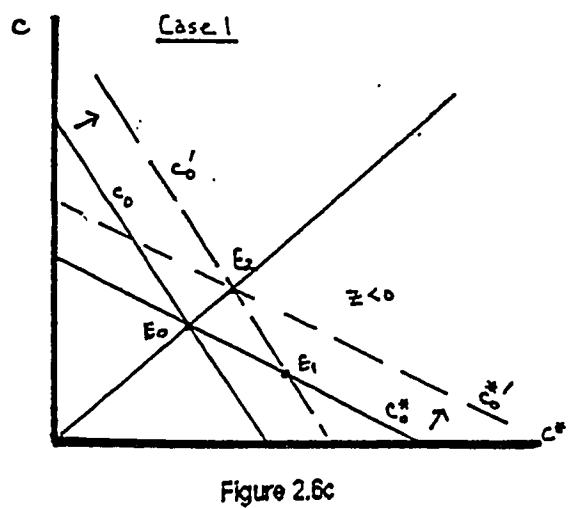


Figure 2.5: Steady-State Consumption Equilibria Space

Long Run Increase In Domestic Public Consumption  $c_g$ Long Run Increase In Domestic Public Capital  $G$ 

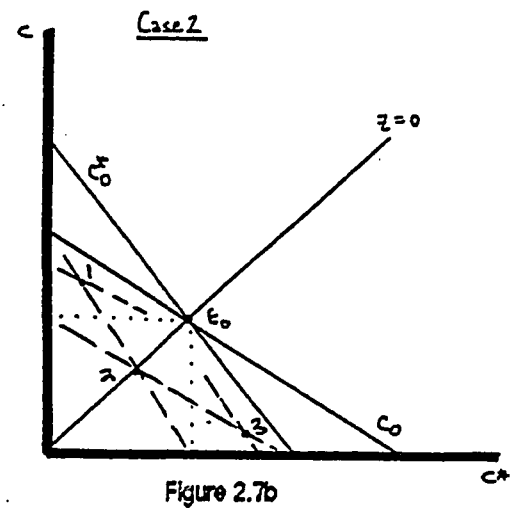
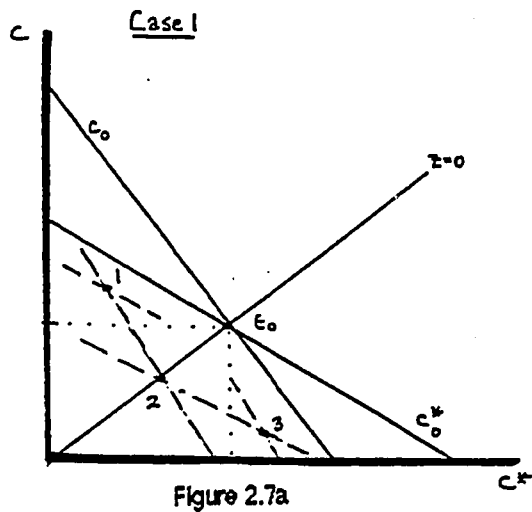
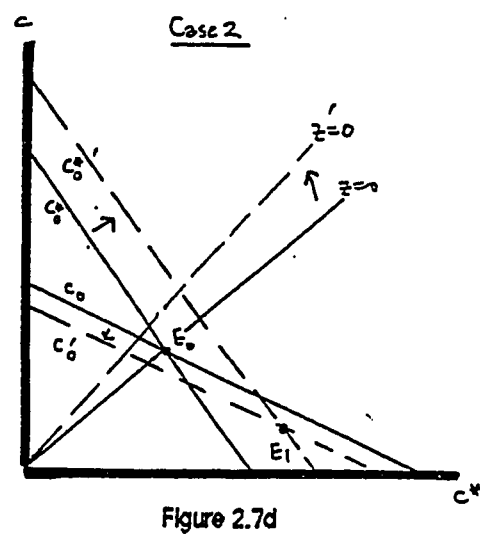
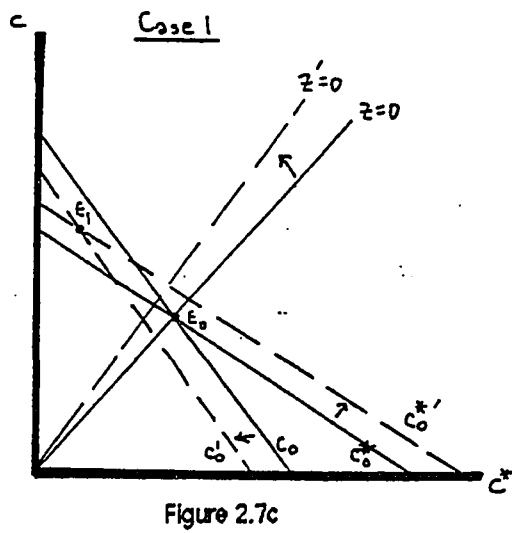
Long Run Increase in World Interest Rate  $r$ Long Run Increase in Foreign Terms of Trade  $p$ 

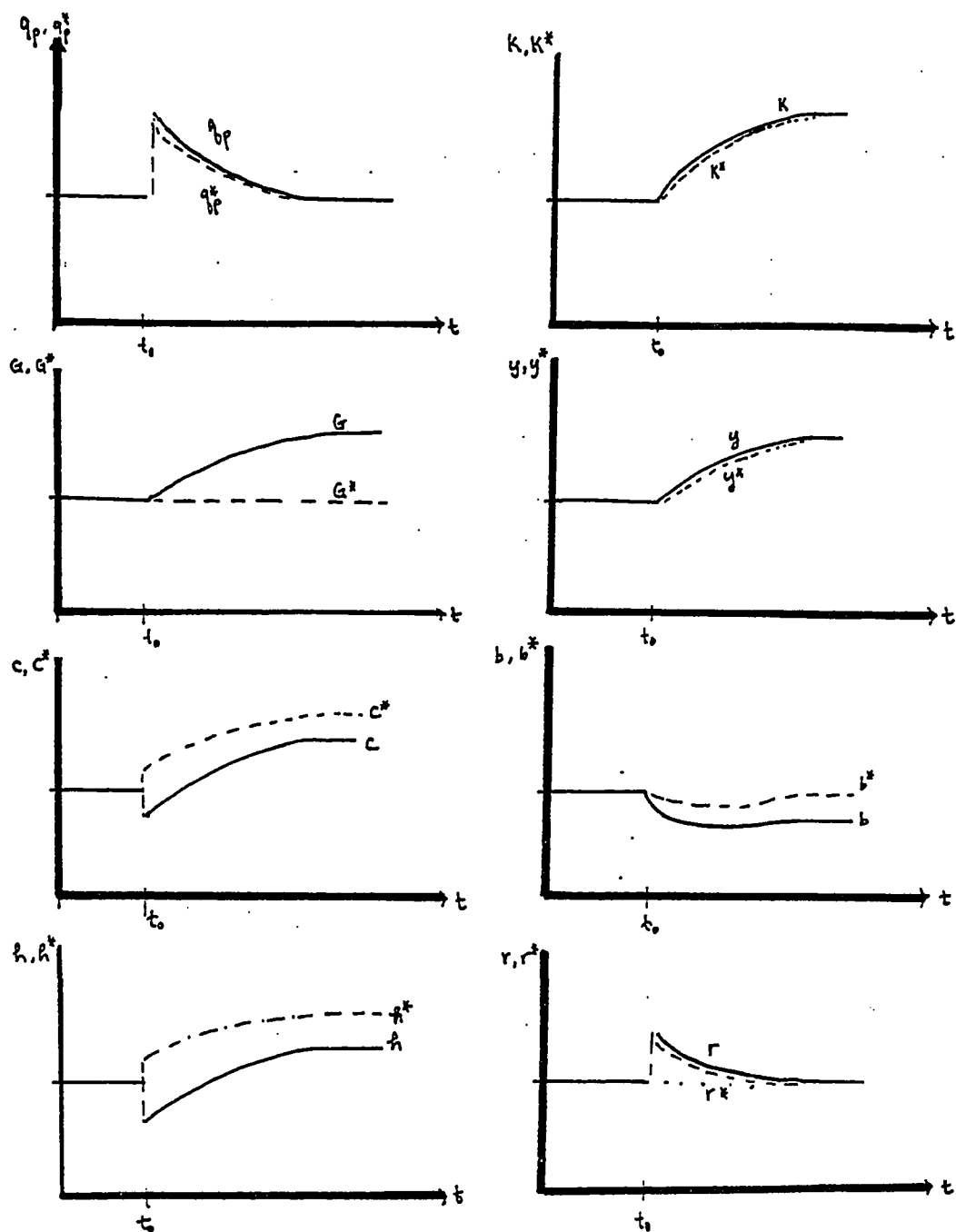
Figure 2.8: Unanticipated Permanent Balanced Budget Increase in  $i_g$ 



Figure 2.8 continued

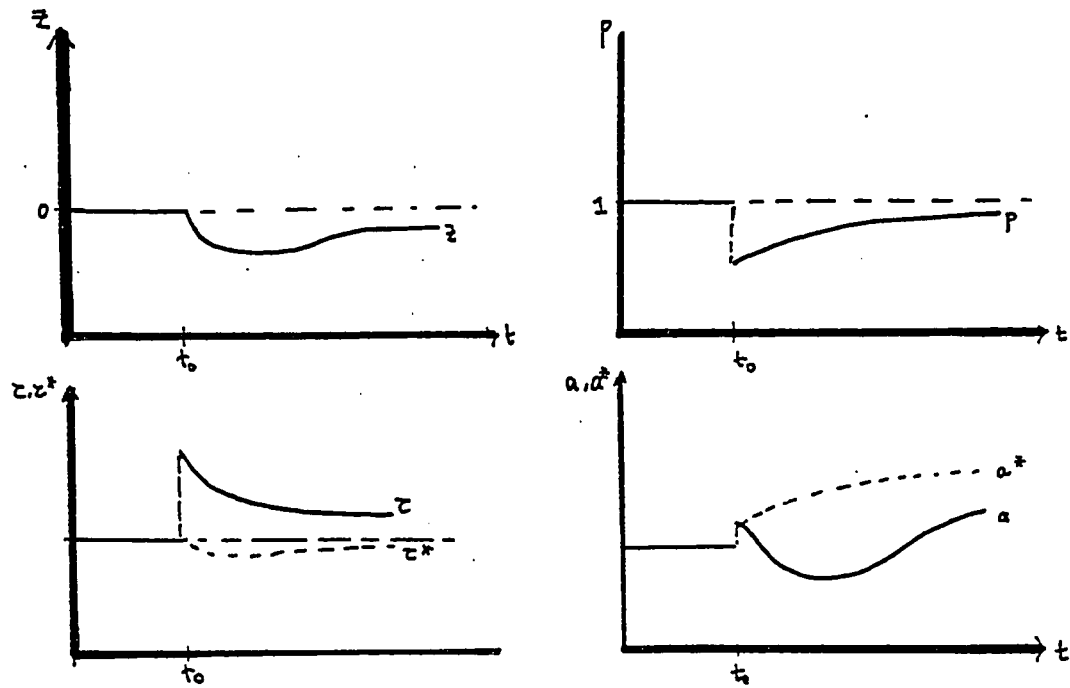


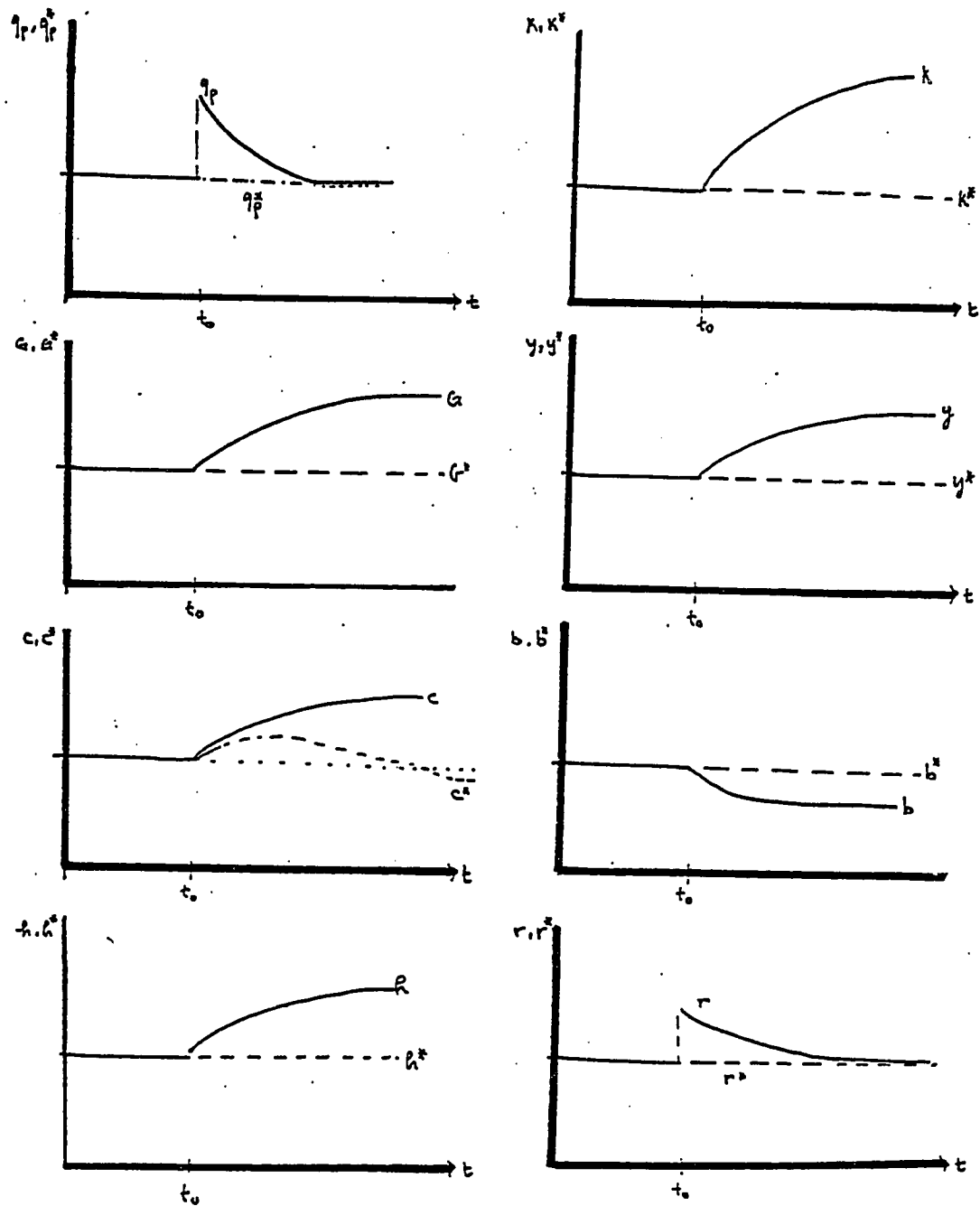
Figure 2.9: Unanticipated Permanent Balanced Budget Increase in  $i_g$  (No Spillovers)

Figure 2.9 continued

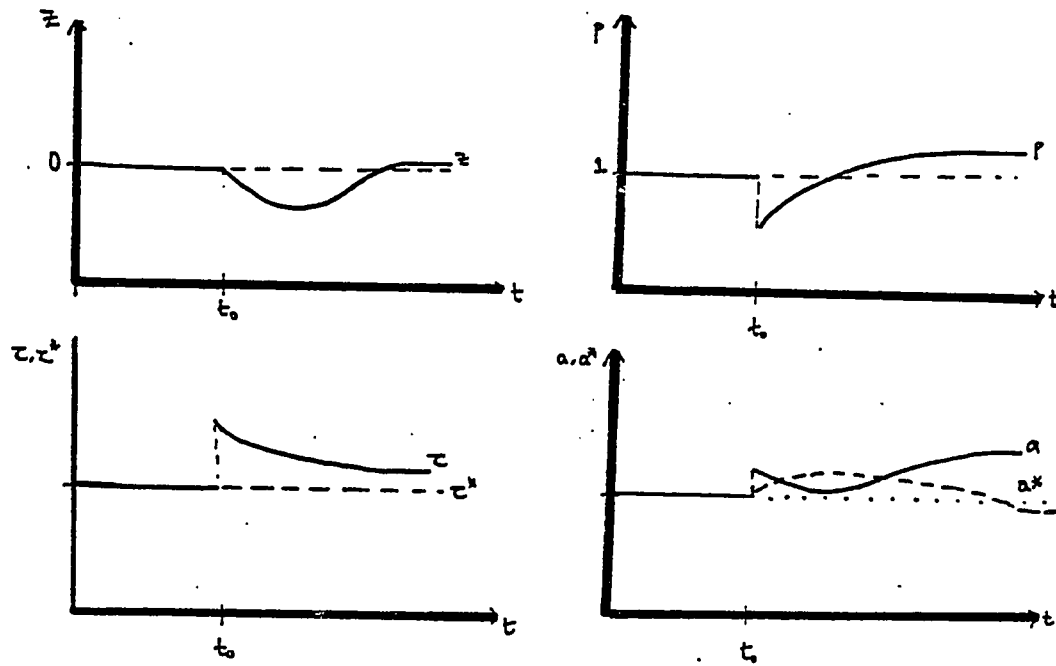


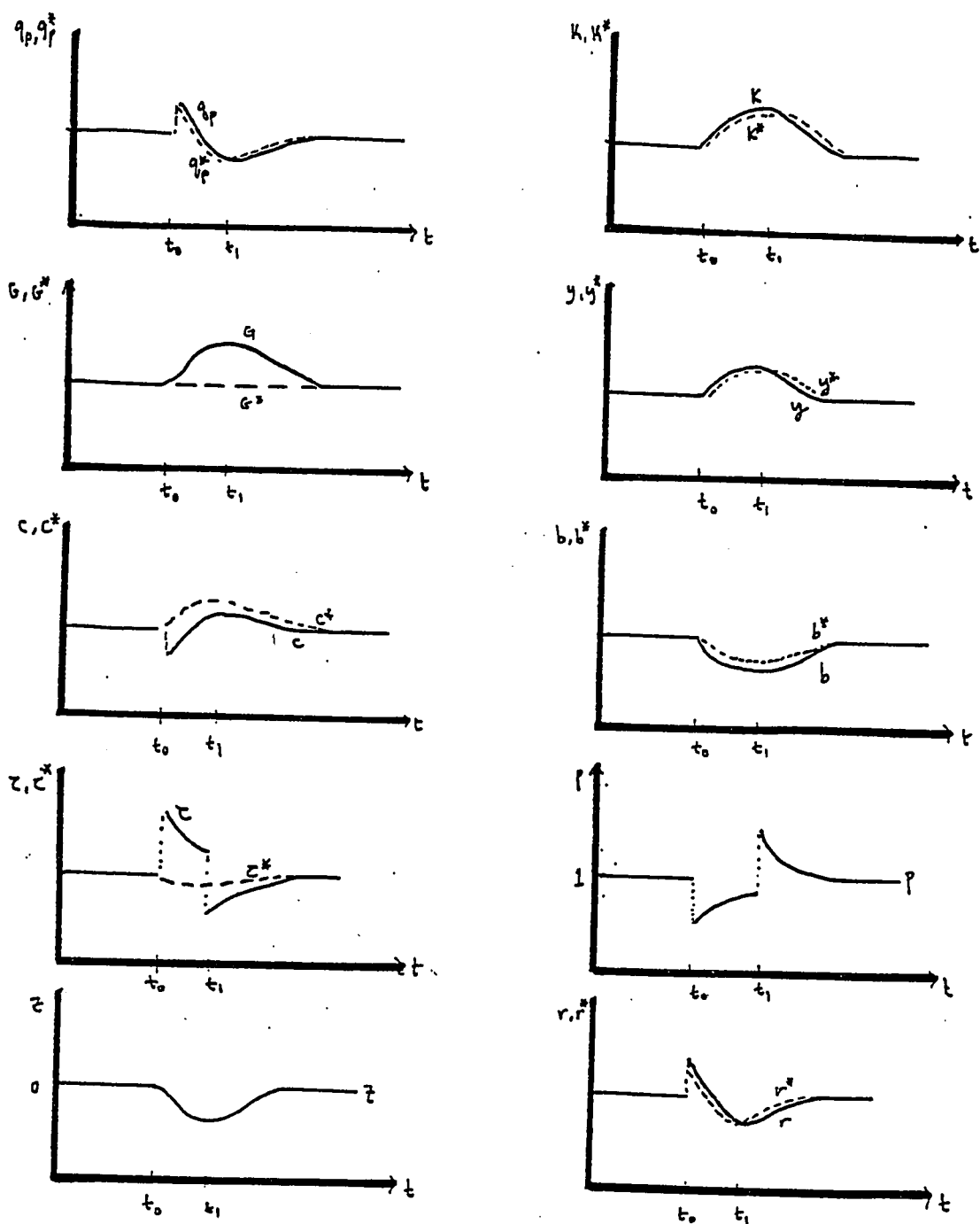
Figure 2.10: Temporary Balanced Budget Increase in  $i_g$ 

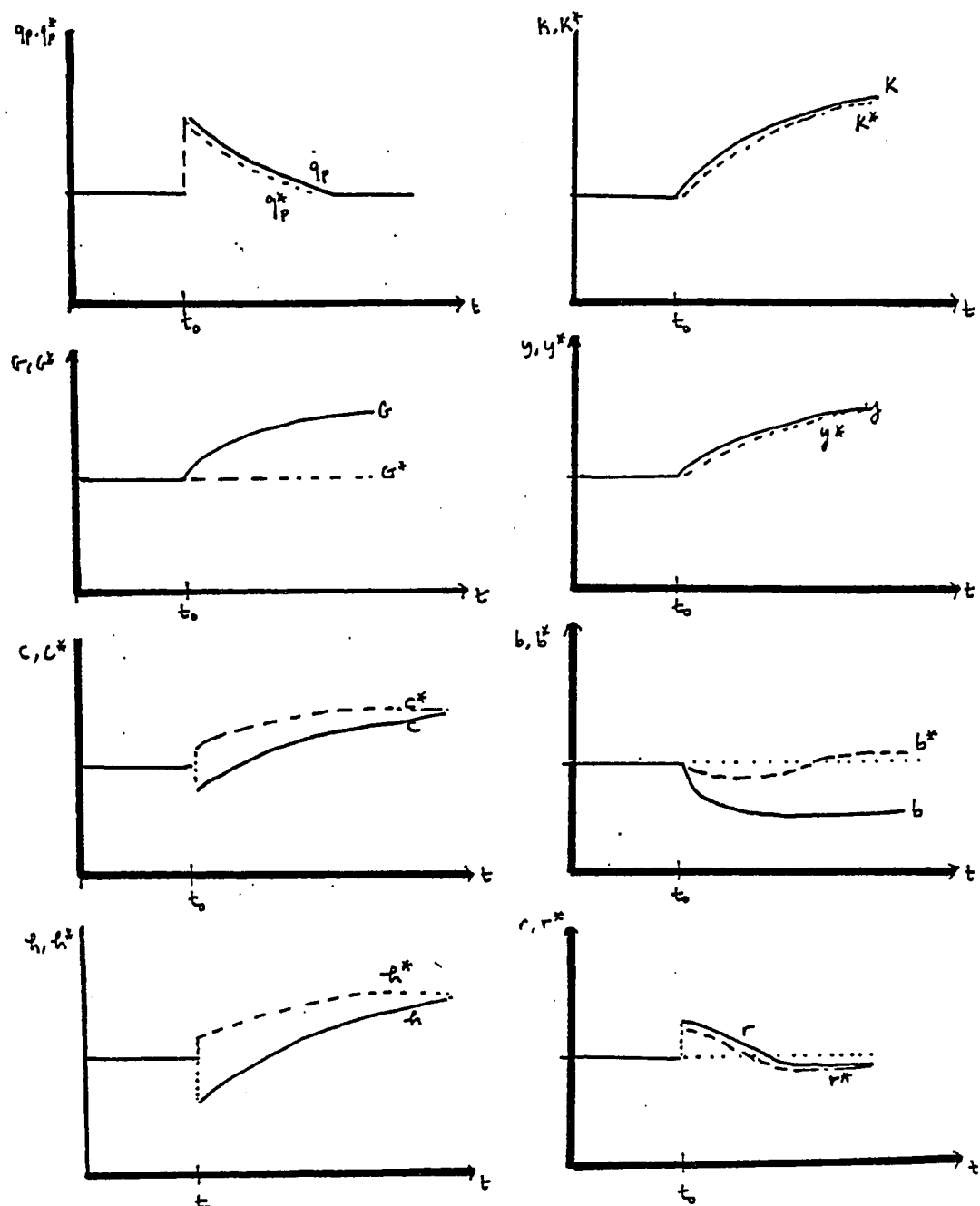
Figure 2.11: Unanticipated Permanent Deficit-Financed Increase in  $i_g$ 

Figure 2.11 continued

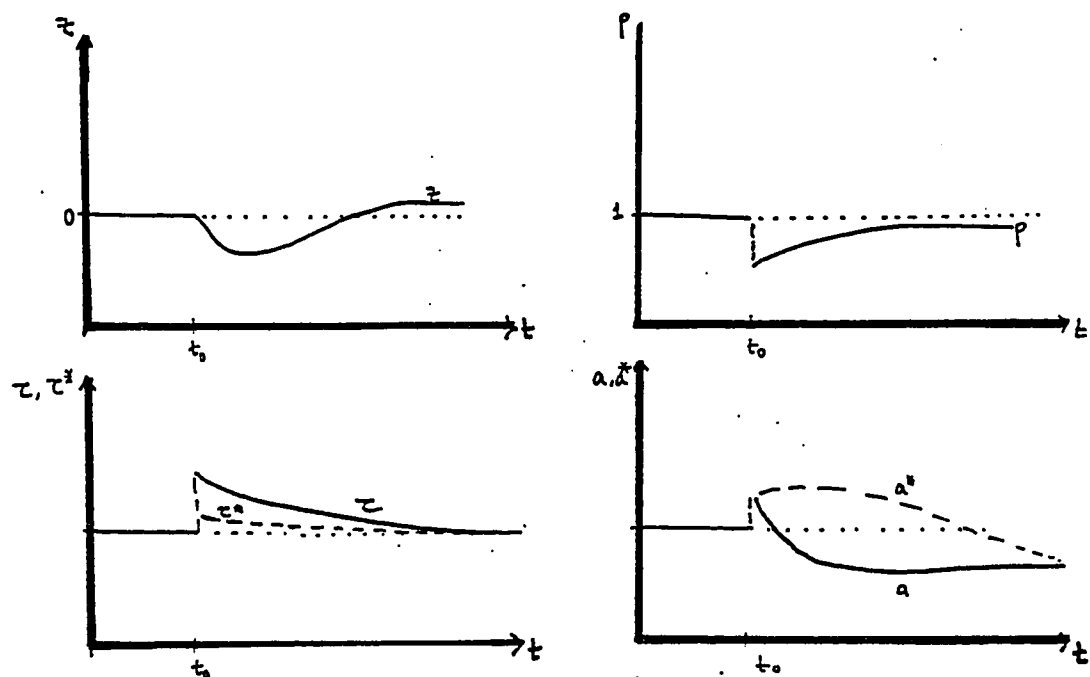


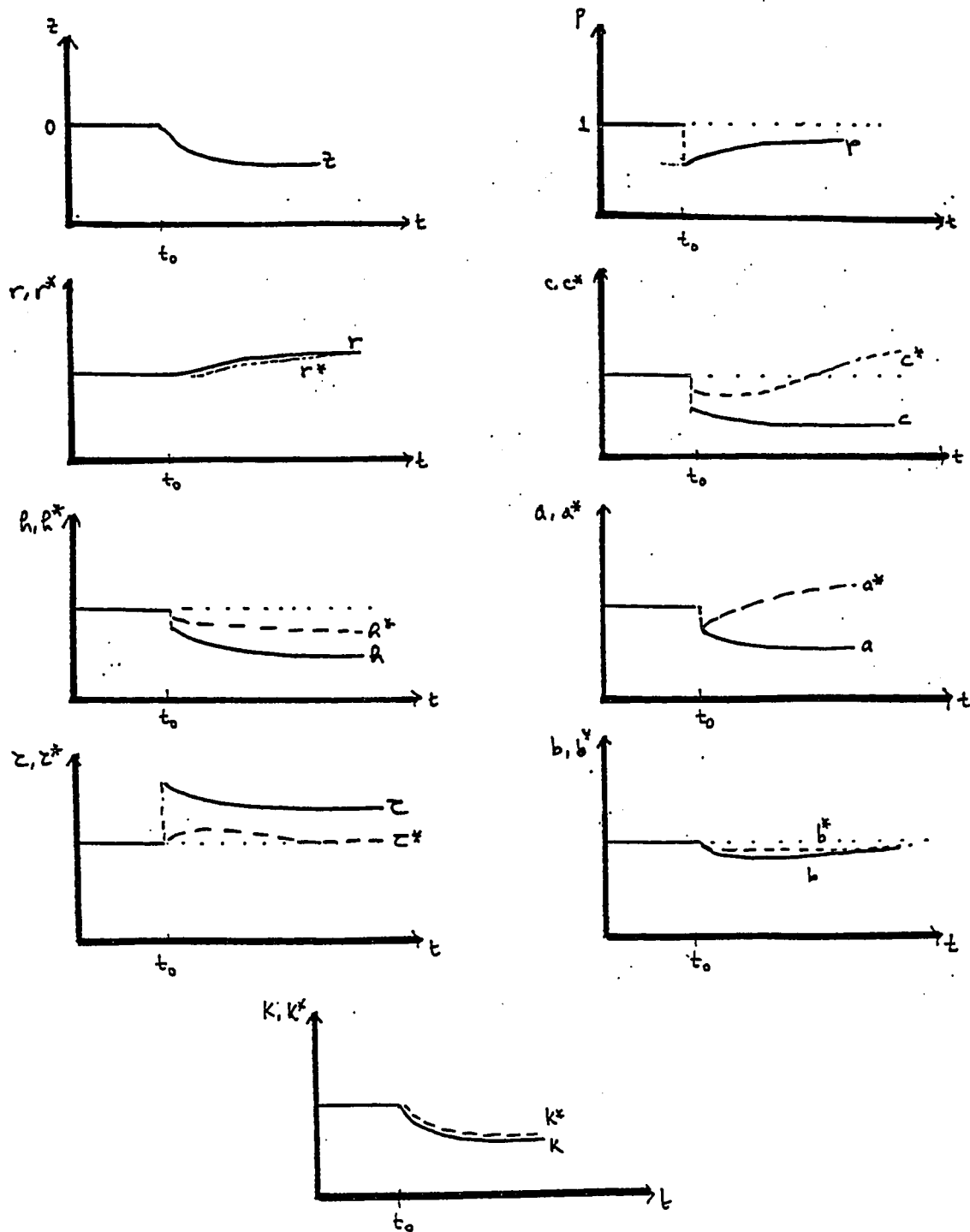
Figure 2.12: Unanticipated Permanent Balanced Budget Increase in  $c_g$ 

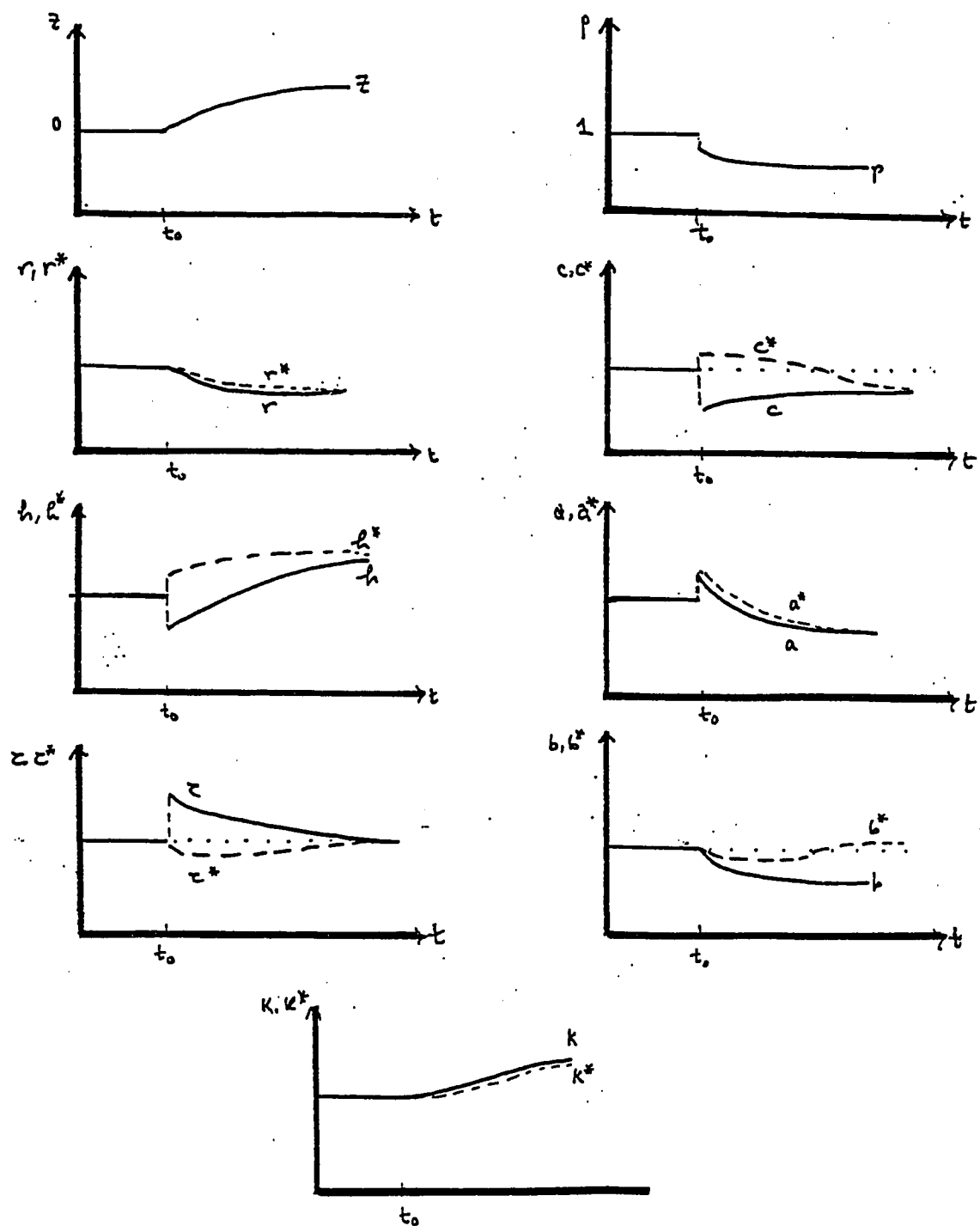
Figure 2.13: Unanticipated Permanent Deficit-Financed Increase in  $c_g$ 



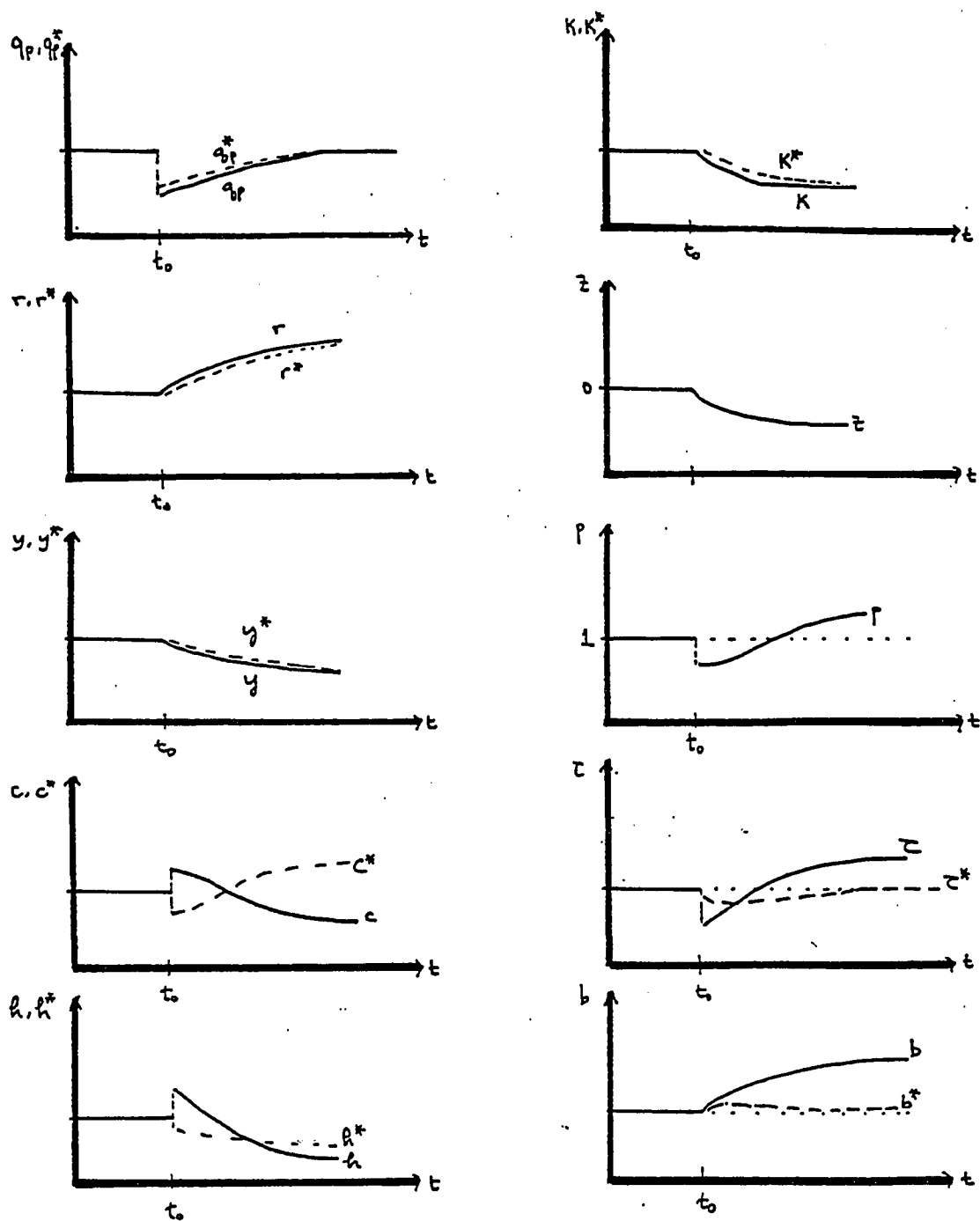
Figure 2.14: Unanticipated Permanent Increase In Long Run  $\tau_0$ 

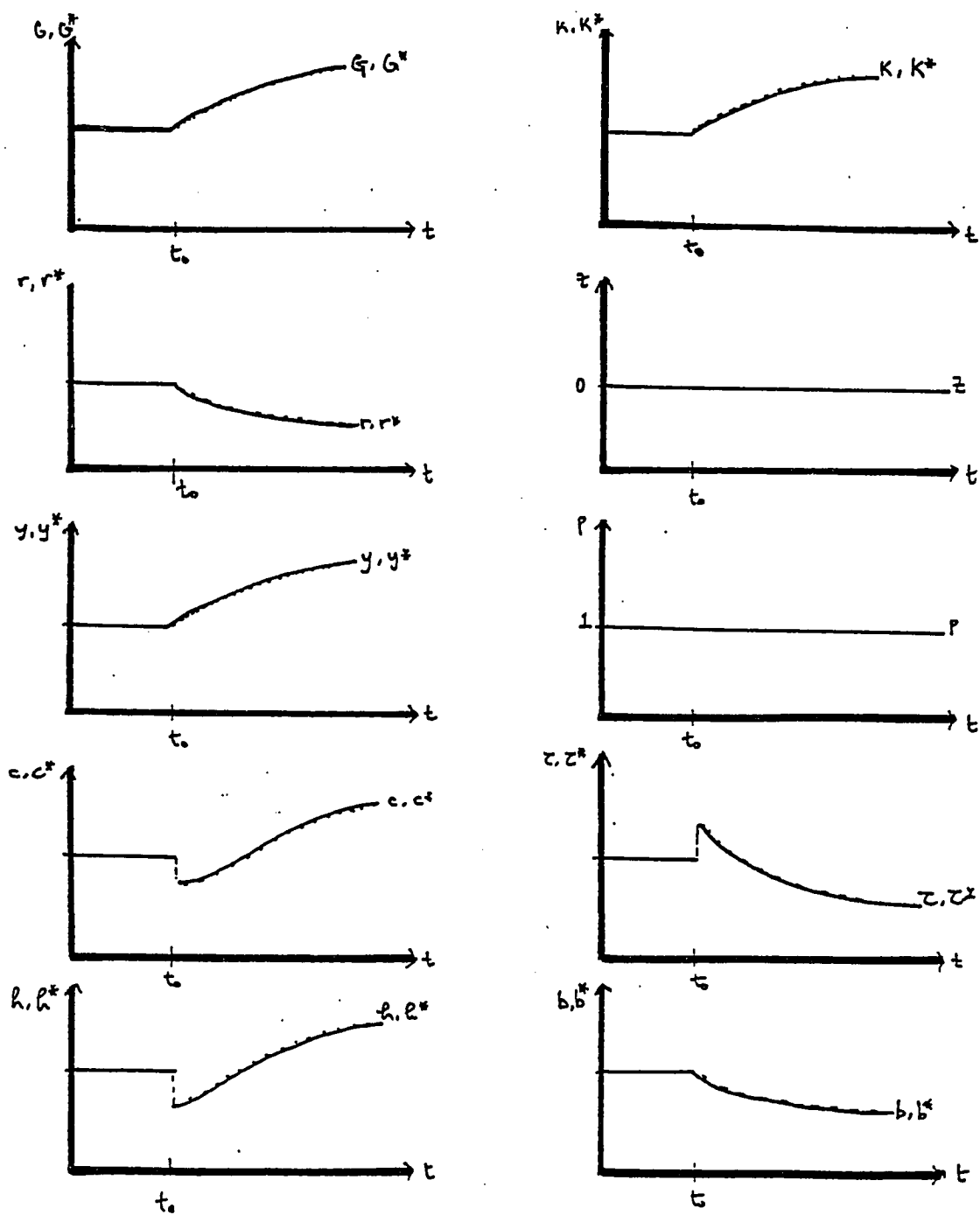
Figure 2.15: Unanticipated Permanent Decreases in Long Run  $\tau_0$  and  $\tau_0^*$ 

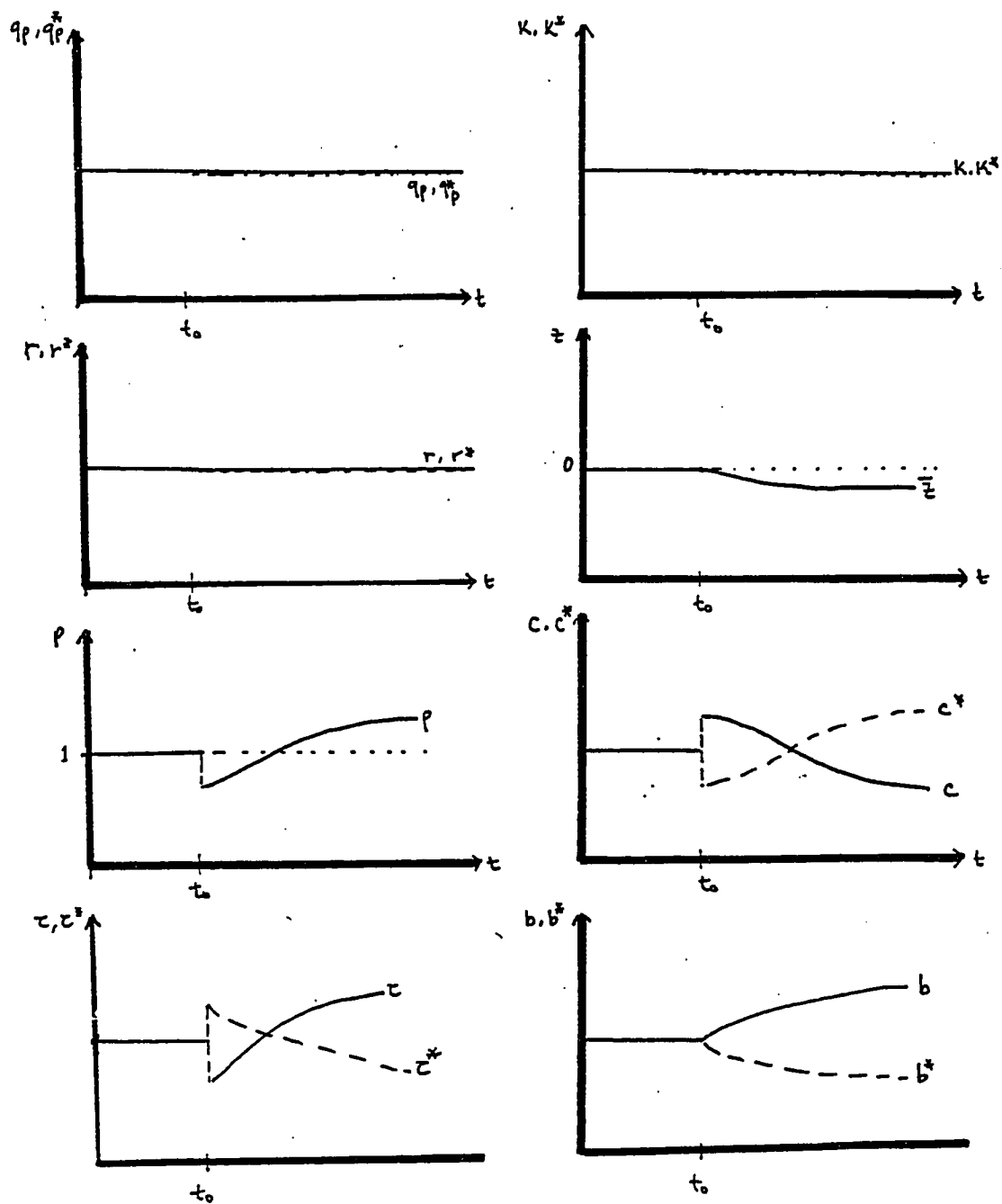
Figure 2.16: Unanticipated Permanent Long Run Increase in  $\tau_0$  and Decrease in  $\tau_0^*$ 

Figure 2.17: Unilateral Conversion to Consumption Taxation

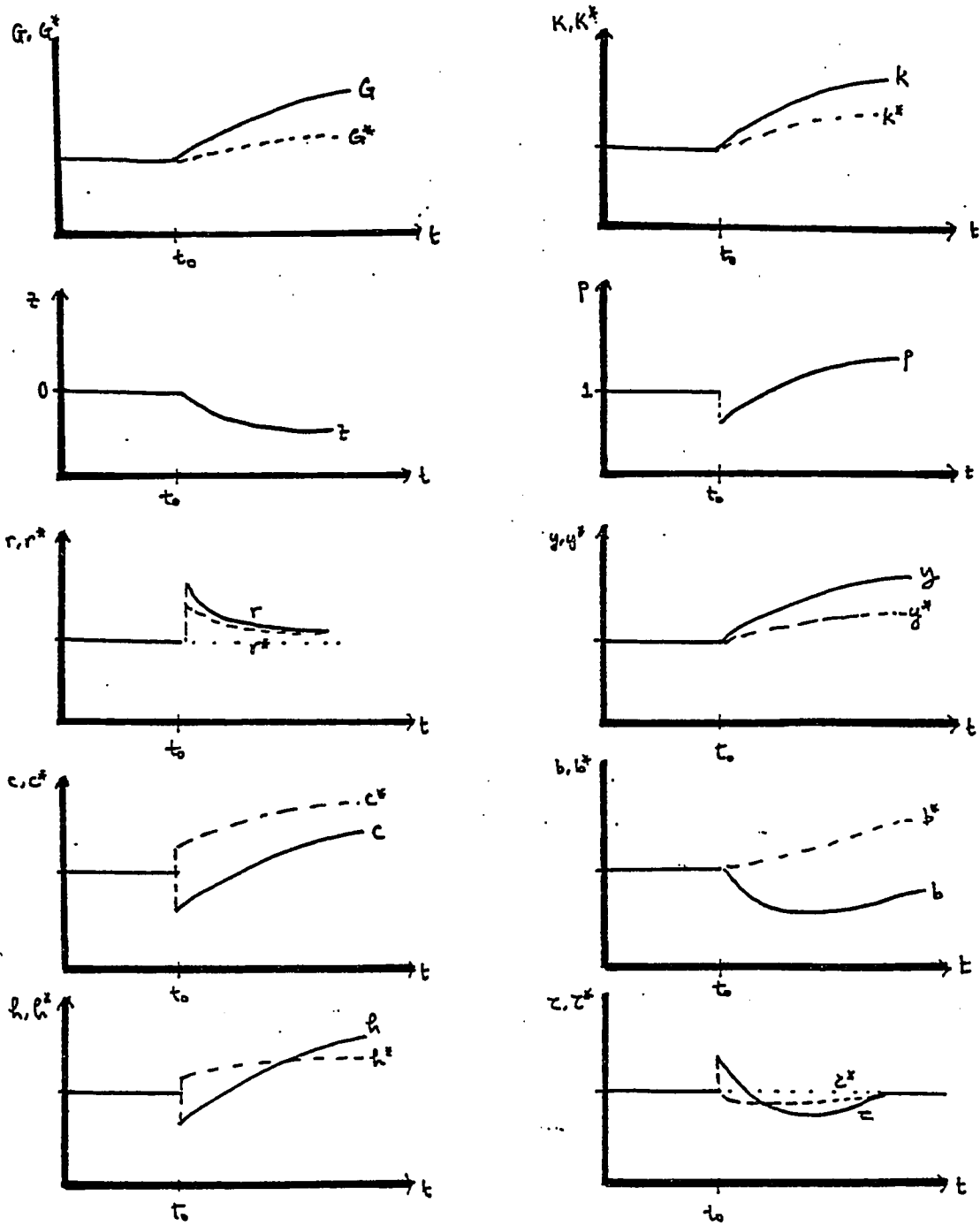


Figure 2.18: Unilateral Conversion to Capital Income Taxation

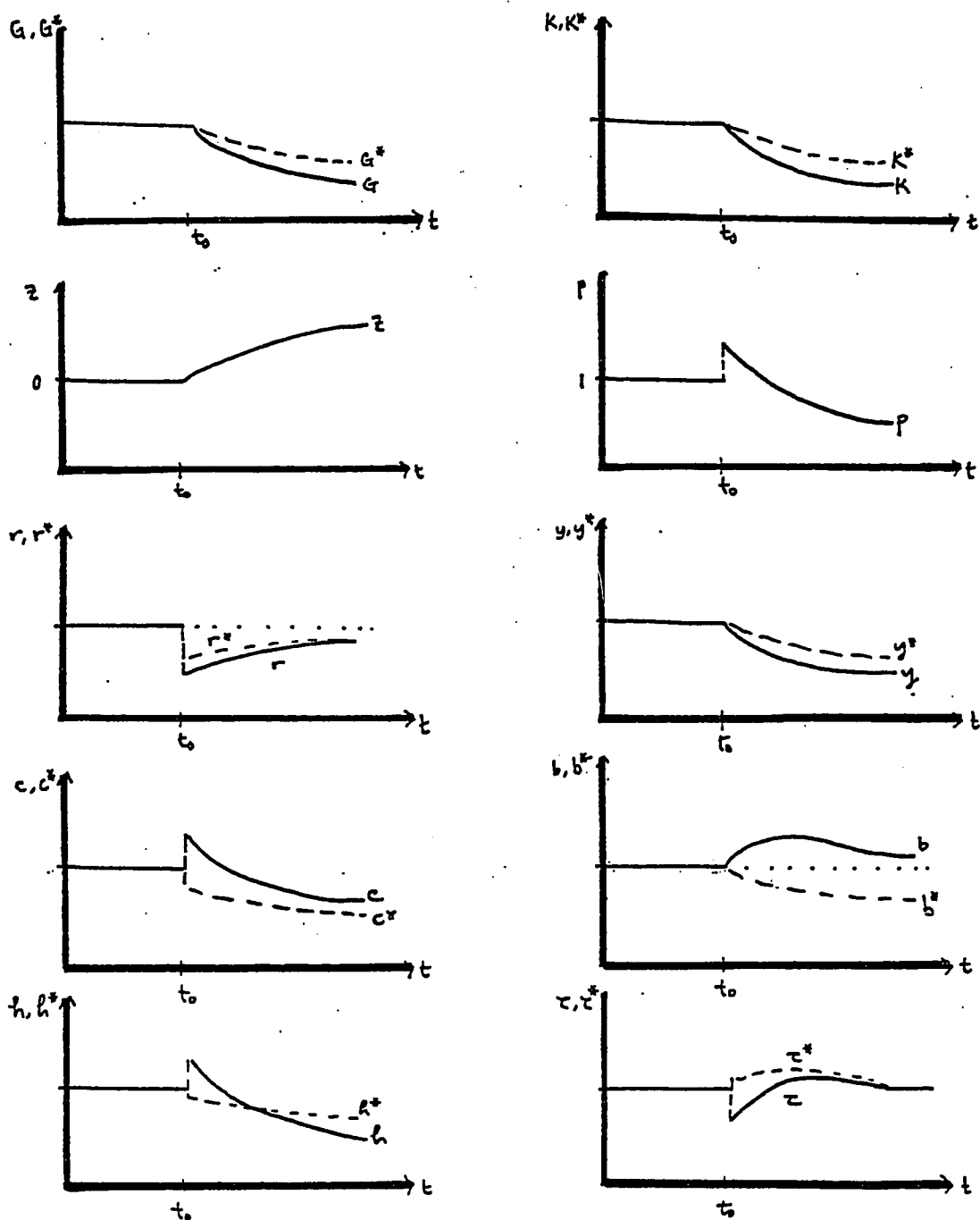


Figure 2.19: Globally Coordinated Conversion to Consumption Taxation

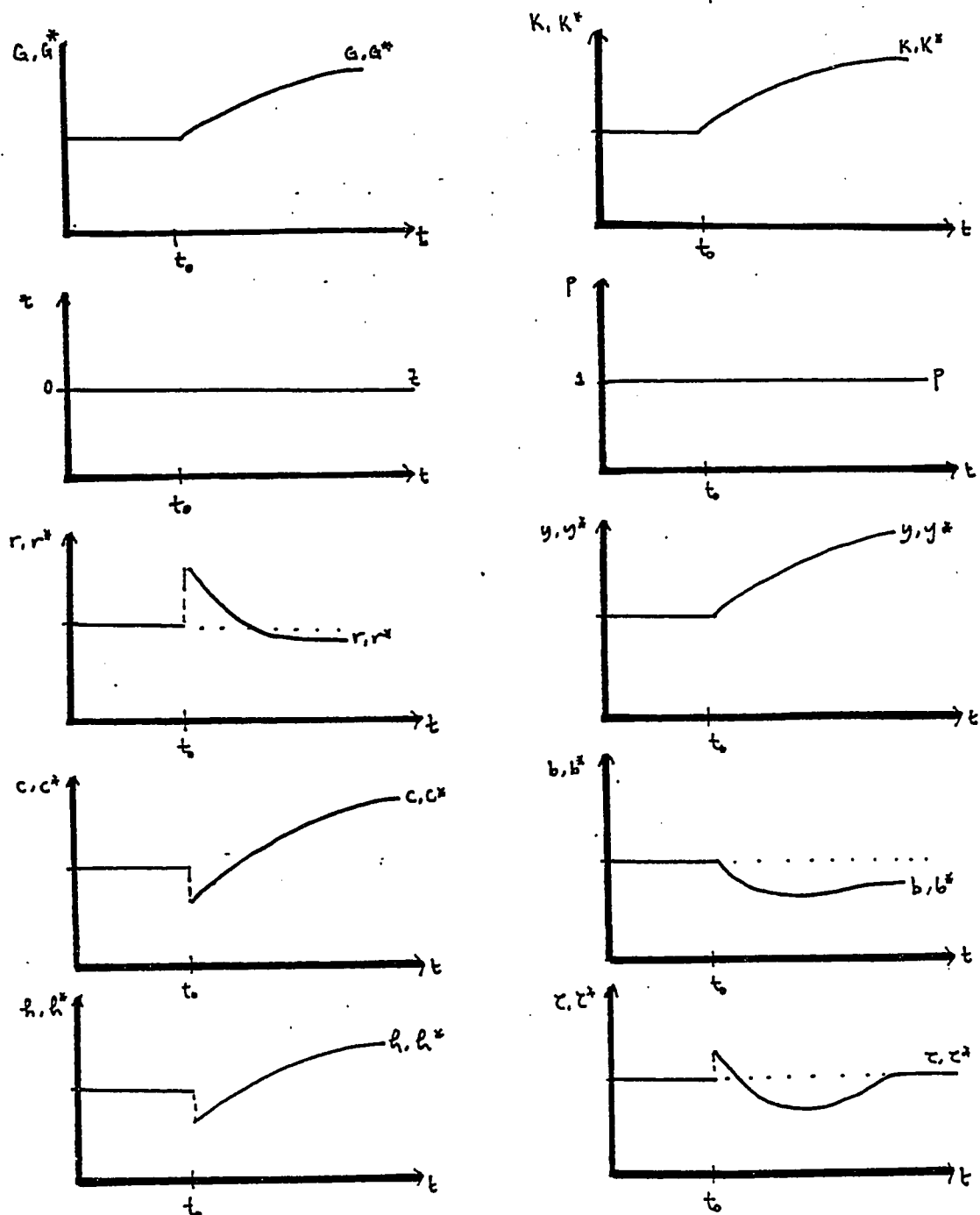
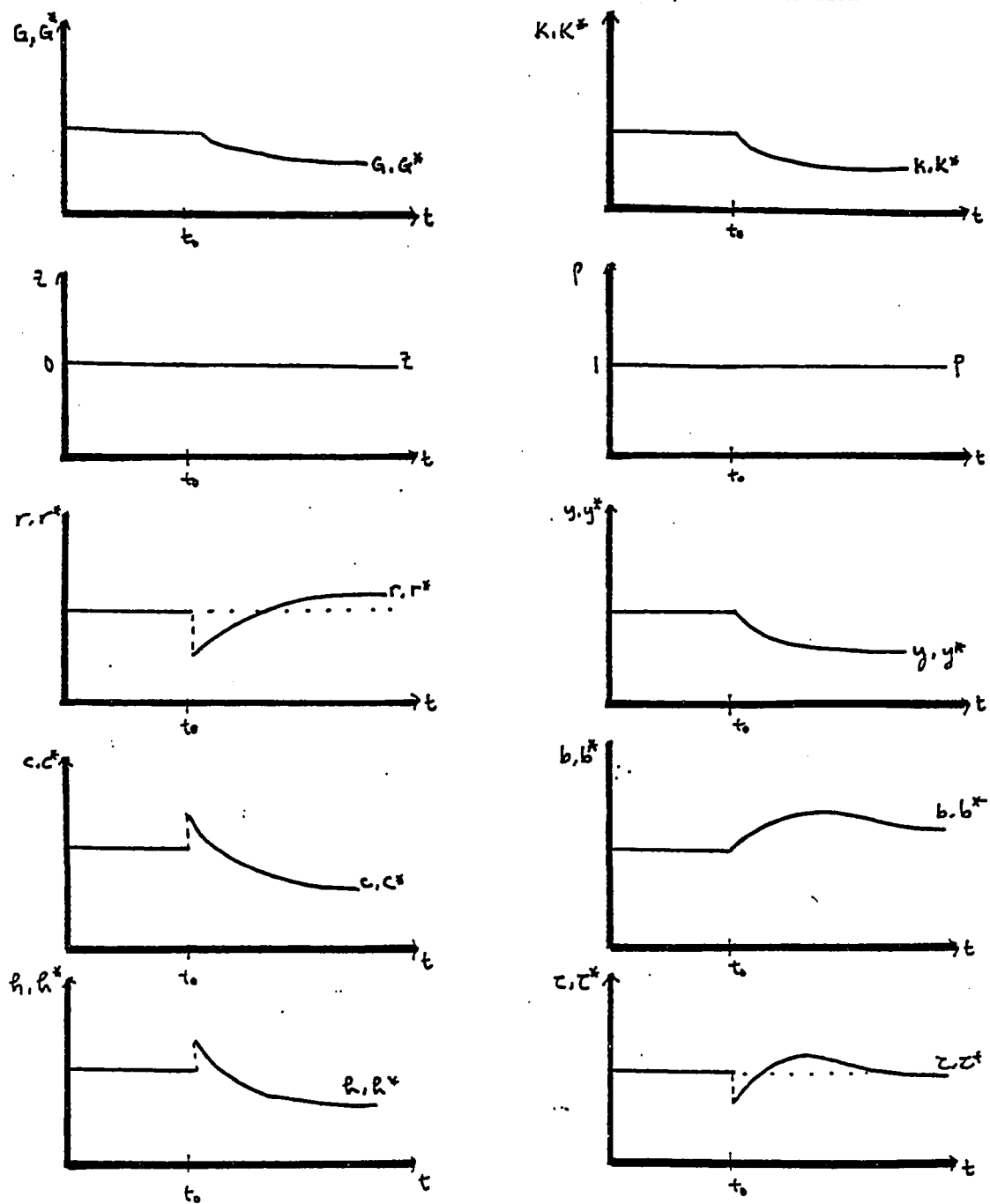


Figure 2.20: Globally Coordinated Conversion to Capital Income Taxation



**Table 2.1: Summary of Two-Country Model****Home Economy****- Firm Sector -**

(i)  $Y = F(G^*, G, K, L)$

(ii)  $\dot{K} = i_p - \delta K$

(iii)  $i_p = \frac{1}{\psi} [q_p - 1]K$

(iv)  $\dot{q}_p = (r + \delta)q_p - (1 - \tau_K)F_K - \frac{\psi}{2} \left( \frac{i_p}{K} \right)^2$

**- Household Sector -**

(v)  $(1 + \tau_c)c = \theta[a + h]$

(vi)  $c_H = \alpha c$

(vii)  $p c_F = (1 - \alpha)c$

(viii)  $a = b + z + q_p K$

(ix)  $\dot{a} = r a + F_L - \tau - (1 + \tau_c)c$

(x)  $\dot{h} = (r + \lambda)h + \tau - F_L$

(xi)  $\dot{\theta} = \theta^2 + \theta \left[ \frac{1}{\sigma} \left( (1 - \sigma)(r + (\alpha - 1)\frac{\dot{p}}{p}) - \rho - \sigma\lambda \right) \right]$

**- Government Sector -**

(xii)  $\dot{G} = i_g - \delta G$

(xiii)  $\dot{a}_g = r a_g + \tau + \tau_c c + \tau_K F_K K - c_g$

where  $a_g = q_g G - b$



**Table 2.1 continued**(xiv) (a)  $i_g$  exogenous

(Exogenous Government)

$$(b) i_g = \frac{1}{\psi}[q_g - 1]G$$

(Endogenous Government)

where

$$\dot{q}_g = (r + \delta)q_g - F_G - \frac{\psi}{2}\left(\frac{i_g}{G}\right)^2 \quad \text{if National Planner}$$

$$\dot{q}_g = (r + \delta)q_g - F_G - pF_G^* - \frac{\psi}{2}\left(\frac{i_g}{G}\right)^2 \quad \text{if Global Planner}$$

$$(xv) \tau = \tau_0 + \eta \dot{b}, \quad \eta < -1$$

- External Balance -

$$(xvi) \dot{z} = rz + c_H^* - pc_F$$

$$(xvii) y = c_H + c_H^* + c_g + J_p + J_g$$

$$\text{where } J_p = i_p + \frac{\psi}{2}\left(\frac{i_p}{K}\right)^2, \quad J_g = i_g + \frac{\psi}{2}\left(\frac{i_g}{G}\right)^2$$

**Foreign Economy**

- Firm Sector -

$$(i)' Y^* = F^*(G, G^*, K^*, L^*)$$

$$(ii)' \dot{K}^* = i_p^* - \delta^* K^*$$

$$(iii)' i_p^* = \frac{1}{\psi^*}[q_p^* - 1]K^*$$

**Table 2.1 continued**

$$(iv)' \dot{q}_p^* = (r^* + \delta^*)q_p^* - (1 - \tau_K^*)F_{K^*}^* - \frac{\psi^* \left( \frac{i_p^*}{K^*} \right)^2}{2}$$

- Household Sector -

$$(v)' (1 + \tau_c^*)c^* = \theta^*[a^* + h^*]$$

$$(vi)' c_H^* = \alpha^* p c^*$$

$$(vii)' c_F^* = (1 - \alpha^*)c^*$$

$$(viii)' a^* = b^* + z^* + q_p^* K^*$$

$$(ix)' \dot{a}^* = r^* a^* + F_{L^*}^* - \tau^* - (1 + \tau_c^*)c^*$$

$$(x)' \dot{h}^* = (r^* + \lambda^*)h^* + \tau^* - F_{L^*}^*$$

$$(xi)' \dot{\theta}^* = \theta^{*2} + \theta^* \left[ \frac{1}{\sigma^*} \left( (1 - \sigma^*)(r^* + \alpha^* \frac{\dot{p}}{p}) - \rho^* - \sigma^* \lambda^* \right) \right]$$

- Government Sector -

$$(xii)' \dot{G}^* = i_g^* - \delta^* G^*$$

$$(xiii)' \dot{a}_g^* = r^* a_g^* + \tau^* + \tau_c^* c^* + \tau_K^* F_{K^*}^* K^* - c_g^*$$

$$\text{where } a_g^* = q_g^* G^* - b^*$$

(xiv)' (a)  $i_g^*$  exogenous

(Exogenous Government)

$$(b) i_g^* = \frac{1}{\psi^*} [q_g^* - 1] G^*$$

(Endogenous Government)

**Table 2.1 continued**

where

$$\dot{q}_g^* = (r^* + \delta^*)q_g^* - F_{G^*}^* - \frac{\psi^*}{2} \left( \frac{i_g^*}{G^*} \right)^2 \quad \text{if National Planner}$$

$$\dot{q}_g^* = (r^* + \delta^*)q_g^* - F_{G^*}^* - pF_{G^*}^* - \frac{\psi^*}{2} \left( \frac{i_g^*}{G^*} \right)^2 \quad \text{if Global Planner}$$

$$(xv)' \tau^* = \tau_0^* + \eta^* \dot{b}^*, \quad \eta^* < -1$$

- External Balance -

$$(xvi)' \dot{z}^* = r^* z^* + c_F - \frac{c_H^*}{p}$$

$$(xvii)' y^* = c_F + c_F^* + c_g^* + J_p^* + J_g^*$$

$$\text{where } J_p^* = i_p^* + \frac{\psi^*}{2} \left( \frac{i_g^{*2}}{K} \right), \quad J_g^* = i_g^* + \frac{\psi^*}{2} \left( \frac{i_g^{*2}}{G} \right)$$

**Global Linkages**

$$(xviii) \dot{r} = r^* + \frac{\dot{p}}{p}$$

$$(xix) z + pz^* = 0$$

**Table 2.2. Long Run Tax Increase in Home Country**  
(ie.  $\tau_0$  raised by 25% of initial steady-state level)

|    | <u>Ratio of Final Steady-State Values to Initial</u> |                         |                       |
|----|------------------------------------------------------|-------------------------|-----------------------|
|    | <u>Exogenous Government</u>                          | <u>National Planner</u> | <u>Global Planner</u> |
| K  | 0.994                                                | 0.9896                  | 0.9789                |
| K* | 0.994                                                | 0.9896                  | 0.9789                |
| y  | 0.998                                                | 0.9943                  | 0.9884                |
| y* | 0.998                                                | 0.9943                  | 0.9884                |
| c  | 0.98                                                 | 0.9759                  | 0.99                  |
| c* | 1.018                                                | 1.0138                  | 0.99                  |

**Table 2.3. Long Run Tax Decrease in Both Countries**  
(ie.  $\tau_0, \tau_0^*$  cut by 25% of initial steady-state level)

|    | <u>Ratio of Final Steady-State Values to Initial</u> |                         |                       |
|----|------------------------------------------------------|-------------------------|-----------------------|
|    | <u>Exogenous Government</u>                          | <u>National Planner</u> | <u>Global Planner</u> |
| K  | 1.013                                                | 1.02                    | 1.04                  |
| K* | 1.013                                                | 1.02                    | 1.04                  |
| y  | 1.005                                                | 1.011                   | 1.02                  |
| y* | 1.005                                                | 1.011                   | 1.02                  |
| c  | 1.0024                                               | 1.008                   | 1.012                 |
| c* | 1.0024                                               | 1.008                   | 1.012                 |

**Chapter 3:**  
**Optimal Public Consumption and  
Government Size in Selected OECD Economies**

## **I. Introduction**

This chapter is an empirical study of public consumption expenditures in selected OECD countries. The primary purpose is to test whether the data conform to a particular characterization of optimal fiscal behaviour. It is hypothesized that governments choose public consumption streams optimally over time and are constrained only by society's permanent income. Governments in general possess advantages in the capital market that private agents do not possess since the institution of government is longer-lived and the tax base of the public authority includes not only the resources of private agents currently alive but also of those yet to be born. This enables governments (especially of developed economies) to trade intertemporally at superior terms of trade.

In this type of environment an analogue of the permanent income hypothesis for government consumption can be formulated. The government is assumed to choose a stream of public consumption expenditures to maximize a utilitarian social welfare function - that is, a weighted aggregate of individual utilities of private agents alive and to be born - subject to a national resource constraint. The conditions for the optimality of public consumption choices can then be derived and their restrictions tested empirically. The outcome of a test of whether governments behave according to this normative characterization is of interest in itself. There are, however, a number of other factors which motivate this study.

First, numerous empirical studies have tested "Wagner's hypothesis"<sup>1</sup> - the hypothesis that increased economic development (higher incomes or per capita incomes) will expand the public sector's involvement in society, as measured by a greater level of government expenditures or share of government expenditures in gross national product. According to the hypothesis, increased public sector activity will result because of the complex social infrastructure, regulations, and laws, needed to cope with the changing economic order. This prediction of increased public sector activity is a distinct strand of thought at odds with both the Marxian view that the state will gradually wither away and with the view that public intervention diminishes as the market economy evolves. A popular approach to testing Wagner's hypothesis has been to examine whether the elasticity of public spending with respect to national income exceeds unity.

One criticism against these studies is that the hypothesis may have been incorrectly interpreted and formulated. In an intertemporal framework, consumption functions take as their arguments "wealth", not "income", as in static, Marshallian demand functions, unless agents are liquidity constrained at points in time. Wagner's law regarding public consumption is the counterpart of Engel's Law regarding private consumption. Previous empirical studies have specified an aggregate demand function for public expenditures and have included current GNP or GDP as an argument. If instead society's demand for public goods were to depend on lifetime wealth, this should explain why current GNP or GDP is not a significant variable in that demand function.

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<sup>1</sup> See Diamond-Tait (1988), Lindauer (1988), Lybeck (1986), Ram (1987), and an earlier study by Peacock-Wiseman (1961).

A second motivation for this chapter is to contrast government consumption behaviour with private consumption behaviour. Earlier, Hall (1978) formulated a permanent income model of private consumption behaviour under rational expectations. If the model were valid, the discounted marginal utility of private consumption should follow a random walk. Subsequent work has been mixed as to whether private consumption does exhibit this property. Flavin (1981), Hayashi (1985), Deaton (1987), Campbell-Deaton (1989), and Campbell-Mankiw (1989), for example, detect the presence of liquidity constraints so that current income does have explanatory power in private consumption. A difficulty with validating the permanent income hypothesis for private consumption is that the assumptions behind the hypothesis place strong restrictions on the data: individuals are not infinitely-lived, the aggregate private economy does not act as one representative individual nor is it characterized by Ricardian equivalence. In contrast, the institution of government is long-lived, represents itself as one national agent, and is not liquidity constrained (at least in developed economies). Thus tests used to reject the permanent income hypothesis for private consumption ought not to reject the hypothesis for public consumption<sup>1</sup> since the conditions under which the hypothesis is derived for public consumption appear to be less restrictive<sup>2</sup>.

A third motivation for this study concerns the recent literature on government and economic growth. Models of endogenous growth theory

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<sup>1</sup> The tests popularized by Hansen (1982), Hansen-Singleton (1982), and by Hall (1978), for example.

<sup>2</sup> These tests may still reject the hypothesis for public consumption if public and private goods consumption are not separable in the private utility function and private consumption is liquidity constrained. In the empirical section I investigate this separability issue further.



permit public policy variables to play an important role in influencing long term growth rates as opposed to having effects only on transitional growth rates as in earlier theories of growth. Several theoretical analyses show that increased public consumption may lower long run growth. Increased public consumption is argued to generate effects which lead to the displacement of private capital especially needed for future growth. In addition, the increased tax burden associated with increased government size is argued to create long-lasting disincentive effects which also discourage growth. Empirical studies by Barro (1981,1989), Grier-Tullock (1989), and Aschauer (1989a) find that public consumption either has negative output effects or is insignificant. What is not clear from the empirical studies is which variable is causing what: is it the supply side that is being captured (that is, the impact of public consumption on national output) or the demand side (that is, the impact of GNP on public consumption demand)? If it is the demand function that is being captured, the insignificance of the relationship between public consumption and GNP is consistent with the permanent income hypothesis which states that public consumption choices depend on lifetime wealth, not current income. A negative relationship between public consumption and national income also arises if public consumption is used as a "Keynesian" stabilization policy device.

To test a permanent income theory of public consumption, I have organized this chapter as follows: in Section II, I provide some background statistics and a brief literature review. In section III, I present a simple intertemporal model of optimal government consumption choice and derive the optimality conditions necessary for social welfare maximization. In section IV, the methodology and data are described. In

section V, the main empirical results are presented: first I conduct specification tests on the social welfare function; secondly, I conduct tests of overidentification to determine whether the restrictions imposed by optimization hold; thirdly, I re-examine Wagner's Hypothesis using two measures of permanent income. Section VI concludes along with some thoughts for extensions.

## **II. Background and Literature Review**

The recent growth in government size in the OECD is well documented in a recent study by Saunders-Klau (1985). The picture painted in their study is one of considerable increases in government size (defined as the share of total government expenditures in GDP) over the past two decades, especially in European countries<sup>1</sup>. Government size in Canada and the United States, however, is relatively smaller and has grown at a relatively slower pace. Another observation is that a large part of this growth is attributable to the growth in income maintenance (pensions, family allowances, sickness benefits) and welfare state expenditures (merit goods, education, health, housing), and public debt interest. Expenditures on public goods (defense, general economic services) as proportions of GDP have also increased but not as rapidly, and have even declined as a percentage of national income in some countries such as the U.S. and the United Kingdom<sup>2</sup>.

In this chapter, the focus is on a narrower component of government expenditures: namely public consumption. In figures 3.1-3.5,

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<sup>1</sup> Even more so for the Scandinavian economies.

<sup>2</sup> This information is presented in Tables 8 and 9 of Saunders-Klau (1985).

I plot the ratio of public consumption spending to national income for five OECD countries: the U.S., Canada, United Kingdom, France, and Italy. Public consumption expenditures refer here to expenditures on public goods, merit goods, and economic services<sup>1</sup>. All variables are in real per capita terms<sup>2</sup>. U.S. data are in constant 1982 dollars while Canadian data are in 1981 dollars. Data for France, Italy, and the U.K. are in constant 1985 prices.

The figures show that the share of public consumption in GNP or GDP has tended to be steady in recent years. The figures for Canada and the U.S. show that the share is much less steady if defence expenditures are not netted out. These figures preclude the possibility of finding a high elasticity of public consumption with respect to either GNP or GDP.

Thus an alternative hypothesis to explore is the possibility that public consumption grows at a rate roughly equal to the long run growth of an economy. This possibility is raised by two sources. One is a survey of government expenditure planning and management in the OECD prepared by the OECD<sup>3</sup>. The other is a collection of recent studies on 'government and growth' which has questioned the use of current income measures. I turn first to the survey.

A key impression made in the survey is that budgetary planners try to be "forward-looking".

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<sup>1</sup> By economic function, public consumption includes: public order and safety; defense; education, health; welfare services; housing and community amenities; recreational, cultural, and religious affairs; economic services in Energy, Agriculture, Forestry, Fishing, Mining, Manufacturing, Construction, and Transportation.

<sup>2</sup> Variables are divided by Population 15 years and over.

<sup>3</sup> see OECD (1987) cited in the reference.

"Most countries develop multi-year projections of the financial consequences of some, if not all, expenditure programmes or investment plans. A few countries develop detailed, annually updated, multiyear expenditure and financial plans, providing targets for the preparation of future years' departmental or programme budgets" (p.11)

Over this horizon, projections are made as to prospective resources available for the expenditures - projections about the trend in national income and fiscal revenues, and about other relevant matters.

Recent academic work is also sensing some limitations with the use of current income measures to explain government size. Alt-Chrystal (1979) employ permanent income (created as a distributed lag on past levels of income with exponentially declining weights) as a determinant of government growth. Their motivation is based on a consumption-smoothing hypothesis:

"Governments plan expenditure to grow in proportion to expected national income. These plans are sticky and are revised when it is clear that trend income has changed. There will, however, be random deviations of [government expenditures less transfers] from plan and of [income] from its expected level, but these deviations will be uncorrelated. This formulation is exactly that invented by Friedman ... to explain the relationship between aggregate consumption and disposable income" (p.132)

Borooah (1988) builds on Alt-Chrystal to examine whether governments target consumption expenditure to "expected" growth in national income. The methodology is based on Flavin (1981). Government consumption is postulated to be proportional to permanent income, where permanent income is the present discounted value of current and future income. The income process is modelled as a first-order autoregressive process for most countries and up to a fourth-order process

for a few other countries. Consequently, the idea is to test whether current government consumption deviates from last period's amount only by the contemporaneous revision in permanent income. This test is essentially, like Flavin's, a test of excess sensitivity, and the results are, like Flavin's, rejections of the permanent income hypothesis. A problem not investigated by Borooah, however, is the consequence of a unit root in the income process. The literature on private consumption has for some time recognized that in this situation standard testing procedures are biased towards finding excess sensitivity. In another line of research, Campbell-Deaton (1989) have discovered the paradox that in the presence of a unit root, permanent income will be noisier than measured income, making observed consumption "too smooth" from the point of view of the permanent income hypothesis.

Courakis et al. (1990) also introduce permanent income into their analysis of government growth in Greece and Portugal. Permanent income is measured as a weighted average of current and past values of measured income with geometrically declining weights. The aim of their paper is not to test a theory of optimizing governments, as in this paper, but to test Wagner's hypothesis in the traditional fashion - namely through the "elasticities" formulation. That is they evaluate whether the elasticity of various components of public expenditures with respect to their definition of permanent income exceeds unity.

A recent study by Henrekson (1989) is critical of the econometric methodologies used in previous tests of Wagner's Law, particularly the failure to deal with issues involving "non-stationarities". Henrekson (1989) argues that earlier empirical results tend to be spurious because tests of Wagner's hypotheses have been carried out on variables that are not

cointegrated. His tests find that government exhaustive expenditures and GDP for Sweden are not cointegrated. However this does not rule out that these variables can be functionally related even if they are not cointegrated.

### **III. An Intertemporal Model of Government Consumption Choice**

The government is postulated to be a utilitarian planner who seeks to maximize by choice of a stream of public consumption expenditures the discounted weighted sum of private utilities<sup>1</sup> of agents currently alive and of those to be born, subject to the national resource constraint. The private agents in turn are postulated to derive satisfaction from the consumption of both private and public goods. Public consumption expenditures benefit private agents through the provision and upkeep of law and order, health and social services, parks and recreation, environmental regulation, postal services, and other public goods and services. Private agents also benefit from direct purchases by the government on their behalf, for example through public housing and food stamp programs.

To analyze optimal fiscal policy, a tractable social welfare function needs to be specified. I first postulate a private utility function with public and private consumption as arguments, and where each agent in the economy is indexed by date of birth  $v$  - ie.  $u(c(t,v),g(t))$ , where  $c(t,v)$  and  $g(t)$  denote private consumption and per capita public consumption respectively. The same amount of per capita supply of public goods is

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<sup>1</sup> Each of which is a multi-period (discounted) utility stream.

available to all agents, hence  $g$  does not depend on  $v$ <sup>1</sup>. Next, the benevolent social planner aggregates the welfare of all  $v$  agents (currently alive and to-be-born) and maximizes the present discounted value of this aggregate welfare over time subject to the national resource constraint<sup>2</sup>. The command solutions are time-consistent (which depends on the social objective function treating all agents alive and to-be-born symmetrically). In addition, the planner employs intercohort lump-sum transfers-cum-taxes to distribute welfare optimally amongst agents at each point in time using the planner's social discount rate to weight different cohorts. In this way the full command optimum can be implemented. Appendix 3.1 formalizes this process and arrives at the following social welfare function for the planner at time  $t=0$ :

$$(1) \quad SW_0 = E_0 \sum_{s=0}^{\infty} V(c(s), g(s)) \left( \frac{1+n}{1+\rho} \right)^s$$

where  $E_0$  is the expectations operator conditional on information at time  $t=0$ . Lowercase letters  $c(s)$  and  $g(s)$  denote real aggregate per capita magnitudes of private and public consumption respectively;  $n$  denotes the growth rate of population, and  $\rho$  is the planner's time preference rate. The aggregate utility function in (1), ie.  $V(c(s), g(s))$ , is the result of maximizing an indirect intratemporal utility function across agents of different ages and substituting the resulting first-order necessary conditions back into the indirect utility function.

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<sup>1</sup> Thus no agent is excluded from enjoying the per capita supply of public goods. It is assumed though that the supply of public goods is congestible, which is why it is divided by population.

<sup>2</sup> To be specified as the GDP identity for an open-economy.

The social planner faces the following national resource constraint:

$$(2) \quad z(t+1) = \left( \frac{1+r(t+1)}{1+n} \right) (z(t) + y(t) - g(t) - c(t) - i(t))$$

$$(2a) \quad y(t) = f(k(t))$$

$$(2b) \quad (1+n)k(t+1) = (1-\delta_K)k(t) + i(t)$$

where  $z$  represents the real per capita stock of external assets (if  $z > 0$ ) or debt (if  $z < 0$ ),  $y$  the real per capita domestic output,  $k$  the real per capita stock of physical capital,  $i$  real per capita private investment, and  $\delta_K$  the depreciation rate. The real interest rate is given by  $r$ .  $f$  is a neoclassical production function with  $f' > 0$ ,  $f'' < 0$ . In this model, I abstract from public investment. The conditions for optimal public consumption would be unchanged. The added complication is that an optimizing government is likely to choose investment expenditures optimally as well, as in the previous chapter. I wish in this chapter to focus primarily on public consumption.

I assume the nation is solvent, and require the following transversality condition to hold:

$$(3) \quad \lim_{T \Rightarrow \infty} z(T) \left[ \prod_{j=t+1}^T \left( \frac{1+r(j)}{1+n} \right) \right]^{-1} = 0.$$



Maximizing (1) subject to (2), by choice of  $\{c(s)\}$ ,  $\{g(s)\}$ , and  $\{z(s)\}$ , for  $s \in [0, \infty)$  gives the following necessary conditions:

$$(4a) \quad E_0 \left\{ \left( \frac{V_c(c(s+1), g(s+1))}{V_c(c(s), g(s))} \right) \left( \frac{1+r(s+1)}{1+\rho} \right) - 1 \right\} = 0$$

$$(4b) \quad E_0 \left\{ \left( \frac{V_g(c(s+1), g(s+1))}{V_g(c(s), g(s))} \right) \left( \frac{1+r(s+1)}{1+\rho} \right) - 1 \right\} = 0$$

$$(4c) \quad E_0 \{ V_c(c(s+1), g(s+1)) - V_g(c(s+1), g(s+1)) \} = 0$$

for  $s \in [0, \infty)$ , where the expectations operator is conditional on information at time  $t=0$  and  $V_g$ ,  $V_c$  are the partial derivatives of  $V$  with respect to government consumption and private consumption respectively. Condition (4a) indicates that at the margin the government cannot alter expected welfare by reducing a unit of private goods consumption at time  $t$  to increase private goods consumption by  $\frac{(1+r(t+1))}{(1+\rho)}$  at time  $t+1$  (where  $\rho$  is the social planner's rate of discount). Condition (4b) is a similar condition with respect to public goods consumption. Condition (4c) is a static condition stating that the planner cannot raise expected welfare at the margin by reducing one unit of private consumption (thereby decreasing welfare by  $V_c$ ) and spending the proceeds on public consumption (thereby raising welfare by  $V_g$ ).

Since all relevant information has been incorporated in determining the optimal path of public consumption, any source of deviation from the optimal "marginal" rule must be due to unanticipated news or information, for example about permanent national income, that was not known earlier

when expectations were formed and plans made. Thus the system (4a,b,c) holds for the intertemporally optimizing government constrained only by the permanent income of the nation. It would not hold under circumstances in which governments were liquidity constrained or rationed by other national governments. Nor would it hold if government objective functions were based not on valuing the consumption of private agents but on some other target objectives or normative criteria, or if governments behaved irrationally or exogenously to economic circumstances. It would also not hold if public consumption were not "separable" from private consumption in the utility function  $V$  and if the private sector did not exhibit life-cycle behaviour.

The necessary conditions can be rewritten to allow for "errors":

$$(4a)' \left\{ \left( \frac{V_c(c(s+1), g(s+1))}{V_c(c(s), g(s))} \right) \left( \frac{1+r(s+1)}{1+\rho} \right) - 1 \right\} = \varepsilon_{1t+1}$$

$$(4b)' \left\{ \left( \frac{V_g(c(s+1), g(s+1))}{V_g(c(s), g(s))} \right) \left( \frac{1+r(s+1)}{1+\rho} \right) - 1 \right\} = \varepsilon_{2t+1}$$

$$(4c)' \{ V_c(c(s+1), g(s+1)) - V_g(c(s+1), g(s+1)) \} = \varepsilon_{3t+1}$$

where  $E_0(\varepsilon_{1t+1}|\Omega_0)$ ,  $E_0(\varepsilon_{2t+1}|\Omega_0)$ , and  $E_0(\varepsilon_{3t+1}|\Omega_0)$  have mean zero conditional on  $\Omega_0$ , the government's information set at time 0. These orthogonality conditions are used to test the overidentifying restrictions imposed by the theoretical framework. The system given by (4a)', (4b)' and (4c)' can be estimated by generalized method of moments after specifying some functional form for  $V$ . The instruments used are variables

assumed to be in the government's information set at  $t=0$ . The permanent income hypothesis is rejected if the errors are correlated with the instruments.

#### **IV. Methodology and Data**

The methodology is based on Hansen (1982), and applications can be found in Hansen-Singleton (1982), Mankiw-Rotemberg-Summers (1985), and Fair (1987). Hansen-Singleton perform tests on a nonlinear system such as (4a'-c') to determine whether the disturbance terms are truly orthogonal to elements of the decision maker's information set by expanding the information set to see if that helps explain the system. Hall's (1978) approach assumes quadratic utility and a constant real interest rate, and tests whether adding additional regressors helps explain the (linear) consumption Euler condition. Both the linear and nonlinear approaches are thus similar in spirit. Both are tests of whether forecasting errors are correlated with information available at the date expectations are formed and optimal plans made. The empirical section will report the results of both approaches, as applied to our government Euler conditions.

Estimating and testing the Euler conditions directly has the advantage of not requiring the econometrician to specify the entire stochastic environment facing the government, nor to solve and estimate all the equilibrium conditions.

The Hansen (1982) approach estimates the parameters of the first-order conditions by a nonlinear instrumental variables procedure. To illustrate briefly, let  $\varepsilon = \varepsilon(X_{t+1}, \varpi)$  be a function mapping  $R^k \times R^p$  into  $R^m$ , where  $X_{t+1}$  is a  $k$ -dimensional vector of variables at time  $t+1$  and  $\varpi$  is a  $p$ -

dimensional vector of parameters. Assume that  $E_t \varepsilon(X_{t+1}, \varpi) = 0$ . Thus  $\varepsilon$  corresponds to the disturbance vector  $(\varepsilon_{1t+1} \varepsilon_{2t+1} \varepsilon_{3t+1})'$  in system (4a'-c'), and  $m$  is the number of equations in that system. Let  $s(X_{t+1}, \varpi, Z_t) = \varepsilon(X_{t+1}, \varpi) \otimes Z_t$  be the Kroneker product of  $\varepsilon$  and a  $q$ -dimensional vector of instruments  $Z_t$ . The  $s$  function maps  $R^k \times R^p \times R^q$  into  $R^h$  where  $h = m \times q$ , ie. the number of equations times the number of instruments.

If  $Z_t$  is in the agent's information set, an implication of the permanent income hypothesis is that  $E[s(X_{t+1}, \varpi, Z_t)] = 0$ , where  $E$  is the unconditional expectations operator. Here  $E[s(X_{t+1}, \varpi, Z_t)] = 0$  represents a set of  $h$  population orthogonality conditions. Hence define:

$$S_T(\varpi) = \frac{1}{T} \sum_{t=1}^T s(X_{t+1}, \varpi, Z_t)$$

and

$$J_T(\varpi) = S_T(\varpi)' W_T S_T(\varpi)$$

where  $W_T$  is a symmetric  $h \times h$  positive definite weighting matrix. The estimation strategy is to choose  $\varpi$  which minimizes the criterion function  $J_T$ .  $W_T$  defines the metric by which  $S_T$  is made close to zero as possible<sup>1,2</sup>.

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<sup>1</sup> If the errors  $\varepsilon$  are conditionally homoskedastic,  $W_T$  is the inverse of the variance-covariance matrix of  $S_T$ . Otherwise  $W_T$  is more involved. See Hansen (1982). The Rats Package, Version 3.1, provides GMM estimation and tests of overidentification.

<sup>2</sup> To deal with autoregressive errors, I follow Fair (1987) and Gallant (1987) in rewriting the nonlinear function  $y = f(x; \varpi) + \varepsilon$  in first differences, where  $\varpi$  is the vector of parameters: ie.  $y - \alpha y_{-1} = f(x; \varpi) - \alpha f(x_{-1}; \varpi) + \varepsilon - \alpha \varepsilon_{-1}$ , where  $\alpha$  is an AR(1) coefficient. The instrument set must then be moved backward in time as the order of the autoregressive

The estimation procedure sets the  $p$  linear combinations of  $h$  orthogonality conditions to zero. However, as long as  $h > p$ ,  $(h - p)$  conditions are not set to zero in estimation, but if the model's restrictions are valid they should be close to zero. I regard the 'restricted' case to be one where not all  $(h - p)$  conditions are set to zero. An alternative 'unrestricted' procedure would be to estimate all the orthogonality conditions, equation by equation using ordinary least squares, which would set all the  $h$  orthogonality conditions to zero. The value of the criterion function,  $J_T$ , is zero under the unrestricted case and nonzero under the restricted.

It follows that a test of the overidentifying restrictions can be carried out by realizing that the minimized value of the criterion function,  $J_T$ , times the sample size  $T$  is asymptotically distributed as a chi-square random variable with degrees of freedom equal to  $(h - p)$ . If the over-identifying restrictions are not valid, the J-statistic - defined as  $J_T$  times  $T$  - will tend to be high, exceeding the critical value of the chi-square at the appropriate degrees of freedom, and thereby cast doubt on the restrictions the first-order conditions place on the data.

The remainder of this section will discuss the data. Five OECD countries are studied, two of which (namely Canada and the United States) are examined in more detail - that is, I examine different levels of government and the consequences of including and netting out defense expenditures from public consumption. The other three countries are: France, Italy, and the United Kingdom. Data for Canada are taken from Statistics Canada, and data for the U.S. are from N.I.P.A. (ie. National

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process increases or else the instrument set will not be truly predetermined. (cf. Fair 1987).

Income and Product Accounts) and from the Survey of Current Business. Data for France, Italy, and the United Kingdom are from the OECD's Quarterly National Accounts. Interest rate and population data are from the IMF's International Financial Statistics.

The sample period for Canada and the U.S. is 1957 (1st quarter) to 1988 (4th quarter). The sample period for the U.K. is 1965 (1st quarter) to 1990 (4th quarter), for France it is 1966 (1st quarter) to 1990 (3rd quarter), and for Italy it is 1973 (1st quarter) to 1990 (3rd quarter).

The three-month Treasury bill rate is used for Canada, the U.S., and the U.K., while a 6-month rate is used for Italy and France. The real interest rate  $r$  is obtained from:

$$(1 + r) = \frac{1 + R}{1 + \pi},$$

where  $R$  is the nominal rate, and  $\pi$  the inflation rate. All other data are converted into real<sup>1</sup> and per capita terms. I obtained quarterly population data by simple linear interpolation, using the reported annual figures as mid-year values.

Private consumption is net of durables and semi-durables. For the U.S. and Canada, government consumption can be broken down into defense and non-defense expenditures, and by level: ie. Federal versus State-Local (or Provincial). The national income/output measure used for

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<sup>1</sup> Own price deflators are used: that is, government consumption is deflated by the government consumption deflator and private consumption is deflated by its own price deflator. As mentioned earlier, U.S. data are in constant 1982 prices, Canadian data in constant 1981 prices, and French, Italian, and U.K. data are in constant 1985 prices.

France, Italy, and the U.K. is GDP, while for Canada and the U.S., I use GNP. The results are insensitive as to which of the two income measures is used.

## **V. Empirical Results**

This section is organized into three parts: the first part examines various functional forms for  $V(c(t), g(t))$  and conducts likelihood ratio tests to see which versions are rejected. In particular I test whether  $g(t)$  and  $c(t)$  are non-separable in private utility functions. The second part conducts tests of overidentification to determine whether the restrictions imposed by the permanent income hypothesis hold. I also perform the same tests on a quadratic utility, constant real interest version as used in previous work. The third part re-examines the "Wagner" hypothesis using a measure of permanent income.

### **(i) Tests of Specification:**

I first examine the issue of non-separability between private and public goods consumption in private utility. Following Mankiw-Rotemberg-Summers (1985) and MaCurdy (1981), I adopt the functional form below:

$$(5) \quad V(c(s), g(s)) = \frac{1}{1-\gamma} \left[ \frac{c(s)^{1-\sigma_1} - 1}{1-\sigma_1} + \Gamma \frac{g(s)^{1-\sigma_2} - 1}{1-\sigma_2} \right]^{1-\gamma}$$

In this specification, additive separability is the special case where  $\gamma=0$ . As in the Appendix, I interpret  $\Gamma$  to be the relative utility weight of public goods consumption in terms of utility from private goods consumption. A CES function of  $c(s)$  and  $g(s)$  results when  $\sigma_1=\sigma_2$ . Log utility functions of  $c(s)$  and  $g(s)$  result when  $\sigma_1=\sigma_2=1$  and  $\gamma=0$ . If  $\gamma=0$ , the intertemporal elasticities of substitution of private and public goods consumption are  $\frac{1}{\sigma_1}$  and  $\frac{1}{\sigma_2}$  respectively. If  $\sigma_1=\sigma_2=\bar{\sigma}$  and  $\gamma$  is nonzero,  $\frac{1}{\gamma}$  is the intertemporal elasticity of substitution of the composite (public and private) good, and  $\bar{\sigma}$  is the elasticity of substitution of private goods consumption for public goods consumption. The function in (5) is concave (locally) if all three parameters -  $\sigma_1, \sigma_2$ , and  $\gamma$  - are non-negative.

Table 3.1 presents the unconstrained nonlinear estimation of (5) in column 1 and the constrained estimation, with  $\gamma = 0$  imposed, in column 2. The LR statistic is the test statistic used to test the constraint that  $\gamma = 0$ , and is computed as:

$$LR = 2(L_1 - L_0)$$

where  $L_1$  is the value of the logarithm of the likelihood of the unconstrained model and  $L_0$  the value of the same for the constrained model. LR is distributed asymptotically as a chi-square variable with degrees of freedom equal to the number of constraints.



The likelihood ratio tests suggest that we cannot reject separability. The chi-square value at the 95 percent point for 1 constraint is 3.841, and the 99 percent point is 6.635. I henceforth impose this constraint throughout. If non-separability were imposed, the utility function would be non-concave as the value of  $\gamma$  is negative, though insignificant. Imposing separability leaves the values and significance of the other parameters virtually unchanged. The discount factor  $\frac{1}{1+\rho}$  ranges from 0.91 for Italy to 0.95 for the U.S. Thus the social planner's time preference rate  $\rho$  varies from 0.099 to 0.053. The U.S. government is relatively the most patient, and the Italian government the least patient. The relative utility weight of public goods consumption  $\Gamma$  is significantly less than one, implying that 1 "util" from public goods consumption provides less felicity to private agents than 1 "util" from private goods consumption. The intertemporal substitution elasticities are greater than one, since  $\sigma_1$  and  $\sigma_2$  range in the neighborhood of 0.29 to 0.43, with the intertemporal substitution elasticity of private goods consumption being somewhat greater than that of public goods consumption.

Next I investigate whether  $\sigma_1$ ,  $\sigma_2$ , or both should equal unity. Acceptance of any of these constraints would imply some log utility representation. Table 3.2 presents the results of these specification tests. Column 1 is the estimation of equation (6a) below, column 2 the estimation of equation (6b), and column 3 the estimation of (6c):

$$(6a) \quad V(c(s), g(s)) = \log(c(s)) + \Gamma \frac{g(s)^{1-\sigma_2} - 1}{1 - \sigma_2}$$

$$(6b) \quad V(c(s), g(s)) = \frac{c(s)^{1-\sigma_1} - 1}{1 - \sigma_1} + \Gamma \log(g(s))$$

$$(6c) \quad V(c(s), g(s)) = \log(c(s)) + \Gamma \log(g(s))$$

Again the likelihood ratio (LR) test statistics are included. This time all the constraints imposed are decisively rejected. (The chi-square value at the 95 percent point for 2 constraints, pertaining to the column 3 results, is 5.991, and at the 99 percent point is 9.21). When the constraints are imposed there is a tendency for the relative utility weight of public consumption to fall, except in column 2 for the U.S. and U.K. The discount factor values are only slightly changed, if at all.

An alternative functional form is one which has been adopted in Aschauer (1985), Barro (1981), and Kormendi (1983) in their analyses of direct crowding out:

$$(7) \quad V(c(s), g(s)) = \frac{[c(s) + \theta g(s)]^{1-\sigma_1} - 1}{1 - \sigma_1} + \Gamma \frac{g(s)^{1-\sigma_2} - 1}{1 - \sigma_2}$$

where  $(c(s) + \theta g(s))$  represents the level of effective private consumption. The  $\theta$  parameter indicates how closely public goods consumption substitutes for a unit of private goods consumption. If  $\theta=0$  public goods consumption does not substitute for private goods consumption. Table 3.3 reports the estimation of (7). The  $\theta$  value is very small in magnitude and insignificant. I also list the LR test statistic for testing the constraint that  $\theta$

= 0. The LR values are quite low indicating that the hypothesis of  $\theta = 0$  cannot be rejected. I therefore find very little empirical support for the specification in (7). Note that  $\theta = 0$  also implies separability.

## (ii) Tests of Overidentification

### **A. Nonlinear Case**

In this subsection I report the results of testing the permanent income hypothesis of government behaviour using the welfare specification in (5) with  $\gamma = 0$  imposed. The nonlinear system (4a'-c') is estimated simultaneously. Although one of the three equations in (4a-c) can be derived from the other two, I estimate all three equations in (4a'-c') to recover the utility function parameters, since none of the equations in (4a'-c') is likely to replicate the data exactly.

Before proceeding I describe the instruments used to represent the government's information set. The instruments are formed from lagged values of income  $y$ , private consumption  $c$ , government consumption  $g$ , and the real interest rate  $r$ . There are three sets of instruments called NLAG2, NLAG4, and NLAG6, each corresponding to a different maximum length of lag:

NLAG2 - the longest lag is 2 periods. This set contains: constant term,  $g(-1)$ ,  $g(-2)$ ,  $c(-1)$ ,  $c(-2)$ ,  $y(-1)$ ,  $y(-2)$ ,  $1+r(-1)$ ,  $1+r(-2)$ .

NLAG4 - the longest lag is 4 periods. This set contains: constant term,  $g(-1)$ ,  $g(-2)$ ,  $g(-3)$ ,  $g(-4)$ ,  $c(-1)$ ,  $c(-2)$ ,  $c(-3)$ ,  $c(-4)$ ,  $y(-1)$ ,  $y(-2)$ ,  $y(-3)$ ,  $y(-4)$ ,  $1+r(-1)$ ,  $1+r(-2)$ ,  $1+r(-3)$ ,  $1+r(-4)$ .

NLAG6 - the longest lag is 6 periods. This set contains: constant term,  $g(-1)$ ,  $g(-2)$ ,  $g(-3)$ ,  $g(-4)$ ,  $g(-5)$ ,  $g(-6)$ ,  $c(-1)$ ,  $c(-2)$ ,  $c(-3)$ ,  $c(-4)$ ,  $c(-5)$ ,  $c(-6)$ ,  $y(-1)$ ,  $y(-2)$ ,  $y(-3)$ ,  $y(-4)$ ,  $y(-5)$ ,  $y(-6)$ ,  $1+r(-1)$ ,  $1+r(-2)$ ,  $1+r(-3)$ ,  $1+r(-4)$ ,  $1+r(-5)$ ,  $1+r(-6)$ .

In NLAG2 there are a total of 9 instruments, in NLAG4 there are 17 instruments, and in NLAG6 there are 25 instruments. The more instruments there are, the more orthogonality conditions and overidentifying restrictions there are.

Tables 3.4A-3.4C contain the results. Table 3.4A focuses on the U.S., analyzing various measures of government consumption by defense and non-defense, and by level of government. Table 3.4B does the same for Canada, and Table 3.4C reports the results for the United Kingdom, France, and Italy using only the total government consumption measure.

The J-statistics are compared to the critical chi-square values below:

**Critical Values of the Chi-Square:**

| DF | Upper 5% | Upper 2.5% | Upper 1% |
|----|----------|------------|----------|
| 23 | 35.17    | 38.08      | 41.64    |
| 40 | 55.76    | 59.34      | 63.69    |
| 50 | 67.5     | 71.42      | 76.15    |
| 70 | 90.53    | 95.02      | 100.43   |
| 80 | 101.88   | 106.63     | 112.34   |

D.F. - degrees of freedom

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The degrees of freedom equal the number of overidentifying restrictions; that is,  $(h - p)$ , where  $h$  is the product of the number of instruments  $q$  and the number of equations  $m$ , and  $p$  is the number of parameters estimated. Here  $m=3$  and  $p=4$ . Thus there are 23 degrees of freedom when we employ NLAG2, 47 degrees of freedom when we employ NLAG4, and 71 degrees of freedom when we employ NLAG6.

In Table 3.4A the J-statistics are low compared to the appropriate chi-square values, except in the case where total U.S. government spending is involved and the instrument set is NLAG6. The J-statistic for this case is 102.6 and the chi-square value for the nearest degrees of freedom (ie. 70) is 100.43 at the 99 percent point. It appears that we are at the borderline of deciding whether to reject the overidentifying restrictions at this level of significance<sup>1</sup>. Thus in the vast majority of cases the permanent income hypothesis for public consumption is not rejected for the U.S. As far as the estimates go, note that the relative utility weight of public consumption is greatest when defense expenditures are netted out of total public consumption and when only state-local expenditures are included.

In Table 3.4B, again the overidentifying restrictions are not rejected in a majority of cases. There are close rejections at the 95 percent point when NLAG2 is involved, but not at the 99 percent point. The relative utility weight of public goods consumption is greatest in Canada when total government consumption is involved. The utility weight is next highest when defense expenditures are netted out of total government consumption, and third highest when total provincial consumption spending is involved.

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<sup>1</sup> I have not computed the chi-square value when there are 71 degrees of freedom.

Thus in Canada and the U.S., public goods provision at the provincial or state-local level adds significantly to private and social welfare.

In Table 3.4C the overidentifying restrictions are rejected for the United Kingdom with all three instrument sets - NLAG2, NLAG4, and NLAG6. There does not appear to be evidence that the U.K. government behaves according to the permanent income model. There are also rejections for France and Italy when NLAG2 is involved. For the other two instrument sets, there are no rejections. France and Italy therefore provide modest evidence of governments optimizing with respect to public consumption constrained only by permanent income.

For these European OECD economies, total public consumption tends to have a relatively high and significant utility weight, ranging from 0.73-0.79 for France and Italy and 0.88-0.92 for the U.K. One oddity, however, is that while the estimated values for the social discount factor  $\frac{1}{1+\rho}$  change negligibly for most countries as we alter the estimation particulars, it is not the case for the U.K. The discount factor varies from 0.51 to 0.84, conflicting with the estimates shown in previous tables.

## **B. Linear Case**

I now examine the permanent income hypothesis for government consumption from another angle. Some studies on private consumption behaviour have postulated a quadratic utility function and imposed the assumption that the real interest rate is constant (for example Hall (1978)). This framework yields linear first-order conditions. In this section I will

also adopt these two assumptions and conduct another set of tests of overidentification.

Assume  $r$  is constant and that the private utility function is:

$$(8) \quad V(c(s), g(s)) = -\frac{1}{2}(\bar{c} - c(s))^2 - \frac{\Gamma}{2}(\bar{g} - g(s))^2 + \frac{\Omega}{2}(\bar{g} - g(s))(\bar{c} - c(s))$$

In what follows I set  $\Omega = 0$  (ie. I impose separability). In earlier tests I could not reject the null hypothesis  $\Omega = 0$ . Substituting (8) into (1) (with separability imposed), and then maximizing (1) subject to (2) gives us the following first-order conditions:

$$(9a) \quad E_0 [g(s+1) - \bar{g} \left( \frac{r - \rho}{1 + r} \right) - \left( \frac{1 + \rho}{1 + r} \right) g(s)] = 0$$

$$(9b) \quad E_0 [c(s+1) - \bar{c} \left( \frac{r - \rho}{1 + r} \right) - \left( \frac{1 + \rho}{1 + r} \right) c(s)] = 0$$

$$(9c) \quad E_0 [c(s+1) - (\bar{c} - \Gamma \bar{g}) - \Gamma g(s+1)] = 0$$

where the actual values must obey:

$$(9a)' \quad c(s+1) = \phi_0 + \phi_1 c(s) + \varepsilon_{1t+1}$$

$$(9b)' \quad g(s+1) = \phi_2 + \phi_1 g(s) + \varepsilon_{2t+1}$$

$$(9c)' \quad c(s+1) = \beta_0 + \beta_1 g(s+1) + \varepsilon_{3t+1}$$

$$\text{or } (9c)'' \quad \Delta c(s+1) = \beta_1 \Delta g(s+1) + \varepsilon_{3t+1}'$$

where  $\phi_0 = \bar{c} \left( \frac{r - \rho}{1 + r} \right)$ ,  $\phi_1 = \left( \frac{1 + \rho}{1 + r} \right)$ ,  $\phi_2 = \bar{g} \left( \frac{r - \rho}{1 + r} \right)$ ,  $\beta_0 = (\bar{c} - \Gamma \bar{g})$ , and  $\beta_1 = \Gamma$ , and where  $E_0(\varepsilon_{1t+1}|\Omega_0) = E_0(\varepsilon_{2t+1}|\Omega_0) = E_0(\varepsilon_{3t+1}|\Omega_0) = E_0(\varepsilon_{3t+1}'|\Omega_0) = 0$ , given information at time 0,  $\Omega_0$ .

If  $c$  and  $g$  are close to steady-state and are near the satiation levels  $\bar{c}$ ,  $\bar{g}$  in steady-state, we should expect  $\beta_0$  close to zero and  $\frac{\phi_2}{\phi_0} = \Gamma$ . If  $r = \rho$ , where  $\rho$  is the planner's discount rate, we should expect  $\phi_0 = \phi_2 = 0$  and  $\phi_1 = 1$  - that is, we should obtain driftless random walk processes for  $c$  and  $g$ .

I estimate (9a)', (9b)', and (9c)" as a system by GMM. I estimate (9c)" instead of (9c)' as I find that  $c$  and  $g$  are integrated processes<sup>1</sup>. I also add a constant term  $\beta_0'$  to (9c)" with the expectation that  $\beta_0'$  is close to zero if the government is optimizing according to the theory and if the quadratic utility specification (with constant real interest rates) is valid. As before I employ three different instrument sets: INST2 contains 2 lagged values of  $g$ ,  $y$ ,  $c$ , and a constant term; INST4 contains 4 lagged values of  $g$ ,  $y$ ,  $c$ , and a constant term; and INST6 contains 6 lagged values of  $g$ ,  $y$ ,  $c$ , and a constant term. There are 7 instruments in INST2, 13 in INST4, and 19 in INST6. There are five parameters to estimate, and three equations. Thus the relevant chi-square degrees of freedom associated with INST2, INST4, and INST6 are 16, 34, and 52 respectively.

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<sup>1</sup> As Stock-Watson (1988) note, the coefficient in a regression of one integrated variable on another will be consistent if the two variables are cointegrated (though the asymptotic theory is nonstandard). The error term in the regression equation will be stationary but may not be serially uncorrelated nor independent of the regressors. In Park (1990a) I could not reject the hypothesis of no cointegration between the levels of  $c$  and  $g$  for a sample of OECD economies.



The following chi-square values should aid in our evaluation of the overidentification tests in this section:

**Critical Values of the Chi-Square:**

| DF | Upper 5% | Upper 2.5% | Upper 1% |
|----|----------|------------|----------|
| 16 | 35.17    | 38.08      | 41.64    |
| 30 | 43.77    | 46.97      | 50.89    |
| 40 | 55.76    | 59.34      | 63.69    |
| 50 | 67.5     | 71.42      | 76.15    |
| 60 | 79.1     | 83.3       | 88.4     |

D.F. - degrees of freedom

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Table 3.5 reports the main results for this section. Only the total government consumption measure is utilized. Again the overidentifying restrictions are rejected by the U.K. data. The restrictions are also rejected for Italy, typically at the 97.5 percent point. They are not rejected however for the U.S., Canada, and France. This table therefore provides support for the permanent income model of government consumption for just a subset of countries in the sample.

The parameter estimates however are not consistent with what we have seen in previous tables. For example there are many instances in which the utility weight  $\Gamma$ , played by  $\beta_1$ , is negative. Moreover the value of  $\phi_1$  is near unity if not equal to unity, implying  $r = \rho$ , yet the values of

$\phi_0, \phi_2$  are small but not zero. The theory under quadratic utility with a constant real interest rate implies  $\beta_0'$  zero but this too is not found.

In conclusion, while the parameter estimates in this linear framework conflict with those obtained in the nonlinear one and with some of the framework's own theoretical restrictions imposed by quadratic utility and a constant real interest rate, the overidentification tests are not rejected in a majority of cases, lending some support for the hypothesis that governments select public consumption optimally, constrained only by society's permanent income.

### (iii) Tests of "Wagner's" Hypothesis

If the permanent income hypothesis for government consumption is valid, tests of Wagner's hypothesis of "growing" government size in response to rising incomes must be evaluated in the context of permanent income. In this section I therefore provide estimates of the elasticities of government consumption with respect to two measures of permanent income and evaluate whether Wagner's hypothesis can be verified in a permanent income framework. One measure of permanent income is the traditional (ad-hoc) one based on an exponentially weighted average of current income and past levels of income. Another one is defined as that constant level of income whose present discounted value equals the present discounted value of future expected income. The first measure is backward-looking and the second forward-looking. Both measures of permanent income are considered in order to evaluate Wagner's hypothesis from different angles.

I will look at two versions of "Wagner's" hypothesis, each corresponding to a different measure of government size - ie. relative size (which is the ratio of government spending to GDP) or absolute size (which is the level of government spending).

Uppercase letters denote aggregate variables, N denotes population, Y current income, Y\* permanent income, and T represents a time trend. A subscript of (-1) denotes a lagged variable:

Version 1:  $\log G = \alpha_0 + \alpha_1 \log Y^* + \alpha_2 T + \mu$

Version 2:  $\log\left(\frac{G}{Y^*}\right) = \alpha_0 + \alpha_1 \log\left(\frac{Y^*}{N}\right) + \alpha_2 T + \mu$

where  $\mu$  is the error term.

In Versions 1-2, the parameter  $\alpha_1$  is the constant elasticity of the spending measure vis-a-vis the income measure. For example under Version 2, if  $\alpha_1 > 1$ , an increase in permanent income per capita will be associated with a more than proportionate increase in the ratio of government spending to permanent income. If  $\alpha_1 < 1$ , an increase in permanent income per capita will be associated with a less than proportionate increase in the ratio of government spending to permanent income. Under Version 1, if  $\alpha_1 > 1$ , an increase in permanent income will be associated with a more than proportionate increase in the level of government spending, and a rise in the ratio of government spending to permanent income. If  $\alpha_1 < 1$ , an increase in permanent income will be associated with a fall in the share of government spending in permanent

income. Whichever version it is that better captures Wagner's hypothesis, the hypothesis will be supported if estimated values of  $\alpha_1$  exceed unity.

- Backward-Looking Measure of Permanent Income -

Here the following definitions of permanent income and permanent income per capita are used respectively:

Definition 1:  $\log Y^* = \lambda \log Y^*_{-1} + (1 - \lambda) \log Y$

Definition 2:  $\log \left( \frac{Y^*}{N} \right) = \lambda \log \left( \frac{Y^*_{-1}}{N} \right) + (1 - \lambda) \log \left( \frac{Y}{N} \right)$

Thus the permanent income measures are derived from a weighted average of current and past values of income with geometrically declining weights  $0 \leq \lambda \leq 1$ . The fraction of current income that constitutes permanent income is  $(1 - \lambda)$ .

Substituting definition 1 into version 1 and definition 2 into version 2, I obtain the following two equations to be estimated:

Version 1

$$(10a) \log G = (\alpha_2 \lambda + \alpha_0 (1 - \lambda)) + \lambda \log G_{-1} + \alpha_1 (1 - \lambda) \log Y + \alpha_2 (1 - \lambda) T + \mu$$

Version 2

$$(10b) \log \left( \frac{G}{N} \right) = (\alpha_2 \lambda + \alpha_0 (1 - \lambda)) + \lambda \log \left( \frac{G}{N} \right)_{-1} + (1 + \alpha_1) (1 - \lambda) \log \left( \frac{Y}{N} \right) + \alpha_2 (1 - \lambda) T + \mu$$

I estimate each of these versions by nonlinear instrumental variables. The equations are linear in variables but nonlinear in parameters. Instruments are needed because the error term is not necessarily orthogonal to one of the regressors, namely income. I use the constant term, trend, and 1-period lagged values of the income and government consumption measures as instruments. The results are reported in Tables 3.6A-3.6B. I examine total government consumption only.

Table 3.6A focuses on Version 1 of the Wagner hypothesis. There is support for the hypothesis only for total Canadian government consumption. The elasticity value,  $\alpha_1$ , for the U.K. is high enough to support the hypothesis except that the coefficient is insignificant. For the U.S. and Italy the elasticity values are negative but insignificant. The elasticity parameter is significant for France but its value does not exceed unity. Italy is the only country for which there is a significant positive trend in the ratio of total public consumption to GDP. The values of  $\lambda$  suggest that a small proportion of current income constitutes permanent income.

In Table 3.6B I focus on Version 2. The Wagner elasticities,  $\alpha_1$ , are quite insignificant, except in the case of Italy, although the elasticity is negative for Italy indicating that the share of public consumption in GDP has fallen by more than the rise in permanent income per capita. The values of  $\lambda$  again suggest that a small proportion of current income constitutes permanent income. I find that the trend term is weak (except in the case of Italy) which denies evidence of an autonomous rise in the share of public consumption to GDP over time.

- Forward-Looking Measure of Permanent Income -

Here the measure of permanent income (GNP/GDP) is given by<sup>1</sup>:

$$(11) \log Y^*(t) = \frac{r}{1+r} \sum_{i=0}^{\infty} \frac{E_t \log Y(t+i)}{(1+r)^i}$$

or

$$(11)' \Delta \log Y^*(t) = \frac{r}{1+r} \sum_{i=0}^{\infty} \frac{E_t \log Y(t+i) - E_{t-1} \log Y(t+i)}{(1+r)^i}$$

where  $r$  is the constant real interest rate and  $E_t$  is the expectation operator based on information available at time  $t$ .

One must also specify income processes for GDP/GNP. Following Campbell-Deaton (1989), I use:

$$(12) \Delta \log Y(t) = \eta_0 + \eta_1 \Delta \log Y(t-1) + \varepsilon(t)$$

and

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<sup>1</sup> I work with logs in (11) so that the measure of permanent income can easily be substituted into the Wagner's Law equations (Versions 1 and 2). Otherwise some transformations of variables and/or linearizations of (11) are required. All possible approaches were considered, and all involve approximations of some sort.

$$(12)' \quad \Delta \log \frac{Y(t)}{N(t)} = \eta_0 + \eta_1 \Delta \log \frac{Y(t-1)}{N(t-1)} + \varepsilon'(t)$$

Table 3.7 presents the estimates of (12) and (12)'.

Campbell-Deaton (1989) and Flavin (1981) show that if income  $Y(t)$  follows  $A(L) Y(t) = \text{constant} + \varepsilon(t)$ , where  $A(L) = 1 - (1 + \eta_1)L + \eta_1 L^2$ , and  $L$  is the lag operator, then (11)' becomes:

$$(11)'' \quad A\left(\frac{1}{1+r}\right) \Delta \log Y^*(t) = \frac{r}{1+r} \varepsilon(t)$$

Thus substituting (12) into (11)' gives:

$$(13) \quad \Delta \log Y^*(t) = \frac{1+r}{(1 - \eta_1 + r)} \varepsilon(t)$$

which can be used with Version 1 of Wagner's Law<sup>1</sup>:

$$\text{ie. } \Delta \log G(t) = \alpha_2 + \alpha_1 \Delta \log Y^*(t) + \mu$$

I estimate the elasticity  $\alpha_1$  by first saving the "innovations"  $\varepsilon(t)$  from (12) and then running a regression on:

$$\Delta \log G(t) = \alpha_2 + \hat{\alpha}_1 \varepsilon(t) + \mu$$

and inferring the elasticity  $\alpha_1$  from  $\hat{\alpha}_1 = \alpha_1 \left[ \frac{1+r}{1 - \eta_1 + r} \right]$ , given the value of  $\eta_1$  estimated from (12) and an assumed value for  $r$ .

---

<sup>1</sup> The constant term is now  $\alpha_2$  after first-differencing the trend term in the original level formulation.

For Version 2 of Wagner's Law I replace  $\log Y$  in (11) by  $\log \frac{Y}{N}$  and work with the income process given by (12)'. Rewriting version 2 as:

$$\Delta \log \left( \frac{G}{Y^*} \right) = \alpha_2 + \alpha_1 \Delta \log \left( \frac{Y^*}{N} \right) + \mu$$

or

$$\Delta \log \left( \frac{G}{N} \right) = \alpha_2 + (1 + \alpha_1) \Delta \log \left( \frac{Y^*}{N} \right) + \mu$$

I use the estimated value of  $\eta_1$  and the innovation series  $\varepsilon'(t)$  from (12)' to run a regression on:

$$\Delta \log \left( \frac{G}{N} \right) = \alpha_2 + (1 + \hat{\alpha}_1) \varepsilon'(t) + \mu$$

and infer the elasticity value for  $\alpha_1$  from  $(1 + \hat{\alpha}_1) = (1 + \alpha_1) \left[ \frac{1 + r}{1 - \eta_1 + r} \right]$  given an assumed value for  $r$ .

Table 3.8 contains estimates of the elasticity parameter  $\alpha_1$  for both versions of Wagner's Law. I assume two alternative values for the real interest rate: 5% and 10%. For version 1, all the elasticity values lie under one so that there is no support for the hypothesis. The results are fairly insensitive to the value of the real interest rate assumed. For version 2, I obtain negative elasticities. This implies that as permanent income per capita increases, the share of government consumption in permanent income falls. This too goes against Wagner's hypothesis.



## VI. Conclusions

This chapter has developed a simple permanent income model of optimal public consumption choice, and has estimated and tested the orthogonality conditions implied by the approach using data from selected OECD countries. As in the literature on the permanent income approach to private consumption, the tests were based on ascertaining whether one could improve upon the forecast errors of the optimality conditions using information which the government should have had available to it when expectations were formed and plans made.

In contrast to the literature on private consumption, however, this chapter has focused on an agent, namely the government, who better fits the assumptions required for the permanent income hypothesis to hold. That is, the government qualifies as an infinitely-lived, representative national agent, who faces no liquidity constraints. The expectation then is that the overidentification tests should less likely reject the permanent income hypothesis for public consumption than reject that for private consumption. The permanent income framework seriously undermines the use of current income measures to assess hypotheses about government size. Tests of Wagner's hypothesis, for instance, have focused on the sensitivity of government expenditure growth to increases in current income (or per capita current income). Under the permanent income model of public consumption, however, there should be little or no relationship between public consumption and current income.

The results in the empirical section of this chapter have given support to the permanent income approach to public consumption choice. In a majority of cases, the overidentifying restrictions could not be rejected

in the nonlinear and linear models. The U.K. data however consistently rejected the hypothesis. This suggests that one should analyze government behaviour on a country by country basis rather than seek a common paradigm for all governments. The empirical section also reconsidered Wagner's hypothesis in light of our findings that public consumption is constrained by permanent income. Using both backward and forward-looking measures of permanent income and two versions of Wagner's hypothesis, I could not find evidence to support the claim that government size is forever growing, where government size is proxied either by the share of government consumption in permanent GDP or by the level of government consumption.

For extensions it would be interesting to develop a comparison between the permanent income model of public consumption and the tax-smoothing model as theories about optimal fiscal behaviour. It is not inconsistent for a government to be interested in both public consumption and taxation-smoothing. However, each may lead to different welfare outcomes. For example, it can be argued that in an economy populated by a number of liquidity constrained individuals a cyclical downswing will affect private consumption less adversely under public consumption smoothing than under taxation-smoothing<sup>1</sup>. The reason is that a government which keeps public consumption in line with permanent income can alter its taxation-borrowing profile when times are "bad" by borrowing from the unconstrained private agents and providing the liquidity constrained individuals a tax cut. The tax cut will cushion the impact of the recession on the consumption of liquidity constrained

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<sup>1</sup> I borrow an argument made in Buiter (1983a).

individuals and have little (or no) impact on individuals who consume out of permanent income. This is not the case when the government adheres to a policy of keeping expected (marginal) tax rates constant, as under a policy of taxation-smoothing.

There are several other extensions worth mentioning. First, Wagner's hypothesis may be rejected for public consumption but this is not to say that there cannot be support for it with respect to other components of government expenditures, such as transfer payments. The rise in transfer payments is well documented and there are several factors behind this development, ranging from the use of transfers as a stabilization policy device to their use as an income redistributive device. The 'Wagner' elasticities for transfer payments may exceed unity, but it is more likely that the growth in transfer payments is sensitive to 'current' (transitory) income rather than to permanent income. A second line of work would be to test other hypotheses about the nature and pattern of government growth<sup>1</sup>. Thirdly, it would be interesting to perform the permanent income hypothesis tests on developing country public consumption data. I suspect that the overidentifying restrictions will be overwhelmingly rejected since many developing country governments appear to operate under "liquidity" or borrowing constraints.

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<sup>1</sup> For example, hypotheses related to the political and social factors behind government growth. See Roubini-Sachs (1989)

### Appendix 3.1

This Appendix describes the derivation of the "utilitarian" social welfare function used in the main text. Individual utilities are aggregated so as to obtain a tractable intertemporal social welfare function.

The optimizing government framework is similar to that studied in the preceding chapter. The key differences are that I now work in discrete-time, focus on the optimal choice of public consumption rather than of public investment, and allow for population growth. The aggregate private economy is non-Ricardian because of two features: (i) the existence of a positive birth rate and (ii) no operative bequest motives. The basic idea is that the consumption of private agents is constrained by their human and non-human wealth. The government, however, has the advantage of being able to tax this wealth as well as the human wealth of agents to be born. The resource base of the government grows at a rate exceeding that of those currently alive by the 'birth' rate of new agents. By altering its intertemporal mix of taxes, the government can postpone taxes to later generations.

To simplify matters, I assume no death rate but only a positive birth rate. Let  $n$  be the birth and population growth rate and  $N(t)$  the population size at time  $t$ , normalized to be of size unity at  $t=0$ . Thus  $N_t = (1+n)^t$ . For each variable  $x(t,v)$ ,  $t$  indexes time and  $v$  the vintage of an agent (ie. the time of birth). Uppercase  $X(t)$  will represent the aggregate magnitude and lowercase  $x(t)$  will represent the aggregate per capita magnitude:

$$X(t) = \sum_{v=-\infty}^t x(t,v)[N(v)-N(v-1)] \text{ or } \sum_{v=-\infty}^t x(t,v)\left[\left(\frac{n}{1+n}\right)(1+n)^v\right]$$

and  $x(t) = \frac{X(t)}{N(t)}$ , where  $X(t)$  sums across agents of all existing vintages, weighted by each cohort's size.

Each individual born at time  $v$  maximizes at time  $t=0$ :

$$(A1) \quad u(0,v) = \sum_{s=0}^{\infty} \left(\frac{1}{1+\delta}\right)^s u(c(v,s), g(s))$$

where  $\delta$  is the private time preference rate,  $g(s)$  the consumption of public goods per capita, and  $c(s,v)$  private consumption. The consumption of public goods does not depend upon  $v$ , individual vintages. The supply of public goods may be congestible (hence  $G$  is divided by population  $N$ ).

The social planner maximizes at time  $t=0$ :

(A2)

$$SW_0 = \frac{\{\text{Welfare of those Alive}\} \text{ plus } \{\text{Welfare of those to-be Born}\}}{N(0)}$$

$$\begin{aligned}
&= \frac{1}{N(0)} \left[ \sum_{v=-\infty}^0 [N(v)-N(v-1)] \left(\frac{1}{1+\rho}\right)^v \sum_{s=0}^{\infty} \left(\frac{1}{1+\delta}\right)^{s-v} u(c(s,v),g(s)) \right. \\
&\quad \left. + \sum_{v=1}^{\infty} [N(v)-N(v-1)] \left(\frac{1}{1+\rho}\right)^v \sum_{s=v}^{\infty} \left(\frac{1}{1+\delta}\right)^{s-v} u(c(s,v),g(s)) \right]
\end{aligned}$$

The inner summations represent private intertemporal discounted utilities and the outer summations aggregate across individuals. Note that the discounting of private welfare occurs down to the date of birth and not to present time. The latter procedure would induce time-inconsistency<sup>1</sup>. The social planner uses the social discount rate,  $\rho$ , to tilt the intratemporal distribution of resources among those alive and those yet to be born.

Changing the order of summation gives:

$$(A2)' \quad SW_0 = \sum_{s=0}^{\infty} \left[ \sum_{v=-\infty}^s \left(\frac{n}{1+n}\right) (1+n)^{v-s} \left(\frac{1+\rho}{1+\delta}\right)^{s-v} u(c(s,v),g(s)) \right] \left(\frac{1+n}{1+\rho}\right)^s$$

To solve this, I divide the problem into two steps: (1) optimize the discounted welfare over time, and (2) distribute the maximized welfare across agents optimally at each point in time. Underlying step (2) is the

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<sup>1</sup> Calvo-Obstfeld (1988) point out that discounting to present time would make planners treat everyone alive as if they had just been born. Thus, those alive at the start of a plan and those yet to be born would be treated asymmetrically by the welfare criterion. Subsequent reoptimization would result in an allocation of aggregate consumption among new agents that would differ from an initial plan.

assumption that the planner possesses the instruments required to effect intratemporal redistributions between each and every individual. In particular the planner is assumed to be able to impose time-varying and vintage-specific lump sum taxes and transfers.

The two-step maximization process just described involves rewriting (A2)' as:

$$(A3) \quad SW_0 = \sum_{s=0}^{\infty} V(c(s), g(s)) \left( \frac{1+n}{1+\rho} \right)^s$$

where

$$(A4) \quad V(c(s), g(s)) = \max_{v=-\infty}^s \sum_{v=-\infty}^s \left( \frac{n}{1+n} \right) (1+n)^{v-s} \left( \frac{1+\rho}{1+\delta} \right)^{s-v} u(c(s, v), g(s))$$

subject to

$$(A4a) \quad C(s) = N(s) \quad c(s) = (1+n)^s c(s, v)$$

and

$$(A4b) \quad C(s) = \sum_{v=-\infty}^s \left( \frac{n}{1+n} \right) (1+n)^v c(s, v)$$

by choice of  $\{c(s,v)\}$  for all  $v$  and time  $s \in [0, \infty)$ .

It is easiest to solve the intra-cohort problem first (ie. step 2 mentioned above and expressed in (A4), (A4a), and (A4b)) and substitute the first-order conditions back into (A4). The first-order conditions derived from maximizing the indirect utility function (A4) yield 'intratemporal' Euler conditions, where the variable index is  $v$ , vintage, not time. The solution to (A4), (A4a), and (A4b) is:

$$(A5) \quad \frac{U_c(c(s,v),g(s))}{U_c(c(s,v-1),g(s))} = \frac{(1 + \rho)}{(1 + \delta)} \text{ for all } v \in [s, -\infty)$$

where  $U_c$  is the partial derivative of  $U$  with respect to  $c$ . Note that an egalitarian distribution results if  $\rho = \delta$ .

### Illustration

I show how  $V(c(s),g(s))$  in (A3) appears once the first-order conditions in (A5) are substituted back into (A4). Suppose:

$$(A6) \quad u(c(s,v),g(s)) = \frac{c(s,v)^{1-\sigma_1} - 1}{1-\sigma_1} + d \frac{g(s)^{1-\sigma_2} - 1}{1-\sigma_2}$$

for each agent of vintage  $v$ . This is one of the functional forms investigated in the main text. The parameter  $d$  weights a unit of 'utils' (or felicity) from public goods in terms of felicity from private goods. The first-order conditions corresponding to (A5) are:



$$(A6a) \quad \frac{c(s,v-1)}{c(s,v)} = \left( \frac{1+\rho}{1+\delta} \right)^{\frac{1}{\sigma_1}}$$

or

$$(A6b) \quad \frac{c(s,v)}{c(s,s)} = \left( \left( \frac{1+\rho}{1+\delta} \right)^{\frac{1}{\sigma_1}} \right)^{s-v}$$

Substituting (A6b) into (A4a), using (A4b), gives

$$(A7) \quad c(s,s) = c(s) \left( \frac{1+n}{n} \right) (1-k)$$

$$\text{where } k = \left[ (1+n) \left( \frac{1+\delta}{1+\rho} \right)^{\frac{1}{\sigma_1}} \right]^{-1}$$

It is assumed that  $k < 1$ , which holds if

$$(1+n) > \left( \frac{1+\rho}{1+\delta} \right)^{\frac{1}{\sigma_1}}$$

for  $n > 0$ . This places restrictions on the discount rates  $\rho$ ,  $\delta$  given  $n$  and the degree of intertemporal substitution of private goods consumption,  $\frac{1}{\sigma_1}$ .

If there is no population growth ( $n=0$ ), we are back to

the representative agent case - or case of a fixed number of identical individuals. There is no reason to tilt the intratemporal distribution away from the egalitarian one of  $\rho = \delta$ . For  $n > 0$ , the social discount rate is bounded by  $\rho < (1 + n)^{\sigma_1}(1+\delta) - 1$ . Typically the higher the  $\rho$ , the greater the weight given to older generations, or agents alive. If  $\rho$  is too high,  $k > 1$ , and the consumption of the new born will be negative - ie.  $c(s,s) < 0$ .

Substituting (A7) back into (A6b) gives:

$$(A8) \quad c(s,v) = (1 + n)^{s-v} k^{s-v} (1 - k) \frac{1+n}{n} c(s)$$

and substituting (A8) into (A6) gives:

$$(A9) \quad u(c(s,v),g(s)) = \Theta \frac{c(s)^{1-\sigma_1} - 1}{1-\sigma_1} + d \frac{g(s)^{1-\sigma_2} - 1}{1-\sigma_2}$$

$$\text{where } \Theta = [(1 + n)^{s-v} k^{s-v} (1 - k) \frac{1+n}{n}]^{1-\sigma_1}$$

Now substitute (A9) into (A4):

$$(A10) \quad V(c(s),g(s)) = \Gamma_1 \frac{c(s)^{1-\sigma_1} - 1}{1-\sigma_1} + \Gamma_2 \frac{g(s)^{1-\sigma_2} - 1}{1-\sigma_2}$$

where

$$\Gamma_1 = \left( \frac{n}{(1+n) - \left( \frac{1+\rho}{1+\delta} \right)^{\frac{1}{\sigma_1}}} \right)^{\sigma_1}, \quad \Gamma_2 = d \left( \frac{n}{(1+n) - \left( \frac{1+\rho}{1+\delta} \right)} \right)$$

and  $\Gamma_2 = \Gamma_1 d$ .

Once again if  $k < 1$ , for all values of  $\sigma_1 \in [0, \infty)$ ,  $\Gamma_1, \Gamma_2 > 0$ . Substituting (A10) into (A3) gives us the social planner's intertemporal welfare function. We can rewrite (A10) as:

$$(A10)' \quad V(c(s), g(s)) = \Gamma_1 \left( \frac{c(s)^{1-\sigma_1} - 1}{1-\sigma_1} + \Gamma \frac{g(s)^{1-\sigma_2} - 1}{1-\sigma_2} \right)$$

where  $\Gamma = \frac{\Gamma_2}{\Gamma_1} = d$ . Since welfare is ordinally measured, I drop  $\Gamma_1$  in (A10)' in the main text.  $\Gamma$  is interpreted as the relative utility weight of public goods consumption.

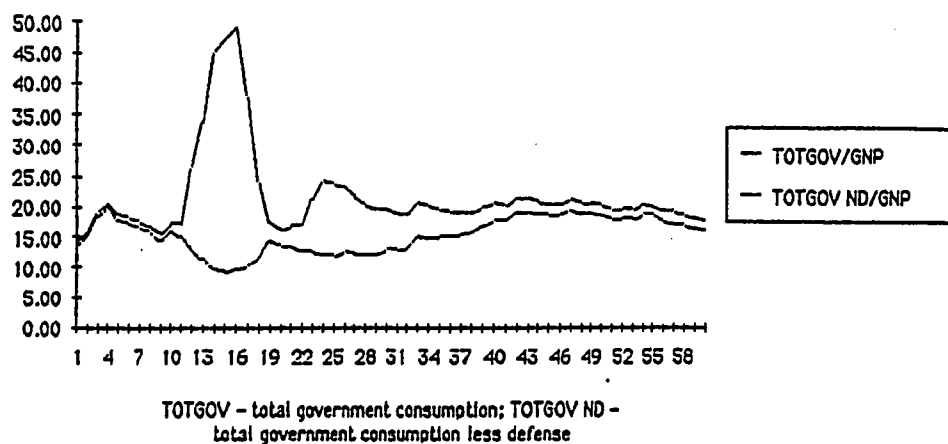
Note that the solutions derived thus far pertain to the command optima. To decentralize a command optimum the social planner requires access to unrestricted age- and time-dependent lump sum taxes and transfers to implement the desired plan. The planner needs these instruments not only to ensure that intratemporal distributions at each point in time are optimal, but also to ensure that the level and path of aggregate private consumption and saving in the market economy correspond to the

social planner's desired choices<sup>1</sup>. The fact that the decentralization requirements can be stringent does not deny the usefulness of the utilitarian aggregate welfare framework as a normative benchmark for the evaluation of public policy choices, even if governments do not behave according to the welfare framework outlined. However the purpose of the tests in this chapter is to determine whether government consumption plans "reveal" choices consistent with this welfare framework.

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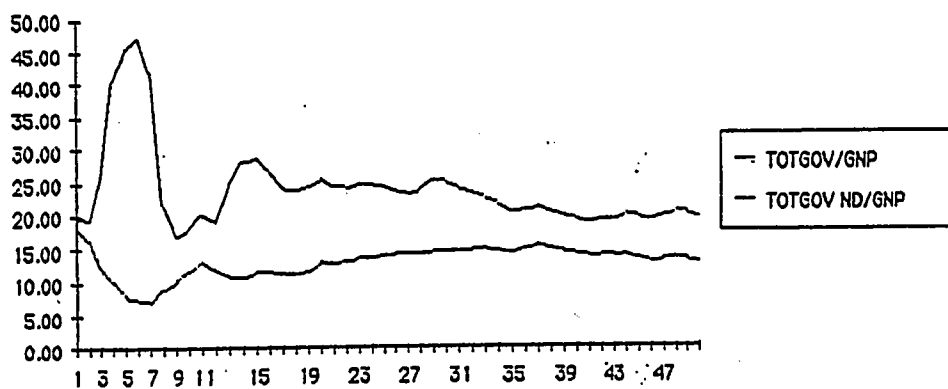
<sup>1</sup> The fiscal choices for taxes, transfers, and public consumption must also be feasible in terms of satisfying the government's intertemporal budget constraint.

FIGURE 3.1: CANADA 1929-1988



Source: Statistics Canada

FIGURE 3.2: USA 1939-1988



see Notes to end of Figure 3.1

Source: U.S. National Income and Product Accounts

FIGURE 3.3: FRANCE 1960-1988

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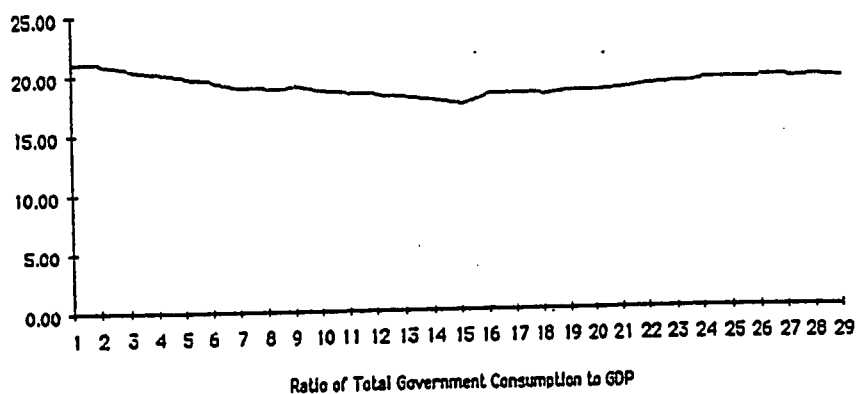


FIGURE 3.4: ITALY 1960-1988

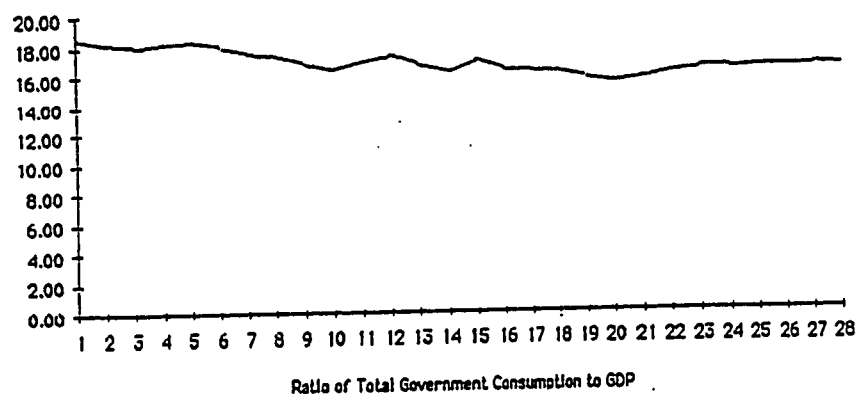
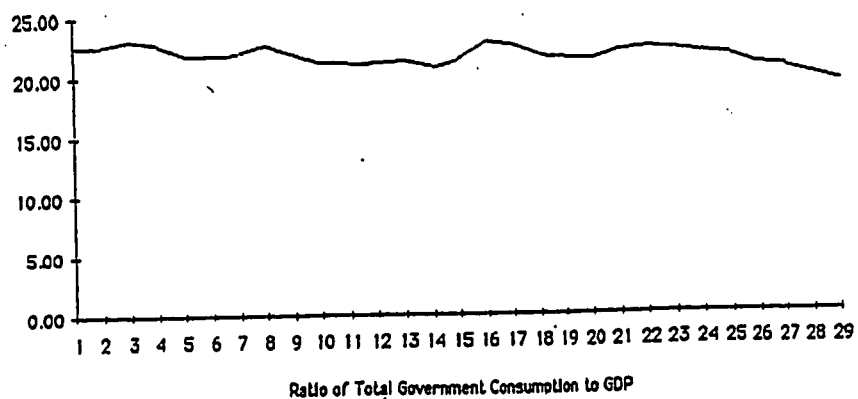


FIGURE 3.5: UNITED KINGDOM 1960-1988



Source: OECD National Accounts

**Table 3.1: Test of Separability**

|                 | <u>U.S.A. (N=127)</u> |                 | <u>Canada (N=127)</u> |                  | <u>U.K. (N=103)</u> |                  |
|-----------------|-----------------------|-----------------|-----------------------|------------------|---------------------|------------------|
|                 | (1)                   | (2)             | (1)                   | (2)              | (1)                 | (2)              |
| $\sigma_1$      | 0.29<br>(0.026)       | 0.3<br>(0.026)  | 0.36<br>(0.054)       | 0.37<br>(0.051)  | 0.31<br>(0.033)     | 0.31<br>(0.033)  |
| $\sigma_2$      | 0.38<br>(0.053)       | 0.38<br>(0.054) | 0.43<br>(0.055)       | 0.43<br>(0.052)  | 0.39<br>(0.033)     | 0.39<br>(0.033)  |
| $\Gamma$        | 0.82<br>(0.02)        | 0.82<br>(0.02)  | 0.8<br>(0.034)        | 0.8<br>(0.032)   | 0.84<br>(0.036)     | 0.84<br>(0.036)  |
| $\frac{1}{1+p}$ | 0.95<br>(0.003)       | 0.95<br>(0.002) | 0.94<br>(0.004)       | 0.94<br>(0.0024) | 0.93<br>(0.0026)    | 0.93<br>(0.0024) |
| $\gamma$        | -0.31<br>(0.31)       | -----           | -0.73<br>(0.57)       | -----            | -0.027<br>(0.098)   | -----            |
| LR              | 1.48                  |                 | 3.5                   |                  | 0.1                 |                  |

Table 3.1 continued

|                    | <u>France (N=98)</u> |                  | <u>Italy (N=70)</u> |                  |
|--------------------|----------------------|------------------|---------------------|------------------|
|                    | (1)                  | (2)              | (1)                 | (2)              |
| $\sigma_1$         | 0.34<br>(0.1)        | 0.34<br>(0.1)    | 0.3<br>(0.07)       | 0.31<br>(0.08)   |
| $\sigma_2$         | 0.43<br>(0.09)       | 0.43<br>(0.09)   | 0.39<br>(0.066)     | 0.4<br>(0.07)    |
| $\Gamma$           | 0.8<br>(0.08)        | 0.8<br>(0.08)    | 0.8<br>(0.063)      | 0.8<br>(0.068)   |
| $\frac{1}{1+\rho}$ | 0.93<br>(0.0033)     | 0.93<br>(0.0023) | 0.91<br>(0.005)     | 0.91<br>(0.0032) |
| $\gamma$           | 0.034<br>(0.27)      | ----             | 0.61<br>(0.57)      | ----             |
| LR                 |                      | 0.14             |                     | 1.04             |

Notes: N is the number of observations, LR is the likelihood ratio test statistic, and heteroskedastic-consistent standard errors are in parentheses.



**Table 3.2: Test of Log Utility**

|                    | U.S.A. (N=127)  |                 |                 | Canada (N=127)  |                 |                 | U.K. (N=103)    |                 |                  |
|--------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|------------------|
|                    | (1)             | (2)             | (3)             | (1)             | (2)             | (3)             | (1)             | (2)             | (3)              |
| $\sigma_1$         | 1               | 0.5<br>(0.02)   | 1               | 1               | 1.03<br>(0.023) | 1               | 1               | 0.52<br>(0.08)  | 1                |
| $\sigma_2$         | 0.37<br>(0.14)  | 1               | 1               | 0.51<br>(0.05)  | 1               | 1               | 0.36<br>(0.06)  | 1               | 1                |
| $\Gamma$           | 0.2<br>(0.03)   | 1.02<br>(0.04)  | 0.39<br>(0.04)  | 0.3<br>(0.034)  | 0.45<br>(0.02)  | 0.47<br>(0.02)  | 0.22<br>(0.016) | 1.09<br>(0.16)  | 0.44<br>(0.04)   |
| $\frac{1}{1+\rho}$ | 0.94<br>(0.002) | 0.96<br>(0.002) | 0.96<br>(0.002) | 0.94<br>(0.003) | 0.95<br>(0.003) | 0.95<br>(0.003) | 0.93<br>(0.003) | 0.93<br>(0.003) | 0.94<br>(0.0028) |
| LR                 | 1273.4          | 1417.8          | 1563.6          | 828.8           | 1081.6          | 1082.6          | 935.8           | 1074.4          | 1160.6           |

Table 3.2 continued

|                 | <u>France (N=98)</u> |                  |                  | <u>Italy (N=70)</u> |                  |                  |
|-----------------|----------------------|------------------|------------------|---------------------|------------------|------------------|
|                 | (1)                  | (2)              | (3)              | (1)                 | (2)              | (3)              |
| $\sigma_1$      | 1                    | 1.008<br>(0.02)  | 1                | 1                   | 0.97<br>(0.03)   | 1                |
| $\sigma_2$      | 0.91<br>(0.013)      | 1                | 1                | 0.54<br>(0.12)      | 1                | 1                |
| $\Gamma$        | 0.38<br>(0.04)       | 0.4<br>(0.014)   | 0.41<br>(0.012)  | 0.22<br>(0.03)      | 0.38<br>(0.025)  | 0.36<br>(0.01)   |
| $\frac{1}{1+p}$ | 0.94<br>(0.0023)     | 0.93<br>(0.0024) | 0.93<br>(0.0024) | 0.91<br>(0.0035)    | 0.91<br>(0.0032) | 0.91<br>(0.0032) |
| LR              | 752.8                | 790.6            | 791.2            | 813.8               | 817.2            | 820.4            |

See Notes to end of Table 3.1.

**Table 3.3: Test of Substitutability**

|                 | <u>U.S.A. (N=127)</u> | <u>Canada (N=127)</u> | <u>U.K. (N=103)</u>  | <u>France (N=98)</u>  | <u>Italy (N=70)</u>   |
|-----------------|-----------------------|-----------------------|----------------------|-----------------------|-----------------------|
| $\sigma_1$      | 0.3<br>(0.03)         | 0.37<br>(0.05)        | 0.31<br>(0.033)      | 0.34<br>(0.1)         | 0.3<br>(0.075)        |
| $\sigma_2$      | 0.38<br>(0.05)        | 0.42<br>(0.05)        | 0.39<br>(0.033)      | 0.42<br>(0.09)        | 0.4<br>(0.07)         |
| $\Gamma$        | 0.82<br>(0.02)        | 0.78<br>(0.03)        | 0.84<br>(0.036)      | 0.78<br>(0.08)        | 0.81<br>(0.068)       |
| $\frac{1}{1+p}$ | 0.95<br>(0.002)       | 0.95<br>(0.0023)      | 0.93<br>(0.0024)     | 0.93<br>(0.0024)      | 0.91<br>(0.0033)      |
| $\theta$        | -0.000001<br>(0.0002) | 0.000063<br>(0.00041) | 0.000028<br>(0.0003) | 0.000028<br>(0.00031) | -0.000007<br>(0.0002) |
| LR              | 0.0108                | 2.3                   | 2.02                 | 2.34                  | 1.24                  |

See Notes at end of Table 3.1.

**Table 3.4A: Tests of Overidentification**

USA (N=120)

**(1) Total Government Consumption**

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$     | $\sigma_2$      | $\Gamma$       | J-Statistic |
|-------|--------------------|----------------|-----------------|----------------|-------------|
| NLAG2 | 0.94<br>(0.004)    | 0.29<br>(0.05) | 0.37<br>(0.02)  | 0.55<br>(0.2)  | 21.9        |
| NLAG4 | 0.93<br>(0.0037)   | 0.3<br>(0.013) | 0.37<br>(0.045) | 0.52<br>(0.15) | 56.4        |
| NLAG6 | 0.92<br>(0.0037)   | 0.3<br>(0.022) | 0.34<br>(0.068) | 0.42<br>(0.13) | 102.6       |

**(2) Total Government Consumption less Defense**

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$     | $\sigma_2$     | $\Gamma$       | J-Statistic |
|-------|--------------------|----------------|----------------|----------------|-------------|
| NLAG2 | 0.93<br>(0.0037)   | 0.29<br>(0.02) | 0.43<br>(0.03) | 0.94<br>(0.29) | 21.36       |
| NLAG4 | 0.93<br>(0.0037)   | 0.29<br>(0.02) | 0.42<br>(0.04) | 0.88<br>(0.26) | 49.8        |
| NLAG6 | 0.92<br>(0.0037)   | 0.3<br>(0.02)  | 0.42<br>(0.04) | 0.85<br>(0.24) | 62.4        |

**(3) Total Federal Government Consumption**

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$      | $\sigma_2$     | $\Gamma$        | J-Statistic |
|-------|--------------------|-----------------|----------------|-----------------|-------------|
| NLAG2 | 0.94<br>(0.0038)   | 0.3<br>(0.04)   | 0.47<br>(0.04) | 0.6<br>(0.05)   | 19.8        |
| NLAG4 | 0.93<br>(0.0038)   | 0.32<br>(0.036) | 0.47<br>(0.04) | 0.58<br>(0.04)  | 36.9        |
| NLAG6 | 0.93<br>(0.0038)   | 0.29<br>(0.04)  | 0.46<br>(0.05) | 0.61<br>(0.055) | 64.8        |

Table 3.4A continued

## (4) Total Federal Government Consumption less Defense

| Instrument |                    |                |                 |                 |             |
|------------|--------------------|----------------|-----------------|-----------------|-------------|
| Set        | $\frac{1}{1+\rho}$ | $\sigma_1$     | $\sigma_2$      | $\Gamma$        | J-Statistic |
| NLAG2      | 0.95<br>(0.002)    | 0.36<br>(0.12) | 0.63<br>(0.023) | 0.23<br>(0.06)  | 21.9        |
| NLAG4      | 0.95<br>(0.002)    | 0.36<br>(0.11) | 0.64<br>(0.021) | 0.23<br>(0.058) | 42.9        |
| NLAG6      | 0.94<br>(0.002)    | 0.34<br>(0.1)  | 0.64<br>(0.021) | 0.24<br>(0.057) | 59.1        |

## (5) Total State-Local Government Consumption

| Instrument |                    |                |                |                 |             |
|------------|--------------------|----------------|----------------|-----------------|-------------|
| Set        | $\frac{1}{1+\rho}$ | $\sigma_1$     | $\sigma_2$     | $\Gamma$        | J-Statistic |
| NLAG2      | 0.94<br>(0.0037)   | 0.31<br>(0.02) | 0.45<br>(0.09) | 0.67<br>(0.028) | 26.1        |
| NLAG4      | 0.93<br>(0.0037)   | 0.3<br>(0.014) | 0.46<br>(0.06) | 0.68<br>(0.02)  | 50.1        |
| NLAG6      | 0.92<br>(0.0037)   | 0.3<br>(0.012) | 0.46<br>(0.05) | 0.68<br>(0.02)  | 65.7        |

Notes: N is the number of observations, and heteroskedastic-consistent standard errors are in parentheses.

**Table 3.4B: Tests of Overidentification**

Canada (N=120)

## (1) Total Government Consumption

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$      | $\sigma_2$      | $\Gamma$       | J-Statistic |
|-------|--------------------|-----------------|-----------------|----------------|-------------|
| NLAG2 | 0.92<br>(0.005)    | 0.35<br>(0.013) | 0.43<br>(0.043) | 0.81<br>(0.02) | 24.6        |
| NLAG4 | 0.9<br>(0.005)     | 0.35<br>(0.013) | 0.42<br>(0.044) | 0.81<br>(0.02) | 37.8        |
| NLAG6 | 0.9<br>(0.005)     | 0.35<br>(0.014) | 0.42<br>(0.05)  | 0.8<br>(0.02)  | 59.7        |

## (2) Total Government Consumption less Defense

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$      | $\sigma_2$      | $\Gamma$       | J-Statistic |
|-------|--------------------|-----------------|-----------------|----------------|-------------|
| NLAG2 | 0.91<br>(0.005)    | 0.42<br>(0.024) | 0.44<br>(0.07)  | 0.7<br>(0.03)  | 35.1        |
| NLAG4 | 0.9<br>(0.005)     | 0.43<br>(0.024) | 0.44<br>(0.074) | 0.7<br>(0.03)  | 53.7        |
| NLAG6 | 0.9<br>(0.005)     | 0.43<br>(0.02)  | 0.43<br>(0.077) | 0.69<br>(0.03) | 68.4        |

## (3) Total Federal Government Consumption

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$      | $\sigma_2$      | $\Gamma$        | J-Statistic |
|-------|--------------------|-----------------|-----------------|-----------------|-------------|
| NLAG2 | 0.92<br>(0.0047)   | 0.34<br>(0.017) | 0.55<br>(0.013) | 0.45<br>(0.015) | 37.8        |
| NLAG4 | 0.91<br>(0.0047)   | 0.34<br>(0.016) | 0.55<br>(0.012) | 0.44<br>(0.014) | 51.6        |
| NLAG6 | 0.91<br>(0.0047)   | 0.34<br>(0.015) | 0.55<br>(0.013) | 0.44<br>(0.014) | 66.0        |

Table 3.4B continued

## (4) Total Federal Government Consumption less Defense

## Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$     | $\sigma_2$      | $\Gamma$        | J-Statistic |
|-------|--------------------|----------------|-----------------|-----------------|-------------|
| NLAG2 | 0.91<br>(0.0047)   | 0.36<br>(0.05) | 0.64<br>(0.024) | 0.3<br>(0.028)  | 33.9        |
| NLAG4 | 0.9<br>(0.0047)    | 0.35<br>(0.05) | 0.64<br>(0.024) | 0.3<br>(0.027)  | 50.4        |
| NLAG6 | 0.87<br>(0.0046)   | 0.36<br>(0.05) | 0.64<br>(0.024) | 0.29<br>(0.026) | 66.9        |

## (5) Total State-Local Government Consumption

## Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$      | $\sigma_2$     | $\Gamma$        | J-Statistic |
|-------|--------------------|-----------------|----------------|-----------------|-------------|
| NLAG2 | 0.91<br>(0.0047)   | 0.46<br>(0.026) | 0.46<br>(0.07) | 0.61<br>(0.027) | 32.4        |
| NLAG4 | 0.9<br>(0.0047)    | 0.47<br>(0.025) | 0.46<br>(0.07) | 0.6<br>(0.026)  | 53.7        |
| NLAG6 | 0.88<br>(0.0047)   | 0.47<br>(0.025) | 0.45<br>(0.08) | 0.6<br>(0.026)  | 70.2        |

Notes: N is the number of observations, and heteroskedastic-consistent standard errors are in parentheses.

**Table 3.4C: Tests of Overidentification**

(1) U.K. (N=96): Total Government Consumption

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$      | $\sigma_2$      | $\Gamma$        | J-Statistic |
|-------|--------------------|-----------------|-----------------|-----------------|-------------|
| NLAG2 | 0.84<br>(0.0074)   | 0.27<br>(0.01)  | 0.35<br>(0.011) | 0.88<br>(0.02)  | 73.2        |
| NLAG4 | 0.67<br>(0.0072)   | 0.22<br>(0.014) | 0.3<br>(0.017)  | 0.92<br>(0.03)  | 95.4        |
| NLAG6 | 0.51<br>(0.0066)   | 0.19<br>(0.017) | 0.25<br>(0.02)  | 0.92<br>(0.034) | 106.8       |

(2) France (N=91): Total Government Consumption

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$      | $\sigma_2$      | $\Gamma$        | J-Statistic |
|-------|--------------------|-----------------|-----------------|-----------------|-------------|
| NLAG2 | 0.92<br>(0.0057)   | 0.35<br>(0.015) | 0.42<br>(0.011) | 0.78<br>(0.02)  | 50.1        |
| NLAG4 | 0.93<br>(0.0057)   | 0.35<br>(0.014) | 0.42<br>(0.01)  | 0.79<br>(0.018) | 59.1        |
| NLAG6 | 0.93<br>(0.0057)   | 0.34<br>(0.014) | 0.41<br>(0.01)  | 0.79<br>(0.018) | 70.2        |

(3) Italy (N=63): Total Federal Government

Instrument

| Set   | $\frac{1}{1+\rho}$ | $\sigma_1$      | $\sigma_2$      | $\Gamma$        | J-Statistic |
|-------|--------------------|-----------------|-----------------|-----------------|-------------|
| NLAG2 | 0.9<br>(0.0013)    | 0.33<br>(0.011) | 0.37<br>(0.08)  | 0.75<br>(0.02)  | 42.9        |
| NLAG4 | 0.9<br>(0.0012)    | 0.32<br>(0.01)  | 0.37<br>(0.08)  | 0.75<br>(0.02)  | 47.4        |
| NLAG6 | 0.9<br>(0.0013)    | 0.32<br>(0.01)  | 0.33<br>(0.083) | 0.73<br>(0.017) | 83.1        |

See Notes at end of Table 3.4B



**Table 3.5: Quadratic Utility and Overidentification Tests**

| (1) USA    | $\phi_0$         | $\phi_1$           | $\phi_2$           | $\beta_0'$        | $\beta_1$          | J     |
|------------|------------------|--------------------|--------------------|-------------------|--------------------|-------|
| INST2      | 0.034<br>(0.014) | 1<br>(0.0022)      | 0.0083<br>(0.0067) | 0.034<br>(0.0037) | -0.043<br>(0.39)   | 18.45 |
| INST4      | 0.034<br>(0.015) | 1<br>(0.0022)      | 0.0083<br>(0.0067) | 0.034<br>(0.0032) | -0.0051<br>(0.198) | 33.11 |
| INST6      | 0.034<br>(0.015) | 1<br>(0.002)       | 0.0083<br>(0.0067) | 0.035<br>(0.003)  | -0.14<br>(0.19)    | 44.75 |
| (2) Canada | $\phi_0$         | $\phi_1$           | $\phi_2$           | $\beta_0'$        | $\beta_1$          | J     |
| INST2      | 0.03<br>(0.013)  | 1<br>(0.0026)      | 0.013<br>(0.0073)  | 0.029<br>(0.0055) | 0.085<br>(0.39)    | 24.6  |
| INST4      | 0.03<br>(0.013)  | 1<br>(0.0026)      | 0.013<br>(0.0073)  | 0.03<br>(0.0045)  | 0.045<br>(0.2)     | 35.24 |
| INST6      | 0.03<br>(0.013)  | 1<br>(0.0026)      | 0.013<br>(0.0073)  | 0.03<br>(0.0035)  | 0.029<br>(0.17)    | 49.8  |
| (3) U.K.   | $\phi_0$         | $\phi_1$           | $\phi_2$           | $\beta_0'$        | $\beta_1$          | J     |
| INST2      | 0.037<br>(0.071) | 0.9999<br>(0.0098) | 0.016<br>(0.028)   | 0.08<br>(0.025)   | -2.7<br>(1.3)      | 105.4 |
| INST4      | 0.037<br>(0.071) | 0.9999<br>(0.0098) | 0.015<br>(0.028)   | 0.078<br>(0.021)  | -2.6<br>(1.06)     | 126.5 |
| INST6      | 0.037<br>(0.07)  | 0.9999<br>(0.0098) | 0.015<br>(0.028)   | 0.068<br>(0.018)  | -2.01<br>(0.8)     | 136.7 |

Table 3.5 continued

| (4) France | $\phi_0$         | $\phi_1$           | $\phi_2$          | $\beta_0'$        | $\beta_1$       | J     |
|------------|------------------|--------------------|-------------------|-------------------|-----------------|-------|
| INST2      | 0.037<br>(0.018) | 0.9999<br>(0.0031) | 0.017<br>(0.0077) | 0.038<br>(0.011)  | -0.08<br>(0.61) | 15.35 |
| INST4      | 0.037<br>(0.018) | 0.9999<br>(0.0031) | 0.017<br>(0.0077) | 0.03<br>(0.0066)  | 0.46<br>(0.32)  | 31.95 |
| INST6      | 0.037<br>(0.018) | 0.9999<br>(0.0031) | 0.017<br>(0.0077) | 0.025<br>(0.007)  | 0.71<br>(0.33)  | 43.73 |
| (5) Italy  | $\phi_0$         | $\phi_1$           | $\phi_2$          | $\beta_0'$        | $\beta_1$       | J     |
| INST2      | 0.041<br>(0.048) | 1<br>(0.006)       | 0.011<br>(0.017)  | 0.04<br>(0.013)   | 0.33<br>(0.73)  | 49.9  |
| INST4      | 0.041<br>(0.048) | 1<br>(0.006)       | 0.0011<br>(0.017) | 0.041<br>(0.0099) | 0.23<br>(0.53)  | 58.1  |
| INST6      | 0.04<br>(0.048)  | 1<br>(0.006)       | 0.011<br>(0.017)  | 0.042<br>(0.0079) | 0.11<br>(0.35)  | 72.6  |

Notes: Heteroskedastic-Consistent Standard Errors in Parentheses. Number of observations for the U.S. and Canada is 122, for the U.K. 98, for France 93, and for Italy 65.

**Table 3.6A. Wagner's Law Version 1**

(1) U.S.A. (N=127)

|         | $\alpha_0$     | $\alpha_1$    | $\alpha_2$       | $\lambda$        | Adj.R-Sq |
|---------|----------------|---------------|------------------|------------------|----------|
| Total G | 14.9<br>(30.8) | -1.3<br>(4.6) | 0.014<br>(0.034) | 0.986<br>(0.025) | 0.99     |

(2) Canada (N=127)

|         | $\alpha_0$      | $\alpha_1$     | $\alpha_2$          | $\lambda$       | Adj.R-Sq |
|---------|-----------------|----------------|---------------------|-----------------|----------|
| Total G | -4.45<br>(1.92) | 1.64<br>(0.41) | -0.0078<br>(0.0046) | 0.91<br>(0.044) | 0.99     |

(3) U.K. (N=103)

|         | $\alpha_0$      | $\alpha_1$    | $\alpha_2$         | $\lambda$       | Adj.R-Sq |
|---------|-----------------|---------------|--------------------|-----------------|----------|
| Total G | -6.92<br>(18.7) | 1.94<br>(3.1) | -0.0084<br>(0.019) | 0.97<br>(0.023) | 0.99     |

(4) France (N=98)

|         | $\alpha_0$      | $\alpha_1$     | $\alpha_2$         | $\lambda$      | Adj.R-Sq |
|---------|-----------------|----------------|--------------------|----------------|----------|
| Total G | 0.026<br>(2.43) | 0.73<br>(0.39) | 0.0019<br>(0.0023) | 0.93<br>(0.05) | 0.99     |

(5) Italy (N=70)

|         | $\alpha_0$     | $\alpha_1$     | $\alpha_2$         | $\lambda$     | Adj.R-Sq |
|---------|----------------|----------------|--------------------|---------------|----------|
| Total G | 5.47<br>(2.35) | -0.1<br>(0.34) | 0.0073<br>(0.0027) | 0.8<br>(0.08) | 0.99     |

Notes: N is the number of observations, and heteroskedastic-consistent standard errors are in parentheses.

**Table 3.6B. Wagner's Law Version 2**

(1) U.S.A. (N=127)

|         | $\alpha_0$    | $\alpha_1$    | $\alpha_2$        | $\lambda$        | Adj.R-Sq |
|---------|---------------|---------------|-------------------|------------------|----------|
| Total G | 3.63<br>(9.1) | -2.5<br>(5.4) | 0.0096<br>(0.026) | 0.986<br>(0.024) | 0.98     |

(2) Canada (N=127)

|         | $\alpha_0$      | $\alpha_1$    | $\alpha_2$          | $\lambda$      | Adj.R-Sq |
|---------|-----------------|---------------|---------------------|----------------|----------|
| Total G | -1.64<br>(1.67) | 0.1<br>(0.89) | -0.0015<br>(0.0065) | 0.94<br>(0.04) | 0.99     |

(3) U.K. (N=103)

|         | $\alpha_0$    | $\alpha_1$     | $\alpha_2$         | $\lambda$       | Adj.R-Sq |
|---------|---------------|----------------|--------------------|-----------------|----------|
| Total G | -3.2<br>(7.7) | 1.002<br>(3.7) | -0.0086<br>(0.022) | 0.98<br>(0.023) | 0.99     |

(4) France (N=98)

|         | $\alpha_0$     | $\alpha_1$      | $\alpha_2$        | $\lambda$       | Adj.R-Sq |
|---------|----------------|-----------------|-------------------|-----------------|----------|
| Total G | -0.85<br>(1.2) | -0.35<br>(0.51) | 0.002<br>(0.0024) | 0.94<br>(0.054) | 0.99     |

(5) Italy (N=70)

|         | $\alpha_0$     | $\alpha_1$     | $\alpha_2$        | $\lambda$      | Adj.R-Sq |
|---------|----------------|----------------|-------------------|----------------|----------|
| Total G | 1.05<br>(1.12) | -1.1<br>(0.44) | 0.0066<br>(0.003) | 0.82<br>(0.08) | 0.98     |

Notes: N is the number of observations, and heteroskedastic-consistent standard errors are in parentheses.

**Table 3.7. Income Processes**(A) Aggregate Income:  $\Delta \log y(t) = \eta_0 + \eta_1 \Delta \log y(t-1) + \varepsilon(t)$ 

| <u>(1) U.S.A. (N=126)</u> | <u><math>\eta_0</math></u> | <u><math>\eta_1</math></u> | <u>S.E.R.</u> |
|---------------------------|----------------------------|----------------------------|---------------|
|                           | 0.0054                     | 0.298                      | 0.0099        |
|                           | (0.0011)                   | (0.085)                    |               |
| <u>(2) Canada (N=126)</u> | <u><math>\eta_0</math></u> | <u><math>\eta_1</math></u> | <u>S.E.R.</u> |
|                           | 0.0086                     | 0.179                      | 0.01          |
|                           | (0.0013)                   | (0.088)                    |               |
| <u>(3) U.K. (N=102)</u>   | <u><math>\eta_0</math></u> | <u><math>\eta_1</math></u> | <u>S.E.R.</u> |
|                           | 0.0067                     | -0.215                     | 0.014         |
|                           | (0.0015)                   | (0.098)                    |               |
| <u>(4) France (N=97)</u>  | <u><math>\eta_0</math></u> | <u><math>\eta_1</math></u> | <u>S.E.R.</u> |
|                           | 0.011                      | -0.37                      | 0.012         |
|                           | (0.0014)                   | (0.095)                    |               |
| <u>(5) Italy (N=69)</u>   | <u><math>\eta_0</math></u> | <u><math>\eta_1</math></u> | <u>S.E.R.</u> |
|                           | 0.008                      | 0.021                      | 0.012         |
|                           | (0.0018)                   | (0.12)                     |               |

Table 3.7 continued

(B) Per-Capita Income:  $\Delta \log \frac{y(t)}{N(t)} = \eta_0 + \eta_1 \Delta \log \frac{y(t-1)}{N(t-1)} + \varepsilon(t)$

|                    |          |          |        |
|--------------------|----------|----------|--------|
| (1) U.S.A. (N=126) | $\eta_0$ | $\eta_1$ | S.E.R. |
|                    | 0.0033   | 0.299    | 0.0099 |
|                    | (0.001)  | (0.085)  |        |
| (2) Canada (N=126) | $\eta_0$ | $\eta_1$ | S.E.R. |
|                    | 0.0055   | 0.199    | 0.01   |
|                    | (0.0011) | (0.087)  |        |
| (3) U.K. (N=102)   | $\eta_0$ | $\eta_1$ | S.E.R. |
|                    | 0.006    | -0.213   | 0.014  |
|                    | (0.0014) | (0.098)  |        |
| (4) France (N=97)  | $\eta_0$ | $\eta_1$ | S.E.R. |
|                    | 0.0096   | -0.39    | 0.011  |
|                    | (0.0013) | (0.094)  |        |
| (5) Italy (N=69)   | $\eta_0$ | $\eta_1$ | S.E.R. |
|                    | 0.0074   | 0.003    | 0.012  |
|                    | (0.0017) | (0.118)  |        |

Notes: N is the number of observations and SER the standard error of regression.

**Table 3.8. Wagner's Law and Permanent Income Innovations**Estimated Elasticities:

|                   | <u>Version 1</u> |              | <u>Version 2</u> |              |
|-------------------|------------------|--------------|------------------|--------------|
| (1)U.S.A. (N=126) | <u>r=0.05</u>    | <u>r=0.1</u> | <u>r=0.05</u>    | <u>r=0.1</u> |
| Total G           | 0.114            | 0.116        | -0.886           | -0.886       |
|                   |                  |              |                  |              |
| (2)Canada (N=126) | <u>r=0.05</u>    | <u>r=0.1</u> | <u>r=0.05</u>    | <u>r=0.1</u> |
| Total G           | 0.18             | 0.181        | -0.83            | -0.83        |
|                   |                  |              |                  |              |
| (3) U.K. (N=102)  | <u>r=0.05</u>    | <u>r=0.1</u> | <u>r=0.05</u>    | <u>r=0.1</u> |
| Total G           | 0.276            | 0.274        | -0.717           | -0.719       |
|                   |                  |              |                  |              |
| (4) France (N=97) | <u>r=0.05</u>    | <u>r=0.1</u> | <u>r=0.05</u>    | <u>r=0.1</u> |
| Total G           | 0.243            | 0.24         | -0.754           | -0.757       |
|                   |                  |              |                  |              |
| (5) Italy (N=69)  | <u>r=0.05</u>    | <u>r=0.1</u> | <u>r=0.05</u>    | <u>r=0.1</u> |
| Total G           | 0.3708           | 0.371        | -0.63            | -0.63        |

Notes: N is the number of observations.

**Chapter 4:**  
**International Spillovers of Public Investment  
and OECD Economic Growth**



## **I. Introduction**

Fiscal policy between open interdependent economies has been the subject of much research. Nevertheless, conflicting views about the international spillover effects and transmission mechanisms of fiscal policy persist, and have resulted in competing debates about optimal government policy in domestic and international affairs.

This chapter is motivated by similar criticisms of previous work that were raised in Chapter 2. First, as emphasized earlier, the literature is dominated by studies which focus on the financing side of fiscal policy - taxation and budgetary deficits. The studies that do focus on fiscal expenditures, however, rarely distinguish amongst public consumption, transfer payments, and public investment. Expenditures are often modelled as having no direct effects on private marginal utility or production, again largely to focus on the effects of "financing" an exogenous level of expenditures.

Another criticism of existing works is their reliance on price variables as the primary means by which fiscal policies spillover to other countries<sup>1</sup>. Fiscal policies of large, or semi-large, economies are argued to impose externalities on other countries by altering the terms of trade, exchange rate, or interest rate - all of which are price variables. As pointed out in Chapter 2 there are two kinds of externalities: pecuniary and technological. Pecuniary externalities work through the price system, whether through the prices of factor supplies or outputs. When markets are competitive, pecuniary externalities are simply the outcomes of normal

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<sup>1</sup>See for example Devereux (1987b), Frenkel and Razin (1987), Ihori (1987), and Van Wijnbergen (1986) to name a few.

marketplace interactions. They do not constitute sufficient grounds for social intervention. Price changes effect distributional changes, but do not by themselves reflect market failure. In contrast, an efficiency-based case for government intervention can be made when "technological externalities" are involved. These occur when the production activities of an agent indirectly affect the production of other agents other than by influencing the latters' input costs or output prices. In the presence of these kinds of externalities the private market will generally not lead to socially optimal outcomes - the first fundamental welfare theorem fails.

In an open-economy context the significance of distinguishing between these two types of externalities is that when fiscal policies generate technological externalities, international coordination by public authorities is needed from an allocative efficiency point of view. Internationally enjoyed public goods provision would tend to be inefficient in a world of non-cooperative nationalistic governments. By focusing on the pecuniary channels of fiscal policy transmission, existing work has not made a strong normative case in favour of global fiscal policy coordination (on both the expenditure and financing sides) nor has it analyzed at the positive level the full range of interdependencies defined by fiscal policy.

The purpose of this chapter is to investigate empirically whether fiscal expenditures do give rise to technological externalities. Two kinds of government investment expenditures are likely candidates for generating technological externalities. First, government spending on R&D is likely to be an important source of international spillover as both domestic and foreign producers can potentially benefit from the increased stock of global knowledge. Second, public infrastructure (for example, transportation, communication, irrigation, and energy) aids private production directly

and is likely to generate external effects at the national level. Since infrastructure is largely non-traded, technological spillovers should tend to be confined within a domestic economy. For some neighboring economies, however, it is conceivable that certain types of infrastructure (such as roads, telecommunication, and irrigation systems) have cross border spillover effects of the technological variety<sup>1</sup>.

This chapter does not argue that public investments have no pecuniary effects - as much work has shown how such investments can lower the costs of production for private producers<sup>2</sup> - nor does this chapter try to compare which of the two types of externalities predominates. The interest here is limited to showing that, in addition to any possible pecuniary effects, there can also be technological externalities associated with fiscal expenditures - that private producers can and do regard public infrastructure and R&D as "inputs" for which they do not make direct compensation (or at least payments commensurate with the marginal benefits derived from them). While it is possible to exact indirect compensation from domestic resident producers through the general tax system, it is somewhat more difficult to do so from foreign non-resident producers.

In the next section, I review some previous literature and discuss some trends and facts about public investment in research and physical capital. In section III, I derive a simple empirical specification outlining

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<sup>1</sup> The World Bank Development Report (1988) documents several cases in developing country regions where rural infrastructures (such as irrigation systems) of certain political units provide uncompensated economic benefits to neighbouring political units.

<sup>2</sup> Murphy, Schleifer, and Vishny (1989) show how pecuniary externalities arising from public works projects, such as public railroads, can expedite economic development.

how public R&D affects private sector productivity and research. The data are discussed in section IV, and the empirical results and tests are presented in section V.

Using OECD data in an aggregate growth-accounting framework, I look for evidence of technological externalities based on whether one country's public sector R&D affects foreign production possibilities<sup>1</sup>. Infrastructure is also studied to assess the degree to which public capital has effects on growth beyond that coming from private factor supplies (such as private capital and labour). Both the U.S. and a panel data of OECD countries are investigated. I also disaggregate public R&D by socio-economic objective and analyze how components of public R&D affect private sector productivity. Overall I find that there exist global spillover effects which spur private sector productivity growth worldwide. When public R&D variables are considered independently of private R&D variables, the former are significant. Once private R&D variables are included, the significance of public R&D diminishes, indicating that the more important source of private sector productivity growth is private R&D; however, public R&D indirectly affects private sector productivity by stimulating private R&D, which is the effect captured when public R&D is considered in isolation of private R&D. Finally, Section VI contains some concluding thoughts.

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<sup>1</sup> The focus therefore is on spillovers into production rather than into research.

## **II. Recent Trends and Previous Literature**

There are very few econometric studies on public investment. Nevertheless some interesting works have appeared recently, employing both time-series as well as cross-sectional data. My general impression is that cross-sectional results tend to show little or no evidence that public investments affect productivity growth, while time-series studies show the opposite. For example, cross-sectional studies by Barro (1989), Diamond (1989), and Grier and Tullock (1989) find either a negative relationship between public investment and economic growth or an insignificant one. I question whether dynamic issues can be studied using cross-sectional data. It is not straightforward to give dynamic interpretations to cross-sectional results. If, for example, at some point in time, countries with relatively higher growth rates have relatively lower shares of public investment in GDP, this need not suggest that public investment slows down growth but that faster growing economies devote a smaller share of GDP to investment. Similarly, slower growing economies may be devoting a larger share of GDP to investment to generate future growth. Thus cross-sectional results do create some ambiguities. A time dimension is needed in general in order to study dynamic multiplier effects. Indeed time-series studies by Aschauer (1989a), Holtz-Eakin (1988), or the pooled time-series-cross-sectional study by Ram (1986), do find significant relationships between government investment and economic growth.

Studies by Griliches-Lichtenberg (1984) and Bartelsman (1990) find that public R&D either is less significant than private R&D for private sector productivity or exerts a negative influence on productivity. Some conjectures are that public R&D diverts resources from the more efficient

private sector, that public R&D funds less productive activities (environment, defence, and higher education in non-science and engineering fields), or that whatever benefits public R&D spending yields, the economic benefits are offset by the distortionary effects of taxes required to finance public investment. Another explanation may be that these studies are at the industry (or firm) level. Private R&D is likely to be more significant at the disaggregated level because it is more specific, or tailored, to the needs of the firm or industry. Government R&D in contrast tends to be targetted towards areas of national priority.

The methodological approach adopted here is very similar to that found in the industrial organization literature specializing in the analysis of either intra-industry or inter-industry spillovers of R&D<sup>1</sup>. The key change is in the economic unit defined; instead of firms or industries, I work with countries. Just as a firm or industry can be affected not only by its own R&D but also by the R&D of other firms or industries, a country can be affected not only by own-country R&D but also by foreign national R&D or global R&D.

It is helpful to know some characteristics of these types of investments and their recent trends before exploring the empirical effects of public investment in R&D and infrastructural capital on productivity growth. In charts 4.1A and 4.1B, I provide a breakdown of the U.S. capital stock in 1985 and 1965. I exclude consumer durables in the measure of capital stock used, though data on them are collected and reported by the BEA. The various percentages are calculated from figures that are net of

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<sup>1</sup> See work by Griliches (1979), Levin and Reiss (1984), Griliches and Lichtenberg (1984), Jaffe (1986), Bernstein and Nadiri (1988), and Bartelsman (1990).

depreciation and discards and are based on constant 1982 dollar values. According to 1985 figures, 73% of the nation's capital stock is private, down from 77% in 1965. Roughly half of the private stock of capital is residential. Of the stock of public capital, 69% is state and local in 1985, up from 63% in 1965. Only a small percentage of the state and local capital is equipment; most are structures, of which highways and streets constitute the largest share. Of the stock of federal government capital, 56% is equipment in 1985, up from 45% in 1965. A large percentage of federal equipment is military in nature. The rest of federal government capital consists of structures, of which 31% are military facilities in 1985, down from 43.2% in 1965. Thus using 1985 figures, one finds that 63% of federal government capital is military, while 20% of total government capital is military. The proportions in 1965 are very comparable as 61% of federal government capital was military, while 22% of total government capital was military.

Charts 4.2A and 4.2B show the shares of private and public gross fixed investments in GDP during 1965, 1975, and 1985 for selected OECD countries. As 1975 is a period shortly after the first OPEC oil shock, the ratios of investment to GDP vary somewhat across the countries. For instance, for some countries the fall in output is larger, so that the ratio of investment to output rises. Nonetheless the general pattern is that the shares of private and public investment in GDP are smaller in 1985 than they were in 1965; the one exception is U.S. private investment, which has a slightly higher share in GDP in 1985 than it did in 1965.

I turn now to public R&D. Much of the world's R&D activities originates in five countries (not considering the Soviet Union). In order of rank, they are: the U.S., Japan, West Germany, France, and the United

Kingdom<sup>1</sup>. The major G-7 countries (including Italy and Canada) account for roughly 91% of total world R&D. According to charts 4.3A and 4.3B, roughly 35% to 55% of R&D in each country is funded by the public sector. Exceptions can be found in Japan and Greece; in Japan, the government funds less than 30% of domestic R&D, while in Greece the government funds more than 75% of domestic R&D. For Japan, this figure may be an underestimate of the actual involvement of the Japanese government in funding R&D since the Japanese government is an important contributor of R&D funds outside its border, for example it helps fund the R&D of Japanese multinationals abroad and contributes to international projects.

Charts 4.3C to 4.3F provide some idea of how much each of the major R&D countries contributes to the global pool of R&D capital. Appendix 4.3 outlines how R&D stocks are constructed from R&D flow data (using the perpetual inventory method). By global stocks of R&D, I am referring to the aggregate stocks of the ten countries listed in Charts 4.3C-4.3F. All stocks are denominated in real 1985 U.S. dollars at PPP rates of exchange<sup>2</sup>. Chart 4.3C indicates that in 1975, the U.S. stock of public R&D capital accounted for 63% of global public R&D capital; this share drops to 58% in 1985 as indicated in Chart 4.3D. The relative shares among the remaining nine countries do not change significantly over this period. Chart 4.3E indicates that the U.S. stock of private R&D capital accounted for 52% of global private R&D capital in 1975 while somewhat less, namely 48%, in 1985. Over this period, the Japanese stock of private

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<sup>1</sup> See the NSF's Science and Engineering Indicators, various issues.

<sup>2</sup> PPP rates deal with prices of outputs. It would be desirable in the future to create and explore the use of an "input" version of PPP rates.



R&D capital posts the most noticeable gain in relative share of global R&D, going from 15% in 1975 to 19% in 1985.

Chart 4.4 shows where the government funds get directed; in the U.S., 23% of government funds are channelled to higher education and 47% to private industry. In Japan, 41% goes to higher education and 49% to public enterprises who then perform the R&D tasks directly. In the Netherlands and Belgium, more than half of public funds goes to higher education. In Greece and Portugal, less than 2% of public funds goes to industry; most of the R&D is performed in these countries by the public sector.

Chart 4.5 shows the trend in public R&D funding as a share of total government expenditures. Public R&D investment is quite a small percentage of total government expenditures and could only have marginally contributed to the rise in government expenditures that have been behind the large increases in budgetary deficits in the OECD. That governments have been scaling back public investment in research is evident in this chart, particularly since total government expenditures have been trending upwards during this period while the share of public R&D funding in total government expenditures has been falling during the mid 1970s and rising somewhat thereafter.

Chart 4.6 provides information on how governments allocate their R&D funds by socio-economic objectives. In the U.S. and U.K., governments have devoted more than half of their R&D funds towards defence; in France, about a third is devoted to defence. The Japanese government devotes most of its funds towards economic development (namely agriculture and industrial production and technology). It also devotes a large share towards research on Energy, which is expected since

Japan is a country heavily dependent on oil imports. Funding for Space research has declined markedly between 1975 and 1985 within the Major 5 (and the OECD at large).

### **III. An Empirical Specification**

This section builds on the aggregate production function approach of Solow (1957) and modifies it along the lines suggested in the recent endogenous growth literature<sup>1</sup> - that is, the recent approaches permit the total factor productivity term to depend on "publicly" available inputs. The term "endogenous growth model" refers to the fact that the determinants of productivity shifts can be found "within" the model, and not from events or influences exogenous to the model. The model I have in mind is one in which government policies can potentially affect long term growth rates (as well as levels) by directly affecting the production function of the private sector.

The term "publicly" available refers to the fact that there are externalities (or uncompensated interdependencies) generated in the activities of economic agents which affect aggregate and individual production possibilities. If these spillovers are positive it is quite likely that there will be an inefficient provision of that activity from a social point of view. Hence a number of "public" inputs comes from the government sector itself, such as infrastructural services (provision of roads, airports, postal services, sewerage, and so forth) and research (on areas such as health, environment, energy, defense, space) which affects the

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<sup>1</sup> See for example Romer (1987), Barro (1990), and Sala-i-Martin (1990).

productivity growth not only of the nation conducting the research but also of the international community.

Governments both perform and fund R&D. I focus on "funding". One view is that government research funds help facilitate technological progress directly by initiating the advancement of knowledge in areas the private sector overlooks and indirectly by stimulating private research through increasing the returns or productivity of research activity. Another view is that governments divert resources that would otherwise be earmarked for private research, and thereby crowd out private investment. The financing of government activity itself is also argued to be distortionary and to absorb private savings, so that in the long run the positive benefits from government spending are diminished<sup>1</sup>. The net "reduced-form" impacts are thus ambiguous and model-dependent. One key ingredient in any structural model designed to explain growth is the production function, and the empirical nature of this function must be investigated. Towards that end I examine government spending on two kinds of "publicly" available inputs for private production - namely the 'stock' of knowledge created by government funding of research (basic and applied) and the stock of government infrastructural capital.

Public investments affect private sector output directly and indirectly. The direct effect is due to the presence of public capital in the private sector production function. The indirect effect arises because public investment can stimulate private investment and augment the stock of private capital which is also in the private sector production function.

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<sup>1</sup>That is, the positive effects of fiscal spending on growth may be outweighed in the long run by a higher tax burden associated with a larger size of government.

Consequently this section will consist of two parts. In the first part I specify the aggregate private production function and its associated growth accounting equation. In the second part I derive investment equations for the private sector.

### (i) Aggregate Private Sector Production Function

Consider the following standard aggregate production function:

$$(1) \quad Y(t) = A(t)F(K_p(t), L(t))$$

where  $F$  is a well-behaved neoclassical production function.  $K_p(t)$  is the stock of private capital, and  $L(t)$  the stock of private labour (measured in hours). Output  $Y(t)$  is interpreted to be private gross domestic product.

It is the  $A(t)$  term which is of interest, and I model it as follows:

$$(1a) \quad A(t) = A(K_g(t), \Psi(t), \dots)$$

where (1b)  $\Psi(t) = \Psi(\text{R\&D stock})$

represents the state of knowledge and is indexed by the stock of R&D capital accumulated, part of which is created by the public sector and part by the private sector<sup>1</sup>. Another part can be due to R&D capital created

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<sup>1</sup> The part created by the private sector is "publicly" available unless patents and other regulations prevent its widespread use, although in some cases a "disclosure" clause may allow knowledge spillovers to occur but prevent other firms or agents from applying the knowledge.

abroad.  $K_g(t)$  is the stock of physical public capital. In  $A(t)$ , the remaining dots suggest that there are other factors which can shift the production function but are not being modelled here, such as regulation, human capital, and institutional attributes. Note that to the extent that some R&D knowledge may already be embodied in the private inputs  $K_p(t)$  and  $L(t)$ , one could interpret the significance of the  $\Psi(t)$  term as indicating the presence of excess social returns to R&D (not already captured in the private inputs).

Consider now a multi-country setting. Let  $R$  denote the stock of knowledge created by the domestic country's R&D activities and  $R^*$  the stock of knowledge created by the foreign countries' R&D activities. For now we assume that  $R$  and  $R^*$  are national stocks of knowledge and that each is the sum of public and private R&D stocks. A special feature about "knowledge" is that it is nonrival and nonexcludable. The  $R$  and  $R^*$  created are, in principle, globally available inputs. Both domestic production and foreign production are affected by  $R$  and  $R^*$ <sup>1</sup>. In addition,

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<sup>1</sup> The model abstracts from how knowledge spillovers occur. There are several ways: private firms can learn by observing the products or processes that result from R&D. There are several intergovernmental cooperative forums, exchanges, or ventures which permit more direct learning. Many government agencies and departments engage in, or encourage some form of international cooperation (NSF, 1984), largely through bilateral arrangements. More than 20 U.S. Federal agencies are in the process of conducting scientific and technological activities with more than 30 nations. The key sectors of cooperation are Space, Energy, Health, Agriculture, and Basic Science and Engineering. There are also several global multilateral efforts facilitated by the United Nations, for example, the World Climate Research Program. The OECD (1985) report documents several collaborative international efforts in high technology such as RACE (R&D in Advanced Communication Technologies for Europe), the EUREKA project (in optic, laser technology, and artificial intelligence), and SDI (Strategic Defense Initiative), and the report

I assume that the aggregate stock of foreign knowledge is just the sum of the individual foreign country<sup>1</sup> R&D stocks - weighted or unweighted:

$$(1c) \quad R(t)^* = \sum_{i=1}^N \gamma_i R_i(t)^*, \quad 0 \leq \gamma_i \leq 1$$

where the weight  $\gamma_i$  is a country-specific technical distance term to reflect the fact that knowledge inputs from the rest of the world are not perfect substitutes from the point of view of the domestic economy. One way to construct these technical distance terms, following Jaffe (1985), is to define a vector of characteristics (such as the functional composition of R&D, or number of patents applied for in one country by another) and identify two countries as being technologically close neighbours if they have a similar vector of characteristics. However our data set is incomplete in the sense that such vectors can only be created for a subset of countries. For simplicity I therefore assume  $\gamma_i = 1$  for all  $i = 1, \dots, N$ . It is hoped that this assumption is not too restrictive to impose among the industrialized countries of the OECD<sup>2</sup>. At this stage of development of empirical work

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highlights the role of governments in initiating and funding the collaborative efforts.

<sup>1</sup> Here "rest of the world" refers to the rest of the partner OECD economies.

<sup>2</sup> Another view is that countries have comparative advantages in different types of research. This should however be reflected in the functional composition of R&D amongst defense, space, energy, and so forth, which I do consider in the empirical section. I concentrate not so much on the identity of the countries responsible for producing the most productive research but on the productivity effect of total or of a particular type of research, however the research effort by country came about. However if the criticism is that a dollar of research in the U.S. is not equivalent (in a productivity sense) to a dollar of research in another OECD country, again the problem is to construct a (correct) index of productivity weights to

there are as yet insufficient criteria to determine which measures of  $\gamma_i$  are best.

It is important to be reminded that  $R^*$  represents to the domestic economy the stock of knowledge created abroad;  $R^*$  for instance does not contain  $R$ , the domestically created stock of knowledge. Moreover  $R^*$  typically varies by country. For the United States,  $R^*$  is very small compared to  $R$ , while for smaller economies like Luxembourg and Ireland,  $R^*$  is very large.

Continuing with the model's specification, I parameterize the  $\Psi(t)$  to be:

$$(2) \Psi(t) = \zeta R(t)^{\theta} R(t)^{* \theta^*} e^{\alpha t},$$

or

$$(2)' \Psi(t) = \zeta (R(t) + R(t)^*)^{\tilde{\theta}} e^{\alpha t}$$

where I assume all other sources of technical change are contained in the constant term,  $\zeta$ , multiplied by a time trend  $e^{\alpha t}$ . In specification (2),  $R$  and  $R^*$  may have different output elasticities. Both are "essential" to production nonetheless. In specification (2)',  $R$  and  $R^*$  are perfectly substitutable and either is not essential if there is a positive stock of the other. I can parameterize the  $A(t)$  to be:

$$(3) A(t) = \Psi(t) K_g(t)^{\beta_g} = \zeta R(t)^{\theta} R(t)^{* \theta^*} e^{\alpha t} K_g(t)^{\beta_g}$$

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attach to each country's R&D expenditures. Again there is not enough work in this area to guide us in the construction and application of these weights.

where (2) is used.

I specialize the production function  $F$  to be Cobb-Douglas:

$$\text{i.e. } F(K_P(t), L(t)) = K_P(t)^{\beta_P} L(t)^{\beta_L}$$

(and  $\beta_P + \beta_L = 1$  of course implies constant returns to scale in  $K_P$  and  $L$ ).

Collecting (1) to (3) and all the definitions and parameterizations, we arrive at the following production function<sup>1</sup>:

$$(4) \quad Y(t) = \zeta e^{\alpha t} R(t)^{\theta} R(t)^{* \theta^*} K_g(t)^{\beta_g} K_P(t)^{\beta_P} L(t)^{\beta_L}, \quad \zeta > 0.$$

In the empirical section I assume a multiplicative error term, and multiply the RHS of the above by the capacity utilization rate - ie.  $cu(t)^{\delta}$ , where  $cu$  denotes capacity utilization and  $\delta$  is a parameter - in order to adjust for under or over-utilization of resources (as occurs during business cycles). The purpose of this is to try to get the "economy" on the production frontier, so that we can isolate the effects of public spending on growth from its effects on cycles. Otherwise we will not be correctly inferring the "efficiency" effects of public spending, or for that matter, of anything else we model in the  $A(t)$  term.

Equation (4) represents the basic aggregate production function. In order to obtain the aggregate growth accounting counterpart, we take the

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<sup>1</sup> The output elasticity of each RHS variable is given by the corresponding exponent of that variable.



logs of both sides of (4), and time-differentiate (using (1c)). This gives us<sup>1</sup>:

$$(4)' \quad \frac{\Delta Y(t)}{Y(t)} = \alpha + \theta \frac{\Delta R(t)}{R(t)} + \theta^* \frac{\Delta R(t)^*}{R(t)^*} + \beta_g \frac{\Delta K_g(t)}{K_g(t)} + \beta_p \frac{\Delta K_p(t)}{K_p(t)} + \beta_L \frac{\Delta L(t)}{L(t)}$$

or

$$(4)'' \quad \frac{\Delta Y(t)}{Y(t)} = \alpha + \theta \frac{\Delta R(t)}{R(t)} + \theta^* \frac{\left( \sum_{i=1}^N \Delta R_i(t)^* \right)}{R(t)^*} + \beta_g \frac{\Delta K_g(t)}{K_g(t)} + \beta_p \frac{\Delta K_p(t)}{K_p(t)} + \beta_L \frac{\Delta L(t)}{L(t)}$$

or

$$(4)''' \quad \frac{\Delta Y(t)}{Y(t)} = \alpha + \theta \frac{\Delta R(t)}{R(t)} + \theta^* \left( \sum_{i=1}^N s_i \frac{\Delta R_i(t)^*}{R_i(t)^*} \right) + \beta_g \frac{\Delta K_g(t)}{K_g(t)} + \beta_p \frac{\Delta K_p(t)}{K_p(t)} + \beta_L \frac{\Delta L(t)}{L(t)}$$

where  $\Delta$  is the difference operator, and

$$(5) \quad s_i = \frac{R_i(t)^*}{R(t)^*},$$

is the share of foreign country  $i$ 's stock of knowledge in the aggregate stock of foreign knowledge,  $i = 1, \dots, N$ .

From (4)' the output elasticity of  $R^*$  is  $\theta^*$ , while from (4)''' the output elasticity of  $R_i(t)^*$  is  $\theta^* s_i$ . The output elasticities of individual

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<sup>1</sup> Recall that for some variable  $X$ ,  $\Delta \log X = \frac{\Delta X}{X}$ .

foreign country knowledge stocks can thus vary depending on individual country shares in the world stock of "foreign" knowledge. While our empirical section concentrates on finding the aggregate elasticity  $\theta^*$ , we can infer the individual elasticities from information about the  $s_i$ 's<sup>1</sup>.

Under constant returns to scale<sup>2</sup>, we can rewrite (4)' as

$$(4)'''' \frac{\frac{\Delta Y(t)}{L(t)}}{\frac{Y(t)}{L(t)}} = \alpha + \theta \frac{\Delta R(t)}{R(t)} + \theta^* \frac{\Delta R(t)^*}{R(t)^*} + \beta_g \frac{\Delta K_g(t)}{K_g(t)} + \beta_p \frac{\frac{\Delta K_p(t)}{L(t)}}{\frac{K_p(t)}{L(t)}}$$

We can also define the change in total factor productivity (TFP) as:

$$(6) \Delta TFP(t) = \frac{\Delta Y(t)}{Y(t)} - \beta_p \frac{\Delta K_p(t)}{K_p(t)} - \beta_L \frac{\Delta L(t)}{L(t)}$$

or

$$(6)' \Delta TFP(t) = \alpha + \theta \frac{\Delta R(t)}{R(t)} + \theta^* \frac{\Delta R(t)^*}{R(t)^*} + \beta_g \frac{\Delta K_g(t)}{K_g(t)}$$

Instead of estimating the output elasticities, one can estimate the marginal products of the inputs. Griliches-Lichtenberg (1984) argue that constraining the output elasticity of the common R&D stock to be the same across the  $i$  sectors (in a panel data or pooled sample) is likely to be

<sup>1</sup> For some idea of the size of the shares, see Charts 4.3C to 4.3F.

<sup>2</sup> One might be able to address the issue of "congestion" effects of public capital via a simple test of constant returns to scale; if there are congestion effects, it might be likely that the constant returns to scale assumption should be applied across public capital and the two private inputs. Otherwise, if the private inputs alone satisfy constant returns, there is increasing returns to all three inputs, inclusive of government capital. See also Aschauer (1989a) for a similar discussion.

bothersome given long term differences in R&D intensities across  $i$ . The authors therefore impose the marginal product of R&D to be the same across  $i$ . The justification for this is that if the marginal products were unequal, research resources would be shifted until the last dollar of research resource is equally productive at the margin in all sectors.

In our formulation the marginal products of foreign  $R_1^*, \dots, R_N^*$  with respect to domestic output are equalized, while we recall that the domestic output elasticities of  $R_1^*, \dots, R_N^*$  are allowed to vary. From (4), the marginal product of  $R^*$  on domestic  $Y$  is:

$$\frac{\Delta Y(t)}{\Delta R^*(t)} = \frac{\theta^* Y}{R^*}$$

while from (4)" the marginal product of  $R_i^*$  on domestic  $Y$  is:

$$\frac{\Delta Y(t)}{\Delta R_i^*(t)} = \frac{\theta^* Y}{R^*} = s_i \frac{\theta^* Y}{R_i^*}$$

Similarly,  $\frac{\Delta Y(t)}{\Delta R_j^*(t)} = \frac{\theta^* Y}{R^*} = s_j \frac{\theta^* Y}{R_j^*}$ . Thus the marginal products of  $R_1^*, \dots, R_N^*$  are equalized (this follows from the fact that they are all perfect substitutes when  $\gamma_i = 1$ )<sup>1</sup>. Then, defining  $\rho^*$  as the common marginal product of foreign R&D stocks and  $\rho$  as the marginal product of the domestic R&D stock, we can substitute the output elasticities  $\theta, \theta^*$ :

$$\theta = \frac{\Delta Y}{\Delta R} \frac{R}{Y} = \rho \frac{R}{Y} \quad \text{and} \quad \theta^* = \frac{\Delta Y}{\Delta R^*} \frac{R^*}{Y} = \rho^* \frac{R^*}{Y}$$

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<sup>1</sup> This requires that  $(s_j / s_i) = (R_j^* / R_i^*)$  which is true by definition since  $(s_j / s_i) = (R_j^* / R^*) / (R_i^* / R^*) = (R_j^* / R_i^*)$ .

into (4)' to get:

$$(7) \frac{\Delta Y(t)}{Y(t)} = \alpha + \rho \frac{\Delta R(t)}{Y(t)} + \rho^* \frac{\Delta R(t)^*}{Y(t)} + \beta_g \frac{\Delta K_g(t)}{K_g(t)} + \beta_p \frac{\Delta K_p(t)}{K_p(t)} + \beta_L \frac{\Delta L(t)}{L(t)}$$

$$\text{where } \rho = \theta \frac{Y}{R} \text{ and } \rho^* = \theta^* \frac{Y}{R^*}$$

There is no consensus as to whether the pooled sample should have equal output elasticities or equal marginal products. Different researchers may wish to calculate the marginal productivity rather than the output elasticity of research capital, and vice versa. I would argue that in a open-economy context it is harder to rationalize the equalization of marginal products in the sense of:

$$(8) \frac{\Delta Y^i}{\Delta R_i} = \frac{\Delta Y^j}{\Delta R_j} = \frac{\Delta Y^k}{\Delta R_k} \text{ for } i \neq j \neq k$$

ie. that country i's stock of R&D capital is equally productive at the margin not only in country i but also in every other country. This is harder to rationalize in the absence of a benevolent global social planner or firm. If the marginal products are not equal, there is no mechanism for eliminating international divergences in marginal productivities even if research resources are perfectly internationally mobile. Researchers in country i are interested in  $\frac{\Delta Y^i}{\Delta R_i}$ , as this represents the benefit to country i; they are not necessarily concerned with how  $\Delta R_i$  affects other countries. Condition

(8) is a global social optimality condition and is not likely to be fulfilled under decentralized world trading conditions.

Up to now I have not specified the composition of  $R$  and  $R^*$  in terms of how much of the stock of national R&D is publicly financed R&D capital and how much is privately financed. To distinguish between public and private stocks of R&D capital, we rewrite (2) as:

$$(2)'' \Psi(t) = \zeta R_g(t)^{\theta_g} R_g(t)^{* \theta_{g^*}} R_p(t)^{\theta_p} R_p(t)^{* \theta_{p^*}} e^{\alpha t}$$

where  $R_g(t)$ ,  $R_g(t)^*$ ,  $R_p(t)$ ,  $R_p(t)^*$  are the R&D capital stocks created by the domestic government, foreign government, domestic private sector, and foreign private sector respectively. Their output elasticities are  $\theta_g$ ,  $\theta_{g^*}$ ,  $\theta_p$ ,  $\theta_{p^*}$ , respectively, and can be estimated from equation (4)' revised:

$$(9) \frac{\Delta Y(t)}{Y(t)} = \alpha + \theta_g \frac{\Delta R_g(t)}{R_g(t)} + \theta_{g^*} \frac{\Delta R_g(t)^*}{R_g(t)^*} + \theta_p \frac{\Delta R_p(t)}{R_p(t)} + \theta_{p^*} \frac{\Delta R_p(t)^*}{R_p(t)^*} \\ + \beta_g \frac{\Delta K_g(t)}{K_g(t)} + \beta_p \frac{\Delta K_p(t)}{K_p(t)} + \beta_L \frac{\Delta L(t)}{L(t)}$$

or

$$(10) \frac{\Delta TFP(t)}{TFP(t)} = \alpha + \theta_g \frac{\Delta R_g(t)}{R_g(t)} + \theta_{g^*} \frac{\Delta R_g(t)^*}{R_g(t)^*} + \theta_p \frac{\Delta R_p(t)}{R_p(t)} + \theta_{p^*} \frac{\Delta R_p(t)^*}{R_p(t)^*} + \beta_g \frac{\Delta K_g(t)}{K_g(t)}$$

where TFP is as defined in equation (6)<sup>1</sup>.

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<sup>1</sup> Despite the disappearance of the trend term in (9) and (10) after first differencing, I find the presence of a trend in the panel data first-difference results, indicating the presence of a quadratic trend term in levels.

A brief word should be devoted to the issue of the length of the lag between R&D spending and its effect on productivity. The production function developed above assumes that  $R(t)$  and  $R(t)^*$  are the productive stocks of knowledge as of time  $t$ . The  $R(t)$ ,  $R(t)^*$ 's are each proxied by the cumulative sum of past R&D expenditures, adjusted for the rate of obsolescence (namely a geometric rate of 0.1). However there should be a lag of length say  $j$ , where  $j$  is the length of time it takes for the R&D spending to add to the stock of productive knowledge capital. The length of  $j$  depends on whether the R&D is predominantly basic, applied, or development-oriented. In the empirical section, I adopt  $j = 3$  years<sup>1</sup>.

Another important consideration is that the short and long-run effects of public investment on productivity growth may be outweighed in the long run (if not sooner) by higher distortionary taxes required to finance public investment. To investigate this issue I also allow for capital income taxes (or corporate profits taxes) in the aggregate production function framework in conjunction with the public investment variables. Appendix 4.1 motivates how distortionary taxes, such as a capital income tax, may be introduced into this framework. First, higher capital income tax rates in the aggregate economy tend to discourage investment because the returns to investment tend to fall. Secondly, within an economy, higher capital income tax rates may affect different sectors of the economy

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<sup>1</sup> A recent study by the U.S. Bureau of Labor Statistics (1989) discusses the issue of R&D lag effects and lag structures, and finds that a lag of  $j = 3$  might be apposite. The mean lag for basic research appears to be 5 years and applied research 2 years. A weighted average would then be 3 or 4 years. When I experimented with  $j=1$ , I found R&D having strong significance on growth and productivity. However, this might be too short a time period for R&D actually to affect 'supply', and might better represent the 'demand' effects of government spending.

differently. Differential tax treatment results in divergences in the marginal productivities of capital across sectors. Resources are induced to flow to less heavily taxed sectors and not necessarily to ones that are economically more productive. The aggregate economy becomes less efficient and operates below its potential.

To estimate the effects of capital income tax rate  $\tau_k$  on productivity growth, I simply append to (4)', (4)''', or (9), the term " $\phi_K \tau_k$ ", where  $\phi_K$ , expected to be negative, is the percentage change in aggregate output per basis point change in  $\tau_k$  (Appendix 4.1 derives  $\phi_K$  explicitly).

### (ii) Private Investment Equations

This section outlines parsimonious models of how public investment may influence private investment. First I discuss the influences on private R&D investment, and I later turn to the influences on private capital investment. The models have simple demand-supply interpretations, and variants have been used earlier by Aschauer (1989b), Levy-Terleckyj (1983), and Lichtenberg (1987).

#### Private R&D Investment Equation

Let  $\phi$  denote the return to private investment in R&D - say the marginal product of, or marginal return to, private R&D capital. Let  $X$  be an index of conventional (private) inputs and  $A$  an index of total factor productivity. As in the earlier section, output  $Y$  is produced from:

$$Y = A \cdot F(X)$$

where  $F_X > 0$ ,  $F_{XX} < 0$ .

Let  $R$  be an index of R&D stocks. Now  $A$  is a function of  $R$  - ie.  $A = A(R, K_g)$ . Thus  $Y = Y(R, X, \dots)$ . Totally differentiating the  $Y$  function:

$$\frac{\Delta Y}{Y} = \frac{R}{Y} \frac{Y_R}{R} \frac{\Delta R}{R} + \frac{X}{Y} \frac{Y_X}{X} \frac{\Delta X}{X}$$

where  $Y_R$  and  $Y_X$  are the marginal products of  $R$  and  $X$  respectively.

$$\text{Thus } \phi = \frac{\Delta Y}{\Delta R} = Y_R(R, X).$$

+ -

A shorthand expression is that  $\phi = \phi(Y, R)$  - ie.  $\phi$  is positively related to output (as sales or market size enhances the return to R&D investment) and negatively to the existing stock of R&D capital, as a larger  $R$  leads to diminishing marginal returns. The latter holds if second order conditions hold (ie.  $Y_{RR} < 0$ ). Note that the marginal product schedule  $\phi = \phi(Y, R)$  defines a firm's demand for  $R$ .

Next we assume that private investment in R&D is a positive function of  $\phi$  and in general equilibrium is also a function of investment in R&D by other sectors - namely the public sector. The impact of public R&D spending on private R&D investment is ambiguous for two reasons. On the one hand, with binding (or near binding) resource constraints, public R&D and private R&D compete for scarce resources. Increases in public R&D spending may raise the prices of factor inputs and thereby raise the marginal cost of research. On the other hand, research by the public sector creates "spillovers" by adding to the stock of knowledge which may



enhance the productivity of private R&D factor inputs and thereby lower the marginal cost of research. Let  $R_p$  denote the index of private R&D capital and  $R_g$  the index of public R&D capital. The above remarks suggest that:

$$\Delta R_p = \Delta R_p (\Delta R_g, \phi)$$

where  $\Delta$  is the differential operator. Collecting our equations we get:

$$(11) \text{ Supply: } \Delta R_p = \Delta R_p (\Delta R_g, \phi)$$

$$(12) \text{ Demand: } \phi = \phi (Y, R)$$

where the theoretically relevant  $R$  in equation (12) is some combination of  $R_p$  and  $R_g$ , depending on their relative productivities in producing output  $Y$ .

Equating demand and supply, by eliminating  $\phi$ , gives:

$$(13) \quad \Delta R_p = \Delta R_p (\Delta R_g, Y, R) = \Delta R_p (\Delta R_g, Y, R_p, R_g)$$

In the empirical section, I take one-period lagged values of  $Y$  and  $R_p$ <sup>1</sup>.

The timing between  $\Delta R_g$  and its subsequent effect on  $\Delta R_p$  has been raised in the literature. Levy-Terleckyj find very small or no lags. I find very little change in the results when going from using contemporaneous to

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<sup>1</sup> Moreover I have tried both  $R_p$ ,  $R_g$  but have found that using  $R_p$  alone produced the better results largely because, as found in the empirical section, the significance of  $R_g$  falls when  $R_p$ ,  $R_g$  are entered together in the production function of  $Y$ .

using one-period lagged values of  $\Delta R_g$ . There are some arguments as to why  $\Delta R_g$  and  $\Delta R_p$  may be studied contemporaneously: (i) if firms plan R&D expenditures given their expectations about future government R&D spending, and if they assume a particular process for future government R&D spending (for example a random walk or simple first-order autoregressive process), the current value of the public R&D variable should enter equation (13) above; (ii) if public R&D crowds out private R&D, the financial and resource pressures arising from the financing of public spending by increased borrowing (or taxation) and from the increased competition for scarce R&D inputs in the factor markets should be felt on private R&D during the same period in which public R&D is incurred, particularly if the period under consideration is a year; (iii) if firms conduct R&D under government contract, the knowledge gained can be usefully applied to contemporaneous private R&D activities.

An issue is raised by Lichtenberg (1987) as to whether the output or sales variable  $Y$  should be separated into purchases by government and purchases by the non-government sector. The reasoning is that the government is as much a customer of R&D as it is a sponsor of research. It is likely that the former role (demand channel) is more important than the latter role (technological channel). When company R&D is regressed against federal R&D, Government Consumption, and Sales to the Non-Government Sector, the federal R&D variable is no longer significant while Government Consumption is quite significant, thus favouring the demand interpretation of the stimulus to company R&D.

While I believe this to be an important insight I do not net out government purchases from  $Y$ . In my empirical results, the output variable does attain the correct sign (though not always), but it is very

often not significant. Thus the distinction between government and non-government purchases of  $Y$  does not play a significant role in my estimations. I also work with R&D stocks and changes in the stocks, not with R&D flows. The difference is that the stock measures take into account depreciation and obsolescence rates, so that changes in the stock represent net investment, not gross as is the case with flow data. Finally, government consumption tends to be smooth over time, exhibiting significant fluctuations only in war periods, during which times the fluctuations arise primarily because of changes in defense spending - see Chapter 3. Government consumption is thus a much less important source of output fluctuations than is non-government demand. Coupled with the fact that R&D outlays are a relatively small fraction of government consumption, this suggests that government demand variations are not likely to be a leading source of stimulus to the growth of company R&D stock.

### Private Capital Investment Equation

The development of the private capital investment equation is quite analogous. Let  $\phi' = AF_K$  be the marginal product of private capital  $K_p$ . In general equilibrium, private investment  $\Delta K_p$  bears the following relationship with other types of investment expenditures:

$$(14) \Delta K_p = \Delta K_p(\Delta K_g, \Delta R_g, \Delta R_p, \phi')$$

Again other types of investment expenditures have ambiguous effects on private investment because on the one hand they may compete with private capital investment for limited resources, and on the other hand they may enhance the productivity of private factors of production and stimulate private capital accumulation.

Substituting

$$(15) \quad \phi' = AF_K = \phi'(K_p, K_g, Y)$$

into (14) yields a private capital investment equation that can be estimated:

$$(16) \quad \Delta K_p = \Delta K_p(\Delta K_g, \Delta R_g, \Delta R_p, K_p, K_g, Y)$$

In the empirical section,  $K_p$ ,  $K_g$ , and  $Y$  are all lagged one-period.

In the empirical section I assume equation (13) to be linear:

$$(13)' \quad \Delta \log R_p(t) = v_0 + v_1 \Delta \log R_g(t) + v_2 \log Y(t-1) \\ + v_3 \log R_p(t-1) + v_4 \log R_g(t-1)$$

I also estimate the private capital investment equation (16) in linear form:

$$(16)' \quad \Delta \log K_p(t) = \eta_0 + \eta_1 \Delta \log K_g(t) + \eta_2 \Delta \log R_g(t) + \eta_3 \Delta \log R_p(t) \\ + \eta_4 \log Y(t-1) + \eta_5 \log K_p(t-1) + \eta_6 \log K_g(t-1)$$

In the empirical analysis I also add the capacity utilization rate to equations (13)' and (16)', again to control for cyclical influences.

Appendix 4.2 describes the stock-flow dynamics underlying these investment equations.

#### **IV. Data Description**

The OECD panel data set begins in 1970 and ends in 1987. Data for the U.S. begin in 1948 and end in 1988. Data used in this chapter come from several sources. For the U.S. I relied largely on BEA sources for capital stock data, NIPA for national accounts data, and the BLS for data on the labour force and on multifactor productivity. Statistics Canada is a convenient all-purpose source for most of my Canadian data. R&D data are obtained from the OECD's Science and Technology Indicators databank. The National Science Foundation provides more detail on U.S. R&D, and Eurostat provides additional detail on R&D in the European Economic Community - particularly on the composition of public R&D. The OECD is also a source of data on taxation, national accounts, total factor productivity, and labour force statistics. Work hours are from the International Labour Office (ILO).

The mnemonics used for the variables throughout the empirical section are given in Table 4.1. Private output  $Y$  refers to the output of private industries, including agriculture. The private capital variable excludes residential capital and the public capital variable excludes military capital. Labour supply is measured in terms of total hours worked per year. To make the data sets of different countries comparable, I convert variables into real 1985 U.S. dollars using the purchasing power parity rates, which eliminate differences in price levels between countries (that is, the PPP rates are those rates which enable a consumer to purchase the same

basket of goods and services in all countries with a given sum of money). I denote changes in these variables by  $\Delta$ ; since the variables are in logs, the 'changes' can be interpreted as growth rates.

The global stocks of private and public R&D are constructed from ten countries: Belgium, Canada, France, West Germany, Japan, Italy, Netherlands, Sweden, the United Kingdom, and the United States. These countries provide the bulk of world R&D. I also disaggregate Global R&D for each of these countries into Own R&D capital and Other R&D capital. For each country, Other R&D consists of the own R&D capital of the other nine countries. All R&D stocks are created from flow data using the perpetual inventory method described in Appendix 4.3 and assuming a 3 year lag and 0.1 depreciation rate.

The capacity utilization rate (defined as the ratio of the index of industrial production to plant and equipment capacity) is employed to control for cyclical fluctuations. The idea is to keep the "economy" on the production frontier so that we can interpret the output elasticities as the effects of changes in the inputs on trend or "long run" output. Alternatively, one can use other measures of cycles (such as the workweek of capital, coincident indicators, or the ratio of actual to potential GDP) which may, in any event, be highly correlated with our capacity utilization index. One can also time-average the data so as to eliminate short term cycles. However, this procedure reduces degrees of freedom and introduces moving average errors.

The capital income tax rate  $\tau_k$  is obtained by taking the ratio of corporate profits taxes to private GDP. Data on an alternative tax base, namely corporate profits less depreciation (ie. taxable income), are not fully available for all countries.

The measure of total factor productivity is based on the Divisia Index approach; TFP growth in other words is the difference between the output growth rate and an index of input growth rates. It is a measure or proxy for technical change. To obtain the index of input growth rates, one can weight the inputs either by the factor shares in income or by the factor shares in cost. I used the TFP measures of the OECD which give roughly a 0.3 weight to private capital and 0.7 to labour; the exact weights vary across countries. Weighting by factor cost shares is the preferred approach when dealing with imperfect competition and mark-up pricing behavior; in an extension of this chapter I hope to compare alternative measures of TFP (following Bartelsman (1990)). For now, as my concern is the relationship between public investment and some measure of multifactor productivity, I have chosen a relatively easy to construct measure of TFP based on constant returns and perfect competition. Use of this measure also facilitates comparison with other studies on the determinants of productivity that measure TFP in a similar fashion.

## **V. Empirical Results**

The organization of this section is as follows. It consist of two subsections: the first subsection presents production function estimates for the U.S., including the results of tests of stationarity, constant returns, and causality. I also control for distortionary taxes, such as capital income taxes, and consider an alternative specification of the production function which postulates that there are external effects to private physical capital.

The second subsection reports on the panel data results using two sets of pooled data: in "Pool 19" all the OECD countries are pooled except for

Iceland, Portugal, Spain, Switzerland, and Turkey. In "Pool 10" the countries that are pooled are the ten leading R&D countries mentioned in Section IV. This subsection conducts tests of the constant returns hypothesis, the common factor restrictions, the homogeneity assumption in the panel data, and the direction of causality. Like the first subsection, it too considers the effects of capital income taxes and examines an alternative source of external effects. Finally in this subsection I also disaggregate and investigate public R&D by socio-economic objective, and investigate the effects of public R&D and capital investment on private R&D and capital investment.

#### (i) U.S.A. Results

The U.S. results are presented in Tables 4.2A, 4.2B, 4.3A, and 4.3B. Before discussing them, I report on some preliminary tests. The purpose of this subsection is to establish the "productivity effects" of fiscal variables. The next subsection explores the "international technological spillover" effects of fiscal variables.

#### (a) Tests of Non-Stationarity

Regressions such as  $Y(t) = \alpha + \beta X(t) + \gamma t + u(t)$ , where  $\{Y\}$ ,  $\{X\}$  are non-stationary, will be spurious if the error  $u(t)$  is non-stationary. It is important therefore to take proper precaution against deriving spurious correlations from non-stationary data. One route is first-differencing of the data so that the resulting error term is stationary. First-differencing, however, has disadvantages in that we lose information about low



frequency relationships in the sample. On the other hand, improper detrending (of variables that are truly difference-stationary) can lead to psuedo-periodic behaviour in the residuals.

I therefore follow Nelson-Plosser (1982) in testing whether the data are better represented as trend-stationary (TS) or difference-stationary (DS). Given:

$$\Delta z(t) = \mu + \rho z(t-1) + \gamma t + e(t),$$

the DS specification is preferred if the null hypothesis  $H_0: \rho = 1$  and  $\gamma = 0$  is not rejected. Otherwise if  $\rho < 1$ , the TS specification is preferred. The relevant (critical) t-distribution for testing the null hypothesis  $H_0$  is presented in Fuller (1976). The results of the Nelson-Plosser test for  $z = (Y, K_p, K_g, L, RD_p, RD_g)$  are reported in the table below:

| <u>U.S.A.</u>                                    | <u>t-statistic</u> | <u>No.of Observations.</u> |
|--------------------------------------------------|--------------------|----------------------------|
| Private Output Y                                 | -2.43              | 39                         |
| Private Capital $K_p$                            | -3.47              | 48                         |
| Public Capital $K_g$                             | -2.76              | 48                         |
| Private Capital $K_p$ net of Residential Capital | 0.17               | 39                         |
| Public Capital $K_g$ net of Military Capital     | 2.48               | 39                         |
| Hours Worked L                                   | -3.28              | 39                         |
| Private R&D Capital $RD_p$                       | 0.22               | 39                         |
| Public R&D Capital $RD_g$                        | -0.56              | 31                         |

For samples of 25 observations, the 1% critical value is -4.38 and the 5% critical value is -3.6. For samples of 50 observations, the 1%

critical value is -4.15 and the 5% critical value is -3.5<sup>1</sup>. The t-statistics reported in the above table do not exceed these critical values, and thus we do not reject the hypothesis that these variables are difference-stationary. Thus in most of my estimations I express variables in first-differences. In the empirical section I do report some level estimations for interest.

#### (b) Test of Constant Returns

I test whether the labour hours and private capital elasticities sum to unity using the usual F-test. The stock of government capital is always present when doing this test. It is also possible to test whether there are constant returns to all of these three inputs. The results are that we cannot reject the null hypothesis of constant returns among private capital and labour hours, but that we can reject the null that all three inputs (public capital, private capital, and labour hours) exhibit constant returns. For the U.S. I get an F-statistic of 1.76 for the test on private inputs and 5.07 for the test on all three inputs; the critical 5%  $F(1,35)$  is 4.12, where the numerator has 1 degree of freedom because there is only one constraint, and the denominator has 35 degrees of freedom because the sample size is 40 and the number of RHS variables is 5, including the capacity utilization rate.

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<sup>1</sup> From Table 8.5.2. in Fuller (1976), p. 373.

(c) Productivity Impacts of Public Investment

Table 4.2A contains both level and first-difference estimations. Equation (1) is the unconstrained production function in levels, while equation (2) corrects for serial correlation in (1). Equation (3) imposes constant returns to (1) and equation (4) corrects for serial correlation in (3). Note that public capital is quite significant throughout, as Aschauer (1989a) finds using level estimation.

Equation (5) adds to equation (4) private capital as an additional regressor. I have called this a test of Romer's (1987) hypothesis of external effects to private capital. Romer argues that each firm  $i$  produces according to

$$Y_i = \kappa^\eta K_i^\beta L_i^{1-\beta}$$

$$\text{where } \kappa = \sum_{i=1}^N \frac{K_i}{N}$$

is the aggregate average stock of private capital and  $N$  is the number of firms. The justification for it is that increased knowledge and experience are embodied in each new machine; this external effect is essentially ignored by private producers. (Note that we divide by  $N$  to avoid scale effects and to incorporate the possibility of congestion; also, not dividing by  $N$  will lead to the implication that larger economies or industries grow faster which is controversial). If in equilibrium all  $N$  firms are identical, so that  $K_i = K$ ,  $L_i = L$ , the aggregate production function is:

$$\frac{Y}{L} = K^\eta \left( \frac{K}{L} \right)^\beta$$

and if  $\eta + \beta > 1$ , the aggregate economy will exhibit increasing returns to capital. However, estimates for equation (7) argue against this specification; in fact  $\eta < 0$ .

Equation (6) controls for taxes on capital income. The influence is rather weak. Since  $\tau_k$  ranges from 0 to 1, or 0 to 100%, a 1 percentage point increase in the capital income tax rate (from say 30% to 31%) will reduce next period's productivity growth by 0.0023%. Thus it may be conjectured that a 1% expansion in the stock of public capital, financed entirely by capital income taxes, will on net improve productivity because the capital income tax rate has to rise by 100 percentage points to cause productivity to fall by 23% and thereby offset the 22% increase in productivity induced by a 1% increase in the stock of public capital.

Equation (7) is the first-difference estimation of (3). Public capital is now insignificant. Private investment is thus a stronger source of productivity growth for the U.S. than is public investment. Later it will be argued that public investment can affect productivity growth through its stimulative effect on private capital investment. Equation (8) is the first-difference estimation of (6). Again the capital income tax rate is weakly significant.

Table 4.2B introduces R&D investments into the growth accounting framework (ie. first difference estimations). The interesting outcome is that the public R&D variable is significant (at the 90% confidence level) whenever it enters without the private R&D variable but insignificant whenever the private R&D variable is included. This pattern also emerges in the panel data regressions. It appears that private R&D investment is more stimulative for private sector productivity than is public R&D. But this need not rule out public R&D having an important role to play in

promoting private sector productivity growth, for public R&D has the potential to stimulate private R&D investment. I turn to this issue later.

Another observation worth noting from Table 4.2B is that capital income taxes are now significant (at least at the 95% confidence level) and their coefficients larger in magnitude. Private capital investment now exerts a weak influence amidst the presence of capital income taxes. Public capital investment (omitted) was also insignificant and negative in sign. This turnaround in results is attributable to the presence of R&D factors of production. In the absence of R&D factors, there was an omitted variable problem: the physical stocks of capital were proxying for them. When the R&D factors enter explicitly, the distortionary effect of capital income taxes on discouraging capital accumulation and productivity growth is more visible.

#### (d) Impact of Public Investment on Private Investment

Table 4.3A presents the results of estimating the private R&D and capital investment equations. In the private capital investment equation, public capital investment exerts a weak influence on private capital investment, but the lagged stock of public capital exerts a positive influence. The last effect appears because a higher stock of public capital raises the marginal product of private capital, and hence encourages private capital accumulation. If there were strong resource crowding out pressures, the coefficient on  $\Delta K_g$  would have been negative; instead it takes on a very small positive value. The lagged stock of private capital is negative indicating that a higher stock of private capital depresses its own marginal product.

In the private R&D investment equation, private R&D investment is stimulated not only by public R&D investment but also by the lagged stock of public R&D capital. Public R&D both raises the marginal product of private R&D capital and encourages private research activity by generating knowledge spillovers which reduce the marginal cost of private research. Thus one reason why public R&D has a negative and weak influence on productivity growth when examined together with private R&D is that the influence of public R&D on productivity growth operates indirectly through its stimulative impact on private R&D activity. Indeed when public R&D is examined without private R&D, the public R&D variable picks up the direct effects of private R&D on productivity growth.

#### (e) Possibility of Reverse Causation

As Chapter 3 pointed out, public spending may react to current or permanent income. One interpretation of the Wagner's Law hypothesis is that public expenditure is a superior good. Thus one criticism of the results in Tables 4.2A, 4.2B may be that we have been estimating not the supply response of output to public investment, but the demand for public investment. That is, a higher national income raises the nation's demand for public services.

In Table 4.3B I examine a reverse specification in which productivity growth is a determinant of public investment (or of changes in the ratio of public capital to output). The results do not indicate that there is a significant positive income elasticity of public capital investment demand. The coefficient of  $\Delta YL$  is weak and in some cases has a negative sign.

## **(ii) OECD Panel Data Results<sup>1</sup>**

The panel data results are presented in Tables 4.4A-4.4F, Tables 4.5A-4.5C, Tables 4.6A-4.6B, Table 4.7, Table 4.8, and Tables 4.9A-4.9D. Again I begin by discussing some preliminary tests before reporting on the main results.

### **(a) Tests of Constant Returns**

The constant returns hypothesis is again not rejected. For the sample of countries represented in Pool 19, the F-statistic for testing the null hypothesis of constant returns to scale in private capital and labour hours is 2.18, below the critical F value of  $F_{0.05}(1, 279) = 3.88$ . There are 279 degrees of freedom in the denominator because the sample size is 285 and there are 6 RHS variables in the unconstrained model, including the capacity utilization rate and Trend. For Pool 10, the F-statistic for testing the same null hypothesis is 1.25 which is also below the critical  $F_{0.05}(1, 144) = 3.912$ . There are 144 degrees of freedom in the denominator because the sample size is 150 and there are 6 RHS variables.

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<sup>1</sup> The U.S. is also included in the panel data.

### (b) Tests of Dynamic Specification

A key feature of the tables I present in this subsection is that the estimated equations are in first-differences. From an aggregate growth accounting framework, this type of specification is just a definition. I wish however to examine some of the statistical aspects of this specification.

Table 4.4A provides a test of the common factor restrictions embedded in our first-difference formulation. There are several ways to model the underlying dynamics. For simplicity consider a pair of variables  $z, x$ . Three possible equations are:

$$(I) \quad z(t) = \alpha_0 + \alpha_1 x(t) + \alpha_2 x(t-1) + \alpha_3 z(t-1) + \varepsilon(t)$$

$$(II) \quad z(t) = \alpha_0' + \alpha_1' x(t) + \mu(t), \quad \mu(t) = \rho \mu(t-1) + \varepsilon(t)$$

$$(III) \quad \Delta z(t) = \alpha_0'' + \alpha_1'' \Delta x(t) + \varepsilon(t)'$$

where  $\varepsilon(t)$  is white noise. (I) is the unconstrained version in which the lagged levels are modelled. (II) is a static version of (I) in which the dynamics are assumed to be in the autoregressive error term. (III) is our version in which a common factor of unity is imposed on (I). The asymptotic likelihood ratio test statistic for testing the restricted models is:

$$2\{\log L_{\text{unrestricted}} - \log L_{\text{restricted}}\} \sim \chi^2(k-1) \text{ -- Chi-Square Distribution}$$

where  $\log L$  is the value of the log likelihood, and  $k$  is the number of restrictions. In Table 4.4A,  $k=6$  (ie. restrictions are imposed on the  $k$  lagged variables). The chi-square value at the 95% point for 5 degrees of



freedom is 11.07. The Chi-square statistic for testing (III) versus (I) in Pool 19 is 7.4. Thus specification (III) is not rejected. It is also not rejected for Pool 10. On the other hand (II) is resoundingly rejected for both the Pool 10 and Pool 19 samples. The implication is that it is not valid to reduce a dynamic growth accounting model to a static model in levels with an autoregressive disturbance term. On the other hand (III) captures the dynamics as well as (I). Indeed in Table 4.4A, the coefficient of each variable tends to be of the same magnitude as the coefficient of the corresponding lagged variable, but with the opposite sign. Note also the coefficient value on lagged output, ie. 0.982, which is close to unity<sup>1</sup>.

An alternative dynamic formulation is the "error correction mechanism" (ECM). Instead of using the capacity utilization rate to control for business cycles, I allow the ECM specification to capture the short and long run dynamics.

Again for a set of variables ( $z$ ,  $x_1$ ,  $x_2$ ), the ECM specification is:

$$(I)' \Delta z(t) = \alpha_0 + \alpha_1 \Delta x_1(t) + \alpha_2 \Delta x_2(t) \\ - \alpha_3 z(t-1) + \alpha_4 x_1(t-1) + \alpha_5 x_2(t-1) + \varepsilon(t)'$$

or

$$(II)' \Delta z(t) = \alpha_0 + \alpha_1 \Delta x_1(t) + \alpha_2 \Delta x_2(t) \\ - \alpha_3 ( z(t-1) - \frac{\alpha_4}{\alpha_3} x_1(t-1) - \frac{\alpha_5}{\alpha_3} x_2(t-1) ) + \varepsilon(t)'$$

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<sup>1</sup> The value of the quadratic trend term is positive whereas in later Tables the linear trend term is negative. An explanation for this could be that as we first-difference, the constant term in model (I) vanishes, and the lost information induces changes in the coefficient values of the new constant term and linear trend term in the first-difference formulations.

where  $\alpha_3 > 0$  leads to "error-correcting" adjustments whenever  $z$ ,  $x_1$ ,  $x_2$  stray apart from one another (assuming also that  $\alpha_1$ ,  $\alpha_2 > 0$ ). In stationary-state,

$$z^* - \frac{\alpha_4}{\alpha_3} x_1^* - \frac{\alpha_5}{\alpha_3} x_2^* = \left( \frac{\alpha_0}{\alpha_3} \right)$$

where the long run  $z$ -elasticities of  $x_1^*$ ,  $x_2^*$  are given by  $\frac{\alpha_4}{\alpha_3}$ ,  $\frac{\alpha_5}{\alpha_3}$  respectively.

The ECM does not describe what the "economic" forces (eg. prices or government policies) are which lead to adjustments in the quantities. Moreover it is not always clear as to what is causing what. The standard interpretation is that "forecast errors" cause adjustments which lead the economy back to equilibrium. Another interpretation is that the short run forecast errors are the result of agents making adjustments toward long run equilibrium. In any case the ECM estimation result is (for Pool 19)<sup>1</sup>:

$$\begin{aligned} \Delta Y = & 0.0038 + 0.19\Delta K_g + 0.44\Delta K_p + 0.48\Delta L \\ & (0.01) \quad (0.07) \quad (0.08) \quad (0.07) \\ & + 0.0043K_g(-1) + 0.008K_g(-1) + 0.00011L(-1) - 0.0192Y(-1) \\ & (0.0036) \quad (0.01) \quad (0.0013) \quad (0.012) \\ \text{D.W.} = & 1.4, \text{ Adjusted R-Sq} = 0.723 \end{aligned}$$

The lagged terms have the correct sign but are largely insignificant. Moreover in stationary-state, the above results in:

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<sup>1</sup> In this regression I assume that the disturbance terms do not depend on individual countries or on time. Otherwise there is little work (to date) integrating the theory of ECM to panel data that would be of any guide.

$$Y^* = 0.198 + 0.224K_g^* + 0.42K_p^* + 0.006L^*$$

which yields long run elasticities different from the ones obtained using the growth accounting framework (with the capacity utilization rate absorbing short run cycles). Note especially the output elasticity of L, hours worked<sup>1</sup>.

I thus draw two conclusions. First, I find that the first difference formulation used so far does not benefit from the introduction of lagged levels to have deviations "steered" into place since the lags are weakly significant. Secondly, I find that the ECM formulation is not easily reconciled with the framework used thus far.

### (c) A Test of External Effects to Private Physical Capital

Finally I have examined the "Romer" specification for the Pool 19 sample:

$$\Delta YL = 0.026 + 0.142\Delta K_g - 0.49\Delta K_p + 0.7\Delta KL + 0.051\Delta CU - 0.00085T$$

(0.004)   (0.054)   (0.1)   (0.07)   (0.008)   (0.0004)

$$D.W. = 1.82, \text{ Adjusted } R\text{-}Sq = 0.33$$

Again, there is little support for the view that there are external effects to private capital. In fact the total output elasticity of private capital is  $-0.49 + 0.7 = 0.21$  which is somewhat smaller than the coefficient values of  $\Delta KL$  that we have been obtaining. Our specification in other words attributes a larger output elasticity to private capital than does the Romer specification.

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<sup>1</sup> Furthermore, when I impose constant returns to the ECM formulation, the output elasticity of private capital is rather large, namely, 0.6.

(d) Effect of Public R&D on Productivity Growth

In the following Tables I present OLS, GLS, and FE (Fixed Effects) estimates. I shall briefly review the assumptions underlying the validity of these alternative estimators. The first point to note is that I do not model specific individual-effects variables though it is possible to include some, such as proxies for institutional or legal differences pertaining to production. Thus since individual-effects variables are absent, I assume that individual effects are entirely unobservable and reflected in the error term:  $\varepsilon_{it} = \mu_{it} + v_i$  where  $v_i$  is that part of the error that varies across countries but not over time.

If the  $v_i$ 's are zero, and the disturbances  $\varepsilon_{it}$  are orthogonal and spherical, OLS is unbiased and efficient. However if the  $v_i$ 's are non-zero, the assumption of sphericity breaks down, and OLS is no longer efficient though it is consistent. If the  $v_i$ 's are fixed constants, FE is preferred; if the  $v_i$ 's are random, then GLS is preferred. Now if the  $v_i$ 's are correlated with the regressors then both the orthogonality and sphericity assumptions break down and neither GLS nor OLS is consistent or efficient. The FE estimation is then preferred for it will yield consistent, though not necessarily efficient, estimates.

A remedy for the correlation between the  $v_i$ 's and the regressors is the Hausman-Taylor IV procedure. The procedure is especially useful when the model contains individual specific variables. The procedure involves selecting a subset of uncorrelated regressors to act as instruments. This is superior to the FE only to the extent that the exogeneity restrictions imposed are valid (that is, we have correctly chosen what are the good instruments and that they outnumber the bad or correlated individual

specific regressors). In our case, we do not have any individual specific variables. The Hausman-Taylor IV will produce better results if at least some of the regressors are uncorrelated with the  $v_i$ 's. Otherwise FE is preferred since performing the IV procedure under the assumption that there are some uncorrelated regressors will induce greater inaccuracies<sup>1</sup>. An advantage of using FE when no individual specific variables are present is that we have no individual specific coefficients to recover, a problem which exists with FE whenever individual specific variables are present.

I return now to the results in the Tables. Tables 4.4B-4.4C present the basic aggregate growth accounting results for Pool 19 and Pool 10 respectively, without any R&D variables. Under Fixed Effects estimation, the coefficient on public capital is quite large in magnitude while the coefficient on KL hardly changes under different estimation procedures. I would argue that this is the result of omitting R&D variables, the consequence of which is that the 'public' physical capital variable ends up capturing most of the unmodelled 'external' effects to productivity growth.

Tables 4.4D-4.4F and Tables 4.5A-4.5C introduce R&D variables. Tables 4.4D-4.4F report the effects on labour productivity while Tables 4.5A-4.5C report the effects on total factor productivity. In each of these Tables I consider two cases: in Case 1 private R&D is omitted; in Case 2 private R&D is included. In Tables 4.4D and 4.5A the pooled sample consists of 19 countries while in Tables 4.4E,F and 4.5B,C the pooled sample consists of 10 countries. Note that the global R&D capital stocks

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<sup>1</sup> An informal way to detect the presence of correlation between the  $v_i$ 's and the regressors is if the OLS and GLS estimates are similar while the FE estimates differ from the OLS/GLS estimates markedly. This will be evident in the tables.

consist only of the R&D stocks of the 10 major contributor countries. Pool 10 may be regarded as a group of "technological" neighbours, while Pool 19 may be regarded as a broader sample of the industrial world at large. Only in the Pool 10 sample can we disaggregate the global stocks of R&D capital by Own National and Other National stocks of R&D.

In Tables 4.4D-4.4F, I find that when public R&D is considered without private R&D, it is typically significant (at about the 80% confidence level in Table 4.4D and at least at the 95% confidence level in Table 4.4E). Once private R&D is controlled for, the significance of public R&D diminishes and the coefficient estimates are often negative. This suggests that private R&D is the more powerful source of productivity growth (though in anticipation of Tables 4.9A-C I would argue that public R&D can still affect production indirectly by stimulating private R&D, just as I have argued in the subsection on the U.S.A. results<sup>1</sup>). In Table 4.4F the stocks of R&D are broken down into Own R&D and Other R&D stocks. The significance of  $OT_g$  under Case 1 (at least at the 80% level of significance) suggests the presence of cross-national spillovers. In Case 2, when private R&D variables are modelled,  $OT_g$  exerts a weak negative influence while  $OW_g$  retains some significance. Note that spillovers can also occur through international private R&D activity, as  $OT_p$  is significant.

Public capital is also significant throughout (at least at the 90% confidence level). Again this is consistent with the findings of empirical work employing time-series data but not with those employing cross-

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<sup>1</sup> The coefficient of public R&D in Case 2 (where both public and private R&D are modelled) can be interpreted as the residual effect of public R&D on growth after its effects on private R&D have been taken into account.

sectional data. The trend term is also significant and negative indicating that trend productivity growth has been declining over the sample period. Tables 4.5A-C are a repeat of Tables 4.4D-F using TFP as the dependent variable. Note that the coefficient estimates in Tables 4.5A-C are quite similar to the corresponding coefficient estimates in Tables 4.4D-F.

I also examine the possibility of reverse causation, running from income variables to government investment:

$$\Delta \frac{K_g}{Y} = -0.016 - 0.00083 T + 0.054 \Delta YL - 0.117 \Delta CU$$

(0.006) (0.0005) (0.042) (0.023)

$$D.W. = 1.3, \text{ Adjusted R-Sq} = 0.2$$

$$\Delta \frac{RD_g}{Y} = 0.009 + 0.0009 T - 0.91 \Delta YL - 0.065 \Delta CU$$

(0.004) (0.0005) (0.042) (0.016)

$$D.W. = 1.4, \text{ Adjusted R-Sq} = 0.8$$

Like the earlier subsection there is not much support for the view that higher productivity raises the demand for public R&D and capital spending. The coefficient of  $\Delta YL$  is either negative or weakly positive.

Throughout this subsection I have assumed that the panel data residuals consist of an individual effect ( $v_i$ ) and not a time effect ( $\delta_t$ ). Indeed an 'Analysis of Variance' shows that this assumption is reasonable. For Pool 19 (with all the variables modelled, including the R&D variables, ie. Table 4.4D, Case 2), we obtain the following residual decomposition:

| Disturbance:               | Sum of Squares | Degrees of Freedom | Mean Square | F-Statistic |
|----------------------------|----------------|--------------------|-------------|-------------|
| Individual ( $v_i$ )       | 0.0237         | 18                 | 0.0013      | 2.46        |
| Time ( $\delta_t$ )        | 0.0097         | 14                 | 0.0007      | 1.3         |
| Joint ( $v_i + \delta_t$ ) | 0.0334         | 32                 | 0.00105     | 1.96        |
| Error ( $\mu_{it}$ )       | 0.1347         | 252                | 0.00053     |             |
| Total                      | 0.1681         | 284                |             |             |

In this table the total sum of squares (TSS) is decomposed into the individual, time, joint, and error components:  $\varepsilon_{it} = 0.1681 = v_i + \delta_t + \mu_{it}$ . Here  $N=19$  (ie. 1973-87) and  $T = 285$ . The F-statistic<sup>1</sup> on  $\delta_t$  indicates that the time effect is relatively weak, and thus I continue not to transform the data to eliminate any time effects in the residuals. Likewise for Pool 10 (with all the variables modelled, ie. Table F, Case 2), the F-statistics for  $v_i$ ,  $\delta_t$ , and  $\mu_{it}$  are 3.14, 1.35, and 2.04 respectively.

#### (e) The Effects of Capital Income Taxation on Productivity Growth

In Table 4.6A the capital income tax rate is entered into the regressions without R&D variables, while in Table 4.6B the R&D variables are present. In the panel data regressions the sign on  $\tau_k(-1)$  is correct and significant at the 80% confidence level. The variable is not as weak as it was in Table 4.2A. In the panel data set regressions, when R&D variables are present, private R&D and public and private physical capital stocks are still significant (at least at the 80% confidence level), in contrast to what was discovered in Table 4.2B. Recall that the physical capital stocks

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<sup>1</sup> The mean square is obtained by dividing the sum of squares by the corresponding degrees of freedom. The F-statistic is obtained by dividing the mean square by the mean square of Error.



diminished in significance in the U.S.A. results when the capital income tax rate and the R&D variables were introduced. This was argued to reflect the negative impact that distortionary taxes had on capital accumulation which could not be captured so long as the R&D variables were absent because the physical stocks of capital were proxying for them. A key difference between the U.S.A. results and the panel data results is that in the panel data regressions, the R&D variables represent the global stocks of knowledge capital, not just the national stocks. Thus in our panel data regressions the global R&D variables are significant enough to raise the marginal productivities of the physical capital inputs.

#### (f) Tests of Heterogeneity in the Panel Data

The panel data results provide some "average" estimate of the various output elasticities. Individual economies will in general differ in structure and behaviour from the average of the pool. The following heterogeneity tests should give us some sense of how restrictive it is to assume that all countries in the pooled sample have identical production functions.

The test procedure is fully outlined in Hsiao (1986). To fix ideas, let the production function be summarized by  $Y = A F(X)$  where  $X$  is a list of factor inputs. Parameterizing  $F(X) = X^\beta$ , and taking logs gives:

$$y = \alpha + \beta x$$

where  $y = \log Y$ ,  $\alpha = \log A$ , and  $x = \log X$ .

First, one runs a regression on the entire pooled sample assuming identical slopes ( $\beta$ ) and intercept ( $\alpha$ ). This is the fully restricted model and let the sum of squared residuals from this be  $SSR_3$ . Secondly one runs

a regression assuming identical slopes only and allowing intercepts to differ by introducing country dummies. This is the partially restricted model and call its sum of squared residuals SSR2. Finally one runs separate regressions for each country. This is the fully unrestricted model. Denote the sum of all the individual sum of squared residuals by SSR1. To test the fully restricted model (identical slopes and intercept), we construct the following test statistic:

$$F3 = \frac{(SSR3 - SSR1)}{(N - 1)(K + 1)} \frac{(NT - N(K+1))}{SSR1}$$

and compare it to the F-distribution with  $((N - 1)(K + 1), NT - N(K+1))$  degrees of freedom. To test the partially restricted model we construct the test statistic:

$$F2 = \frac{(SSR2 - SSR1)}{(N - 1)K} \frac{(NT - N(K+1))}{SSR1}$$

and compare it to the F-distribution with  $((N - 1)K, NT - N(K+1))$  degrees of freedom. If both F3 and F2 are large, the null hypothesis of homogeneous production functions can be rejected.

Table 4.7 summarizes the findings of these tests for a variety of models:

Model A, the input list X consists of { KL }

Model B, the input list X consists of {  $K_g$ , KL }

Model C, the input list X consists of {  $RD_g$ ,  $RD_p$ ,  $K_g$ , and KL }

Model C', the input list X consists of {  $OW_g$ ,  $OT_g$ ,  $OW_p$ ,  $OT_p$ ,  $K_g$ , KL }

The input list is successively expanded in order to see whether the heterogeneity in the intercept becomes less prominent as more factors are taken into account. Table 4.7 makes it clear that heterogeneity in the slopes and intercept begins with Model A. The implication is that we cannot infer individual behaviour from the pooled aggregate. Any predictions based on the pooled results must be in reference to some aggregate average measure<sup>1,2</sup>. Mairesse-Griliches (1988) also find heterogeneity in panel data production functions but argue that the heterogeneity found is not "real" but reflects non-stable production functions. Their time period is 1967-

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<sup>1</sup> Though not pursued here, it would be worth investigating whether the rejections of homogeneity are attributable to measurement errors. The degree to which measurement errors affect our findings depends on how the residual sums of squares behave. It is likely that SSR1, SSR2, and SSR3 are all proportionally affected by measurement errors. If so, since the tests are based on the relative discrepancies between SSR's it is not likely that the outcomes of the tests would change if all of the regressors and dependent variable were measured with more precision.

<sup>2</sup> The rejections of homogeneous production functions should have implications for the literature on economic convergence. Recall  $y = \alpha + \beta x$ . Crucial to predictions of convergence is that "knowledge" spillovers occur perfectly over the globe so that all countries have the same  $\alpha$  or level of technical efficiency. Secondly all countries must have the same  $\beta$  and that this  $\beta$  be less than 1 so that factor  $x$  exhibits diminishing marginal productivity. Thus in steady-state all countries will converge to the same steady-state levels of  $y$  and  $x$  and grow at the same steady-state growth rate, namely zero. If  $\beta = 1$  all countries will grow at the same steady-state growth rate, which will be non-zero, but no catch-up will occur because countries with lower  $y$  are growing at the same rate as countries with higher  $y$ . If countries have the same  $\beta < 1$  but different  $\alpha$ 's, countries will still converge to different levels but share the same steady-state growth rate of zero. If the  $\alpha$ 's and  $\beta$ 's differ across countries, there will be no convergence in either levels or rates. If the heterogeneity tests in this section are valid it is not likely that we will find economic convergence within the OECD.

1979 which of course includes the first OPEC oil crisis, the breakup of Bretton Woods, and the introduction of flexible exchange rates and increased capital mobility. These are events which could have altered production relations fundamentally. In this chapter, the sample period is 1973-1987 which does not quite fully coincide with those events. Nonetheless it would be useful in later work to conduct sub-sample stability tests.

#### (g) Disaggregating Public R&D by Socio-Economic Objective

In Table 4.8 I disaggregate the global stock of public R&D capital,  $RD_g$ , into Defence, Space, Economic Development (in Agriculture and Industry), Health, Energy, Advancement of Knowledge, and Other Research capital (ie. on Infrastructure, Environment, Earth, Civil Aeronautics, and Social Services). Data on the percentage composition of public R&D by socio-economic objective are available at intervals of four years. I therefore had to interpolate the stocks between those intervals (ie. by linear interpolation between intervals). I then converted the flows into stocks via the perpetual inventory method. To make sure the stocks added up to  $RD_g$  exactly, I determined the Other R&D capital stock residually. Each of the component stocks is a global variable; for example Defence R&D capital represents the sum of the Defence R&D capital stocks of the ten major R&D contributor countries. Each nation of course has a different composition of R&D capital by socio-economic objective - see Chart 4.6. To keep matters simple I do not disaggregate them any further into Own and Other National component stocks.

Table 4.8 indicates that Defence, Space, and Health, exert a negative and/or weak influence on productivity growth while the 'Other' category, Energy, and Economic Development exert positive influences. That Space R&D is weakly productive may seem counterintuitive since it is a very science and engineering intensive activity. But upon further reflection one may question why Space research should enhance private sector productivity in the first place. Space research is a public good, the benefits of which are not entirely monetary in nature. The motives for engaging in Space research are not entirely driven by production or profit considerations. Achieving a better understanding of how the universe operates is likely to be an important factor. Moreover, the returns to Space research are very uncertain, and the outcomes of experiments or data from missions will not be known for centuries.

Defence R&D spending is also found not to enhance private sector productivity despite defence research also being a very high-technology oriented activity. Like Space research, Defence research is not immediately tied to the concern of increasing private sector production but to other concerns - noneconomic. However, as the next part of this subsection indicates, we cannot rule out the possibility that defence or space research stimulates private sector research. Thus indirectly Space and Defence research can aid private-sector production.

Recall that the individual public R&D component capital stocks (Defence, Space, and so forth) sum up to the global stock of public R&D capital. I now want to report on the F-test that the coefficients of the individual public R&D component capital stocks add up to the output elasticity of the global stock of public R&D capital. The reason why this should hold can be seen as follows: let  $x_1, \dots, x_7$  denote the individual

stocks - ie.  $RD_g = x_1 + \dots + x_7$ . From the production function

$Y = RD_g^\theta F$ , where  $F$  is an index of the rest of the production function, we obtain:

$$\frac{\Delta Y}{Y} = \theta \frac{\Delta RD_g}{RD_g} + \frac{\Delta F}{F}.$$

$$\text{But } \frac{\Delta RD_g}{RD_g} = \sigma_1 \frac{\Delta x_1}{x_1} + \dots + \sigma_7 \frac{\Delta x_7}{x_7}$$

$$\text{where } \sigma_i = \frac{x_i}{RD_g} \quad \text{for } i = 1, \dots, 7$$

is the share of  $x_i$  in  $RD_g$ . Substituting  $\frac{\Delta RD_g}{RD_g}$  into  $\frac{\Delta Y}{Y}$  gives:

$$\frac{\Delta Y}{Y} = \theta \sigma_1 \frac{\Delta x_1}{x_1} + \dots + \theta \sigma_7 \frac{\Delta x_7}{x_7} + \frac{\Delta F}{F}$$

which shows that  $\theta \sigma_1 + \dots + \theta \sigma_7 = \theta$  since the weights  $\sigma_1, \dots, \sigma_7$  add up to one.

Thus the coefficient of each of the component stocks  $x_1, \dots, x_7$  in Table 4.8 is the output elasticity of the global stock of public R&D capital multiplied by the share of the component stock,  $x_i$ , in the global stock of public R&D capital. The output elasticity of the global public R&D stock for Pool 10 is in the range 0.66-0.68 (see Table 4.4E). The F-statistics for testing the constraint that the global output elasticity equals the sum of the individual output elasticities are 1.03, 1.5, and 1.69 for the OLS, GLS, and FE cases respectively. The critical F value is  $F_{0.05}(1,138) = 3.909$ , where there are 138 degrees of freedom in the denominator because there are 12 RHS variables and 150 observations. Thus the null hypothesis that there exist such coefficient constraints is not rejected.

### (h) Impact of Public Investment on Private Investment

Tables 4.9A-4.9C present the results of estimating equation (13)' from Section III and Table 4.9D presents the result of estimating equation (16)' from Section III. Again OLS, GLS, and FE estimates are given. In Tables 4.9A-4.9C, only the Pool 10 countries are analyzed since it is their R&D expenditure behaviour that is being analyzed. Table 4.9A studies global R&D variables while Table 4.9B studies Own and Other National R&D variables. Tables 4.9C disaggregates the global stock of R&D by socio-economic objective.

The results in Table 4.9A support the view that public R&D significantly stimulates private R&D. Lagged output and the lagged stock of private R&D capital have the correct signs, except in the GLS estimation where  $Y(-1)$  is negative, but insignificant.

In Table 4.9B<sup>1</sup>, Other National Public R&D is significant in affecting private R&D investment (according to GLS and FE estimates) at about the 90% confidence level, hence pointing to the presence of positive international spillovers. Note that  $\Delta OT_p$  has a weak and negative influence on Own National Private R&D investment. This indicates that public R&D variables are stronger influences on private research activity<sup>2</sup>.

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<sup>1</sup> In Table 4.9B the original regressions exhibited very strong serial correlation. Rather than use maximum-likelihood methods, I simply corrected the serial correlation by estimating the serial correlation coefficient and transforming the model into a standard variance components model.

<sup>2</sup> In preliminary investigations I found that when public R&D variables are absent, Other National Private R&D does positively stimulate Own National Private R&D investment.

Table 4.9C supports our conjecture earlier that while specific components of public R&D, such as Space and Defence, may not be conducive to private sector productivity growth, they may nonetheless stimulate private R&D significantly, and thereby indirectly affect productivity growth. In fact Space and Defense research are seen to affect private R&D investment quite strongly. Health research is the only item exerting a negative influence on private R&D investment. Energy research does exert a positive influence but rather weakly (at a confidence level of about 70%). Further research on these disaggregated stocks and the institutions behind them should better explain the results.

I carried out another F-test for the constraint that the sum of the coefficients of the component stocks of public R&D, shown in Table 4.9C, be equal to 1.06 (or 1.09 in the case of FE estimation), which is the coefficient of the aggregate stock of public R&D, shown in Table 4.9A. This time the null hypothesis is strongly rejected. The implication is that the component stocks of public R&D capital are not perfect substitutes in the private R&D investment equation, as assumed in Table 4.9A.

The significant negative effect of capacity utilization seems to suggest that private R&D activity is countercyclical. One possible explanation is that during peak capacity periods, firms devote more resources to production rather than to longer term investment. If cyclical events are not associated with trend growth, firms will not interpret cyclical demand fluctuations as necessarily leading to changes in future profitability. Moreover the negative sign may be an indication that private R&D does influence productivity with a lag. The reason is that the capacity utilization rate covaries with shifts in productivity, as seen from Tables 4.4 and 4.5. Yet since the capacity utilization rate enters negatively in Tables 4.9A-



4.9C, it cannot be the case that movements in productivity (as proxied by changes in the capacity utilization rate) and private R&D investment are positively correlated when considered contemporaneously (that is, when R&D investments are assumed to add to the stock of productive knowledge capital with no time lags).

Table 4.9D shows that global public R&D can also stimulate national private capital investment. Public capital investment is significant and the sign on the lagged stock of private capital is negative, except in the case of FE estimation. What is interesting is that private R&D has a negative, but weak, influence on private capital investment. This may be the case if private firms have limited resources with which to allocate between R&D and physical capital investment activities. In this case, private capital investment and private R&D investment are competing. In contrast, public R&D is provided by the government so that the "external" effects of an increased stock of knowledge dominate in raising the marginal productivities of private inputs.

## **VI Concluding Remarks**

In summary I have used OECD data to investigate whether national aggregate production functions are affected by own and cross national public investments in R&D. While the empirical strategy largely revolves around the aggregate growth accounting framework, several variations of it were tried in order to provide different angles from which to assess the spillover issue. In particular, I worked with both pooled OECD and U.S. data sets, tested for alternative specifications of the aggregate production

function, and studied the effects of public R&D and capital investment on labour as well as total factor productivity. I also controlled for capital income taxes, disaggregated public R&D by socioeconomic objective, and studied the effects of public R&D on private R&D investment. The empirical findings are supportive: fiscal investments do appear to have positive effects on national productivity growth as well as on foreign productivity growth through technological channels.

I have found that public R&D investment stimulates private sector output growth primarily by stimulating private R&D investment. Public R&D is significant when it enters the production function without private R&D but is insignificant when private R&D is also entered. This suggests that private R&D is the more important determinant of private-sector output. The fact that public R&D stimulates private R&D investment suggests that when private R&D is omitted from the production function, public R&D proxies for the 'direct' effects of private R&D on private output and the 'indirect' effects of public R&D on private output through the stimulative impacts of public R&D on private R&D investment. When private R&D is included in the production function, the explicit role of private R&D in private output is captured and public R&D has no further additional significance. On net, public R&D stimulates private output, for the following reason. According to Table 4.9A, the public R&D variable stimulates the private R&D variable by a coefficient greater than one. From Case 2 of either Tables 4.4D or 4.4E, we can see that the coefficient of the private R&D variable is more 'positive' than the coefficient of the public R&D variable is 'negative' in the private sector production function. Putting these estimates together, one can infer that the net (direct plus indirect) effect of public R&D capital on private output is 'positive'.

In the remainder of this section I will comment on the implications of the empirical results, and suggest some possible extensions. The first implication is that future open-economy fiscal policy models should endeavour to incorporate these external effects of fiscal investment expenditures as part of a general modelling strategy. Once models have incorporated these features, several important international coordination issues arise. In the introduction I alluded to the fact that public goods provision may be inefficient from a global social welfare planner's point of view if provision is left to non-coordinating national governments. The question remains as to how sovereign nations should coordinate their policies. Another question is whether globally optimal solutions are decentralizable in an international market setting.

There are also important financing issues. How should governments raise and allocate funds for internationally available public goods? Should the "benefits principle" apply? If so, what if the "users" represent the international community at large? The countries that generate most of the external effects may insist that other nations share the costs of public goods activities through appropriate budgetary and financial coordination. There may be concern that some countries prefer to "free ride" by waiting to apply the results of research conducted and financed by other countries. The free-riding countries may be in a position to provide to their own residents the output or services of the R&D effort of other countries at marginal cost. International reform or coordination of taxation should take into account the presence of internationally enjoyed public goods (uncompensated interdependencies). Forcing countries to participate in international financing will of course meet with reprisals if certain

countries feel that they are involuntary beneficiaries of another country's public goods activities, and feel no obligation to pay for activities that are not part of their national agenda. Future work needs to establish clearer principles of public finance amongst sovereign nations.

It is disputable whether any government (especially the U.S. government) can justify its large budgetary deficits on the grounds that it has generated significant external spillovers and that deficit-financing is simply a substitute for not being able to tax the rest of the world. The reason is that R&D funding is a relatively small fraction of total government spending. The size of the deficit is largely related to other sources (relatively low taxes, sluggish economic growth, growth in transfer payments, and other factors). In any event, governments with short run goals are likely to cut public investment items (including public R&D) amidst budgetary problems since the benefits from public investment tend to accrue in the longer term.

In conclusion I believe that there are a number of interesting areas for further research. First, work by Hall (1988) shows that when increasing returns to scale and markup pricing are present, conventional measures of total factor productivity are "too" procyclical. The procyclicality can be construed as evidence against the assumption of perfect competition. It is important therefore to study productivity measurement under imperfect competition in greater detail.

Secondly, it is rather difficult to distinguish growth from business cycles - even conceptually. The focus of this chapter has been on growth. It would also be interesting to study the business cycle effects of public investment. Perhaps the impacts of public investment may be predominantly cyclical phenomena and not aggregate supply reactions.

Some decomposition techniques may be a good first start towards evaluating this issue. Another is to try the duality approach and interpret varying capacity utilization as a proxy for cycles.

Finally, in an open-economy context, another measure of importance is international competitiveness. A nation's competitiveness should be reflected in its real wages and real exchange rate. Public investments which improve productivity and international competitiveness should thus have implications for movements in the real exchange rate. It would be interesting to investigate those implications empirically.

#### **Appendix 4.1: Note on Distortionary Taxation and Productivity**

This Appendix explains how capital income taxes are introduced into the production function and growth accounting equations. To motivate the introduction of taxes I follow the practice in Easterly (1989), Hansson (1984), Park (1990b), and Skinner (1987), of assuming a simple two sector economy. In one sector, factor supplies are taxed, while in the other, factor supplies are either untaxed or relatively lightly taxed. Examples of such two sectors include the rural versus urban sector, manufacturing versus agricultural sector, or corporate versus household sector. The differential treatment of taxes gives rise to distortions since factor resources shift from the taxed to untaxed sector. If the taxed sector were initially more productive, the resource shift should impact negatively on output growth. Even if the two sectors were equally productive, however, if there were initially a tax wedge (so that at the existing distribution of factor resources, one sector had higher marginal productivities of factor inputs than the other sector), a higher tax rate which induced a greater flow of factor resources in a direction against equalizing marginal productivities (or in a direction magnifying differential marginal productivities) would impact negatively on output growth.

Let the superscript  $i$  denote sector  $i$  (where  $i = 1, 2$ ). Each sector's production function is given by:

$$Y^i = A F(K_p^i, L^i)$$

where  $A$  is expression (3) in the main text (ie. an index of other inputs). Aggregate output is given by:

$$Y = Y^1 + Y^2 = A [ F(K_p^1, L^1) + F(K_p^2, L^2) ]$$

where  $F(K_p^i, L^i) = (K_p^i)^{\beta_K} (L^i)^{\beta_L}$ ,  $i = 1, 2$ .

Let  $\tau$  represent a vector of distortionary tax rates, say  $\tau = (\tau_K, \tau_L)$ , where  $\tau_K$  is a tax on capital and  $\tau_L$  a tax on labour. Let  $\omega_K(\tau)$ ,  $\omega_L(\tau)$  be the shares of capital and labour respectively in sector 1, which I assume to be the taxed sector:

$$\begin{aligned} K_p^1 &= \omega_K(\tau) K_p & \text{and} & & L^1 &= \omega_L(\tau) L \\ K_p^2 &= (1 - \omega_K(\tau)) K_p & \text{and} & & L^2 &= (1 - \omega_L(\tau)) L \end{aligned}$$

where  $K_p^1 + K_p^2 = K_p$ ,  $L^1 + L^2 = L$ . Assume  $\frac{\partial \omega_K(\tau)}{\partial \tau_K} < 0$  and  $\frac{\partial \omega_L(\tau)}{\partial \tau_L} < 0$ . Assume also that in the absence of taxes that  $\omega_K(0) = \omega_L(0) = 0.5$ , ie. equal supplies of resources in each sector.

Substituting these expressions into aggregate output:

$$(1) \quad Y = A K_p^{\beta_K} L^{\beta_L} T$$

where (2)  $T = [\omega_K^{\beta_K} \omega_L^{\beta_L} + (1 - \omega_K)^{\beta_K} (1 - \omega_L)^{\beta_L}]$ . Here I omit the tax arguments within the share functions so as to avoid cluttering up the notation.

### Interpretation of (2)

It can be shown that (2) measures the distortions arising from a tax system which drives wedges between the marginal factor productivities of different sectors. I start by log differentiating (2) around some initial steady state - ie.  $Y = Y^*$ ,  $\tau_K = \tau_K^*$ , and  $\tau_L = \tau_L^*$ , etc., where the tax rates are non-zero. This yields:

$$(2)' \text{ dlog } T = \frac{\beta_K \omega_K' \left[ \frac{\omega_K \beta_K \omega_L \beta_L}{\omega_K} - \frac{(1 - \omega_K) \beta_K (1 - \omega_L) \beta_L}{(1 - \omega_K)} \right] d\tau_K}{\omega_K \beta_K \omega_L \beta_L + (1 - \omega_K) \beta_K (1 - \omega_L) \beta_L} \\ + \frac{\beta_L \omega_L' \left[ \frac{\omega_K \beta_K \omega_L \beta_L}{\omega_L} - \frac{(1 - \omega_K) \beta_K (1 - \omega_L) \beta_L}{(1 - \omega_L)} \right] d\tau_L}{\omega_K \beta_K \omega_L \beta_L + (1 - \omega_K) \beta_K (1 - \omega_L) \beta_L}$$

By recognizing that:

$$\omega_K \beta_K \omega_L \beta_L + (1 - \omega_K) \beta_K (1 - \omega_L) \beta_L = \frac{Y}{AK_p \beta_K \beta_L}$$

$$\omega_K \beta_K \omega_L \beta_L = \frac{Y^1}{AK_p \beta_K \beta_L}$$

$$(1 - \omega_K) \beta_K (1 - \omega_L) \beta_L = \frac{Y^2}{AK_p \beta_K \beta_L}$$



we can simplify (2)' to:

$$(2)'' \quad d \log T = \left[ \frac{\partial y^1}{\partial k^1} \frac{\partial k^1}{\partial \tau_k} + \frac{\partial y^2}{\partial k^2} \frac{\partial k^2}{\partial \tau_k} \right] \frac{d\tau_k}{y} + \left[ \frac{\partial y^1}{\partial L^1} \frac{\partial L^1}{\partial \tau_L} + \frac{\partial y^2}{\partial L^2} \frac{\partial L^2}{\partial \tau_L} \right] \frac{d\tau_L}{y}$$

where:

$$\begin{aligned} \frac{\partial y^1}{\partial \tau_k} &= \frac{\partial y^1}{\partial k^1} \frac{\partial k^1}{\partial \tau_k} = \frac{\beta_k \omega_k'}{\omega_k} y^1 < 0 & \frac{\partial y^1}{\partial \tau_L} &= \frac{\partial y^1}{\partial L^1} \frac{\partial L^1}{\partial \tau_L} = \frac{\beta_L \omega_L'}{\omega_L} y^1 < 0 \\ \frac{\partial y^2}{\partial \tau_k} &= \frac{\partial y^2}{\partial k^2} \frac{\partial k^2}{\partial \tau_k} = - \frac{\beta_k \omega_k'}{(1 - \omega_k)} y^2 > 0 & \frac{\partial y^2}{\partial \tau_L} &= \frac{\partial y^2}{\partial L^2} \frac{\partial L^2}{\partial \tau_L} = - \frac{\beta_L \omega_L'}{(1 - \omega_L)} y^2 > 0 \end{aligned}$$

Thus (2)'' has a simple interpretation. As long as taxes drive wedges between the marginal factor productivities of the two sectors, the two inner brackets in (2)'' will be negative in value since the marginal productivity of factor inputs will be higher in the taxed sector - ie. sector 1.

I now take the logs of both sides of (1) to arrive at:

$$(3) \quad \log Y = \text{constant} + \log A + \beta_k \log K_p + \beta_L \log L + \phi_k \tau_k + \phi_L \tau_L$$

where:

$$\text{constant} = \log T^* - \phi_k \tau_k^* - \phi_L \tau_L^*$$

$$\log T^* = \log [\omega_K^{\beta_k} \omega_L^{\beta_L} + (1 - \omega_K)^{\beta_k} (1 - \omega_L)^{\beta_L}]$$

$$\text{and } \phi_k = \left[ \frac{\partial y^1}{\partial k^1} \frac{\partial k^1}{\partial \tau_k} + \frac{\partial y^2}{\partial k^2} \frac{\partial k^2}{\partial \tau_k} \right] \frac{1}{y^*}$$

$$\phi_L = \left[ \frac{\partial y^1}{\partial L^1} \frac{\partial L^1}{\partial \tau_L} + \frac{\partial y^2}{\partial L^2} \frac{\partial L^2}{\partial \tau_L} \right] \frac{1}{y^*}$$

are each evaluated at the initial steady state equilibrium. (For (3), I substituted in the first-order Taylor series expansion of  $\log T$ .)

We can also express (3) in "growth accounting" form:

$$(3)' \Delta \log Y = \Delta \log A + \beta_k \Delta \log K_p + \beta_L \Delta \log L + \phi_k \Delta \tau_k + \phi_L \Delta \tau_L$$

Note the difference between  $\Delta \tau_k = \tau_k - \tau_k(-1)$  and  $d\tau_k = \tau_k - \tau_k^*$  since the latter represents deviations from steady state and the former represents 'change in value' between periods (and similarly for  $\Delta \tau_L$ ,  $d\tau_L$ ).

If we were to impose constant returns, (3)' becomes:

$$(3)'' \Delta \log \frac{Y}{L} = \Delta \log A + \beta_k \Delta \log \frac{K_p}{L} + \phi_k \Delta \tau_k + \phi_L \Delta \tau_L$$

In the empirical investigation I consider only capital income taxes,  $\tau_k$ . The sign of  $\phi_k$  is expected to be negative.

## Appendix 4.2: Note on the Private Investment Equations

### Stock-Flow Dynamics

This Appendix illustrates the stock-flow dynamics that are behind the private investment equations described in the main text. A dynamic version of the system of equations (11) and (12) from the text - or equations (14) and (15) from the text - is:

$$(1) \dot{R} = I(\dots, \phi) - \delta R$$

$$(2) \dot{\phi} = (r + \delta)\phi - Y_R$$

where  $I$  is private investment (either  $\Delta K_p$  or  $\Delta R_p$ ),  $R$  the corresponding stock of capital (either  $K_p$  or  $R_p$ ),  $Y_R$  the marginal product of  $R$  (for output  $Y$ ),  $r$  the interest rate, and  $\delta$  the depreciation rate.  $\phi$  is the present discounted value of marginal products. In steady-state,  $\phi = \frac{Y_R}{(r + \delta)}$ .

Equation (1) is the net investment equation. Equation (2) is a necessary condition for maximizing the value of a firm which faces the following optimization problem:

$$\max_{\dot{R}, R} \int_0^{\infty} e^{-rt} [Y(R(t), \dots) - \phi I(t)] dt$$

subject to (1), where the price of output  $Y$  is normalized to unity.

The supplier of investment goods or research output maximizes:

$$\max_I \int_0^{\infty} e^{-rt} [\phi(t)I(t) - c(I(t))] dt$$

where  $c(\cdot)$  is a cost function such that  $c(0) = c'(0) = 0$ , and  $c' > 0$ ,  $c'' > 0$  for nonzero  $I$ . A necessary condition for the latter maximization problem is:

$$\phi = c'(I)$$

- ie. invest to the point where the marginal benefit equals the marginal cost.

The diagram below depicts the dynamic adjustment of investment to a positive demand shock using (1) and (2). Because the stock of capital is predetermined, there is an immediate jump in  $\phi$  onto the new saddlepath. The adjustment path is given by  $E_0$ - $E_1$ - $E_2$ :

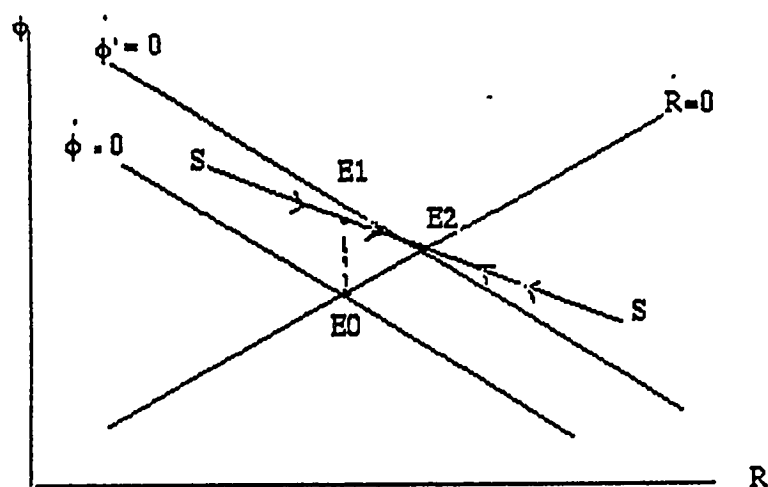


Figure 4.1: Investment Dynamics

### **Appendix 4.3: Note on the Perpetual Inventory Method**

This Appendix outlines how stock data on physical and R&D capital are derived from data on flows<sup>1</sup>. In the absence of a complete set of stock data, researchers and statisticians resort to the perpetual inventory method:

$$K(t+1) = K(t)(1 - \delta) + I(t+1)$$

where  $K(t)$  is the fixed stock of capital at the end of period  $t$ ,  $I(t+1)$  is gross fixed investment during  $(t, t+1)$ , and  $\delta$  is the geometric rate of depreciation. A value for the initial stock,  $K(0)$ , is needed. If there is no datum on  $K(0)$ , one can infer its value by rolling the above equation backwards in time:

$$K(t+1) = I(t+1) + (1 - \delta)I(t) + (1 - \delta)^2 I(t-1) + \dots$$

$$\text{or } K(t+1) = I(t+1) \left( 1 + (1 - \delta) \left( \frac{I(t)}{I(t+1)} \right) + (1 - \delta)^2 \left( \frac{I(t-1)}{I(t+1)} \right) + \dots \right)$$

Let  $(1 + g(t+1)) = \left( \frac{I(t+1)}{I(t)} \right)$ , where  $g(t+1)$  is the growth rate of investment between period  $t$  and  $t+1$ . Suppose  $\bar{g}$  is an historical average of this growth rate. For simplicity let  $g(t) = \bar{g}$  for all  $t$ :

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<sup>1</sup> Physical capital refers to the net fixed stock of reproducible assets. R&D activities comprise creative work undertaken in order to increase the stock of knowledge, including the use of this stock of knowledge to develop new applications. The OECD (1976,1980) manuals describe standardized practices in the measurement of physical and intellectual capital.

$$K(t+1) = I(t+1) \left( 1 + \frac{1 - \delta}{1 + \bar{g}} + \left( \frac{1 - \delta}{1 + \bar{g}} \right)^2 + \dots \right) = I(t+1) \left( \frac{1 + \bar{g}}{\bar{g} + \delta} \right)$$

for which we require  $\bar{g} + \delta > 0$ .

Thus if we have flow data from 1960 to 1989, we can assign a value to the capital stock at 1960 by assuming some reasonable depreciation rate  $\delta$  and investment growth rate  $\bar{g}$  - ie.  $K(1960) = I(1960) \left( \frac{1 + \bar{g}}{\bar{g} + \delta} \right)$ .

This approach can also be used to construct the stock of R&D knowledge capital from R&D flow expenditure data, assuming some rate of depreciation of knowledge<sup>1</sup>. Another consideration is that there are likely to be lags in the creation of productive knowledge from R&D investment spending - ie.  $K(t+1) = K(t)(1-\delta) + I(t+1-j)$ , where  $j$  is the length of time it takes R&D investment to add to the stock of productive knowledge capital. I assume  $j=3$  and that the geometric depreciation rate of R&D capital is 10%<sup>2</sup>.

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<sup>1</sup>The concept of a depreciating stock of knowledge may be difficult to fathom. On the other hand it is difficult to assume a zero depreciation rate to this stock particularly if new technologies are constantly rendering existing ones obsolete. Further discussion is in the U.S. Department of Labor Statistics (1989).

<sup>2</sup> I also assume physical capital depreciates geometrically at 10%. These values are in accord with conventional practice. I have experimented with different values of  $\delta$ ,  $\bar{g}$ , and  $j$ . Holding  $\delta$  fixed, I find that different values of  $\bar{g}$  produce different initial conditions that affect the coefficients from level estimations but not their significance. Difference estimation results change very mildly, if at all. Holding  $\bar{g}$  fixed and varying  $\delta$  will also produce different initial conditions, but varying  $\delta$  between 0.05 and 0.15 basically preserves our results. Different values of  $j$  produce noticeable changes. If  $j > 3$  the significance of R&D investments for productivity growth diminishes; if  $j < 3$ , R&D investments stimulate productivity growth but  $j < 3$  may be too soon for R&D investments to affect production, and we may be capturing demand-side effects instead.

CHART 4.1 A  
Composition of U.S. Capital Stock in 1965

|                                |      |     |       |
|--------------------------------|------|-----|-------|
| Total Net Fixed Capital Stock: | 100% |     |       |
| Private:                       | 73%  |     |       |
| of which, Residential          |      | 51% |       |
| Nonresidential:                |      | 49% |       |
| of which, Equipment            |      |     | 40%   |
| Structures                     |      |     | 60%   |
| Public:                        | 27%  |     |       |
| of which, State and Local:     |      | 63% |       |
| of which, Equipment            |      |     | 4%    |
| of which, Structures:          |      |     | 96%   |
| Educational Buildings          |      |     | 21%   |
| Hospitals                      |      |     | 3%    |
| Highways and Streets           |      |     | 47%   |
| Conservation & Development     |      |     | 2%    |
| Water and Sewer Systems        |      |     | 15%   |
| Other Buildings & Structures   |      |     | 12%   |
| of which, Federal Government:  |      | 37% |       |
| of which, Equipment:           |      |     | 45%   |
| of which, Military             |      |     | 82%   |
| Nonmilitary                    |      |     | 18%   |
| of which, Structures:          |      |     | 55%   |
| of which, Industrial Buildings |      |     | 12%   |
| Educational Buildings          |      |     | 0.3%  |
| Hospitals                      |      |     | 1%    |
| Highways and Streets           |      |     | 3%    |
| Conservation & Development     |      |     | 32%   |
| Military Facilities            |      |     | 43.2% |
| Other Buildings & Structures   |      |     | 8.5%  |

Source: U.S. Bureau of Economic Analysis  
Fixed Reproducible Tangible Wealth in the  
United States 1987

**CHART 4.1B**  
**Composition of U.S. Capital Stock in 1985**

|                                       |             |            |             |
|---------------------------------------|-------------|------------|-------------|
| <b>Total Net Fixed Capital Stock:</b> | <b>100%</b> |            |             |
| <b>Private:</b>                       | <b>77%</b>  |            |             |
| of which, Residential                 |             | <b>47%</b> |             |
| Nonresidential:                       |             | <b>53%</b> |             |
| of which, Equipment                   |             |            | <b>49%</b>  |
| Structures                            |             |            | <b>51%</b>  |
| <b>Public:</b>                        | <b>23%</b>  |            |             |
| of which, State and Local:            |             | <b>69%</b> |             |
| of which, Equipment                   |             |            | <b>6%</b>   |
| of which, Structures:                 |             | <b>94%</b> |             |
| Educational Buildings                 |             |            | <b>19%</b>  |
| Hospitals                             |             |            | <b>3%</b>   |
| Highways and Streets                  |             |            | <b>41%</b>  |
| Conservation & Development            |             |            | <b>2%</b>   |
| Water and Sewer Systems               |             |            | <b>17%</b>  |
| Other Buildings & Structures          |             |            | <b>18%</b>  |
| of which, Federal Government:         |             | <b>31%</b> |             |
| of which, Equipment:                  |             | <b>56%</b> |             |
| of which, Military                    |             |            | <b>88%</b>  |
| Nonmilitary                           |             |            | <b>12%</b>  |
| of which, Structures:                 |             | <b>44%</b> |             |
| of which, Industrial Buildings        |             |            | <b>8%</b>   |
| Educational Buildings                 |             |            | <b>0.4%</b> |
| Hospitals                             |             |            | <b>3%</b>   |
| Highways and Streets                  |             |            | <b>5.3%</b> |
| Conservation & Development            |             |            | <b>43%</b>  |
| Military Facilities                   |             |            | <b>31%</b>  |
| Other Buildings & Structures          |             |            | <b>9.3%</b> |

Source: U.S. Bureau of Economic Analysis  
 Fixed Reproducible Tangible Wealth in the  
 United States 1987



CHART 4.2A  
Share of Private Gross Fixed Investment in Gross Domestic Product

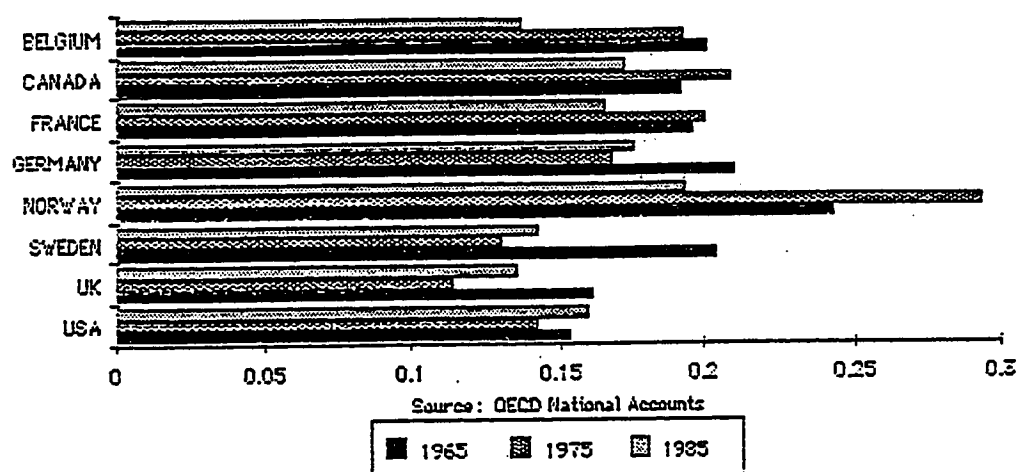


CHART 4.2B  
Share of Government Gross Fixed Investment in Gross Domestic Product

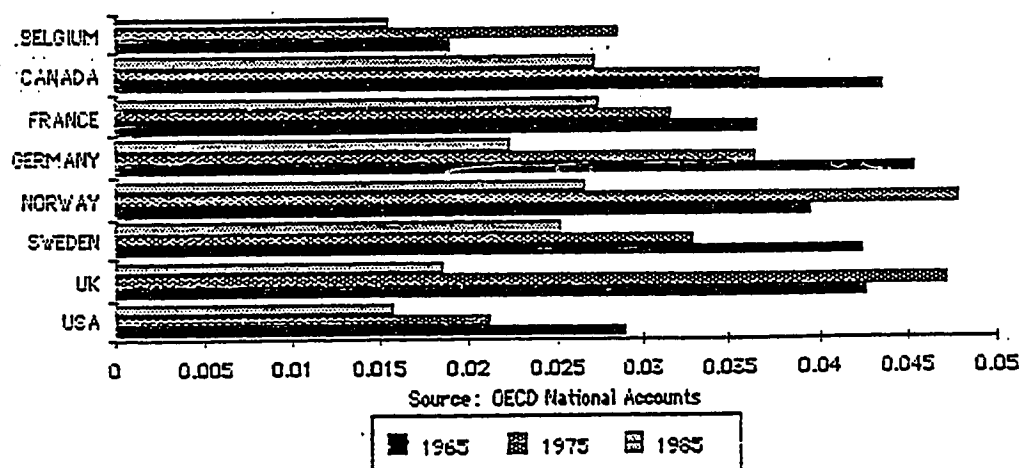


CHART 4.3A  
Share of Government Funding in Gross Domestic Expenditure on R&D - 1975

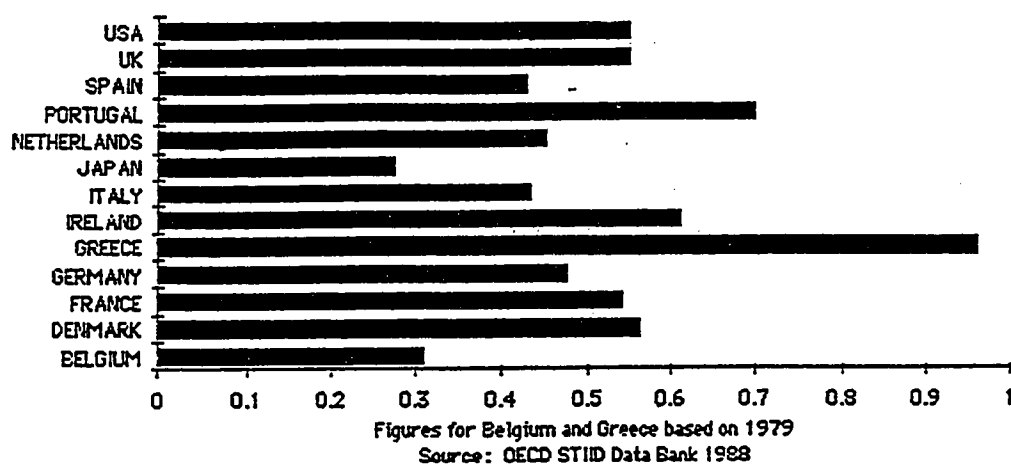


CHART 4.3B  
Share of Government Funding in Gross Domestic Expenditure on R&D - 1985

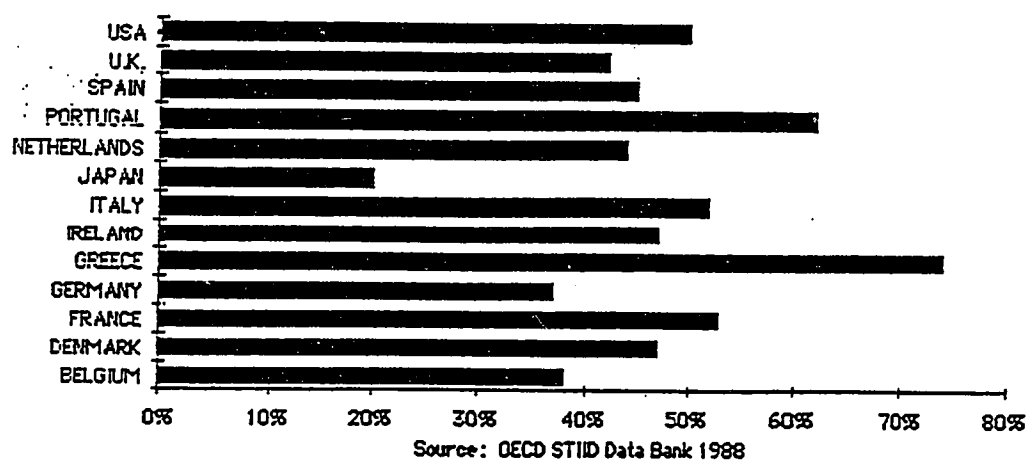


CHART 4.3C  
Shares in Global Public R&D Stock 1975

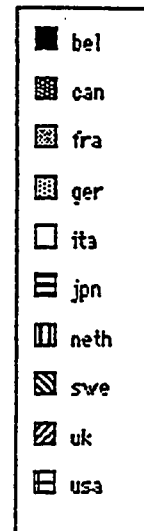
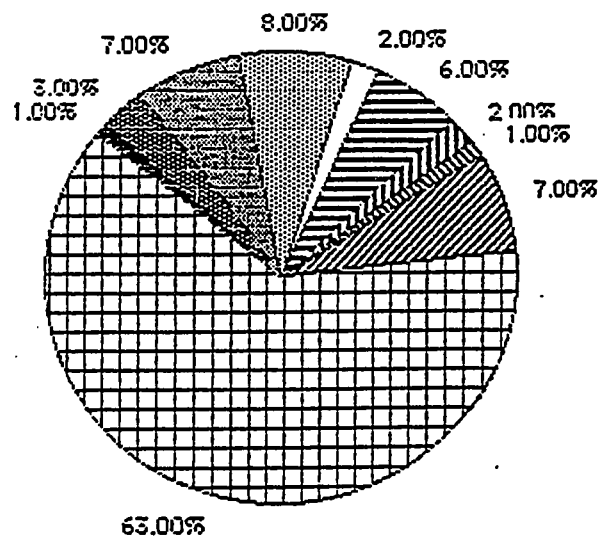
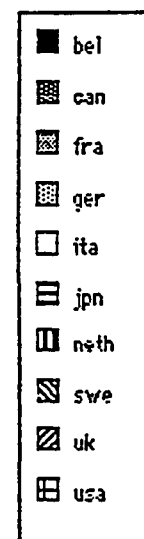
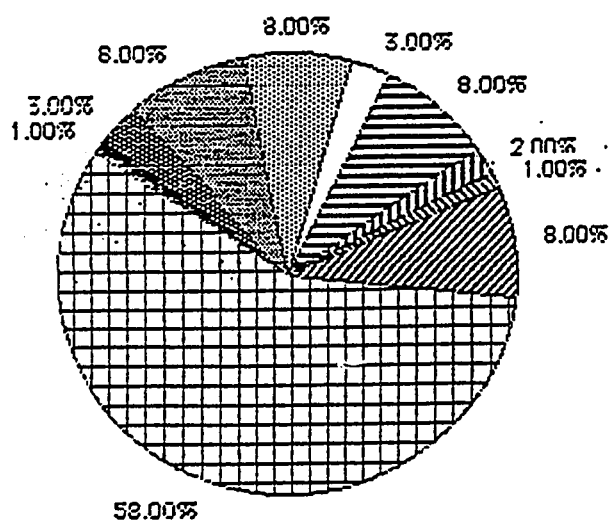


CHART 4.3D  
Shares in Global Public R&D Stock 1985



Source:  
Author's Calculations

CHART 4.3E  
Shares in Global Private R&D Stock 1975

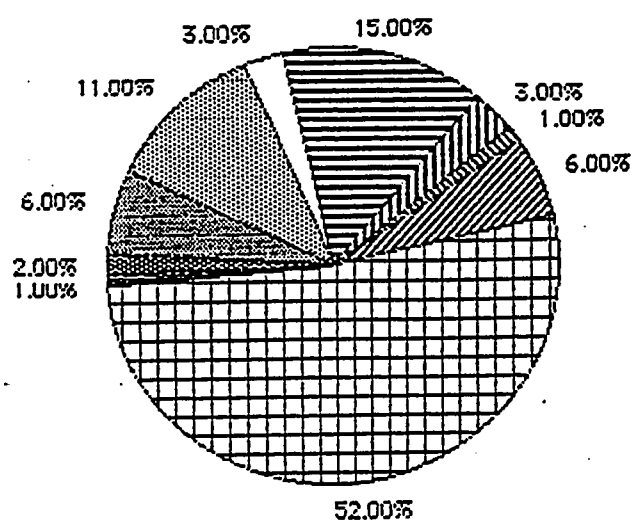
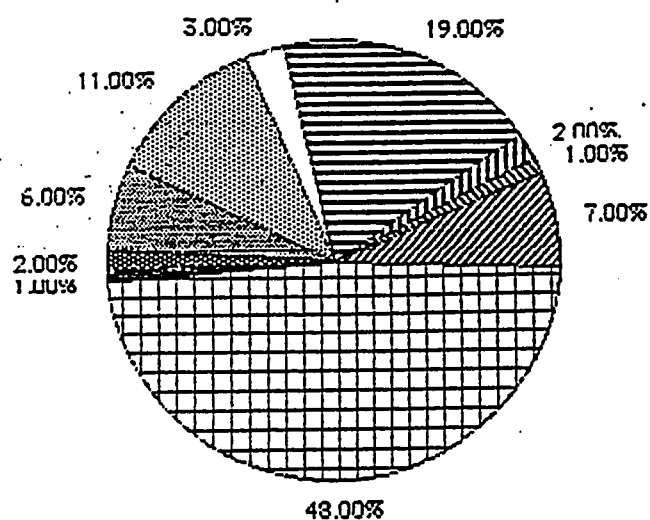
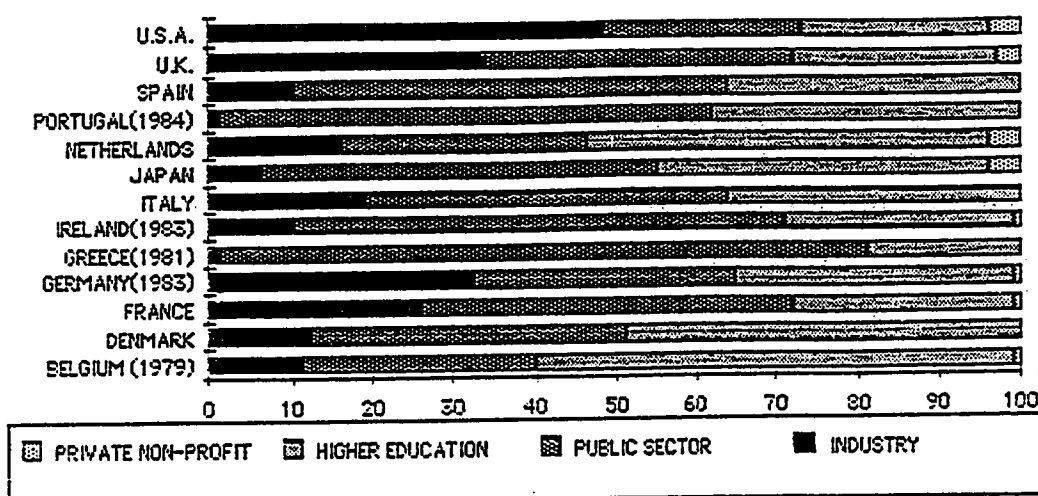


CHART 4.3F  
Shares in Global Private R&D Stock 1985



Source:  
Author's Calculations

CHART 4.4 Public R&amp;D Funding by Performing Sectors - 1985 or nearest year



Source: OECD STIID Data Bank 1988

CHART 4.5

Public R&D Funding as a Percentage of Total Government Expenditures:

| Major R&D   |      |      |      |  |
|-------------|------|------|------|--|
| Performers: | 1970 | 1977 | 1985 |  |
| U.S.A.      | 7.2  | 6.2  | 6.7  |  |
| Japan       | 4.0  | 3.1  | 3.3  |  |
| Germany     | 4.4  | 3.9  | 4.1  |  |
| France      | 6.5  | 4.5  | 5.5  |  |
| U.K.        | 5.7  | 4.2  | 4.0  |  |

| Small R&D   |      |      |      |  |
|-------------|------|------|------|--|
| Performers: | 1970 | 1977 | 1985 |  |
| Austria     | 1.2  | 1.8  | 2.2  |  |
| Norway      | 2.4  | 2.5  | 2.8  |  |
| Denmark     | 1.9  | 1.7  | 2.0  |  |
| Finland     | 1.5  | 1.8  | n.a. |  |
| Ireland     | 1.7  | 1.8  | 1.6  |  |
| Spain       | 0.8  | 1.0  | 1.1  |  |
| Portugal    | 1.0  | 1.3  | n.a. |  |
| Iceland     | 2.2  | 2.2  | 2.4  |  |

| Medium R&D  |      |      |      |  |
|-------------|------|------|------|--|
| Performers: | 1970 | 1977 | 1985 |  |
| Italy       | 2.4  | 2.0  | 2.4  |  |
| Canada      | 4.2  | 2.8  | 2.8  |  |
| Netherlands | 4.3  | 4.0  | 4.3  |  |
| Sweden      | 2.6  | 2.0  | 2.5  |  |
| Australia   | n.a. | 3.2  | 3.0  |  |
| Belgium     | 3.6  | 2.0  | 2.2  |  |

Source: OECD STIID Data Bank (1988)  
and Eurostat (1990)

CHART 4.6

GOVERNMENT R&amp;D FUNDING BY SOCIO-ECONOMIC OBJECTIVES IN THE MAJOR 5 (Percentage Distribution)

|       |      | DEFENCE | SPACE | AGRICULTURE<br>& INDUSTRY | HEALTH | ENERGY | ADVANCEMENT OF<br>KNOWLEDGE | OTHER | TOTAL |
|-------|------|---------|-------|---------------------------|--------|--------|-----------------------------|-------|-------|
| USA   | 1975 | 51      | 14    | 3                         | 12     | 7      | 4                           | 9     | 100   |
|       | 1985 | 66      | 5     | 2                         | 11     | 6      | 4                           | 6     | 100   |
| JAPAN | 1975 | 4.7     | 15    | 42                        | 6.3    | 16     | 3.5                         | 12.5  | 100   |
|       | 1985 | 5       | 12    | 36                        | 5      | 25     | 4                           | 13    | 100   |
| GER   | 1975 | 19      | 7.4   | 13                        | 6      | 18     | 16                          | 20.6  | 100   |
|       | 1985 | 15      | 6     | 21                        | 5      | 22     | 17                          | 14    | 100   |
| FRA   | 1975 | 33      | 6     | 13                        | 4.4    | 9.4    | 17                          | 17.2  | 100   |
|       | 1985 | 35      | 7     | 18                        | 4      | 9      | 18                          | 9     | 100   |
| UK    | 1975 | 53      | 2.5   | 8.2                       | 2.3    | 7      | 14                          | 13    | 100   |
|       | 1985 | 58      | 3     | 16                        | 4      | 4      | 6                           | 9     | 100   |

The OTHER category includes: Earth Exploration, Civil Aeronautics, Environment, Infrastructure, and Social Services

Source: OECD  
STIID

**Table 4.1: Mnemonics/Notations used**

All variables are in real 1985 U.S. dollars. Each variable  $x$  is in logs, implying that  $\Delta x = x - x(-1)$  is the growth rate of  $x$ .

- Y - private sector output
- L - labour hours worked in private sector
- YL - private sector output per labour hours worked
- K<sub>g</sub> - public sector physical capital stock
- K<sub>p</sub> - private sector physical capital stock
- KL - private sector physical capital stock per labour hours worked
  
- CU - capacity utilization rate in industry (or manufacturing)
- $\tau_K$  - capital income tax rate
- TFP - total factor productivity in the private sector
  
- RD<sub>g</sub> - global public sector R&D capital stock
- RD<sub>p</sub> - global private sector R&D capital stock
- OW<sub>g</sub> - own-country national public sector R&D capital stock
- OW<sub>p</sub> - own-country national private sector R&D capital stock
- OT<sub>g</sub> - other-country national public sector R&D capital stock
- OT<sub>p</sub> - other-country national private sector R&D capital stock

(Note that:  $RD_g = OW_g + OT_g$  and  $RD_p = OW_p + OT_p$ )

T - trend term                      C - constant term

**Other Notation:**

- OLS - ordinary least squares estimation
- IV - instrumental variables estimation
- AR - correction for first-order serial correlation
- GLS - generalized least squares estimation
- FE - fixed effects or within estimation

Pool 19 - refers to group of 19 countries pooled (identified in text); size: 285 observations

Pool 10 - refers to smaller group of 10 countries pooled (identified in text); size: 150 observations

Adj R-Sq is the adjusted  $R^2$ , DW the Durbin-Watson statistic, and S.E.R. the standard error of the regression. Standard errors are in parentheses.

**Table 4.2A Public Infrastructural Investment Results - U.S.A. 1948-88**

I. Level Estimations

(1) OLS Dependent Variable: Y

| C      | K <sub>g</sub> | K <sub>p</sub> | L      | CU     |
|--------|----------------|----------------|--------|--------|
| -2.18  | 0.41           | 0.3            | 0.72   | 0.19   |
| (0.17) | (.09)          | (0.12)         | (0.12) | (0.06) |

S.E.R. = 0.01      Adjusted R-Sq=0.999      D.W.=1.17

(2) AR(1) Dependent Variable: Y

| C      | K <sub>g</sub> | K <sub>p</sub> | L      | CU     | Rho=0.65 |
|--------|----------------|----------------|--------|--------|----------|
| -2.05  | 0.30           | 0.5            | 0.49   | 0.22   |          |
| (0.19) | (0.14)         | (0.17)         | (0.16) | (0.07) |          |

S.E.R.=0.0092      Adjusted R-Sq=0.999      D.W.=1.71

(3) OLS Dependent Variable: YL (Constant Returns to Scale Imposed)

| C      | K <sub>g</sub> | KL     | CU     |
|--------|----------------|--------|--------|
| -2.18  | 0.44           | 0.27   | 0.18   |
| (0.17) | (0.08)         | (0.11) | (0.06) |

S.E.R.=0.01      Adjusted R-Sq=0.997      D.W.=1.1

(4) AR(1) Dependent Variable: YL (Constant Returns to Scale Imposed)

| C      | K <sub>g</sub> | KL     | CU     | Rho=0.5 |
|--------|----------------|--------|--------|---------|
| -2.14  | 0.27           | 0.52   | 0.25   |         |
| (0.29) | (0.12)         | (0.18) | (0.08) |         |

S.E.R.=.0092      Adjusted R-Sq=0.998      D.W.=1.67

(5) AR(1) Dependent Variable: YL A Test of the "Romer" Specification

| C      | K <sub>g</sub> | K <sub>p</sub> | KL     | CU      | Rho=0.8 |
|--------|----------------|----------------|--------|---------|---------|
| -1.32  | 0.17           | -0.101         | 0.77   | 0.345   |         |
| (0.18) | (0.11)         | (0.089)        | (0.21) | (0.087) |         |

S.E.R.=0.0104      Adjusted R-Sq=0.995      D.W.=1.52



Table 4.2A continued

| (6) AR(1) Dependent Variable: YL Distortional Taxation Considered |        |        |              |         |          |
|-------------------------------------------------------------------|--------|--------|--------------|---------|----------|
| C                                                                 | $K_g$  | KL     | $\tau_K(-1)$ | CU      | Rho=0.77 |
| -1.13                                                             | 0.22   | 0.48   | -0.23        | 0.22    |          |
| (0.21)                                                            | (0.13) | (0.21) | (0.33)       | (0.087) |          |
| S.E.R.=0.0038      Adjusted R-Sq=0.995      D.W.=1.54             |        |        |              |         |          |

## II. Difference Estimations

| (7) OLS Dependent Variable: $\Delta YL$ (Constant Returns to Scale Imposed) |              |             |             |
|-----------------------------------------------------------------------------|--------------|-------------|-------------|
| C                                                                           | $\Delta K_g$ | $\Delta KL$ | $\Delta CU$ |
| 0.0031                                                                      | 0.022        | 0.7         | 0.303       |
| (0.0042)                                                                    | (0.13)       | (0.18)      | (0.077)     |
| S.E.R.=0.0109      Adjusted R-Sq=0.33      D.W.=1.6                         |              |             |             |

| (8) OLS Dependent Variable: $\Delta YL$ Distortional Taxation Considered |              |             |                     |             |
|--------------------------------------------------------------------------|--------------|-------------|---------------------|-------------|
| C                                                                        | $\Delta K_g$ | $\Delta KL$ | $\Delta \tau_K(-1)$ | $\Delta CU$ |
| 0.0043                                                                   | 0.16         | 0.42        | -0.35               | 0.2         |
| (0.0042)                                                                 | (0.16)       | (0.27)      | (0.34)              | (0.107)     |
| S.E.R.=0.0041      Adjusted R-Sq=0.28      D.W.=1.66                     |              |             |                     |             |

Note: In each equation, the capital stock variables take on beginning-of-period values and hence are predetermined vis-a-vis the dependent variables, which are within-period flows. Standard errors are in parentheses. Rho is the first-order serial correlation coefficient.

**Table 4.2B. Public R&D Investment Results - U.S.A. 1956-87**

|                     | Dependent Variable: $\Delta Y_L$ |                    |                    |                   |                     |                     |
|---------------------|----------------------------------|--------------------|--------------------|-------------------|---------------------|---------------------|
|                     | (i)                              | (ii)               | (iii)              | (iv)              | (v)                 | (vi)                |
| C                   | 0.0035<br>(0.0045)               | -0.006<br>(0.0062) | -0.0077<br>(0.008) | 0.0053<br>(0.004) | -0.0042<br>(0.0057) | -0.0025<br>(0.0076) |
| $\Delta RD_g$       | 0.091<br>(0.059)                 | ---                | -0.032<br>(0.094)  | 0.12<br>(0.054)   | ---                 | 0.031<br>(0.089)    |
| $\Delta RD_p$       | ---                              | 0.25<br>(0.11)     | 0.3<br>(0.18)      | ---               | 0.25<br>(0.1)       | 0.204<br>(0.16)     |
| $\Delta KL$         | 0.46<br>(0.25)                   | 0.395<br>(0.24)    | 0.42<br>(0.25)     | 0.2<br>(0.25)     | 0.216<br>(0.23)     | 0.19<br>(0.24)      |
| $\Delta \tau_K(-1)$ | ---                              | ---                | ---                | -1.47<br>(0.52)   | -1.28<br>(0.5)      | -1.33<br>(0.53)     |
| ACU                 | 0.2<br>(0.107)                   | 0.2<br>(0.095)     | 0.21<br>(0.104)    | 0.11<br>(0.1)     | 0.14<br>(0.09)      | 0.13<br>(0.1)       |
| Adj R-Sq            | 0.24                             | 0.31               | 0.28               | 0.39              | 0.42                | 0.4                 |
| D.W.                | 1.87                             | 2.04               | 2.05               | 1.81              | 1.89                | 1.9                 |
| SER                 | 0.011                            | 0.0105             | 0.0107             | 0.0098            | 0.0096              | 0.0098              |

(standard errors in parentheses)

**Table 4.3A. Investment Equations - U.S.A.****Private Capital Investment**      **Sample: 1948-88**

$$\Delta K_p = 0.72 + 0.008 \Delta K_g - 0.19 K_p(-1) + 0.086 K_g(-1) + 0.019 \Delta CU$$

(0.7)      (0.27)      (0.14)      (0.045)      (0.018)

$$+ 0.0038 T + u_t,$$

(0.0041)

$$\text{where } u_t = 0.6 u_{t-1} + \varepsilon_t, \text{ S.E.R.} = 0.0067 \quad \text{Adj. } R^2 = 0.3$$

(0.13)

**Private R&D Investment**      **Sample: 1956-87**

$$\Delta RD_p = 0.76 + 0.21 \Delta RD_g - 0.45 RD_p(-1) + 0.15 RD_g(-1) - 0.045 \Delta CU$$

(0.1)      (0.062)      (0.14)      (0.04)      (0.01)

$$+ 0.0173 T + u_t,$$

(0.003)

$$\text{where } u_t = 0.82 u_{t-1} + \varepsilon_t, \text{ S.E.R.} = 0.0034 \quad \text{Adj. } R^2 = 0.9$$

(0.1)

(standard errors in parentheses)

**Table 4.3B. Reverse Causation Analysis - U.S.A.****Public Infrastructural Investment****Sample: 1948-88**

$$(1) \quad \Delta K_g = 0.036 - 0.0005 T + 0.025 \Delta YL - 0.001 \Delta CU + u_t,$$

$$(0.011) \quad (0.0004) \quad (0.044) \quad (0.0082)$$

$$\text{where } u_t = 0.92 u_{t-1} + \varepsilon_t, \quad \text{S.E.R.} = 0.0042 \quad \text{Adj. } R^2 = 0.11$$

$$(0.052)$$

$$(2) \quad \Delta\left(\frac{K_g}{Y}\right) = 0.021 - 0.0009 T - 0.31 \Delta YL - 0.42 \Delta CU + u_t,$$

$$(0.007) \quad (0.0003) \quad (0.14) \quad (0.026)$$

$$\text{where } u_t = 0.4 u_{t-1} + \varepsilon_t, \quad \text{S.E.R.} = 0.0103 \quad \text{Adj. } R^2 = 0.8$$

$$(0.14)$$

**Public R&D Investment****Sample: 1956-87**

$$(1) \quad \Delta RD_g = 0.107 - 0.0025 T + 0.154 \Delta YL + 0.12 \Delta CU + u_t,$$

$$(0.03) \quad (0.001) \quad (0.168) \quad (0.034)$$

$$\text{where } u_t = 0.82 u_{t-1} + \varepsilon_t, \quad \text{S.E.R.} = 0.0134 \quad \text{Adj. } R^2 = 0.4$$

$$(0.09)$$

$$(2) \quad \Delta\left(\frac{RD_g}{Y}\right) = 0.097 - 0.003 T - 0.28 \Delta YL - 0.27 \Delta CU + u_t,$$

$$(0.024) \quad (0.0008) \quad (0.2) \quad (0.04)$$

$$\text{where } u_t = 0.7 u_{t-1} + \varepsilon_t, \quad \text{S.E.R.} = 0.0149 \quad \text{Adj. } R^2 = 0.7$$

$$(0.12)$$

(standard errors in parentheses)

**Table 4.4A. Sample Group: Pool 19      Sample Period: 1973-1987****Common Factor Restriction Test****Unconstrained (Levels):**

Dependent Variable: Y

| <u>Independent<br/>Variables</u> | <u>Coefficient<br/>Estimate</u> | <u>Standard<br/>Error</u> |
|----------------------------------|---------------------------------|---------------------------|
| C                                | -0.05                           | (0.038)                   |
| K <sub>g</sub>                   | 0.19                            | (0.066)                   |
| K <sub>p</sub>                   | 0.52                            | (0.076)                   |
| L                                | 0.373                           | (0.072)                   |
| CU                               | 0.058                           | (0.01)                    |
| T <sup>2</sup>                   | 0.0011                          | (0.0096)                  |
| K <sub>g</sub> (-1)              | -0.18                           | (0.066)                   |
| K <sub>p</sub> (-1)              | -0.51                           | (0.074)                   |
| L(-1)                            | -0.374                          | (0.072)                   |
| CU(-1)                           | -0.048                          | (0.0095)                  |
| T <sup>2</sup> (-1)              | -0.0012                         | (0.001)                   |
| Y(-1)                            | 0.982                           | (0.011)                   |
| D.W.                             | 1.7                             |                           |
| Adj R-Sq                         | 0.9999                          |                           |

**Productivity Analyses****Table 4.4B. Sample Group: Pool 19      Sample Period: 1973-1987**

| Independent<br>Variables | Dependent Variable: $\Delta YL$ |                       |                      |
|--------------------------|---------------------------------|-----------------------|----------------------|
|                          | OLS                             | GLS                   | FE                   |
| C                        | 0.015<br>(0.0038)               | 0.0087<br>(0.0028)    |                      |
| $\Delta RD_g$            | ---                             | ---                   | ---                  |
| $\Delta RD_p$            | ---                             | ---                   | ---                  |
| $\Delta K_g$             | 0.295<br>(0.059)                | 0.416<br>(0.074)      | 0.86<br>(0.115)      |
| $\Delta KL$              | 0.46<br>(0.063)                 | 0.47<br>(0.064)       | 0.45<br>(0.066)      |
| $\Delta CU$              | 0.042<br>(0.008)                | 0.041<br>(0.0077)     | 0.038<br>(0.0075)    |
| T                        | -0.00063<br>(0.00038)           | -0.00047<br>(0.00037) | 0.00002<br>(0.00037) |
| D.W.                     | 1.61                            | 1.64                  | 1.66                 |
| Adj R-Sq                 | 0.324                           | 0.34                  | 0.41                 |

(Standard errors are in parentheses)

**Table 4.4C.** Sample Group: Pool 10      Sample Period: 1973-1987

| Independent<br>Variables | Dependent Variable: $\Delta YL$ |                      |                      |
|--------------------------|---------------------------------|----------------------|----------------------|
|                          | OLS                             | GLS                  | FE                   |
| C                        | 0.025<br>(0.004)                | 0.016<br>(0.003)     |                      |
| $\Delta RD_g$            | ---                             | ---                  | ---                  |
| $\Delta RD_p$            | ---                             | ---                  | ---                  |
| $\Delta K_g$             | 0.12<br>(0.056)                 | 0.16<br>(0.073)      | 0.42<br>(0.14)       |
| $\Delta KL$              | 0.32<br>(0.074)                 | 0.34<br>(0.076)      | 0.34<br>(0.08)       |
| $\Delta CU$              | 0.08<br>(0.0134)                | 0.081<br>(0.0131)    | 0.077<br>(0.013)     |
| T                        | -0.0012<br>(0.00039)            | -0.0011<br>(0.00038) | -0.0009<br>(0.00039) |
| D.W.                     | 1.6                             | 1.7                  | 1.85                 |
| Adj R-Sq                 | 0.304                           | 0.32                 | 0.35                 |

(Standard errors are in parentheses)

**Table 4.4D.** Sample Group: Pool 19      Sample Period: 1973-1987Case 1: Private R&D Excluded      -      Global R&D Measure Used

| Independent<br>Variables | Dependent Variable: $\Delta YL$ |                       |                       |
|--------------------------|---------------------------------|-----------------------|-----------------------|
|                          | OLS                             | GLS                   | FE                    |
| C                        | 0.0103<br>(0.005)               | 0.0054<br>(0.0032)    |                       |
| $\Delta RD_g$            | 0.273<br>(0.202)                | 0.244<br>(0.19)       | 0.18<br>(0.16)        |
| $\Delta RD_p$            | ---                             | ---                   | ---                   |
| $\Delta K_g$             | 0.239<br>(0.056)                | 0.33<br>(0.074)       | 0.62<br>(0.115)       |
| $\Delta KL$              | 0.56<br>(0.057)                 | 0.58<br>(0.059)       | 0.56<br>(0.064)       |
| $\Delta CU$              | 0.039<br>(0.0074)               | 0.038<br>(0.00706)    | 0.0356<br>(0.0069)    |
| T                        | -0.00074<br>(0.00043)           | -0.00057<br>(0.00041) | -0.00021<br>(0.00041) |
| D.W.                     | 1.5                             | 1.545                 | 1.6                   |
| Adj R-Sq                 | 0.402                           | 0.42                  | 0.46                  |
| SER                      | 0.025                           | 0.023                 | 0.0229                |

(Standard errors are in parentheses)



**Table 4.4D. continued****Case 2: Private R&D Included - Global R&D Measures Used**

| Independent Variables | Dependent Variable: $\Delta YL$ |                       |                       |
|-----------------------|---------------------------------|-----------------------|-----------------------|
|                       | OLS                             | GLS                   | FE                    |
| C                     | -0.0072<br>(0.0167)             | -0.0056<br>(0.0104)   |                       |
| $\Delta RD_g$         | -0.34<br>(0.424)                | -0.347<br>(0.41)      | -0.43<br>(0.56)       |
| $\Delta RD_p$         | 0.62<br>(0.46)                  | 0.59<br>(0.44)        | 0.61<br>(0.43)        |
| $\Delta K_g$          | 0.24<br>(0.056)                 | 0.33<br>(0.074)       | 0.63<br>(0.115)       |
| $\Delta KL$           | 0.565<br>(0.057)                | 0.584<br>(0.0593)     | 0.55<br>(0.064)       |
| $\Delta CU$           | 0.0406<br>(0.0075)              | 0.0396<br>(0.0072)    | 0.037<br>(0.007)      |
| T                     | -0.00103<br>(0.0005)            | -0.00085<br>(0.00041) | -0.00049<br>(0.00048) |
| D.W.                  | 1.5                             | 1.55                  | 1.6                   |
| Adj R-Sq              | 0.403                           | 0.423                 | 0.461                 |
| SER                   | 0.024                           | 0.023                 | 0.0229                |

(Standard errors are in parentheses)

**Table 4.4E.** Sample Group: Pool 10      Sample Period: 1973-1987Case 1: Private R&D Excluded      -      Global R&D Measure Used

| Independent<br>Variables | Dependent Variable: $\Delta YL$ |                       |                       |
|--------------------------|---------------------------------|-----------------------|-----------------------|
|                          | OLS                             | GLS                   | FE                    |
| C                        | 0.015<br>(0.005)                | 0.01<br>(0.0036)      |                       |
| $\Delta RD_g$            | 0.68<br>(0.29)                  | 0.67<br>(0.28)        | 0.66<br>(0.28)        |
| $\Delta RD_p$            | ---                             | ---                   | ---                   |
| $\Delta K_g$             | 0.0988<br>(0.0549)              | 0.13<br>(0.072)       | 0.35<br>(0.138)       |
| $\Delta KL$              | 0.36<br>(0.072)                 | 0.374<br>(0.073)      | 0.372<br>(0.077)      |
| $\Delta CU$              | 0.0687<br>(0.012)               | 0.069<br>(0.012)      | 0.0656<br>(0.012)     |
| T                        | -0.0017<br>(0.00045)            | -0.00168<br>(0.00043) | -0.00146<br>(0.00044) |
| D.W.                     | 1.63                            | 1.743                 | 1.91                  |
| Adj R-Sq                 | 0.353                           | 0.37                  | 0.39                  |
| SER                      | 0.0192                          | 0.0185                | 0.0183                |

(Standard errors are in parentheses)

**Table 4.4E. continued****Case 2: Private R&D Included - Global R&D Measures Used**

| Independent Variables | Dependent Variable: $\Delta YL$ |                     |                      |
|-----------------------|---------------------------------|---------------------|----------------------|
|                       | OLS                             | GLS                 | FE                   |
| C                     | -0.00415<br>(0.0178)            | -0.0033<br>(0.0119) |                      |
| $\Delta RD_g$         | -0.014<br>(0.29)                | -0.016<br>(0.64)    | -0.05<br>(0.64)      |
| $\Delta RD_p$         | 0.7<br>(0.61)                   | 0.7<br>(0.58)       | 0.72<br>(0.58)       |
| $\Delta K_g$          | 0.102<br>(0.055)                | 0.136<br>(0.072)    | 0.358<br>(0.138)     |
| $\Delta KL$           | 0.35<br>(0.072)                 | 0.363<br>(0.074)    | 0.36<br>(0.077)      |
| $\Delta CU$           | 0.0707<br>(0.0125)              | 0.071<br>(0.012)    | 0.068<br>(0.0123)    |
| T                     | -0.0021<br>(0.00053)            | -0.002<br>(0.00051) | -0.0018<br>(0.00052) |
| D.W.                  | 1.62                            | 1.73                | 1.9                  |
| Adj R-Sq              | 0.35                            | 0.37                | 0.398                |
| SER                   | 0.0192                          | 0.0185              | 0.0183               |

(Standard errors are in parentheses)

**Table 4.4F. Sample Group: Pool 10      Sample Period: 1973-1987****Case 1: Private R&D Excluded      -      National R&D Measures Used**

| Independent<br>Variables | Dependent Variable: $\Delta YL$ |                      |                      |
|--------------------------|---------------------------------|----------------------|----------------------|
|                          | OLS                             | GLS                  | FE                   |
| C                        | 0.0113<br>(0.0056)              | 0.00176<br>(0.0034)  |                      |
| $\Delta OW_g$            | 0.273<br>(0.071)                | 0.43<br>(0.09)       | 0.63<br>(0.114)      |
| $\Delta OT_g$            | 0.222<br>(0.202)                | 0.37<br>(0.235)      | 0.39<br>(0.25)       |
| $\Delta OW_p$            | ---                             | ---                  | ---                  |
| $\Delta OT_p$            | ---                             | ---                  | ---                  |
| $\Delta K_g$             | 0.0903<br>(0.0534)              | 0.135<br>(0.078)     | 0.238<br>(0.127)     |
| $\Delta KL$              | 0.38<br>(0.0696)                | 0.367<br>(0.0694)    | 0.333<br>(0.0705)    |
| $\Delta CU$              | 0.0651<br>(0.012)               | 0.063<br>(0.0113)    | 0.058<br>(0.0112)    |
| T                        | -0.0012<br>(0.0004)             | -0.0013<br>(0.00039) | -0.0012<br>(0.00039) |
| D.W.                     | 1.4                             | 1.544                | 1.75                 |
| Adj R-Sq                 | 0.387                           | 0.44                 | 0.49                 |
| SER                      | 0.0187                          | 0.0173               | 0.0167               |

(standard errors in parentheses)

**Table 4.4F. continued****Case 2: Private R&D Included - National R&D Measures Used**

| Independent<br>Variables | Dependent Variable: $\Delta YL$ |                       |                     |
|--------------------------|---------------------------------|-----------------------|---------------------|
|                          | OLS                             | GLS                   | FE                  |
| C                        | -0.0235<br>(0.0151)             | -0.0136<br>(0.0085)   |                     |
| $\Delta OW_g$            | 0.201<br>(0.08)                 | 0.333<br>(0.099)      | 0.53<br>(0.13)      |
| $\Delta OT_g$            | -0.76<br>(0.45)                 | -0.439<br>(0.485)     | -0.216<br>(0.52)    |
| $\Delta OW_p$            | 0.1084<br>(0.073)               | 0.141<br>(0.078)      | 0.144<br>(0.086)    |
| $\Delta OT_p$            | 1.15<br>(0.47)                  | 0.88<br>(0.47)        | 0.63<br>(0.47)      |
| $\Delta K_g$             | 0.109<br>(0.0534)               | 0.13<br>(0.076)       | 0.187<br>(0.13)     |
| $\Delta KL$              | 0.324<br>(0.074)                | 0.33<br>(0.071)       | 0.32<br>(0.071)     |
| $\Delta CU$              | 0.0655<br>(0.0119)              | 0.064<br>(0.0113)     | 0.0603<br>(0.0112)  |
| T                        | -0.00206<br>(0.00051)           | -0.00194<br>(0.00049) | -0.0017<br>(0.0005) |
| D.W.                     | 1.43                            | 1.54                  | 1.75                |
| Adj R-Sq                 | 0.41                            | 0.45                  | 0.5                 |
| SER                      | 0.0184                          | 0.0171                | 0.0166              |

(standard error in parentheses)

**Total Factor Productivity Analyses****Table 4.5A. Sample Group: Pool 19      Sample Period: 1973-1987****Case 1: Private R&D Excluded      -      Global R&D Measure Used**

| Independent<br>Variables | Dependent Variable: $\Delta TFP$ |                       |                       |
|--------------------------|----------------------------------|-----------------------|-----------------------|
|                          | OLS                              | GLS                   | FE                    |
| C                        | 0.016<br>(0.005)                 | 0.0098<br>(0.0034)    |                       |
| $\Delta RD_g$            | 0.33<br>(0.28)                   | 0.3<br>(0.27)         | 0.2<br>(0.25)         |
| $\Delta RD_p$            | ---                              | ---                   | ---                   |
| $\Delta K_g$             | 0.32<br>(0.056)                  | 0.42<br>(0.07)        | 0.8<br>(0.11)         |
| $\Delta CU$              | 0.033<br>(0.0075)                | 0.032<br>(0.0073)     | 0.029<br>(0.007)      |
| T                        | -0.00117<br>(0.00043)            | -0.00101<br>(0.00041) | -0.00044<br>(0.00041) |
| D.W.                     | 1.614                            | 1.655                 | 1.66                  |
| Adj R-Sq                 | 0.176                            | 0.192                 | 0.26                  |
| SER                      | 0.0255                           | 0.0245                | 0.0236                |

(Standard errors are in parentheses).

**Table 4.5A. continued****Case 2: Private R&D Included - Global R&D Measures Used**

| Independent Variables | Dependent Variable: $\Delta TFP$ |                      |                      |
|-----------------------|----------------------------------|----------------------|----------------------|
|                       | OLS                              | GLS                  | FE                   |
| C                     | -0.0059<br>(0.0173)              | -0.0053<br>(0.0116)  |                      |
| $\Delta RD_g$         | -0.44<br>(0.64)                  | -0.46<br>(0.62)      | -0.57<br>(0.59)      |
| $\Delta RD_p$         | 0.77<br>(0.58)                   | 0.77<br>(0.56)       | 0.78<br>(0.55)       |
| $\Delta K_g$          | 0.32<br>(0.056)                  | 0.43<br>(0.07)       | 0.83<br>(0.11)       |
| $\Delta CU$           | 0.035<br>(0.0076)                | 0.034<br>(0.0074)    | 0.032<br>(0.007)     |
| T                     | -0.0015<br>(0.0005)              | -0.0014<br>(0.00049) | -0.0008<br>(0.00048) |
| D.W.                  | 1.62                             | 1.66                 | 1.66                 |
| Adj R-Sq              | 0.178                            | 0.195                | 0.265                |
| SER                   | 0.025                            | 0.0245               | 0.0236               |

(Standard errors are in parentheses)

**Table 4.5B.** Sample Group: Pool 10      Sample Period: 1973-1987Case 1: Private R&D Excluded      -      Global R&D Measure Used

| Independent<br>Variables | Dependent Variable: $\Delta TFP$ |                       |                      |
|--------------------------|----------------------------------|-----------------------|----------------------|
|                          | OLS                              | GLS                   | FE                   |
| C                        | 0.0166<br>(0.005)                | 0.011<br>(0.0036)     |                      |
| $\Delta RD_g$            | 0.69<br>(0.29)                   | 0.688<br>(0.28)       | 0.68<br>(0.28)       |
| $\Delta RD_p$            | ---                              | ---                   | ---                  |
| $\Delta K_g$             | 0.112<br>(0.0525)                | 0.15<br>(0.07)        | 0.38<br>(0.133)      |
| $\Delta CU$              | 0.0677<br>(0.012)                | 0.068<br>(0.012)      | 0.0639<br>(0.0121)   |
| T                        | -0.0018<br>(0.00044)             | -0.00176<br>(0.00043) | -0.0015<br>(0.00044) |
| D.W.                     | 1.68                             | 1.79                  | 1.97                 |
| Adj R-Sq                 | 0.236                            | 0.253                 | 0.29                 |
| SER                      | 0.0192                           | 0.0185                | 0.01833              |

(Standard errors are in parentheses)



**Table 4.5B. continued****Case 2: Private R&D Included - Global R&D Measures Used**

| Independent Variables | Dependent Variable: $\Delta TFP$ |                    |                      |
|-----------------------|----------------------------------|--------------------|----------------------|
|                       | OLS                              | GLS                | FE                   |
| C                     | -0.0045<br>(0.0178)              | -0.0037<br>(0.012) |                      |
| $\Delta RD_g$         | -0.053<br>(0.66)                 | -0.064<br>(0.64)   | -0.096<br>(0.636)    |
| $\Delta RD_p$         | 0.74<br>(0.61)                   | 0.75<br>(0.58)     | 0.78<br>(0.577)      |
| $\Delta K_g$          | 0.11<br>(0.0524)                 | 0.15<br>(0.069)    | 0.386<br>(0.133)     |
| $\Delta CU$           | 0.07<br>(0.0125)                 | 0.0702<br>(0.012)  | 0.066<br>(0.0122)    |
| T                     | -0.0021<br>(0.00052)             | -0.002<br>(0.0005) | -0.0019<br>(0.00051) |
| D.W.                  | 1.66                             | 1.78               | 1.95                 |
| Adj R-Sq              | 0.238                            | 0.256              | 0.296                |
| SER                   | 0.0192                           | 0.0185             | 0.0183               |

(Standard errors are in parentheses)

**Table 4.5C.** Sample Group: Pool 10      Sample Period: 1973-1987**Case 1: Private R&D Excluded      -      National R&D Measures Used**

| Independent<br>Variables | Dependent Variable: $\Delta TFP$ |                       |                      |
|--------------------------|----------------------------------|-----------------------|----------------------|
|                          | OLS                              | GLS                   | FE                   |
| C                        | 0.0134<br>(0.0053)               | 0.0025<br>(0.0033)    |                      |
| $\Delta OW_g$            | 0.269<br>(0.071)                 | 0.44<br>(0.09)        | 0.636<br>(0.113)     |
| $\Delta OT_g$            | 0.193<br>(0.22)                  | 0.376<br>(0.235)      | 0.39<br>(0.25)       |
| $\Delta OW_p$            | ---                              | ---                   | ---                  |
| $\Delta OT_p$            | ---                              | ---                   | ---                  |
| $\Delta K_g$             | 0.108<br>(0.0513)                | 0.153<br>(0.076)      | 0.252<br>(0.124)     |
| $\Delta CU$              | 0.0638<br>(0.012)                | 0.062<br>(0.0112)     | 0.058<br>(0.011)     |
| T                        | -0.0013<br>(0.0004)              | -0.00134<br>(0.00038) | -0.0012<br>(0.00039) |
| D.W.                     | 1.45                             | 1.579                 | 1.77                 |
| Adj R-Sq                 | 0.27                             | 0.34                  | 0.41                 |
| SER                      | 0.0187                           | 0.0173                | 0.0167               |

(standard error in parentheses)

**Table 4.5C. continued****Case 2: Private R&D Included - National R&D Measures Used**

| Independent Variables | Dependent Variable: $\Delta TFP$ |                     |                      |
|-----------------------|----------------------------------|---------------------|----------------------|
|                       | OLS                              | GLS                 | FE                   |
| C                     | -0.024<br>(0.0149)               | -0.0139<br>(0.0085) |                      |
| $\Delta OW_g$         | 0.195<br>(0.078)                 | 0.33<br>(0.098)     | 0.53<br>(0.13)       |
| $\Delta OT_g$         | -0.79<br>(0.44)                  | -0.469<br>(0.479)   | -0.22<br>(0.52)      |
| $\Delta OW_p$         | 0.117<br>(0.068)                 | 0.148<br>(0.077)    | 0.146<br>(0.085)     |
| $\Delta OT_p$         | 1.18<br>(0.46)                   | 0.91<br>(0.46)      | 0.64<br>(0.47)       |
| $\Delta K_g$          | 0.114<br>(0.051)                 | 0.134<br>(0.074)    | 0.193<br>(0.13)      |
| $\Delta CU$           | 0.0652<br>(0.0119)               | 0.064<br>(0.011)    | 0.0599<br>(0.011)    |
| T                     | -0.0021<br>(0.00049)             | -0.002<br>(0.00047) | -0.0017<br>(0.00049) |
| D.W.                  | 1.44                             | 1.56                | 1.76                 |
| Adj R-Sq              | 0.3                              | 0.36                | 0.42                 |
| SER                   | 0.0183                           | 0.01708             | 0.01656              |

(standard error in parentheses)

**Table 4.6A.** Sample Group: Pool 10      Sample Period: 1973-1987Distortionary Taxation Considered - R&D Variables Excluded

| Independent<br>Variables | Dependent Variable: $\Delta YL$ |                       |                       |
|--------------------------|---------------------------------|-----------------------|-----------------------|
|                          | OLS                             | GLS                   | FE                    |
| C                        | 0.024<br>(0.0039)               | 0.015<br>(0.0028)     |                       |
| $\Delta RD_g$            | ---                             | ---                   | ---                   |
| $\Delta RD_p$            | ---                             | ---                   | ---                   |
| $\Delta K_g$             | 0.11<br>(0.056)                 | 0.15<br>(0.075)       | 0.39<br>(0.14)        |
| $\Delta KL$              | 0.35<br>(0.076)                 | 0.374<br>(0.079)      | 0.39<br>(0.083)       |
| $\Delta \tau_K(-1)$      | -0.001<br>(0.00065)             | -0.00116<br>(0.00063) | -0.00117<br>(0.00063) |
| $\Delta CU$              | 0.078<br>(0.0134)               | 0.079<br>(0.013)      | 0.076<br>(0.0133)     |
| T                        | -0.0012<br>(0.00039)            | -0.0011<br>(0.00038)  | -0.00083<br>(0.00039) |
| D.W.                     | 1.56                            | 1.66                  | 1.82                  |
| Adj R-Sq                 | 0.31                            | 0.33                  | 0.36                  |
| SER                      | 0.0198                          | 0.01906               | 0.0188                |

(Standard errors are in parentheses)

**Table 4.6B.** Sample Group: Pool 10      Sample Period: 1973-1987Distortionary Taxation Considered - R&D Variables Included

| Independent Variables | Dependent Variable: $\Delta YL$ |                       |                      |
|-----------------------|---------------------------------|-----------------------|----------------------|
|                       | OLS                             | GLS                   | FE                   |
| C                     | -0.0073<br>(0.018)              | -0.0055<br>(0.012)    |                      |
| $\Delta RD_g$         | -0.06<br>(0.68)                 | -0.056<br>(0.65)      | -0.086<br>(0.64)     |
| $\Delta RD_p$         | 0.8<br>(0.6)                    | 0.8<br>(0.59)         | 0.82<br>(0.58)       |
| $\Delta K_g$          | 0.11<br>(0.055)                 | 0.15<br>(0.074)       | 0.396<br>(0.14)      |
| $\Delta KL$           | 0.35<br>(0.075)                 | 0.36<br>(0.077)       | 0.37<br>(0.082)      |
| $\Delta \tau_K(-1)$   | -0.00109<br>(0.00064)           | -0.00119<br>(0.00062) | -0.0012<br>(0.00062) |
| $\Delta CU$           | 0.079<br>(0.0133)               | 0.08<br>(0.013)       | 0.077<br>(0.013)     |
| T                     | -0.0021<br>(0.00053)            | -0.00207<br>(0.00052) | -0.0018<br>(0.00052) |
| D.W.                  | 1.63                            | 1.75                  | 1.93                 |
| Adj R-Sq              | 0.34                            | 0.36                  | 0.39                 |
| SER                   | 0.0194                          | 0.0187                | 0.0184               |

(Standard errors are in parentheses)

**Table 4.7: Tests of Heterogeneity in Panel Data****Pool 19 Sample (N=19, T=15, NT=285)**

|          |            |   |                     |
|----------|------------|---|---------------------|
| Model A: | $F_3=4.59$ | > | $F_{0.1} (54,228)$  |
|          | $F_2=4.94$ | > | $F_{0.1} (36,228)$  |
| Model B: | $F_3=5.2$  | > | $F_{0.1} (72,209)$  |
|          | $F_2=4.87$ | > | $F_{0.1} (54,209)$  |
| Model C: | $F_3=4.16$ | > | $F_{0.1} (108,171)$ |
|          | $F_2=3.65$ | > | $F_{0.1} (90,171)$  |

**Pool 10 Sample (N=10, T=15, NT=150)**

|           |           |   |                    |
|-----------|-----------|---|--------------------|
| Model A:  | $F_3=6.2$ | > | $F_{0.1} (27,120)$ |
|           | $F_2=7.5$ | > | $F_{0.1} (18,120)$ |
| Model B:  | $F_3=6.3$ | > | $F_{0.1} (36,110)$ |
|           | $F_2=6.4$ | > | $F_{0.1} (27,110)$ |
| Model C': | $F_3=3.5$ | > | $F_{0.1} (72,70)$  |
|           | $F_2=2.7$ | > | $F_{0.1} (63,70)$  |

Notes: N is the number of countries, T the number of time periods (ie. 1973-1987), and NT the number of observations. Models A, B, C, and C' are described in the text.  $F_3$  is the test statistic for testing the hypothesis of common slopes and intercept, and  $F_2$  is the test statistic for testing the hypothesis of common slopes only.  $F_{0.1}(k_1,k_2)$  indicates the F-distribution value at the upper 1% point for given degrees of freedom  $k_1, k_2$ .

**Table 4.8.** Sample Group: Pool 10      Sample Period: 1973-1987Public R&D Capital by Socio-Economic Objective and Productivity

| Independent<br>Variables | Dependent Variable: $\Delta YL$ |                   |                  |
|--------------------------|---------------------------------|-------------------|------------------|
|                          | OLS                             | GLS               | FE               |
| C                        | -0.0045<br>(0.065)              | 0.0015<br>(0.025) |                  |
| $\Delta DEF$             | -0.57<br>(0.46)                 | -0.66<br>(0.4)    | -0.79<br>(0.4)   |
| $\Delta S$               | 0.59<br>(0.47)                  | 0.14<br>(0.32)    | 0.21<br>(0.31)   |
| $\Delta ECO$             | 0.59<br>(0.47)                  | 0.75<br>(0.42)    | 0.97<br>(0.42)   |
| $\Delta H$               | -0.84<br>(0.92)                 | -0.78<br>(0.8)    | -0.75<br>(0.78)  |
| $\Delta E$               | 0.67<br>(0.45)                  | 0.6<br>(0.42)     | 0.52<br>(0.4)    |
| $\Delta KN$              | 0.3<br>(0.84)                   | 0.2<br>(0.73)     | 0.1<br>(0.7)     |
| $\Delta OTHER$           | 0.54<br>(0.24)                  | 0.55<br>(0.2)     | 0.58<br>(0.2)    |
| $\Delta K_g$             | 0.14<br>(0.054)                 | 0.25<br>(0.07)    | 0.38<br>(0.086)  |
| $\Delta KL$              | 0.38<br>(0.07)                  | 0.36<br>(0.077)   | 0.34<br>(0.08)   |
| $\Delta CU$              | 0.083<br>(0.017)                | 0.076<br>(0.015)  | 0.069<br>(0.015) |

**Table 4.8 continued**

|          |                    |                    |                    |
|----------|--------------------|--------------------|--------------------|
| T        | 0.0019<br>(0.0015) | 0.0017<br>(0.0013) | 0.0016<br>(0.0013) |
| D.W.     | 1.2                | 1.6                | 1.75               |
| Adj R-Sq | 0.76               | 0.71               | 0.7                |
| SER      | 0.0219             | 0.0189             | 0.0185             |

Notes: Standard errors are in parentheses. DEF is the log of the global stock of Defence R&D Capital, S Space R&D Capital, ECO Economic Development (ie. Agriculture and Industry) R&D Capital, H Health R&D Capital, E Energy R&D Capital, KN Advancement of Knowledge R&D Capital, and OTHER is the stock of R&D Capital from remaining research activities, such as research on Infrastructure, the Environment, Civil Aeronautics, Earth Exploration, and Social Services.  $\Delta$  is the difference operator. (In 1985, Global Defence R&D capital constituted approximately 42.4% of the Global stock of Public Sector R&D capital, Space 12%, Economic Development 10%, Health 8.3%, Energy 9.8%, Advancement of Knowledge 9%, and OTHER 8.6%).



**Table 4.9A.** Sample Group: Pool 10      Sample Period: 1973-1987Estimation of Private R&D Equation      Global R&D Measures Used

| Independent<br>Variables | Dependent Variable: $\Delta RD_p$ |                     |                     |
|--------------------------|-----------------------------------|---------------------|---------------------|
|                          | OLS                               | GLS                 | FE                  |
| C                        | 0.28<br>(0.177)                   | 0.66<br>(0.415)     |                     |
| $\Delta RD_g$            | 1.06<br>(0.063)                   | 1.06<br>(0.063)     | 1.09<br>(0.068)     |
| Y(-1)                    | 0.2E-5<br>(0.2E-3)                | -0.9E-5<br>(0.8E-4) | 0.0065<br>(0.0046)  |
| $RD_p (-1)$              | -0.044<br>(0.031)                 | -0.044<br>(0.031)   | -0.048<br>(0.031)   |
| $\Delta CU$              | -0.0028<br>(0.0017)               | -0.0028<br>(0.0017) | -0.0026<br>(0.0018) |
| T                        | 0.0024<br>(0.0014)                | 0.0024<br>(0.0014)  | 0.0024<br>(0.0014)  |
| D.W.                     | 1.4                               | 1.4                 | 1.4                 |
| Adj R-Sq                 | 0.894                             | 0.894               | 0.895               |
| SER                      | 0.0026                            | 0.0026              | 0.0026              |

(Standard errors are in parentheses)

**Table 4.9B.** Sample Group: Pool 10      Sample Period: 1973-1987

Estimation of Private R&D Equation      National R&D Measures Used

| Independent<br>Variables | Dependent Variable: $\Delta OW_p$ |                     |                    |
|--------------------------|-----------------------------------|---------------------|--------------------|
|                          | OLS                               | GLS                 | FE                 |
| C                        | 0.042<br>(0.0143)                 | 0.013<br>(0.006)    |                    |
| $\Delta OW_g$            | 0.41<br>(0.11)                    | 0.23<br>(0.14)      | 0.153<br>(0.14)    |
| $\Delta OT_g$            | 0.43<br>(0.42)                    | 0.57<br>(0.36)      | 0.67<br>(0.32)     |
| $\Delta OT_p$            | -0.75<br>(0.39)                   | -0.36<br>(0.31)     | -0.18<br>(0.27)    |
| Y(-1)                    | 0.0015<br>(0.001)                 | 0.0022<br>(0.002)   | 0.043<br>(0.009)   |
| $OW_p$ (-1)              | 0.0015<br>(0.0017)                | -0.0025<br>(0.0046) | -0.06<br>(0.017)   |
| $\Delta CU$              | -0.027<br>(0.015)                 | -0.029<br>(0.012)   | -0.024<br>(0.01)   |
| T                        | 0.0015<br>(0.0006)                | 0.001<br>(0.0005)   | 0.003<br>(0.00092) |
| Adj R-Sq                 | 0.74                              | 0.66                | 0.66               |
| SER                      | 0.024                             | 0.019               | 0.016              |
| RHO                      | 0.76<br>(0.038)                   | 0.74<br>(0.048)     | 0.7<br>(0.058)     |

(Standard errors are in parentheses)

**Table 4.9C.** Sample Group: Pool 10      Sample Period: 1973-1987Estimation of Private R&D Equation - Public Sector R&D Capital by Type

| Independent<br>Variables | Dependent Variable: $\Delta RD_p$ |                     |                     |
|--------------------------|-----------------------------------|---------------------|---------------------|
|                          | OLS                               | GLS                 | FE                  |
| C                        | 1.45<br>(0.133)                   | 8.4<br>(0.77)       |                     |
| $\Delta DEF$             | 0.46<br>(0.02)                    | 0.46<br>(0.02)      | 0.46<br>(0.02)      |
| $\Delta S$               | 0.22<br>(0.016)                   | 0.22<br>(0.016)     | 0.22<br>(0.016)     |
| $\Delta ECO$             | 0.51<br>(0.02)                    | 0.51<br>(0.02)      | 0.51<br>(0.02)      |
| $\Delta H$               | -0.22<br>(0.041)                  | -0.22<br>(0.041)    | -0.22<br>(0.042)    |
| $\Delta E$               | 0.34<br>(0.031)                   | 0.34<br>(0.031)     | 0.34<br>(0.032)     |
| $\Delta KN$              | 0.71<br>(0.04)                    | 0.71<br>(0.04)      | 0.71<br>(0.041)     |
| $\Delta OTHER$           | 0.0266<br>(0.0015)                | 0.0265<br>(0.0015)  | 0.0266<br>(0.0016)  |
| $Y(-1)$                  | -0.6E-6<br>(0.3E-4)               | 0.13E-6<br>(0.3E-5) | -0.2E-3<br>(0.5E-3) |
| $RD_p(-1)$               | -0.25<br>(0.023)                  | -0.25<br>(0.023)    | -0.25<br>(0.024)    |
| $\Delta CU$              | -0.0016<br>(0.0008)               | -0.0015<br>(0.0007) | -0.0016<br>(0.0008) |

**Table 4.9C continued**

|          |                    |                    |                    |
|----------|--------------------|--------------------|--------------------|
| T        | 0.0137<br>(0.0011) | 0.0137<br>(0.0011) | 0.0137<br>(0.0011) |
| D.W.     | 2.5                | 2.5                | 2.5                |
| Adj R-Sq | 0.98               | 0.98               | 0.98               |
| SER      | 0.000953           | 0.000954           | 0.000988           |

**Table 4.9D.** Sample Group: Pool 10      Sample Period: 1973-1987Estimation of Private Capital Investment Equation

| Independent<br>Variables | Dependent Variable: $\Delta K_p$ |                      |                       |
|--------------------------|----------------------------------|----------------------|-----------------------|
|                          | OLS                              | GLS                  | FE                    |
| C                        | 0.0176<br>(0.0116)               | 0.0026<br>(0.0042)   |                       |
| $\Delta K_g$             | 0.45<br>(0.046)                  | 0.31<br>(0.07)       | 0.27<br>(0.085)       |
| $\Delta RD_g$            | 0.77<br>(0.59)                   | 0.76<br>(0.47)       | 0.79<br>(0.46)        |
| $\Delta RD_p$            | -0.25<br>(0.48)                  | -0.27<br>(0.38)      | -0.3<br>(0.37)        |
| Y(-1)                    | 0.018<br>(0.0065)                | 0.042<br>(0.011)     | 0.043<br>(0.0075)     |
| $K_p(-1)$                | -0.016<br>(0.0066)               | -0.00067<br>(0.011)  | 0.008<br>(0.015)      |
| $\Delta CU$              | 0.043<br>(0.0094)                | 0.042<br>(0.0075)    | 0.043<br>(0.0075)     |
| T                        | 0.37E-4<br>(0.54E-3)             | 0.17E-4<br>(0.44E-3) | -0.29E-4<br>(0.45E-3) |
| Adj R-Sq                 | 0.45                             | 0.28                 | 0.26                  |
| SER                      | 0.0146                           | 0.0115               | 0.0113                |

(Standard errors are in parentheses)

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